

My 4F / Four Gods / v3.45

完全再現・研究復元書

Frozen Core → v3.45 → 4F → TEL-DM / NDL → Four Gods Mapping → Leveraged Asset R5 → Booster R6

Snapshot date: 2026-09-08 / Development cutoff: 2026-07-31

目的 この1冊だけを読めば、ゼロから研究の系譜・凍結ルール・採否理由・公式検証値・Streamlit再構築手順・現在のProduction/Challenger・次の研究判断まで復元できる状態を残す。

0. この復元書の位置づけ

本書は、Frozen Core / v3.45 完全再現・復元書を起点に、その後に確立した My 4F、TEL-DM、NDL、Four Gods 対応、資産置換実験、v3.45 Booster の 4F 移植までを一続きの研究記録として固定する。ここに記載した「Frozen」「Official」は、当該時点でユーザーが Streamlit 上で実行して出力した ZIP を公式検証結果とする。ローカル計算は実装デバッグにのみ使い、公式成績の代替にしない。

- Development cutoff は原則 2026-07-31。これより後のデータは過去へ戻して閾値・lookback・weight を調整しない。
- 2026-08 以降は My 4F / v4.03 系の prospective OOS。Frozen Core/v3.45 系は復元書上の規約に従い 2026-09 以降を Forward OOS として追跡する。
- Production と Challenger を分離する。DEV で良かった Challenger は、過去へ戻って救済せず Forward で観察する。
- 研究の失敗は削除しない。棄却ルールと理由を残し、同じ探索を再開しない。
- 今後の app.py は情報保持を優先し、コード長の増加を許容する。完了済み研究 renderer・系譜・監査情報を軽量化目的だけで削らない。
- ユーザーの運用は、生成された app.py をコピーし GitHub へアップロードする方式を基本とする。

0.1 一目で分かる現在地

層	現在の位置づけ	状態
Production	Frozen 4F EW = MOM/MR/VIX/TEL-DM 各 25%	維持
Weight overlay	v4.03 TEL Price12M inverse weighting	Frozen / OOS monitor
High-firepower challenger	TQQQ-MOM 4F	研究 Challenger
Booster challenger	4F + v3.45 Attack75 Selector	最重要 Challenger 候補
Aggressive booster	4F + Promotion100	高 CAGR・tail risk 増
Post-hoc structural challenger	Fixed VIX Tilt 50/25/25	Forward でのみ観察

1. 研究原則・データ規約

- 未来情報禁止。シグナル確定日と holding 開始日を分離する。
- 同じ検証でパラメータを多数試して最良値だけを採用しない。事前固定値を使う。
- WF / fixed eras / CPCV / LOO / best-month removal / stress / execution sensitivity を、必要性に応じて full-sample と併記する。
- CAGR の上昇と risk-adjusted の改善を分離する。『リターンをレバレッジで買っただけ』かを常に監査する。
- 相関・クラスター一致はルール同一性の証明ではない。return-engine / role analogy と exact rule reconstruction を区別する。
- Bam 公開情報は authoritative prior として使うが、公開されていない詳細ルールは推測として明示する。

1.1 主要評価指標

CAGR / Annualized Vol / Daily & Monthly MaxDD / Sharpe / Sortino / Calmar / active return / Information Ratio。Target 側と自作側で定義が異なる場合は直接比較しない。

2. Frozen Core / v3.45 完全仕様

この章は 2026-09-06 版『Frozen Core / v3.45 完全再現・復元書』を基礎仕様として継承する。

Sleeve	Model	Weight
MOM	QQQ-MOM	25%
MOM	QLD-MOM	25%
MR	Weekly MR 24M	12.5%

MR	Quarterly MR 24M	12.5%
VIX	Old VIX 15d	6.25%
VIX	Old VIX 20d	6.25%
VIX	New VIX 20d	12.5%

2.1 Frozen Core signal

Model	Risk/Model asset	Absolute momentum / gate	Out-of-market	Frequency
QQQ-MOM	QQQ	LQD 11M, most recent 1M excluded	GLD	Monthly
QLD-MOM	QLD	LQD 11M, most recent 1M excluded	GLD	Monthly
Weekly MR24M	XLU	SPXL 504 trading days	TQQQ	Weekly
Quarterly MR24M	XLU	SPXL 24 calendar months	TQQQ	Quarterly
Old VIX15d	SPXL	^VIX 15 trading days	GLD	Quarterly
Old VIX20d	SPXL	^VIX 20 trading days	GLD	Quarterly
New VIX20d	TQQQ/TECL	^VIX20d; TQQQ vs TECL relative mom	GLD	Quarterly

2.2 Adaptive Safe Haven → Accelerator / Brake → Turbo → Attack → v3.45

- Adaptive Safe Haven: 最終 holding が GLD の portion だけ、prior month-end GLD12M と TLT12M を比較し強い方へ置換。ALL sleeves 対象。
- A Gate: prior month-end TLT 3M total return > 0。
- A Gate 有効時のみ MOM risk-on を 1 段上げる: QQQ→QLD、QLD→TQQQ。
- VIX30 Brake: 日次 close ^VIX>30。次 open から追加 L2 のみ解除し、Adaptive ALL へ戻す。
- Inflection = XLK 1M - XLK 3M。current inflection > prior-month inflection を improving とする。
- Turbo = allow_L2 & improving。
- Attack selector: prior month-end XLK6M > QQQ6M なら TECL、それ以外 TQQQ。
- Attack #1: Turbo 中 75% selected attack asset + 25% proportional non-MOM。
- v3.45 Promotion: Turbo & TECL selected の期間だけ 100% TECL。新 indicator/lookback/threshold は追加しない。

```
A = (TLT_3M > 0)
Brake = (VIX > 30)
allow_L2 = A & ~Brake
```

```
if allow_L2:
    QQQ-MOM: QQQ -> QLD
    QLD-MOM: QLD -> TQQQ
```

```
Inflection_t = XLK_1M_t - XLK_3M_t
Turbo = allow_L2 & (Inflection_t > Inflection_(t-1))
choose_TECL = (XLK_6M > QQQ_6M)
```

```
if not Turbo:
    v345 = A+VIX30 Control
elif Turbo and not choose_TECL:
    v345 = 75% TQQQ + 25% proportional non-MOM
elif Turbo and choose_TECL:
    v345 = 100% TECL
```

Metric	Adaptive ALL	Attack #1	v3.45
CAGR	34.1347%	49.1884%	52.8826%
Vol	22.9211%	28.9259%	30.8324%
Daily MaxDD	-27.4997%	-33.5621%	-33.5621%
Monthly MaxDD	-16.0658%	-19.8574%	-23.6960%
Sharpe0	1.3988	1.5311	1.5344
Sortino0	1.7643	2.0041	2.0020

v3.45 vs Attack: CPCV Active 28/28 positive、Sortino 21/28、Sharpe 21/28。Promotion100 約9.5%の月。Promotion autopsy は 100% winner 16/20=80%。ただし Promotion100 は risk concentration source。

3. My 4F: Frozen production architecture

My 4F は、Frozen Core の 7 sleeve を Family 単位に再構成し、独立した TEL-DM を加えた 4-family architecture である。Production の基準は各 family 25%の monthly rebalanced EW。

Family	構成	役割の現時点解釈
MOM	QQQ-MOM + QLD-MOM	Offense / trend
MR	Weekly + Quarterly MR24M	Shield / defensive counter-reversal
VIX	Old15 + Old20 + New20	Offense / convex reversal-like attack
TEL-DM	Treasury–Equity Layered Dual Momentum	Independent macro diversifier / Suzaku lineage

3.1 Frozen 4F headline

Metric	Frozen 4F EW
CAGR	38.28%
Vol	23.27%
Daily MaxDD	-36.77%
Monthly MaxDD	-19.49%
Sharpe0	1.513
Sortino0	1.961

注: 検証モジュール・データ規約の違いにより過去の別モジュールでは小差がある。小差を消すために戦略ルールを変えない。

4. TEL-DM と v4.03

4.1 TEL-DM rule

Treasury–Equity Layered Dual Momentum (TEL-DM): Layer 1 で TMV 3M > TMF 3M なら TMV。そうでなければ Layer 2 で TQQQ 3M > TECL 3M なら TQQQ、そうでなければ TECL。月次 rebalance。

4.2 TEL 逆張り weight signal

v4.02 系列で TEL の Price12M が高いほど翌期 residual が弱いという強い inverse signal を確認。絶対 Price12M inverse は WF 13/13、CPCV 28/28、LOO 15/15 で negative。

Specification	Frozen value
Primary signal	TEL-DM Price12M
z standardization	expanding prior-only mean/sample SD; min 36 prior obs
TEL weight	$25\% \times [1 - 0.5 \times \tanh(z)]$
Bound	12.5% - 37.5%
Other families	remaining weight equally split MOM/MR/VIX
Timing	month-end t observed → calendar month t+1
DEV cutoff	2026-07-31

v4.03a	Control 4F	Overlay
CAGR	44.57%	47.05%
Vol	24.40%	24.89%
Daily MaxDD	-36.77%	-37.85%

Monthly MaxDD	-19.49%	-19.99%
Sharpe	1.634	1.675
Sortino	2.087	2.124
Calmar	2.287	2.354

Placebo v4.03c: 2,000 block permutations で ΔCAGR observed +0.024715、percentile 96.8%、p=0.0325。ここで歴史的チューニングを停止。2026-08 以降は prospective OOS monitor。

5. NDL: family structure discovery

R1 universe: QQQ-MOM, QLD-MOM, Weekly MR24M, Quarterly MR24M, Old VIX15d, Old VIX20d, New VIX20d, TEL-DM。

Finding	Official result
Ward K=4	semantic 4F を完全回収
Pairwise agreement	1.0
Silhouette	K4 = 0.432, max
Within-family corr	0.847
Between-family corr	0.373
Rolling60 exact	84%; TEL singleton 100%; mean ARI 0.885
Rolling36 exact	64.4%; TEL singleton 92.6%; mean ARI 0.806

5.1 Representative falsification / diversification

All-8 EW は非常に強く、単純に代表 1 本へ縮約すると分散効果を失う。Ward clustering は return alpha generator ではなく risk-structure normalizer として解釈する。

R5b で All8→Cluster active -0.50%/yr。VIX 縮小が約-6.27%/yr を失い、TEL 増加が約+5.92%/yr で相殺。COVID では All8 MaxDD -26.9%に対し Cluster -36.8%。相関分散≠tail diversification。

6. Ninjutsu / dynamic selector research

Technique	結果	結論
Bunshin / EW	4F EW strong	支持
Gatekeeper	CAGR/Sharpe/DD 悪化	棄却
Tailwind all-in	MaxDD 約-60.8%	棄却
Nukimi all-in	MaxDD 約-59.3%	棄却
Role-preserving Nukimi	見かけの改善は VIX exposure 寄与	selector alpha 棄却
Kawarimi	active はぼ 0; CPCV14/28	棄却
Acceleration	CAGR ↑ だが DD/コスト依存; best3 removal で負	棄却

Spear 内部 MOM/VIX selector も Price12M/Prior11M winner-take-all は CAGR を上げる一方、MaxDD を-39%〜-41%へ悪化し risk-adjusted が低下。Spear 内部研究は MOM/VIX 50/50 で凍結。

7. MR / Genbu reverse engineering

Frozen Weekly MR24M は単純だが強く、single-strategy modification を複数検証した後に凍結。continuous tanh transplant、sparse early attack、exit buffer などは棄却。

Strategy	CAGR	MaxDD	Sharpe	Sortino
Frozen MR	30.09%	-33.14%	1.069	1.377
Genbu 激攻	39.09%	-58.23%	1.146	—

Genbu 常勝	24.33%	-62.36%	0.783	—
Genbu 鉄壁	32.01%	-36.07%	1.192	1.540

Frozen vs 鉄壁 corr 0.908。172 か月中 165 か月一致。差の 7 か月で Frozen-only attack が大勝するケースがあり、繰り返し可能な構造的改善は確認できず Frozen MR を維持。

7.1 FoF role

MR を 3F ex-MR へ 25%追加すると CAGR は 39.43→38.12%へ少し低下する一方、Vol 27.73→23.48、Monthly MaxDD -23.77→-17.90、Sharpe 1.343→1.502、Sortino 3.264→3.948。MR は crash hedge ではなく Defensive Counter-Reversal Sleeve。

8. Four Gods lineage and role mapping

Bam 本人の authoritative lineage comment: 青龍←DM2、朱雀←DM3、白虎←DM6、玄武←DM7。以前から活躍していた特徴の異なる低相関 DM を理論ベースで再構築したものがシン四神で、これらを素材としてルールベースの動的 portfolio にする。

God	Source lineage	Observed universe	Mapped My family	Pearson
青龍 / Seiryu	DM2	XLU / TQQQ / TECL	MOM	0.729 original
朱雀 / Suzaku	DM3	TMV / TQQQ / TECL	TEL-DM	0.803
白虎 / Byakko	DM6	GLD / TQQQ / TECL	VIX	0.733
玄武 / Genbu	DM7	XLU / TQQQ	MR	0.720

R3 final mapping は 4/4。これは role/return-engine analogy の強い証拠だが、exact rule identity の証明ではない。

8.1 Genbu state structure

Genbu base/defense state は MR との相関約 0.835、attack state では VIX との相関約 0.878、MOM 約 0.718、MR 約 0.631。したがって『MR 的 reversal/base + 独自 attack layer』の像が有力。

8.2 Bam Four Gods philosophy

青龍/朱雀/白虎=offense、玄武=shield を分離し、spear choice と spear-vs-shield allocation を別レイヤーで扱う『矛と盾』思想。Ninjutsu は分身、門番、追い風、抜き身、加速、変わり身。公開定義上、加速は momentum level ではなく momentum の変化、変わり身は strongest+weakest。

9. R4: actual TQQQ substitution

最初の synthetic 3x 案はユーザーが訂正。正式 R4 は actual historical TQQQ を使い、QQQ-MOM の QQQ→TQQQ、QLD-MOM の QLD→TQQQ のみ変更。その他は不変。

Seiryu vs MOM	Original	TQQQ-MOM
Pearson	0.729	0.762
Spearman	0.703	0.735
R ²	0.531	0.580
Family ann vol	21.3%	43.2%
Seiryu ann vol	44.4%	44.4%
Fingerprint distance	2.175	0.821

4F	Frozen	TQQQ-MOM 4F
CAGR	38.28%	43.22%
Vol	23.27%	27.38%
Daily MaxDD	-36.77%	-45.40%
Monthly MaxDD	-19.49%	-26.54%

Sharpe	1.513	1.453
Sortino	1.961	1.836

13-block WF 11/13、CPCV active 28/28 positive。しかし CPCV ΔSortino 2/28 のみ。結論: Seiryu との差の大きな部分は actual leveraged ETF choice で説明されるが、4F としては『return を risk efficiency と引き換えに買った』Challenger。

10. R5: Four Gods × My 4F leveraged asset substitution

各 family の signal/lookback/timing を固定し、God の観測 leveraged universe へ asset だけを寄せる反証実験。

God / Family	Pearson orig→asset	Fingerprint distance orig→asset	判定
Seiryu / MOM	0.729 → 0.762	2.175 → 0.821	大幅改善
Suzaku / TEL	0.803 → 0.803	2.121 → 2.121	positive lineage control; already same universe
Byakko / VIX	0.733 → 0.741	1.691 → 2.175	相関微増だが risk fingerprint 悪化
Genbu / MR	0.720 → 0.720	2.121 → 2.121	actual holding change なし

Portfolio	CAGR	Vol	Daily DD	Monthly DD	Sharpe	Sortino
Frozen 4F	38.28%	23.27%	-36.77%	-19.49%	1.513	1.961
God-Asset 4F	44.25%	28.12%	-45.40%	-26.54%	1.448	1.832

結論: 『全部 3 倍 ETF だから Four Gods になる』は否定。Seiryu では asset choice が大きい。Suzaku/TEL は既に同じ leveraged world。Byakko と Genbu は rule/state/base-asset 差の寄与が大きい。

11. R6: v3.45 Booster → 4F transplant

R6 では Frozen 4F を Control のまま、v3.45 で既に凍結された attack overlay だけを外付け。Adaptive Safe Haven は 4F 内 family の既存 defensive logic との混線を避けるため新規変更としては導入せず、既存 4F をそのまま基礎にした。

Variant	CAGR	Vol	Daily DD	Monthly DD	Sharpe0	Sortino0
Frozen 4F EW	38.28%	23.27%	-36.77%	-19.49%	1.513	1.961
A+VIX30	40.33%	24.25%	-39.10%	-20.97%	1.523	1.973
Turbo75 TQQQ	47.79%	28.95%	-39.10%	-22.42%	1.498	1.904
Attack75 Selector	49.82%	29.21%	-39.10%	-22.42%	1.534	1.955
Promotion100	51.31%	30.32%	-39.10%	-25.23%	1.522	1.930

11.1 Stage attribution

Stage	Mean active / month	CAGR change	ΔSharpe0	ΔSortino0
Accelerator+Brake	+0.156%	38.25→40.31%	-0.016	-0.139
Turbo75 TQQQ	+0.560%	40.31→47.75%	-0.047	+0.007
TECL/TQQQ Selector	+0.134%	47.75→49.79%	+0.019	+0.035
75→100 TECL Promotion	+0.109%	49.79→51.28%	-0.002	-0.083
Full Booster vs Frozen	+0.960%	38.25→51.28%	-0.046	-0.180

11.2 Trigger sanity check

State	Months with state	% months
A Gate	98	53.26%
VIX30 Brake	32	17.39%
allow_L2	95	51.63%
Inflection improving	90	48.91%
Turbo	47	25.54%
Choose TECL	93	50.54%
Promotion100	18	9.78%

元 v3.45 の A Gate 約 52.1%、Turbo 約 25.6%、Promotion100 約 9.5% とほぼ一致し、移植実装の sanity check として強い。

11.3 Robustness

- Promotion100 vs Frozen: WF 10/13 wins。negative block は 1,10,11 で小幅。
- CPCV: Promotion100 active return 28/28 positive。Attack75 は 27/28 positive。
- CPCV risk-adjusted: Attack75 ΔSortino 14/28、Promotion100 13/28。return robustness の方が risk-adjusted robustness より強い。
- Best active month removal: Promotion100 mean active は 0 removals +0.960%/mo、best1 +0.842%、best3 +0.674%、best5 +0.527%、best10 +0.225%。
- Stress: COVID Frozen +18.0% / DD -36.8% に対し Booster +15.9% / DD -39.1%。2025 selloff Frozen -6.5% / DD -23.4%、Attack75 -9.5% / DD -28.2%。

R6 interpretation 常時 TQQQ 化より、状態依存で攻撃力を上げる方が歴史上は高 CAGR かつ DD 悪化が小さい。特に Attack75 Selector は CAGR 49.82%、Sharpe 1.534 で、Full Promotion100 よりバランスが良い。ただしこの移植も同じ DEV 上の評価であり、今後の調整は禁止。

12. 現在の Production / Challenger 体系

Tier	Strategy	Why
Production	Frozen 4F EW	最も解釈可能で、広い研究を通じて構造が安定。
Frozen overlay	v4.03 TEL inverse weight	事前固定済み。2026-08+ prospective only。
Challenger #1	4F + Attack75 Selector	R6 で CAGR 大幅増、full-period Sharpe も維持/改善、Promotion100 より tail balance 良好。
Aggressive Challenger	4F + Promotion100	CAGR 最大だが Monthly DD と risk-adjusted が悪化。
High-firepower asset challenger	TQQQ-MOM 4F	Seiryu gap 説明には有効だが risk efficiency 低下。
Post-hoc Challenger	Fixed VIX Tilt 50/25/25	DEV では強いが post-hoc origin。Forward でのみ評価。

13. 棄却・停止した探索: 再開しない

Area	Rejected / frozen conclusion
Safe haven	XLU 3-way, VIX-only, 6/9/18M sweep を再開しない
v3.45 risk-off	Risk-Off GLD3M/LQD3M, TLT state machine, TLT down-cross, QQQ exhaustion brake を棄却
Turbo rescue	persistence 1M / capped75 persistence / second attack / Turbo-OFF attack opportunity を停止
MR modification	continuous tanh, sparse early attack, exit buffer 等を棄却; Frozen MR 維持
Spear selector	Price12/Prior11 winner-take-all は高 CAGR だが risk-adjusted 悪化; 50/50 freeze
Ninjutsu selectors	Gatekeeper, Tailwind all-in, Nukimi all-in, Kawarimi, Acceleration を棄却
TEL rescue	Prior11M/threshold/strength/weight tuning 禁止; Price12M primary freeze
R5 follow-up	Byakko/Genbu asset substitution を結果後に別 ticker へ救済しない

14. Streamlit app 再構築仕様

今後は『軽量化のため完了済み研究を削る』より『全情報を app.py 内に残し、1 ファイルで復元可能』を優先する。コード長増加を許容する。

- 毎回 app.py を完全置換可能な形で出力し、python -m py_compile を PASS させてから渡す。
- ユーザーは app.py をコピーして GitHub へアップロード/commit し、Streamlit Cloud で実行する。
- 各検証は st.progress で段階表示する。
- 検証結果の元データを ZIP 1 個で export する。iPhone で複数 download button を置かない。
- 重い計算は cache / vectorization / month-end evaluation / lazy computation で抑える。ただし研究情報は削除しない。
- 公式結果はユーザーが Streamlit で実行した ZIP。ZIP 名・主要 CSV・freeze spec を README に残す。
- 実装 bug 修正は research spec 変更と明確に分離する。

14.1 必須データ / ticker

Frozen Core/v3.45 系: QQQ, QLD, LQD, GLD, TLT, XLU, SPXL, TQQQ, TECL, SPY, XLK, ^VIX, TB3MS/必要時^IRX。TEL-DM には TMV/TMF も必要。Four Gods mapping には公開 God monthly series を保持する。

15. 公式ファイル・系譜

Artifact	役割
Frozen_Core_v3_45_Complete_Reproduction_Guide_2026-09-06.pdf	Frozen Core〜v3.45 source-of-truth
fourgods_my4f_role_mapping_r1/r2/r3 ZIP	4 Gods↔My families mapping
my4f_tqqq_substitution_r4.zip	Seiryu/MOM actual TQQQ substitution
fourgods_my4f_leveraged_asset_substitution_r5.zip	4 family God-asset substitution
fof_booster_v345_transplant_r6.zip	v3.45 attack overlay → 4F official result
current app.py	全研究・monitor・export を含む Streamlit 本体

過去の app.py 名や ZIP 名が異なる場合、README 内の spec と DEV cutoff を優先して identity を確認する。

16. ゼロからの完全再構築手順

10. 価格/RF source を揃え、adjusted close を基本に DatetimeIndex を統一する。
11. Frozen Core 7 model を復元し、model weight 合計 100%、MOM50/MR25/VIX25 を監査する。
12. v3.45 source-frozen signals を復元し、spot-check 値(CAGR 約 52.88、Promotion frequency 約 9.5%)で QA する。
13. My family daily return を MOM/MR/VIX へ集約し、TEL-DM を別途構築する。
14. Frozen 4F を MOM/MR/VIX/TEL-DM 各 25% monthly rebalance で構築する。
15. v4.03 TEL Price12M inverse overlay を別系列として構築し、2026-08+を prospective OOS として分離する。
16. NDL clustering / role tables は Family 構造確認用に復元する。selector としては使用しない。
17. God public monthly series を読み、R3 mapping (Seiryu→MOM, Suzaku→TEL, Byakko→VIX, Genbu→MR)を監査する。
18. R4 actual TQQQ substitution を再現し、Seiryu correlation/fingerprint 改善と portfolio risk 悪化を確認する。
19. R5 family-by-family asset substitution を再現し、Byakko/Genbu が単純 asset 置換で近づかないことを確認する。
20. R6 v3.45 attack overlay を Frozen 4F へ外付けし、Frozen→A+Brake→Turbo75→Attack75→Promotion100 を段階監査する。
21. 全 module で DEV cutoff=2026-07-31 を固定し、Forward data を過去の rule 変更には使用しない。
22. ユーザー実行 ZIP を公式記録として保存し、本書の主要値と一致するか監査する。

17. QA / 再現スポットチェック

Check	Expected
Frozen Core weights	MOM50 / MR25 / VIX25
MOM lookback	11M, most recent 1M excluded
Weekly MR	SPXL 504 trading days; SPXL>=RF→XLU else TQQQ
TEL-DM	TMV3M>TMF3M→TMV else TQQQ3M>TECL3M→TQQQ else TECL
v3.45 A	TLT3M>0
v3.45 Brake	VIX>30
Turbo	allow_L2 & XLK(1M-3M) improving
Selector	XLK6M>QQQ6M
Promotion	Turbo & choose_TECL → 100% TECL
Frozen 4F	MOM/MR/VIX/TEL each25% monthly rebalance
R5 Seiryu Pearson	~0.729→~0.762
R5 God-Asset 4F CAGR	~44.25%; Sharpe lower than Frozen
R6 Attack75	CAGR ~49.82%; Daily DD ~-39.10%; Sharpe ~1.534
R6 Promotion100	CAGR ~51.31%; Monthly DD ~-25.23%; Promotion ~9.8% months

18. 今後の研究判断ルール

- Frozen 4F を Production control として常に残す。
- Attack75 Selector を最重要 Booster challenger として固定し、DEV 上で 75%の周辺 sweep や VIX30 周辺 sweep をしない。
- Promotion100 は Aggressive challenger。高 CAGR だけで Production 化しない。
- Forward OOS で Challenger が複数 regime にわたり優位を蓄積したときだけ、別 version として採否を検討する。
- Four Gods reverse engineering は、公開 series/author comment で裏付けられる範囲と推測を区別する。
- Genbu の次の科学的問いは『MR-like base から attack へ移る局面の構造』。ただし結果を見ながら trigger を多数探索しない。
- Dynamic portfolio 研究は、Bam の『offense 選択』と『spear-vs-shield allocation』を別層として扱う。

19. 新しい ChatGPT 会話へ渡す最小ハンドオフ

最重要 この章と本書全体を新しい会話の最初に読み込ませれば、過去の研究コンテキストを大きく失わず再開できる。

Current state: Production=Frozen 4F EW。v4.03 TEL inverse overlay は Frozen で 2026-08+ OOS。Four Gods mapping は Seiryu→MOM, Suzaku→TEL, Byakko→VIX, Genbu→MR。R5 で Seiryu のみ actual TQQQ asset substitution が God へ大幅接近。R6 で v3.45 attack overlay を 4F へ移植し、Attack75 Selector が CAGR49.82%, Sharpe1.534, Daily DD-39.10%、Promotion100 が CAGR51.31%だが Monthly DD-25.23%。Production は変更しない。今後は app.py の長さを許容して情報を削らず、ユーザーがコードをコピーして GitHub へアップロードする。

20. 最終運用思想

Optimize less. Observe more. Freeze what survived.
Change only after sustained forward evidence.

21. Master Bible v1.0 - 完全復旧契約

前 20 章の研究要約を、実装・公式結果・再開手順まで含む復旧パッケージへ昇格する。本書と同梱 app.py があれば、過去チャットが消失しても研究を再開できることを目標とする。

- 文章の source of truth = 本 Master Bible。
- 実装の source of truth = 同梱 app.py。
- 成績の source of truth = ユーザーが Streamlit で実行して生成した official ZIP。
- 三者に差があれば勝手に整合させず、差分を監査する。strategy rule を変更して数値を合わせない。

Field	Value
app.py bytes	1,970,239
app.py lines	42,134
Top-level functions	661
app SHA-256	07c7b149e7ed011b4545b0554d6391effbf522d2a64cef3aa17f63f6f29ba0bf
DEV cutoff	2026-07-31
Latest module	4F Booster R6

22. Canonical implementation map

Canonical function	Snapshot lines	SHA prefix
_build_core_holding_matrix	2796-2829	0c40702eb046f83b
_adaptive_safe_haven_monthly_series	2832-2906	27365d5e42d1c472
_v326_bond_regime_signals	4893-4915	4983253c2f797597
_v332_xlk_inflection_signal	6607-6626	8f7998dd44d930e1
_v335_attack_selector_signal	6876-6895	e8363d143eae6b87
_v355_wf_blocks	9147-9172	85e059df022b93f1
_v326_cpcv_summary	4937-4975	fbfe13661fc377b4
_v389_layered_dm	28264-28302	dadf39fab34c717e
_v390_monthly_rebalanced_ew	28641-28652	034e07634f7cedec
_v403a_monthly_price12	30567-30570	e33ccbd27a28f40e
_v403a_frozen_signal	30573-30619	266df437b15997d6
_v403a_monthly_target_portfolio	30622-30659	51ebb85de35f9a5a
render_v403a_frozen_tel_inverse_overlay	30737-30881	bd1e4caa0587bc0d
render_v403_prospective_oos_monitor	31233-31338	d16f8ae9f43adc0
render_ndl_r2a_ward_structure	31428-31593	8880426888f256c
render_ndl_r8_ninjutsu_family_compatibility	33090-33255	e90bb8540b09abc7
render_fof_s1_shield_role_audit	37701-37874	293cc1a3262b2ca
render_fof_s2_spear_shield_architecture	37441-37557	89f2f3de782b795f
render_fof_s3_spear_internal_selection_audit	37080-37198	014486c89ff0ec03
render_fof_booster_r6	34943-35105	28bef4dcf114e616

行番号はこの snapshot 限定。将来追記で変化してよい。hash は意図しない変更の監査用。

23. Frozen constants / invariants

Invariant	Frozen value
Frozen Core	MOM50 / MR25 / VIX25
MOM	11M, most recent 1M excluded; matched RF compound
Adaptive Safe Haven	GLD final holdings only; GLD12M vs TLT12M; ALL sleeves
Frozen 4F	MOM/MR/VIX/TEL-DM each 25%, monthly rebalance
TEL-DM	TMV3M>TMF3M→TMV; else TQQQ3M>TECL3M→TQQQ else TECL
v4.03 signal	TEL Price12M; expanding prior-only z; min 36 obs
v4.03 TEL weight	25% × [1 - 0.5 × tanh(z)], bound 12.5-37.5%
A Gate	TLT3M > 0
Brake	^VIX > 30
Turbo	A & ~Brake & XLK(1M-3M) improving
Attack selector	XLK6M > QQQ6M => TECL, else TQQQ
Attack intensity	75% attack +25% proportional non-MOM
Promotion	TECL-selected Turbo => 100% TECL

24. Exact strategy implementation contract

24.1 MOM

- QQQ-MOM と QLD-MOM は Frozen Core で各 25%。
- LQD 11M total return (most recent 1M excluded) を同 window RF compound と比較。
- LQD>=RF なら risk-on: QQQ-MOM→QQQ、QLD-MOM→QLD。未満なら GLD (Adaptive Safe Haven 前)。
- 11M/exclude1/RF matching を結果に合わせて変更しない。

24.2 MR

- Weekly MR24M: SPXL trailing 504 trading days >= RF なら XLU、未満なら TQQQ。前週末までの情報で次週最初の取引日から保有。
- Quarterly MR24M: Jan/Apr/Jul/Oct 判定。SPXL 24 calendar months >= RF なら XLU、未満なら TQQQ。
- continuous tanh / sparse early attack / exit buffer 等の棄却済み改変を再導入しない。

24.3 VIX

- Old VIX15d/20d: quarterly. VIX momentum >= matched RF なら SPXL、未満なら GLD。
- New VIX20d: quarterly. VIX20d<RF なら GLD。RF 以上なら TQQQ vs TECL 20d relative momentum winner。
- v3.45 の VIX30 は追加攻撃を止める Brake であり、VIX family signal そのものを変更しない。

24.4 TEL-DM

```
if TMV_3M > TMF_3M:
    holding = TMV
else:
    holding = TQQQ if TQQQ_3M > TECL_3M else TECL
# month-end decision -> next calendar month
```

- TEL は Suzaku 解析を起点に構築されたため、Suzaku↔TEL の高類似性は独立発見ではなく lineage validation。

25. Execution / data contract

Topic	Contract
Price	Adjusted close / total-return-equivalent を基本。v3.45 execution audit では Open も保持。
RF	Local FRED TB3MS 優先。必要な長期のみ^IRX fallback。annualized percent yield を holding-period return へ変換。
Signal lag	利用可能な prior close/month-end/week-end のみ。future information 禁止。
Month-end	原則 month-end t signal → calendar month t+1 holding。
VIX30	raw close >30; next effective row/open から extra aggression を解除。
Missing data	Frozen implementation で許容された forward-fill 以外を勝手に追加しない。
Leveraged ETF	R4/R5 は actual historical ETF。synthetic 3x は official experiment ではない。
Numerical mismatch	vendor/adjustment/start/RF/execution/index mapping を先に監査。rule を変えて一致させない。

26. Validation contract

Audit	Contract
Full sample	Headline descriptive DEV only.
Fixed eras	Predefined broad regimes; result 後に境界変更禁止。

WF	13 chronological blocks.
CPCV	Current 8-group pair combinations = 28.
LOO	Where implemented, leave-one-calendar-year-out sign consistency.
Best-month removal	Active-return extreme dependence check.
Stress	Fixed episodes; post-hoc window rescue 禁止。
Placebo	v4.03c: 2,000 block permutations, seed=40303.
Execution	Open→Open primary where specified; Close→Close sensitivity retained.

27. Research ledger

Study	Hypothesis	Official result	Decision
v4.02 TEL inverse	High TEL Price12M predicts weaker next residual	WF13/13, CPCV28/28, LOO15/15 negative	Freeze inverse mechanism
v4.03	Can inverse TEL weighting improve 4F?	Risk-adjusted improved; placebo ΔCAGR $p \approx 0.0325$	Freeze; prospective only
NDL	Are 8 sleeves really 4 families?	Ward K4 exactly recovers semantic 4F	4-family architecture supported
MR evolution	Can Genbu-inspired tweaks improve Frozen MR?	No robust local improvement	Freeze Frozen MR
Spear selector	Can MOM/VIX winner selection improve?	CAGR ↑ but DD/risk-adjusted worse	50/50 freeze
Ninjutsu	Do Gatekeeper/Tailwind/Nukimi/Kawarimi/Acceleration help?	Most rejected; Nukimi benefit explained by VIX exposure	No selector rescue
B6 VIX tilt	Does VIX50/TEL25/MR25 improve?	Strong DEV but post-hoc	Forward challenger
Four Gods R1-R3	Do Gods map to My families?	4/4: Seiryu→MOM, Suzaku→TEL, Byakko→VIX, Genbu→MR	Role analogy supported
R4	Does actual TQQQ explain Seiryu gap?	Similarity improves; portfolio risk efficiency worsens	Research challenger
R5	Are all God gaps just 3x assets?	No; Seiryu yes largely, Byakko/Genbu no	Reject 'everything is 3x'
R6	Does frozen v3.45 attack layer transfer to 4F?	Attack75 CAGR49.82%, DD-39.10%, Sharpe1.534	Challenger #1

28. Official R6 registry

Variant	CAGR	Vol	Daily DD	Monthly DD	Sharpe0	Sortino0
Frozen 4F EW	38.2752%	23.2670%	-36.7681%	-19.4928%	1.512963	1.961383
A+VIX30	40.3345%	24.2467%	-39.0978%	-20.9718%	1.522688	1.972921
Turbo75 TQQQ	47.7873%	28.9524%	-39.0978%	-22.4153%	1.498016	1.903920
Attack75 Selector	49.8232%	29.2073%	-39.0978%	-22.4153%	1.534498	1.954981
Promotion100	51.3148%	30.3154%	-39.0978%	-25.2307%	1.522165	1.930324

Stage	Mean active/mo	CAGR base→challenger	ΔSharpe	ΔSortino
Accelerator+Brake	+0.1564%	38.248→40.306%	-0.0156	-0.1394
Turbo75	+0.5601%	40.306→47.752%	-0.0474	+0.0067
Selector	+0.1341%	47.752→49.787%	+0.0188	+0.0354
Promotion	+0.1092%	49.787→51.277%	-0.0022	-0.0831
Full Booster	+0.9599%	38.248→51.277%	-0.0464	-0.1804

State	% trading days	% months with state
A Gate	53.50%	53.26%
VIX30 Brake	6.05%	17.39%
allow_L2	49.27%	51.63%
Inflection improving	48.91%	48.91%
Turbo	23.59%	25.54%
Choose TECL	50.65%	50.54%
Promotion100	9.78%	9.78%

- Promotion100 vs Frozen WF = 10/13 wins.
- Promotion100 CPCV active return = 28/28 positive; Attack75 = 27/28 positive.

- Risk-adjusted CPCV is much weaker than active-return robustness; do not conflate the two.
- Best active month removal: Promotion100 mean active remains +0.2248%/mo after removing best 10 active months.
- COVID: Frozen +18.05%, DD -36.77%; Booster +15.88%, DD -39.10%. 2025 selloff: Frozen -6.54%, Attack75 -9.48%.

29. Four Gods evidence hierarchy

Evidence	Allowed conclusion
Bam authoritative comment	青龍←DM2, 朱雀←DM3, 白虎←DM6, 玄武←DM7.
Published monthly series	Observed returns, correlations, visible holdings/universe.
My reconstruction	Role mapping/state interpretation; label as inference.
Not supported as fact	Unpublished exact God triggers/lookbacks/weights.

- Genbu: base/defense state is strongly MR-like; attack state becomes VIX/MOM-like. This supports MR-like base + distinct attack layer, not exact-rule identity.
- Bam's architecture separates offense selection (青龍/朱雀/白虎) from spear-vs-shield allocation (玄武).

30. Rejected ideas registry - do not reopen casually

Area	Rejected idea	Reason
MR	continuous tanh transplant	Return gain came with severe DD/leverage.
MR	sparse early attack / dRet rescue	Era instability.
MR	exit buffer neighborhood	No smooth robustness; turnover/cost.
Spear	Price12/Prior11 winner-take-all	CAGR ↑ but DD/risk-adjusted worse.
Ninjutsu	Gatekeeper / Tailwind / Nukimi all-in	Rejected; all-in variants carry extreme DD.
Ninjutsu	Role Nukimi selector alpha	Fixed VIX tilt explains improvement.
Ninjutsu	Kawarimi	No robust active edge.
Ninjutsu	Acceleration	Best-month/cost/risk dependence.
v3.45	Promotion failure brake	Only 4 failures; overfit risk.
R4	synthetic 3x as official test	Withdrawn after user correction; actual TQQQ is official.

31. Production / challenger governance

Tier	Strategy	Governance
Production	Frozen 4F EW	Default control; no DEV-only promotion.
Frozen overlay	v4.03 TEL inverse	Monitor prospective OOS; no tuning.
Challenger #1	4F + Attack75 Selector	Exact R6 rules frozen; forward observation.
Aggressive	4F + Promotion100	Track separately; tail risk higher.
High-firepower	TQQQ-MOM 4F	Research comparison only.
Post-hoc	Fixed VIX Tilt	Forward evidence required.

32. Streamlit engineering contract

- Information preservation > code shortening. Do not delete completed research modules, lineage notes, frozen specs or audit information merely to reduce file size.
- Runtime optimization should use cache, vectorization, month-end evaluation, fragments/lazy calculation and avoidance of duplicate work.
- Every replacement app.py must pass python -m py_compile before delivery.
- New validation modules include descriptive st.progress stages.
- Export one ZIP containing raw daily/monthly series, state/trigger tables, robustness outputs, implementation audit and README freeze spec.

- User-run ZIP is the official result; local calculations are debugging only.
- Bug fix = implementation correction, not research-spec change, unless explicitly versioned.

33. Brand-new ChatGPT recovery protocol

23. Read this Master Bible completely.
24. Read bundled app.py and verify SHA-256 against MANIFEST.json when using this snapshot.
25. Locate Chapter 22 canonical functions and Chapter 23 invariants.
26. Confirm Production and each Challenger separately.
27. For any study under discussion, read the newest user-run official ZIP; do not replace it with local calculations.
28. Respect DEV cutoff=2026-07-31 and do not retune using later data.
29. Check rejected-ideas registry before proposing a new historical search.
30. Modify the latest full app.py, not a reconstructed shortened version.
31. Before delivery, run `python -m py_compile` and preserve progress/export behavior.
32. When chat context nears its limit, create a handoff with frozen specs, official results, unresolved hypothesis, latest app/ZIP names and exact next step.

34. Current research queue

- Prospective monitor: Frozen 4F, v4.03 TEL inverse, Attack75 Selector, Promotion100.
- Allowed mechanism work: decompose Attack75 using already-frozen states/exposures; do not search new thresholds.
- Four Gods: incorporate new Bam source material when available, preserving DM2/3/6/7 lineage.
- Genbu: only pre-specified tests of MR-like base vs attack-layer mechanism.
- No Production promotion until sustained forward evidence accumulates.

35. Final zero-context handoff

Production=Frozen 4F EW: MOM/MR/VIX/TEL-DM each25%. TEL-DM is TMV-vs-TMF 3M first layer, then TQQQ-vs-TECL 3M second layer. v4.03 inversely weights TEL using frozen Price12M z-score and is prospective after DEV cutoff. Four Gods lineage/mapping: Seiryu/DM2→MOM, Suzaku/DM3→TEL, Byakko/DM6→VIX, Genbu/DM7→MR. R4/R5 show leveraged ETF choice explains much of Seiryu but not all Gods. R6 transplants v3.45 attack logic: $A=TLT3M>0$, $Brake=VIX>30$, $Turbo=A\&\sim Brake\&XLK$ inflection improving, $Selector=XLK6M>QQQ6M$. Attack75: $CAGR\approx 49.82\%$, $Daily\ DD\approx -39.10\%$, $Sharpe\approx 1.534$. Promotion100: $CAGR\approx 51.31\%$, $Monthly\ DD\approx -25.23\%$. Keep Frozen 4F as Production; Attack75 is Challenger #1. Do not tune historical parameters. Bundled full app.py is canonical implementation; user-run ZIPs are official results.

36. 2026-09-10 Research Update - Seiryu / MOM Layer 0 Closure (R41-R68)

Status: OFFICIAL UPDATE. R64, R65, R66, R67 and R68 user-run ZIPs passed their Exactness audits. User-run ZIPs remain the only official research evidence; local calculations remain debugging only.

36.1 Executive conclusion

- Operational Layer-0 Seiryu control remains Frozen11M ordinary MOM.
- The strongest underlying temporal structure is Long Momentum around 10-12M, with the most compelling explicit window being 12-1: price from 12 months ago to 1 month ago.
- 3xMOM 12-1 is retained as an archived high-octane / leverage-overlay candidate, not promoted to the operational Seiryu control.
- Further MOM parameter mining is closed unless genuinely new primary information appears (for example, Bam's exact DM2 scalar, GS architecture, Tier definitions, or exact absolute-momentum/rebalance interaction).

36.2 Lookback terrain established by R41-R54

The 15-horizon atlas repeatedly revealed a three-peak terrain: Ultra approximately 10D, Mid approximately 4-5M, and Long approximately 10-12M. The Long region was the most robust. R47/R52 confirmed that the strength was primarily attributable to the lookback itself rather than the Risk-Off policy. R49 clarified that Frozen11M is not simply 11M raw momentum: it uses recent-month exclusion and matched RF comparison, and is conceptually close to a 12-1 window without being declared mathematically identical.

36.3 Multiple-lookback and temporal-diversification findings (R50-R63)

Temporal diversification is real: different lookbacks are imperfectly correlated and 2P/3P portfolios can improve median risk-adjusted metrics versus generic 1P portfolios. However, the generic claim $3P > 2P > 1P$ was not confirmed after candidate-count adjustment and OOS validation. Long-core weighting reduced equal-weight dilution, but Ultra/Mid satellites did not robustly beat the Long control. Prospective regime proxies such as Prev3MReturn and PriorDrawdown showed some predictive information, yet R62 demonstrated that this did not translate into a robust adaptive portfolio. R63 found no robust incremental residual alpha for Ultra or Mid beyond Long.

36.4 Formal ordinary-MOM closure (R64)

R64 Exactness 18/18 PASS formally froze the ordinary MOM branch. Operational control = Frozen11M exact historical implementation. The 10-12M plateau is structural; Ultra/Mid satellites are not promoted; scalar and adaptive-satellite branches are closed until new information.

36.5 Separate 3xMOM branch (R65-R67)

The user explicitly reopened a distinct 3xMOM lookback branch. R65 established that leverage did not move the optimum to a short horizon: 12M led the full-sample ranking and was most frequently selected train-only in CPCV28 and WF13, while 4M remained a meaningful secondary peak. R66 robustness showed 12M was not a start/end-boundary artifact and that 4M was a genuine secondary peak but not a replacement for Long.

R67 definition	CAGR	Sharpe	Sortino	Calmar
12-1	49.18%	1.077	2.176	0.960
11M raw	47.44%	1.046	2.086	0.926
11-1	45.96%	1.038	2.055	0.897
12M raw	45.26%	1.020	1.984	0.883

R67 Exactness 20/20 PASS explicitly separated 11M raw, 11-1, 12M raw and 12-1. 12-1 ranked first on all four full-sample headline metrics. In train-only selection it was chosen in 50.0% of CPCV28 folds and 61.5% of WF13 folds. Importantly, excluding the most recent month was not universally beneficial: 11M raw -> 11-1 worsened results. Therefore the supported statement is not 'always exclude 1M', but rather that the specific 12-1 temporal window is strong.

36.6 Final leverage duel (R68)

Metric	Frozen11M ordinary MOM	3xMOM 12-1
CAGR	29.41%	49.18%
Sharpe	1.235	1.077
Sortino	2.520	2.176
MaxDD	-23.21%	-51.23%
Calmar	1.267	0.960
Underwater rate	53.6%	58.5%
Max underwater	14 months	15 months

R68 Exactness 17/17 PASS. 3xMOM 12-1 is a powerful return amplifier but not a risk-adjusted-quality improvement. Across the 12 boundary perturbations, 3x won CAGR 12/12 but won Sharpe, Sortino and Calmar 0/12. In Rolling 60M, 3x won CAGR in every window but won Sharpe and Sortino in none. QQQ beta was approximately 0.99 for Frozen11M versus approximately 2.06 for 3xMOM 12-1, supporting the interpretation that much of the extra CAGR is additional market beta rather than improved Seiryu signal quality.

36.7 Frozen Seiryu governance after R68

Role	Strategy / hypothesis	Governance
Operational control	Frozen11M ordinary MOM	RETAIN / FROZEN
Underlying temporal hypothesis	12-1 Long Momentum	SUPPORTED
Aggressive archived variant	3xMOM 12-1	SAVE; do not promote standalone
Secondary peaks	Ultra ~10D; Mid ~4-5M	REAL TERRAIN; DO NOT PROMOTE
Multiple-lookback	Temporal diversification	MECHANISM CONFIRMED; generic superiority not confirmed
Adaptive satellite	Prospective switching to Ultra/Mid	CLOSED FOR NOW

36.8 Open Bam reverse-engineering gaps

- Exact DM2 scalar definition: OPEN - HIGH.
- Exact GS architecture and the reported ~62.9% DM2_GS construction: OPEN - HIGH.
- Tier A/B/C/D definitions: OPEN - MEDIUM.
- Exact Frozen11M versus generic 12-1 mathematical identity: OPEN - LOW.
- Interaction of exact DM2 with absolute momentum and rebalance timing: OPEN - MEDIUM.

36.9 Research stop rule / next-stage use

Do not reopen fine MOM parameter searches (for example 10.5M/11.5M or optimized 4M/12M weights) without genuinely new primary information. At Layer 1 / Four Gods integration, Frozen11M remains the default Seiryu sleeve. 3xMOM 12-1 may be tested only as a pre-frozen high-octane variant where other Gods or defensive logic can potentially control its tail risk; such integration must not retroactively alter the Layer-0 conclusion.

36.10 Official result lineage

- R64 - ordinary MOM research closure / gap audit - Exactness 18/18 PASS.
- R65 - Seiryu 3xMOM lookback grid atlas - Exactness 20/20 PASS.
- R66 - 3xMOM peak robustness audit - Exactness 20/20 PASS.
- R67 - 3xMOM recent-1M exclusion definition audit - Exactness 20/20 PASS.

- R68 - Seiryu final leverage duel - Exactness 17/17 PASS.

Master Bible revision: v1.1 - 2026-09-10. Appended Chapter 36; prior v1.0 content preserved.

37. R75-R88 MOM / Temporal Diversification Research Update

Master Bible revision: v1.2 - 2026-09-10. This chapter appends the research completed after v1.1. Prior chapters and frozen strategy definitions remain authoritative unless explicitly superseded below.

37.1 Canonical MOM definition - non-negotiable

- Frozen Core QQQ-MOM canonical signal/gate remains LQD absolute momentum versus matched-window RF compound. Frozen Core uses LQD 11M with the most recent 1M excluded.
- If LQD $\geq$ matched RF: QQQ-MOM holds QQQ. If LQD $<$ matched RF: the pre-Adaptive-Safe-Haven holding is GLD.
- Pure/original QQQ-MOM lookback research isolates this single QQQ-MOM sleeve: LQD absolute-momentum gate $\rightarrow$ QQQ/GLD. It excludes the QLD-MOM companion sleeve, the 50:50 QQQ/QLD blend, Adaptive Safe Haven GLD $\rightarrow$ TLT, Accelerator/Brake/Turbo/Attack, and quality/metric selectors.
- Do NOT redefine Pure QQQ-MOM as QQQ/CASH or QQQ-vs-GLD relative momentum. Those were transient implementation mistakes and are not canonical.
- Exact Frozen11M versus a generic mathematical '12-1' identity remains OPEN-LOW unless explicitly reconciled by the canonical engine.

37.2 R75-R80 - Peak-3P MOM architecture branch

- R75 official: Peak-Stratified 4-Metric / 15-Pattern selector. Ultra/Mid/Long candidate buckets were combined as 1P/2P/3P. The mechanism was promising but did not establish Bam's exact operator.
- R76 official: direct second-stage quality-score weighting across Ultra/Mid/Long was rejected. This rejects direct quality-as-allocation-weight only; it does not reject inverse-volatility, ERC, or HRB concepts.
- R77 official: quality-window robustness. The 36M + Sharpe+Calmar configuration produced strong ex-post Sharpe results, but 36M is an observed ex-post setting, not a Bam-sourced truth or independently established optimum.
- R78 official: temporal risk construction / HRB mechanism audit. Equal3P slightly beat InverseVol, ERC, and HRB initial-weight proxy. ERC weights were near one-third each, consistent with the supplied Bam/J.P. Morgan evidence that temporal horizons can naturally approach equal holdings. Exact HRB Signal, risk-budget definition, clustering distance/linkage remain unknown.
- R79 official: frozen Peak-3P championship/falsification. Peak-3P showed positive incremental-return diagnostics versus Frozen11M in much of the sample, but the advantage was regime-dependent and walk-forward evidence was weaker. Status: SUPPORTED_ARCHITECTURE; Frozen11M remains operational Seiryu control.
- R80 official: processing-transplant identity audit. R75-R79 Peak-3P candidate series were already generated through the Frozen-derived R41 MOM engine. This discovery means those results cannot by themselves prove that the observed temporal structure belongs to a processing-free/original MOM definition.

37.3 R81-R86 - Pure QQQ-MOM lookback re-baselining

- R81 branch purpose: remove later Frozen additions and re-test lookback structure in the canonical single-sleeve QQQ-MOM architecture: LQD absolute momentum versus matched RF $\rightarrow$ QQQ/GLD.
- R81 early PATCH1/PATCH2 definitions (QQQ/CASH and QQQ-vs-GLD relative momentum) were wrong and must never be used as canonical evidence. The Master Bible definition corrected the branch to LQD absolute momentum versus matched RF.
- R84 dense landscape: preregistered dense daily/monthly lookback mapping was designed as a landscape audit, not a best-parameter promotion exercise.
- R85 official: long-horizon 2-5Y extension. 12M remained the dominant risk-adjusted long peak in the tested common era, while 24M appeared as a secondary local peak. Longer 30M/42M/54M features were weaker. Interpretation: 12M is not merely a grid-edge artifact; 24M is a secondary long-horizon structure, not a replacement.

- R86 official: fixed four-peak temporal-diversification comparison using 10D / 5M / 12-1 / 24M. The best full-sample Sharpe among the tested fixed constructions was the 3P 10D + 12-1 + 24M equal-weight portfolio; adding 5M to make 4P diluted Sharpe. Therefore generic 'more periods is always better' and $4P > 3P > 2P > 1P$ are rejected. A local performance peak is not automatically a useful portfolio component.
- R86 also reinforced that temporal distance matters: the ultra-short 10D component was materially less correlated with the long components than adjacent/mid horizons, supporting a temporal-diversification mechanism rather than simple parameter averaging.

37.4 R87 - RAW versus recent-1M exclusion audit: provisional only

- R87 was requested to separate monthly RAW lookbacks (include the most recent completed 1M) from EX1 lookbacks (exclude the most recent completed 1M), inspect 1M-60M, and test fixed RAW/EX1 combinations.
- 1M_EX1 was explicitly treated as undefined because removing the recent 1M from a 1M measurement window leaves a zero-length window. 1M_RAW was intended to be tested directly.
- R87 PATCH1-PATCH3 contained runtime/index/column-key bugs and are non-official. PATCH4 completed its internal Exactness audit, and PATCH5 corrected the chart x-axis by keeping LookbackMonths numeric through rendering.
- However, R87's newly written explicit RAW/EX1 engine produced materially different results from the R81-R86 canonical Pure engine. The difference is too large to interpret as ordinary sample-period variation.
- Therefore R87 performance rankings, including apparent short-horizon/1M behavior and the 64-combination championship, are PROVISIONAL / NOT PROMOTABLE. They must not be used to revise the canonical lookback conclusion until cross-generation definition reconciliation is complete.
- An important diagnostic pattern was an apparent one-month RAW/EX1 shift equivalence in several horizons. This is evidence that window-length semantics and recent-month exclusion may have been conflated in the explicit R87 implementation, not evidence of a new alpha result.

37.5 R88 - Definition reconciliation and lightweight research protocol

- R88 makes R81 canonical `_r81_pure_daily` the gold standard and compares R87 explicit EX1 against it at preregistered anchors 2M, 3M, 5M, 10M, 11M, 12M, and 24M using holding agreement and monthly-return MaxAbsDiff.
- R88 PATCH1 was superseded before official execution by a lightweight design.
- Current build: R88-PATCH2-LIGHT-PURE-MOM-DEFINITION-RECONCILIATION.
- The lightweight build computes only the seven reconciliation anchors, prepares one shared LQD/QQQ/GLD/open/RF input panel, caches inputs and reconciliation outputs with Streamlit cache, and does NOT rebuild the 1M-60M landscape, 120 RAW/EX1 series, or combination grid.
- R88 has not yet produced an official user-run result ZIP. Until it does, the exact source of the R87/R81 mismatch remains OPEN-HIGH.
- After reconciliation passes, any corrected RAW/EX1 full-grid research should reuse frozen/cached canonical series rather than recomputing the entire candidate family.

37.6 Compute / app engineering lessons now frozen

- Official research evidence continues to require a user-executed Streamlit result ZIP. Assistant-side AST/py_compile checks establish code integrity only, not research findings.
- Reuse frozen/precomputed results by default. Do not regenerate the same Layer-0 candidate families or daily series for every downstream audit.
- Prefer compact frozen inputs, one-pass shared signal panels, vectorization, caching, and lazy computation. Heavy validation should expose progress.
- On 2026-09-10 Streamlit Community Cloud temporarily CPU-throttled the app after repeated heavy RAW/EX1 recomputation. This is an engineering signal to reduce repeated computation, not a strategy result.

- Chart lookback axes must remain numeric through the plotting layer. Converting 1M..60M to categorical strings can cause lexicographic ordering (1M,10M,...,2M).
- Period joins/comparisons must canonicalize to Period[M]. Avoid positional matching between series whose histories may begin on different dates; matched RF windows should use actual period/date alignment.
- Keep the global stable download-button behavior (on_click='ignore' with compatibility fallback).

37.7 Current decision state

- Operational Seiryu control: Frozen11M remains unchanged.
- Peak-3P MOM: SUPPORTED_ARCHITECTURE, but prior R75-R80 evidence is Frozen-processing-derived and cannot substitute for Pure-MOM validation.
- Pure lookback structure: 12M/12-1 region remains the main long-horizon reference; 24M is a secondary long peak; ultra-short temporal diversification remains plausible. Do not promote a new operational lookback from landscape inspection alone.
- R87 RAW/EX1 rankings: PROVISIONAL / HOLD.
- R88 definition reconciliation: OPEN-HIGH and immediate next research task.
- Exact Bam DM2 scalar/operator: OPEN-HIGH.
- Exact GS architecture: OPEN-HIGH.
- Tier A/B/C/D definitions: OPEN-MEDIUM.
- Exact HRB construction details: OPEN.
- Do not resume fine-grained parameter mining until definition reconciliation is complete and/or genuinely new primary information arrives.

37.8 Official / provisional lineage added in v1.2

- Official user-run lineage: R75, R76, R77, R78, R79, R80, R85, R86 (plus earlier official lineage retained in prior chapters).
- R81-R84 are part of the Pure-MOM re-baselining lineage; only results backed by successful user-run Exactness/ZIP evidence should be treated as official in downstream claims.
- R87 PATCH1-PATCH3: failed runtime; non-official. R87 PATCH4/PATCH5: diagnostic output exists, but research conclusions are held provisional because the engine does not reconcile with the R81 gold standard.
- R88 PATCH2 LIGHT: current code state; awaiting user-run ZIP. No research conclusion yet.

Revision v1.3 Addendum — MOM Temporal Consensus / 3vs1 / Production Architecture

記録日: 2026-09-11 Development cutoff: 2026-07-31

位置づけ: 本追補は v1.2 の本文を保持したまま、R89〜R101 の Pure MOM 再基準化、時間的分散、Exact HoldingSignal consensus、3-v-1 Majority、Production Decision Audit、3vs1 × Frozen Implementation を追加する。公式成績はユーザーが Streamlit で実行して返却した ZIP のみを根拠とする。

v1.3での運用規約更新: 完了済み検証コードを app.py に永続保持することは必須としない。公式 ZIP・本Bible・軽量registryで結果と系譜を復元できる場合、完了済み実行UI/巨大payloadは退役可能。現行/次検証のUI、Production/Frozen仕様、監査系譜は保持する。単一 app.py 運用を優先する。

重要: R101は有望だが自動Production昇格ではない。Architecture Gate 8/9 PASS。Productionは現行 Frozen Operational 11Mを維持し、R101をFrozen ChallengerとしてForward OOSへ送る。

11. Pure MOM の再基準化: R89 – R94

11.1 R89 / R90 — canonical temporal landscape

R89で月次1-60MのRAW/EX1を統一定義した。RAW N = latest completed month / N-month-old anchor - 1。EX1 N = one-month-before-latest / same N-month-old anchor - 1。したがって12M_EX1はconventional 12-1、1M_EX1は未定義。共通期間は2014-02〜2026-07の150か月。

代表系列	CAGR	Sharpe	MaxDD	Calmar
13M EX1	23.51%	1.478	-15.89%	1.479
12M EX1	22.94%	1.414	-15.89%	1.444
11M RAW	23.70%	1.413	-15.89%	1.491
13M RAW	22.38%	1.410	-15.89%	1.408
25M RAW	24.01%	1.406	-22.97%	1.045

EX1はCAGR 32/59、Sharpe 28/59、Sortino 28/59、Calmar 33/59でRAWを上回ったに過ぎず、「直近1M除外は常に有利」は棄却。月次地形はMid 5-6M、Long 11-13M、Very Long 24-25M。R90で日次1-60Dを追加し、10-12D Ultra、54-57D Shortも確認。以後はexact winnerではなく代表帯を使い、再fine-tuningしない。

11.2 R93 / R94 — fixed temporal diversification

固定代表を Ultra=10D、Short=1M_RAW、Mid=6M_EX1、Long=12M_EX1、VeryLong=24M_RAW とした。R93ではcanonical Pureにおける1M追加は15/15構成でCAGRを悪化させ、旧engineでの1M diversifier仮説は棄却。

構成	CAGR	Sharpe	MaxDD	Calmar
Long 12M_EX1	22.94%	1.414	-15.89%	1.444
Ultra+Long	21.87%	1.488	-14.38%	1.520
Ultra+Mid+Long	21.53%	1.514	-14.47%	1.488

R94の最終falsificationはPeak3Pが4/7 PASS。Full sampleではSharpe/Calmar改善があるが、WF Sharpe win 40%、Rolling60 Sharpe win 42.86%で、Long/Pure controlのreplacementには昇格せず。時間的分散は主としてDD/tail diversificationとして支持。

12. Dynamic allocation から Exact Signal Consensus へ: R95A – R98

12.1 R95-A — trailing performance allocation は棄却

10D / 6M_EX1 / 12M_EX1 / 24M_RAWを固定し、36M trailing SharpeでTop2またはpositive-Sharpe proportional配分。Top2 Full Sharpe 1.304、Soft 1.467、Long 1.385。WFではTop2/SoftともSharpe win rate 0%。過去に良かった時間軸を追う方式は明確に棄却し、selector windowのgrid searchも行わない。

12.2 R95-B2 — Exact HoldingSignal Majority

R95-Bのreturn-equalityによるstate推定はExactness FAILだったため無効化。R95-B2では6M/12M/24Mを公式R89 HoldingSignalから、10Dをcanonical LQD 10D vs matched RF gateで固定再構築。Exactness 12/12 PASS。

Strategy	CAGR	Sharpe	Sortino	MaxDD	Calmar
Equal4P	20.96%	1.453	2.850	-17.12%	1.224
MajorityOnly_3v1	23.82%	1.503	3.068	-16.41%	1.452
SlowConsensus	21.98%	1.440	2.909	-18.50%	1.188
UltraLead	17.37%	1.149	2.099	-18.09%	0.961

MajorityOnly_3v1: 4本中3本が同一HoldingSignal、1本が反対なら少数派sleeveを除き多数派3本を各1/3。それ以外はEqual4P。実売買では3-v-1時に多数派assetが100%になる。UltraLeadは明確に棄却。

12.3 R96 — Majority mechanism falsification

Exactness 7/7、Falsification Gate 8/8 PASS。3-v-1は150か月中59か月。Equal4P比の平均edge +0.540%/月、勝率62.7%。少数派別edgeは10D +0.375%、6M +0.399%、12M +0.537%、24M +1.727%。QQQ-majority +0.496%、GLD-majority +0.633%。5/5 chronological erasで平均edgeがプラス。特定peak・risk-on方向・単一時代だけの現象という説明を退けた。

12.4 R97 — 一般的なvote monotonicityは部分支持に留まる

QQQ票	月数	平均 QQQ-GLD	QQQ勝率
0	14	-0.29%	64.3%
1	19	-2.49%	36.8%
2	28	+1.05%	67.9%
3	40	+1.68%	67.5%
4	48	+1.54%	62.5%

Falsification Gate 6/8 PASS。vote数とrelative returnの全体的な正関係はあるが、0→4の完全単調構造は成立しない。したがって「票数が多いほど常に強い」「4/4なら追加で攻める」等への一般化は棄却。支持境界は3-v-1という離散構造。

12.5 R98 — Temporal Majority as Noise Filter

Exactness 9/9、Falsification Gate 8/8 PASS。59件の3-v-1後、翌月も現在の多数派が多数派44/59=74.6%、2-v-2が10/59=16.9%、現在の少数派側へ多数派逆転5/59=8.5%。少数派自身が多数派へ再合流30/59=50.8%、多数派3本のうち2本以上が少数派方向へ動くのは5/59=8.5%。さらに翌月に現在多数派方向へ4-v-0が23回、少数派方向へ4-v-0は0回。

現時点の経済的説明: 同一LQD absolute-momentum regimeを異なる時間尺度で観測した際、1本だけ反対になるケースは新regimeの先行検出より時間尺度固有のtemporary noiseであることが多く、多数決がnoise filterとして働く可能性が高い。

13. Pure vs 3vs1 と Production Decision: R99 – R100

13.1 R99 — Pure MOM vs 3vs1 Final Head-to-Head

比較はPure MOM=12M_EX1単独、3vs1 MOM=MajorityOnly_3v1。Exactness 11/11、Falsification Gate 10/10 PASS。

指標	Pure 12M_EX1	3vs1	差
CAGR	22.94%	23.82%	+0.88pt
Sharpe	1.414	1.503	+0.089
Sortino	2.857	3.068	+0.211
MaxDD	-15.89%	-16.41%	-0.51pt
Calmar	1.444	1.452	+0.009

5-block Sharpe win 4/5=80%。Rolling36: CAGR/Sharpe/Calmar win 64.3%/64.3%/69.6%。Rolling60: 69.2%/70.3%/83.5%。Worst Rolling60 Δ Sharpe -0.166。3vs1はPureを常時支配しないが、長期risk-adjusted robustnessで優位。R99後の分類: Pure 12M_EX1=Robust Benchmark、3vs1=Frozen Challenger。

13.2 R100 — Production Decision Audit

Frozen Operational 11M、Frozen Signal-Only 11M、Pure 12M_EX1、3vs1を同時比較。Exactness 12/12 PASS。

戦略	CAGR	Sharpe	MaxDD	Calmar
Frozen Operational 11M	28.03%	1.266	-21.34%	1.314
Frozen Signal-Only 11M	22.50%	1.413	-15.91%	1.414
Pure 12M_EX1	22.94%	1.414	-15.89%	1.444
3vs1 MOM	23.82%	1.503	-16.41%	1.452

3vs1 vs Frozen OperationalはDecision Gate 9/10 PASS。唯一のFAILはFull CAGRで、3vs1 23.82% < Frozen Operational 28.03%。一方Sharpe/Calmar/DDは3vs1が優位。Rolling60 Sharpe winは96.7%。Signal-only比較でも3vs1はCAGR +1.33pt、Sharpe +0.090、Calmar +0.038。

R100判断: Frozen Operationalを3vs1 Signal-Onlyへ全面置換すると火力を失う。そこで「3vs1のsignal quality × 既存Frozen implementation」という固定2層architectureのみを最後に検証する。

14. R101 — 3vs1 Signal × Frozen Implementation

14.1 Architecture

R101は新しいlookback・threshold・weight探索ではない。既に固定済みの3vs1 signal allocatorへ、既に固定済みのFrozen Operational implementationをそのまま載せるarchitecture test。equity exposureにはQQQ/QLD=50/50、defensive exposureには既存Adaptive Safe Haven (GLD/TLT) を適用する。

PATCH2はASH warm-up不足により2014-02～2014-12の11か月が欠落し139か月となったため、final promotion evidenceとして無効。PATCH3でprice warm-upを2012-12-01まで前倒しし、2014-02～2026-07の150か月を完全復元。ルール変更は行っていない。

14.2 PATCH3 official result

Build: R101-PATCH3-150M-WARMUP-EXACTNESS-REPAIR。Exactness 13/13 PASS。

戦略	CAGR	Sharpe	MaxDD	Calmar
Frozen Operational 11M	28.03%	1.266	-21.34%	1.314
Pure 12M_EX1	22.94%	1.414	-15.89%	1.444
3vs1 Signal-Only	23.82%	1.503	-16.41%	1.452
3vs1 × Frozen Impl.	30.45%	1.394	-21.34%	1.427

Hybrid vs Frozen Operational: CAGR +2.42pt、Sharpe +0.128、MaxDDほぼ同一、Calmar +0.113。Rolling36のCAGR/Sharpe/Calmar winは75.7%/79.1%/80.9%。Rolling60は68.1%/70.3%/68.1%。Rolling60 medianはΔ CAGR +3.71pt、Δ Sharpe +0.147、Δ Calmar +0.174。5-block Sharpe win 3/5=60%。

14.3 Architecture Gate と正式判定

Architecture Gateは8/9 PASS。唯一のFAILは「Hybrid Sharpe degradation vs 3vs1 Signal-Only <= 0.10」。実績は1.394 - 1.503 = -0.109で、閾値を0.009超過した。事後的にgateを変更してPASS扱いしない。

正式判定: R101 = Frozen Challenger / NO automatic promotion。現行ProductionをCAGR・Sharpe・Calmarで同時に上回り、MaxDDも同水準という最も強いProduction replacement候補だが、事前Architecture Gate 8/9のため自動昇格しない。2026-08以降をretune禁止のForward OOSとして追跡する。

15. R89 – R101で確定した採否

項目	判定	固定理由
Frozen Operational 11M	Production維持	現行火力。R101の自動昇格条件未達
Pure 12M_EX1	Robust Benchmark	単純・低自由度・高Sharpe
3vs1 Signal-Only	Frozen Challenger	R96/R98 mechanism支持、R99 10/10
3vs1 × Frozen Implementation	最重要 Frozen Challenger	R101 exactness 13/13、Architecture 8/9
1M canonical diversifier	Rejected	R93で15/15 CAGR悪化
Peak3P replacement	Rejected	R94 final promotion 4/7
Trailing Sharpe selector	Rejected	R95-A WF Sharpe win 0%
UltraLead dissent	Rejected	R95-B2で明確悪化
Continuous 0-4 vote confidence	Rejected as general rule	R97 monotonicity不成立
追加lookback/weight fine-tuning	Do not reopen	selection bias防止、DEV cutoff固定

16. 実運用・Forward OOS 規約

1) ProductionはFrozen Operational 11Mを維持する。2) R101を最重要Frozen Challengerとして毎月shadowまたは限定実弾で追跡可能。3)

2026-08以降の結果を使って10D/6M/12M/24M、3-v-1条件、QQQ/QLD 50/50、ASH条件を変更しない。4)
Forwardでは月次signal、3-v-1発生、実配分、turnover、実コスト、Frozenとの差分returnを保存する。5)
数か月のunderperformanceで救済・再最適化しない。

R95-B2～R101の研究全体は、「最良lookbackを探す」段階から「複数時間軸を独立観測器として扱い、3-v-1時のみ多数決で時間軸固有ノイズを除去し、そのsignalへ既存Frozen implementationを載せる」段階へ移行した。今後の主課題は新しいparameter searchではなく、Forward OOSでの再現性確認とProduction Monitor整備である。

Addendum v1.4 — R102 – R106 Temporal MOM / Majority Mechanism

Master Research Bible 追補 | Official Streamlit ZIP evidence only | Development cutoff: 2026-07-31

追補の目的

v1.3 (R89 – R101までを反映) 以降に完了した R102 – R106 を正式系譜へ追加する。ここに記載する成績・Exactness・Gate は、ユーザー実行の公式 Streamlit ZIP を根拠とする。ローカル計算は公式成績の代替としない。

現在の最重要結論

R102 – R103 により、LQD/RF regime signal の時間構造は QQQ 固有ではなく、QLD/TQQQ へ leverage を上げて主要 peak region が再現した。R104 では固定 Full5P Temporal Majority が Long 12M_EX1 を改善。R105 はその改善が方向非対称であることを示し、R106 は risk-on re-entry lag compensation を 12/12 gate PASS で支持した。

Research	正式状態	主要結論
R102	Exactness 17/17 PASS	QLD/TQQQでも 54 – 57D, 11 – 13M, 24 – 25M 等の時間地形が再現
R103	Exactness 13/13 PASS	固定5 peak の cross-leverage robustness。Long 11 – 13M が最強の頑健領域
R104	Exactness 13/13 PASS	Full5P majority が Long を改善。単純EW以上の majority effect
R105 PATCH2	Exactness 14/14 PASS; Gates 6/6	効果は普遍的でなく方向非対称。risk-on majority 側が主因
R106	Exactness 11/11 PASS; Gates 6/6	Long risk-on 復帰遅れを Temporal Majority が補完する機序を支持

A. R102 — Pure QLD/TQQQ MAX HISTORY Dual-Era Landscape

固定定義

Pure QLD-MOM / TQQQ-MOM は、LQD absolute momentum を matched RF と比較し、risk-on 時のみ QLD または TQQQ、defensive は GLD。QLD/TQQQ 自身の momentum は使わない。ASH、3vs1、QQQ/QLD blend 等は除外。Grid は $1D - 60D + 1M - 60M \text{ RAW} + 2M - 60M \text{ EX1} = 179 \text{ series/asset}$ 。

公式Exactness: 17/17 PASS。QLD MAX history は 2006-06、TQQQ は 2010-02 から。2014 common では QQQ と比較可能。

代表領域	QQQ Sharpe 2014 common	QLD Sharpe 2014 common	TQQQ Sharpe 2014 common	解釈
10 - 12D	1.232	1.038	0.931	Ultraは残るがleverageで弱化
54 - 57D	1.383	1.240	1.125	Short region 強く再現
5 - 6M	1.264	1.145	1.050	Mid region 再現
11 - 13M	1.478	1.231	1.115	Long region 最重要
24 - 25M	1.406	1.212	1.105	VeryLong region 再現

MAX/ALL179での注意点

QLD MAX の 11M_RAW は CAGR 33.55%, Sharpe 1.127, MaxDD -33.45%。TQQQ MAX の 25M_RAW は CAGR 51.93%, Sharpe 1.109, MaxDD -51.23%。ただし exact winner を新しい凍結値にしない。54D や 25M は代表点/感度結果であり、理論的な一点最適値ではない。

R102の正式解釈: leverage が上がるほど payoff/risk magnitude は変わるが、peak location は驚くほど維持される。したがって temporal landscape は risk-on asset 固有よりも、共有する LQD/RF regime signal に結び付く可能性が高い。Landscape discovery のみで promotion なし。

B. R103 — Fixed Five-Peak Cross-Leverage Robustness

R102の地形を見た後に winner を選び直さず、事前固定 representative を検証した。Primary: Ultra=10D, Short=54D, Mid=6M_EX1, Long=12M_EX1, VeryLong=24M_RAW。Region sensitivity も事前固定。CPCV-style 8 choose 2、Rolling36/60、WF36を実施。

Exactness 13/13 PASS。これは portfolio construction ではなく、各時間領域が leverage を跨いでも残るかの robustness audit。

Asset / Peak	CAGR	Sharpe	MaxDD	Calmar
QLD 10D	30.54%	1.015	-41.03%	0.744
QLD 54D	22.77%	0.824	-74.60%	0.305
QLD 6M_EX1	28.11%	0.973	-43.54%	0.646
QLD 12M_EX1	32.46%	1.087	-32.23%	1.007
QLD 24M_RAW	28.73%	0.949	-52.52%	0.547
TQQQ 10D	38.11%	0.944	-51.23%	0.744
TQQQ 54D	37.61%	0.960	-52.71%	0.714
TQQQ 6M_EX1	39.85%	0.976	-51.23%	0.778
TQQQ 12M_EX1	46.00%	1.050	-51.23%	0.898
TQQQ 24M_RAW	41.11%	0.940	-66.63%	0.617

Cross-leverage signal consistency (2010-02 – 2026-07)

QLD/TQQQ monthly return correlation は固定 representative ごとに 0.9888 – 0.9951、same-sign は 98.48 – 98.99%。Long 11 – 13M は broad / strong / cross-leverage / CPCV / rolling の総合で最も頑健。Mid 5 – 6M も coherent。Ultra は局所的に uneven、Short は 54 – 57D region として残るが exact 54D は特別でなく、VeryLong は recent instability を含む。

研究判断: Frozen 11M を「QQQだけの偶然」とみなす根拠は弱くなった。R103を理由に 11M_RAW や 25M_RAW へ retune しない。

C. R104 — Fixed Temporal Signal Combination

固定5 peak の exact HoldingSignal を組み合わせ、単純な独立 sleeve portfolio ではなく Temporal Majority を検証。Full5P = 10D / 54D / 6M_EX1 / 12M_EX1 / 24M_RAW。3P/5P は strict majority、majority側 sleeves を等ウェイト。2P tie は固定50/50。各構成に own EW control を置いた。

Exactness 13/13 PASS。

Asset	Strategy	CAGR	Sharpe	MaxDD	Calmar
QLD	Long 12M_EX1	32.53%	1.081	-32.23%	1.009
QLD	Core3P	34.80%	1.115	-32.86%	1.059
QLD	Full5P	36.66%	1.184	-32.23%	1.137
TQQQ	Long 12M_EX1	46.00%	1.050	-51.23%	0.898
TQQQ	Core3P	55.09%	1.162	-51.23%	1.075
TQQQ	Full5P	56.31%	1.206	-51.23%	1.099

Full5P - Long: QLD ΔCAGR +4.14pt, ΔSharpe +0.103, ΔCalmar +0.128。TQQQ +10.30pt, +0.156, +0.201。MaxDD は実質同等。

Majority - own EW: QLD ΔCAGR +7.50pt, ΔSharpe +0.089, ΔCalmar +0.247。TQQQ +13.11pt, +0.116, +0.256。したがって改善は「複数期間をEWしただけ」では説明し切れない。

Full5P robustness vs Long	QLD	TQQQ
Rolling36 Sharpe win	68.7%	71.8%
Rolling60 Sharpe win	85.3%	99.3%
Rolling60 Calmar win	87.6%	100.0%
CPCV Sharpe win	75.0%	96.4%
CPCV Calmar win	71.4%	89.3%

近い時間軸の Ultra+Mid+Long 3P は robustness が弱く、Core3P (10D+12M+24M) は比較的強い。『期間数が多ければ良い』ではなく、離れた時間階層の情報を majority で統合することが重要という仮説が強まった。

D. R105 PATCH2 — Full5P Majority Mechanism Falsification

PATCH1の agree-state return identity は R104 の sleeve-weight semantics と整合しない誤ったExactness前提だったため正式採用せず、PATCH2で R104 official weights × peak returns から Full5P return を完全再構築する hard exactness に修正。

PATCH2 Exactness 14/14 PASS。Full5P再構築誤差は QLD/TQQQ とも約 1e-16、Long identity は0、majority mask/weight sum も一致。

Mechanism summary	QLD	TQQQ
Common months	236	198
Long-dissent events	37	28
Mean edge / dissent month	+2.23%	+4.61%
Dissent win rate	54.1%	60.7%
Majority persists next month	66.7%	63.0%
Long rejoins next month	75.0%	74.1%
Rolling60 median Δ Sharpe	+0.119	+0.120
Rolling60 Sharpe win	85.3%	99.3%

Falsification Gates: QLD 6/8 PASS、TQQQ 6/8 PASS。両assetで FAIL したのは (1) both majority directions mean edge > 0、(2) 5 eras中4以上で positive mean dissent edge。

方向別では QQQ-majority / Long-GLD の risk-on disagreement が主な正のedgeを生み、GLD-majority / Long-QQQ の risk-off disagreement は平均で負。したがって R104 の効果を『普遍的な多数決noise filter』とは一般化しない。

正式分類: PARTIALLY SUPPORTED — DIRECTIONALLY ASYMMETRIC TEMPORAL MAJORITY。次の固定仮説は、Longのrisk-on re-entry lagを多数時間軸が先に検出しているか、である。

E. R106 — Risk-On Re-entry Lag Mechanism Audit

R105で見た方向非対称性を、parameter変更なしで episode 単位に分解。RiskOn = Full5P majority QQQ / Long GLD。対称controlとして RiskOff = Full5P majority GLD / Long QQQ を置く。

Exactness 11/11 PASS。Falsification Gates は QLD 6/6、TQQQ 6/6、合計12/12 PASS。

Risk-on episode	QLD	TQQQ
Episodes	16	11
Mean cumulative edge	+5.93%	+14.97%
Median cumulative edge	+6.84%	+11.99%
Episode win rate	62.5%	72.7%
Long rejoins before reversal	68.8%	63.6%
Median months to Long rejoin	1	1
Rejoin within 2M	62.5%	54.5%
Median majority persistence	5 months	5 months
Transition-onset share	56.3%	54.5%

対称Risk-off control: mean episode cumulative edge は QLD -0.96%、TQQQ -3.24%。Risk-on minus Risk-off はそれぞれ +6.88pt、+18.21pt で gate PASS。

時間順序の解釈

典型的には、複数時間軸の majority が先に QQQ へ転換し、Long 12M_EX1 は GLD に残る。その後 Long は中央値1か月で majority に再合流する一方、majority regime は中央値5か月持続する。これは Long signal が不要という意味ではなく、頑健なLong-horizon signalの弱点である risk-on 復帰の遅さを、Temporal Majority が補完する構造と整合する。

制約: QLDの <=2009 risk-on episodes は平均累積edge -2.37%。激しいbear marketでは早いrisk-on復帰が false start になり得る。万能な危機回避/転換検出器とは扱わない。

正式分類: SUPPORTED MECHANISM — RISK-ON RE-ENTRY LAG COMPENSATION。この限定されたmechanismとして凍結する。

F. v1.4 Research Governance / Current Decision

凍結すべきこと

R102 – R106を見た後に 54D→57D、24M→25M、QLD 11M_RAW、peak weight、vote threshold 等を最適化しない。R103 sensitivity の exact winner は新しい凍結値ではない。Full5PもR104で固定した5 peakのまま扱う。

Production / Challengerへの影響

Layer	v1.4 status
Production	Frozen Operational 11M / Frozen 4F production architecture は維持
Production replacement challenger	R101 3vs1 × Frozen Implementation は引き続き最重要 Frozen Challenger
Temporal mechanism	R106: risk-on re-entry lag compensation = SUPPORTED MECHANISM
Full5P	強い固定 challenger / architecture component。自動promotionなし
Forward boundary	2026-08以降は prospective / forward OOS。DEVへ戻してretuneしない

次に許される研究

新しいlookback探索ではなく、固定architecture testまたはforward OOS monitor。最も自然な次段階は、Full5P signal を Frozen implementation (QQQ+QLD / existing ASH) へ接続し、Frozen Operational 11M および R101 3vs1×Frozen と同一ルールで head-to-head すること。これは parameter mining ではなく、既に固定された signal mechanism の Production implication を確認する試験と位置づける。

v1.4で追加された研究鎖

R102 landscape → R103 fixed peak robustness → R104 fixed signal combination → R105 directional mechanism falsification → R106 risk-on re-entry lag audit。

この鎖の中心的結論は、Long 11 – 13M の頑健性を維持しつつ、その risk-on re-entry lag を異なる時間階層の合意が補完し得る、である。Longを捨てる結論でも、全方向でmajorityを盲信する結論でもない。

End of Addendum v1.4 | Added 2026-09-11

Master Research Bible - v1.5 Addendum

My 4F / Four Gods / v3.45 / Frozen Core

R107-R113: Full5P Temporal Majority, Frozen Implementation Attribution, Defensive Asset Falsification, Final Four Championship

Revision: v1.5 / Updated 2026-09-12

Development cutoff: 2026-07-31. 2026-08 onward remains prospective / forward OOS and must not be used for retuning.

Evidence rule: performance conclusions below are based on user-executed official Streamlit result ZIPs. Local calculations are not substituted for official evidence.

Executive conclusion

研究は、Long MOMの頑健性確認から、複数時間軸の多数決による Full5P Temporal Majority へ進み、さらにrisk-on exposureとdefensive assetを分離して検証した。現時点のpost-hoc championは Full5P × QQQ/QLD 50:50 × GLD。ただしProductionはFrozen Operational 11Mのままで、post-hoc championは2026-08以降のforward OOS対象として扱う。

現在の固定Full5P: 10D / 54D / 6M_EX1 / 12M_EX1 / 24M_RAW。多数決はsignal aggregationであり、過去成績による動的selectorではない。

R107 - Pure QQQ Full5P Replication

R104でQLD/TQQQに確認された5-peak majorityを、パラメータ変更なしでPure QQQへ移植。Exactness 14/14 PASS、Replication Gates 10/10 PASS。

Pure QQQ	CAGR	Sharpe	Sortino	MaxDD	Calmar
Long 12M_EX1	22.94%	1.414	2.857	-15.89%	1.444
Existing 4P Majority	23.82%	1.503	3.068	-16.41%	1.452
Full5P	27.82%	1.617	3.493	-16.54%	1.682
Full5P EW	21.39%	1.510	2.950	-18.33%	1.167

Full5PはLong、既存4P、単純5P EWを上回った。4Pの2-v-2 tie
29か月では54D追加の平均edgeが+1.42%/月で、54Dは単体winnerというよりtie-break informationとして寄与する可能性がある。

R105-R106 mechanism

Leveraged sleeveではFull5Pがrisk-on、LongがGLDに残るdisagreementでedgeが大きく、Longは中央値約1か月で多数派へ再合流。よってrisk-on re-entry lag compensationが支持された。ただしPure QQQではrisk-off側にもedgeがあり、より一般的なTemporal Majority noise-filter mechanismの一部と解釈する。

R108 - Full5P × Frozen Implementation

Full5P signalをFrozen implementationへ接続。Risk-onは50% QQQ + 50% QLD、defensiveは既存Adaptive Safe Haven。Exactness 17/17 PASS、Architecture Gates 14/17 PASS。

Strategy	CAGR	Sharpe	MaxDD	Calmar
Frozen Operational 11M	28.03%	1.266	-21.34%	1.314
R101 3vs1 × Frozen	30.45%	1.394	-21.34%	1.427
Full5P Signal-Only	27.82%	1.617	-16.54%	1.682
R108 Full5P × Frozen	33.02%	1.432	-21.34%	1.547

R108はFrozen比でCAGR +4.98pt、Sharpe +0.166、Calmar +0.234、MaxDDほぼ同一。R101もfull-sampleでは上回ったが、Rolling60 Sharpe win 59.34%、CPCV Sharpe win 57.14%で事前60% gateをわずかに未達。R108は当時の最重要Frozen Challengerだが自動昇格なし。

R109-R110 - ASH Contribution Attribution

R109ではPure QLD/TQQQ Full5Pのdefensive側だけGLD→Frozen ASHへ置換。R110ではR108のrisk-on 50% QQQ + 50% QLDを固定し、defensiveだけASH vs GLDを比較した。

Architecture	GLD CAGR	ASH CAGR	GLD Sharpe	ASH Sharpe	MaxDD
QLD Full5P	36.66%	34.56%	1.184	1.127	-32.23%
TQQQ Full5P	56.31%	54.13%	1.206	1.174	-51.23%
QQQ/QLD 50:50 Full5P	34.54%	33.02%	1.484	1.432	-21.34%

R110ではASHをGLD固定へ戻すだけでCAGR +1.53pt、Sharpe +0.052、Calmar +0.071、MaxDD不変。ASHがTLTを選択したdefensive月がperformanceを削った。したがってR108の改善主因はASHではなく、Full5P signal quality + QQQ/QLD firepower と判断する。

R109/R110はASH自体を一般に否定するものではなく、Full5Pとの組み合わせにおいてGLD固定の方が優位だったという限定された結論である。

R111-R112 - Defensive Asset Falsification

Full5P signalと risk-on 50% QQQ + 50% QLDを固定し、defensive assetだけを変更。探索ではなく少数の事前固定候補による falsification。

Defensive	CAGR	Sharpe	MaxDD	Calmar	判定
GLD	34.54%	1.484	-21.34%	1.619	Champion
BIL	27.67%	1.306	-21.34%	1.296	Reject
IEF	27.58%	1.280	-21.34%	1.292	Reject
XLU	31.77%	1.387	-21.34%	1.489	Reject
GDX	37.92%	1.300	-37.94%	0.999	Return ↑ / Defense ×

BIL/IEFはGLDより約7pt低いCAGRでMaxDD改善なし。「株から逃げれば何でもよい」は支持されない。XLUも総合的にGLD未満。GDXはCAGRを上げたがMaxDDが-37.94%へ悪化し、Gold-related equity riskを持ち込むため退避資産としては不適。

現時点の解釈

Full5P defensive regimeでは、Treasury/cash-like/defensive equity/gold minersより、現物Gold exposureのGLDがrisk-adjusted balanceで最も適合した。R112をもってdefensive-asset searchを終了し、追加候補探索はselection bias回避のため停止する。

R113 - Final Four Strategy Championship

4戦略を同一2014-02～2026-07、150か月の公式共通パネルで比較。Exactness 10/10

PASS。ここでR104のMAX-history値と共通150か月値を明確に分離し、Final Fourでは以下を正式比較値とする。

Strategy	CAGR	Sharpe	Sortino	MaxDD	Calmar
Frozen Operational 11M	28.03%	1.266	2.490	-21.34%	1.314
Full5P × QQQ/QLD 50:50 × GLD	34.54%	1.484	3.014	-21.34%	1.619
Full5P × QLD × GLD	45.25%	1.431	2.788	-32.23%	1.404
Full5P × TQQQ × GLD	61.12%	1.297	2.422	-51.23%	1.193

Risk ladder

Balanced: QQQ/QLD 50:50 - Sharpe/Sortino/Calmar首位、Frozenとほぼ同じMaxDDでCAGR +6.51pt。

Aggressive: QLD - CAGR 45.25%、Sharpe 1.431、MaxDD -32.23%。想定以上に強いhigh-growth variant。

Maximum Firepower: TQQQ - CAGR 61.12%だがMaxDD -51.23%。別risk budgetとして扱う。

Rolling60ではQQQ/QLDのSharpeがQLDに勝つ割合75.8%、TQQQに勝つ割合91.2%。QLDのSharpeはTQQQに91/91

windowで勝利。QLDとTQQQの相関は約0.992で、TQQQは別alphaではなくほぼ同じengineの高レバレッジ版と解釈する。

Current Canonical Architecture & Governance

現在のpost-hoc champion architecture

- Signal source: LQD absolute momentum vs matched RF
- Temporal observations: 10D / 54D / 6M_EX1 / 12M_EX1 / 24M_RAW
- Aggregation: Full5P Temporal Majority
- Balanced risk-on: 50% QQQ + 50% QLD
- Defensive: 100% GLD

Formal status

Role	Strategy	Status
Production	Frozen Operational 11M	UNCHANGED
Predeclared Frozen Challenger	R108 Full5P × Frozen ASH	Frozen / forward OOS
Post-hoc Champion	Full5P × QQQ/QLD 50:50 × GLD	Forward-OOS challenger
Aggressive tier	Full5P × QLD × GLD	Frozen research tier
Maximum-firepower tier	Full5P × TQQQ × GLD	Frozen research tier

開発期間は2026-07-31で終了。2026-08以降はprospective / forward OOS。54D→57D、12M→13M、24M→25M、vote weighting、QQQ/QLD比率、追加defensive assetなどの再探索は行わない。

重要: Post-hoc Championはバックテスト上の最良候補であり、Productionへの自動昇格を意味しない。今後はProduction / R108 / post-hoc Championを並走し、forward OOSで確認する。

Research Lineage R107-R113

ID	Question	Formal conclusion
R107	Full5PはPure QQQでも再現するか	SUPPORTED REPLICATION
R108	Full5P × FrozenはProduction候補か	Strongest predeclared Frozen challenger; 14/17 gates
R109	QLD/TQQQ Full5PでASHはGLDを改善するか	REJECTED
R110	R108のASH単独寄与は正か	NO; GLD fixed superior
R111	BIL/IEFはGLD代替になるか	REJECTED
R112	XLU/GDXはGLD代替になるか	REJECTED as defensive replacements
R113	4 risk tiersの同一期間最終比較	Balanced / Aggressive / Max-firepower ladder confirmed

Source hierarchy

Text authority = this Master Bible. Implementation authority = current/bundled app.py. Performance authority = official user-run Streamlit ZIP. Discrepancyがあればauditし、戦略ルールを結果に合わせて変更しない。

Next phase

Parameter searchではなく、(1) 2026-08以降のforward OOS monitoring、(2) GLD優位のmechanism audit、(3) Production replacement governanceの明文化を優先する。

Master Research Bible v1.6 追補

Byakko / VIX Research Addendum - R114～R128

更新日: 2026-09-12 / DEV cutoff: 2026-07-31

本追補は Master Research Bible v1.5 のR113までの正本に、ユーザー実行済み公式Streamlit ZIPで確定したR114～R128の白虎/VIX研究を追加する。性能値の正本は各公式ZIP、実装の正本は対応するapp.py、文章仕様の正本は本Master Research Bibleとする。

最終結論

Production 4FのVIX sleeveは変更しない。現行Frozen VIX (Old15 25% + Old20 25% + New20 50%)を維持する。Pure OLDでは14DをFrozen benchmark、Pure NEWでは47Dをsingle benchmarkとし、NEW 47D/20M_RAW 50:50 Equal2Pは白虎単体のFrozen Challengerとして保存する。ただしR127で4F Productionへの置換はREJECTED。R128で、現行Frozen VIXが他3神stress時に低相関・高rescue contributionを持つことが確認され、Production維持のmechanismまで説明された。

研究ガバナンス

R114～R128の結果は、ユーザーがStreamlit上で実行して返したZIPのみを公式研究結果とする。ローカルdebugやpy_compileは研究証拠ではない。R122以降もDEV cutoff 2026-07-31を維持し、2026-08以降はforward OOSとして扱う。R125以降、47D/20Mのweightやlookbackを再探索しない。

R114～R128 研究系譜

Run	目的	正式判定
R114	OLD/NEW VIX 1D-60D landscape	OLD $\approx$ 14D、NEW $\approx$ 47-51Dの異なる地形を確認
R115	Pure OLD 179-series landscape	OLD dominant short peak=14D。49D/20-23Mは副峰
R116	OLD fixed peaks 14D/49D/21M robustness	14D standalone優位、slowは低agreementで別情報
R117	Fast/Slow disagreement mechanism	risk-on復帰ではFast優位、risk-off継続ではslow情報が非対称
R118	AsymConfirm architecture	Full改善するがrobustness不足。research challenger
R119	15 intervention months event audit	5 episodes。OnsetよりContinuationでedge集中
R120	risk-off forward path v1	counterfactual観測0件でINCONCLUSIVE
R120-P2	counterfactual return修復	Slow vote 0/1/2でforward edgeが段階的
R121	ordinal information audit	SUPPORTED ORDINAL MECHANISM、LOEO 21/21
R122	OLD final architecture falsification	Fast14 frozen benchmark、R118はchallengerのまま
R123	Pure NEW 179-series landscape	47Dと20-23Mの二峰構造
R124	NEW 47D vs 20M independence	holding agreement $\approx$ 52%、Equal2Pが強い分散効果
R125	Equal2P 50:50 falsification	12/13 gates PASS。Frozen Temporal Challenger
R126	Byakko standalone championship	NEW Equal2Pが白虎単体champion
R127	4F VIX replacement H2H	置換REJECTED: 3/10 gates PASS
R128	portfolio role attribution	Frozen VIXのcrisis diversification / rescue roleを確認

Pure OLD VIX - R115～R122

固定mapping: VIX absolute momentum >= matched RF → SPXL、otherwise → GLD。Quarterly evaluationはJan/Apr/Jul/Oct。R115は1D-60D、1M-60M RAW、2M-60M EX1の179系列。

R115: temporal landscape

ALL179 common eraは2008-11～2026-07の213か月。主な地形はShort 12-15D、Intermediate 47-51D、Long 20-23M。14Dはcommon-era CAGR/Sharpe/Calmarのwinnerで、Pure OLDのdominant standalone peakと判定。RAW vs EX1に普遍的優位はなかった。

Series	CAGR	Sharpe	MaxDD	Calmar
14D	38.70%	1.147	-32.01%	1.209
49D	30.85%	0.927	-41.77%	0.739
21M_RAW	33.92%	0.991	-46.92%	0.723

R116-R117: slow peaksの意味

14D vs 49Dのquarterly signal agreementは約63%、14D vs 21Mは約59%で、slow peaksは単なる14Dのコピーではなかった。disagreement auditでは、14D=GLD / Slow=SPXLの局面でslow側が相対的に強く、14D=SPXL / Slow=GLDではslow側が大きく劣後。これによりFast14はshock detector、slowはrisk-offのpersistence/staleness情報という仮説が形成された。

OLD Asymmetric mechanism - R118〜R122

R118 AsymConfirmの固定ルール

Fast14=SPXLなら即SPXL。Fast14=GLDなら、49Dまたは21M_RAWの少なくとも1本もGLDならGLD。Fast14=GLDでも両slowがSPXLならSPXLを維持する。R117後に1つだけ事前固定し、その後ルールは変更していない。

Strategy	CAGR	Sharpe	MaxDD	Calmar
Fast14	38.70%	1.147	-32.01%	1.209
Sym Majority3P	39.38%	1.109	-32.01%	1.230
AsymConfirm	43.78%	1.186	-33.15%	1.321
Equal3P context	36.38%	1.168	-31.81%	1.144

R119-R121 mechanism evidence

R119では15介入月が5 episode（2016-01〜03、2019-07〜09、2020-04〜06、2020-10〜12、2025-04〜06）に分解された。利益はOnsetではなく Continuationに集中。R120-PATCH2ではFast14=GLDの全38 quarterly anchorを母集団にcounterfactual SPXL/GLD returnを213/213月で復元した。

Slow SPXL votes	Anchors	3M mean SPXL-GLD edge	Median	SPXL win rate
0 (GLD/GLD)	16	-11.06%	-1.08%	43.8%
1 (Mixed)	17	+4.55%	+8.32%	64.7%
2 (SPXL/SPXL)	5	+16.06%	+25.02%	60.0%

R121ではepisode-balancedでも0<1<2を維持し、21 risk-off episodesを1つずつ除外するLOEOで21/21すべてordinal slope/rank relationが正。したがってSUPPORTED ORDINAL MECHANISM。ただし2票群はn=5でsmall-N。

R122 final decision

AsymConfirmはFullでは改善したが、5-block Sharpe、Rolling36/60 Sharpe、Worst Rolling60 Δ SharpeなどがFAILし、最終10 gates中6 PASS。よってPure OLD Frozen benchmark=14D、R118はmechanism-supported research challengerとしてforward OOSへ。DEV期間での追加調整は禁止。

Pure NEW VIX - R123～R125

固定mapping: VIX absolute momentum >= matched RF → TQQQ vs TECLのsame-lookback winner、otherwise → GLD。OLDとは混ぜず独立に179系列を探索。

R123 landscape

ALL179 common eraは2012-10～2026-07の166か月。NEWでは47D付近と20-23M付近の二峰構造。47Dはcommon-era CAGR/Sharpe winner、20M_RAWはCalmarに優れた。RAW/EX1の普遍的優位はなし。

R124 independence

47Dと20M_RAWのquarterly holding agreementは約51.8%、Cramér's $V \approx 0.252$ 。約半分で異なるholdingとなり、regimeごとに優劣が交代。固定50:50 Equal2Pは探索用weightではなく、時間分散の記述的controlとして事前固定。

Strategy	CAGR	Sharpe	Sortino	MaxDD	Calmar
47D	51.34%	1.138	2.123*	-59.85%	0.858
20M_RAW	50.01%	1.137	2.526*	-33.46%	1.495
Equal2P 50:50	52.98%	1.242	3.003*	-29.91%	1.772

* SortinoはR125の最終falsification出力。R124のSortino計算値とは実装差があるため、最終比較ではR125値を採用。

R125 falsification

Equal2Pは13 gates中12 PASS。唯一のFAILは47Dに対する5 chronological blocksのSharpe勝利が2/5だった点。Rolling/CPCV/LOBOは概ね良好。したがってPure NEW single benchmark=47D、Frozen Temporal-Diversification Challenger=47D/20M_RAW 50:50。60:40等のweight探索は禁止。

R126 - Byakko Standalone Championship

OLD14、現行Frozen VIX、NEW47、NEW Equal2Pを同一166か月で比較。

Byakko architecture	CAGR	Sharpe	Sortino	MaxDD	Calmar
OLD14	35.02%	1.146	1.388	-36.57%	0.958
Frozen VIX	43.64%	1.092	1.535	-39.36%	1.109
NEW47	51.34%	1.138	1.556	-59.85%	0.858
NEW Equal2P	52.98%	1.242	1.976	-29.91%	1.772

白虎単体ではNEW Equal2Pが明確なchampion。現行Frozenに対してCAGR約+9.35pt、Sharpe約+0.15、MaxDD約9.45pt改善。NEW47単独は高CAGRだがMaxDD -59.85%で、20M_RAWとの時間分散が47Dのregime failureを大きく緩和した。

ただしR126は白虎単体の比較であり、Production 4Fへの昇格を意味しない。4Fでは他familyとの相関・損失タイミングが重要なため、R127でVIX sleeveだけを置換して直接検証した。

R127 - Production 4F VIX Replacement Head-to-Head

MOM / MR / TEL-DMと各Family 25%を完全固定し、VIX family 25%だけを現行Frozen VIXからNEW Equal2Pへ交換。

4F	CAGR	Sharpe	Sortino	MaxDD	Calmar	Worst Month
Production 4F	39.58%	1.534	3.941	-19.48%	2.031	-13.30%
4F + NEW Equal2P	40.96%	1.519	3.818	-22.74%	1.801	-15.06%

CAGRは+1.39pt上がったが、Sharpe/Sortino/MaxDD/Calmar/Worst Monthはすべて悪化。最終gateは3/10 PASS。5-block Sharpe win=1/5、Rolling36 Sharpe win=40.5%、Rolling60=37.4%、CPCV=40.0%。したがってProduction replacementはREJECTED。

重要なのは、NEW Equal2P自体が弱いのではないこと。白虎単体ではchampionだが、4F内で白虎に要求されるportfolio roleを現行Frozenほど満たさなかった。

R128 - Frozen VIX Portfolio Role Attribution

R127の逆転理由を、新しい戦略を作らずfamily monthly panelで監査。Other3=MOM/MR/TEL-DM合成が悪い月ほど、Frozen VIXとNEW Equal2Pのconditional contributionを比較した。

Condition	Frozen VIX	NEW Equal2P	Frozen advantage
All 166 months	3.80%	4.22%	-0.42pt
Other3 < 0	-0.56%	-1.50%	+0.94pt
Other3 bottom20%	-1.54%	-2.64%	+1.09pt
Other3 bottom10%	-4.32%	-7.60%	+3.28pt
Production worst10	-7.16%	-9.87%	+2.71pt

通常時はNEW Equal2Pが平均returnで上回るが、Other3 stressが深くなるほどFrozen VIXの相対優位が拡大。Other3との相関も全期間でFrozen 0.535 vs NEW 0.637、Other3 bottom20%ではFrozen 0.096 vs NEW 0.333。Frozenはstress時に他3神からより独立する。

特にTEL-DM bottom10%ではFrozen VIX +3.04%に対しNEW Equal2P -0.03%。bottom20%のTEL-DM相関はFrozen -0.094、NEW +0.135。現行Frozen VIXにはcrisis diversification / rescue roleがあり、これがR127で単体championが4Fでは負けた主要mechanismと解釈される。

最終判定

Production VIX = 現行Frozenを維持。 NEW Equal2Pは白虎単体Frozen Challengerとして保存するが、4F Production replacementはREJECTED。 Frozen/NEWの混合weight探索など、R127を見た後の再最適化は行わない。

v1.6 Frozen Registry - 白虎/VIX

項目	v1.6 正式状態
Production 4F VIX	Frozen VIX = Old15 25% + Old20 25% + New20 50%（変更なし）
Pure OLD benchmark	OLD14D
OLD mechanism	Fast14=shock detector; 49D/21M=staleness/persistence information
OLD challenger	R118 AsymConfirm; mechanism-supported, architecture robustness不足; forward OOS only
Pure NEW benchmark	NEW47D
NEW temporal challenger	NEW Equal2P = 47D 50% + 20M_RAW 50%; frozen weight
NEW standalone status	R126 standalone champion
4F replacement status	R127 REJECTED
Frozen VIX portfolio role	R128 crisis diversification / rescue role supported
DEV cutoff	2026-07-31
Forward OOS	2026-08 onward; no DEV retuning

研究上の一般原則

R114～R128から得られた最重要な一般則は、standalone sleeve optimization ≠ portfolio optimization である。候補sleeveはStandalone performance → robustness → mechanism → portfolio-level replacement → stress-conditional contributionまで通して初めてProduction replacementを判断する。

NEW Equal2Pは単体ではFrozen VIXを大幅に上回ったが、4Fでは他3神stress時の補完性を失い、MaxDD/Calmarを悪化させた。したがって4Fの各神は「単体で最強」ではなく、「他3神と組み合わせた時に固有の役割を持つ」ことが重要。今後のFamily研究でもこのR114～R128プロトコルを標準とする。

次候補

新しいparameter searchではなく、MOM/MR/TEL-DM/VIXの4 familyについて、各familyを除いたOther3 stress時のrescue contributionを同一尺度で比較する4F Family Role

Auditが自然な次段階。ただし本v1.6時点では未実施であり、正式結果には含めない。

Master Research Bible v1.7 追補

Byakko / VIX Rebalance Timing & State-Persistence Research - R130～R134

更新日: 2026-09-12 / DEV cutoff: 2026-07-31 (研究ガバナンス上の正本)

本追補は v1.6 の R114～R128 白虎/VIX研究を継承し、ユーザーが Streamlit 上で実行して返した公式ZIP R130～R134を正本として追加する。R129は既存Research Registry上の「4F Family Role Audit」として予約されたままであり、本系列では上書きしない。R130～R134は白虎単体のrebalance timing / mechanism研究であり、Production 4Fのルール変更を目的としない。

最終結論: Production Frozen VIXは変更しない。現行のQuarterly P1 (Jan/Apr/Jul/Oct) を維持する。R130～R134は、Quarterly P1の優位が単純なcalendar anomalyではなく、quarter途中のVIXノイズに反応したrisk-on→defensive switchのwhipsawを避けるstate-persistence mechanismで相当程度説明できることを示した。

38. R130 - Pure Byakko 4-Architecture Rebalance Cadence Audit

固定architectureは OLD14D、現行Frozen VIX、NEW47D、NEW Equal2P。lookback・asset mapping・内部weightを固定し、Weekly / Monthly / 2M / Quarterly / 6Mのみ比較した。2M/Quarterly/6Mは全calendar phaseを監査し、best phaseのex-post採用は禁止。

Architecture	Weekly CAGR	Monthly CAGR	Quarterly P1 CAGR	Q-P1 Sharpe	Q-P1 Sortino	Q-P1 MaxDD
OLD14D	22.86%	29.74%	35.02%	1.146	1.388	-36.57%
Frozen VIX	18.51%	32.62%	43.69%	1.097	1.524	-39.32%
NEW47D	21.34%	34.05%	51.34%	1.138	1.556	-59.85%
NEW Equal2P	28.20%	31.97%	49.19%	1.170	1.843	-29.90%

4 architectureすべてでWeekly/MonthlyがQuarterly P1を上回るという単純な高頻度更新優位は否定された。一方、Quarterly内のphase差が大きく、次のmechanism auditへ進んだ。

39. R131 - Quarterly Phase Mechanism Audit

Architecture	P1 CAGR	P2 CAGR	P3 CAGR	LOYO P1勝利	LOQO active>0
OLD14D	35.02%	25.99%	10.65%	15/15	56/56
Frozen VIX	43.69%	22.54%	19.91%	15/15	56/56
NEW47D	51.34%	12.39%	22.28%	15/15	56/56
NEW Equal2P	49.19%	20.29%	27.70%	15/15	56/56

R130公式Quarterly系列とのidentity bridgeは全12 phase seriesで correlation=1.0、誤差は約1e-16。1年ずつ除外してもP1はSharpe/CAGRでP2・P3双方に15/15勝利し、1 quarterずつ除外してもP1 active meanは56/56で正。したがってP1優位は単一年・単一quarterだけでは説明できない。

VIX最大上昇がquarter第3月に来る場合の平均P1 activeは Frozen VIX +14.11%、NEW47D +11.90%、NEW Equal2P +10.51%と大きかった。単純なGLD exposure量だけでは説明できず、state固定時点の違いが重要と判断。

40. R132 - Quarterly Disagreement / Event Audit

Architecture	P1 active寄与 M1	M2	M3	P1 unique N	P1 unique mean	P1 unique win
OLD14D	48.7%	30.4%	20.8%	11	+13.96%	63.6%
Frozen VIX	58.2%	24.7%	17.1%	10	+11.76%	50.0%
NEW47D	35.3%	28.5%	36.3%	17	+15.61%	76.5%
NEW Equal2P	32.9%	21.5%	45.6%	14	+9.45%	78.6%

OLD/FrozenではP1優位の主寄与はM1の早期更新効果だったが、NEW47D/Equal2PではM3にも大きな寄与が残った。したがってP1 advantageは「early rebalance」だけでなく「signal-state selection / persistence」も含む。

P1だけmore risk-onだったquarterでは、Frozen VIX平均+21.06%、NEW47D +17.63%（勝率83.3%）、NEW Equal2P +14.99%（10/10勝利）。quarter途中のVIX上昇に対してP2/P3がdefensiveへ寄り、P1がrisk-onを維持する局面が主要mechanism候補となった。

41. R133 - Quarterly State Persistence / Whipsaw Audit

switch edge = 「switch後holdingの実現return - 旧holdingを維持した反実仮想return」。21営業日、42営業日、次回同phase更新までを事前固定窓として評価。SwitchEdge<0をwhipsawと定義。

Architecture	P1 MeanEdgeFull	Late(P2/P3)	P1-Late	Late risk-on→defensive	P1 defensive→risk-on
OLD14D	+7.96%	-0.60%	+8.56pt	-7.47%	+11.96%
Frozen VIX	+7.72%	-0.69%	+8.41pt	-8.57%	+18.18%
NEW47D	+9.97%	-1.88%	+11.85pt	-15.38%	+29.38%
NEW Equal2P	+7.49%	-0.49%	+7.98pt	-6.31%	+19.57%

4 architectureすべてでP1 switchは正、P2/P3平均はほぼゼロ～負。特にLate risk-on→defensiveが一貫して負で、Frozen VIXではP2/P3の同transitionのwhipsaw率は概ね65%前後、NEW47Dではさらに強い。一方、P1 defensive→risk-onは大きな正edge。白虎のswitch自体が悪いのではなく、方向とタイミングに非対称性がある。

42. R134 - Temporal Mechanism Robustness Audit

Architecture	P1-Late all DEV	Late R→D edge	LOYO符号維持	LOQO符号維持	Bootstrap P(P1-Late>0)
OLD14D	+8.56pt	-7.47%	15/15	56/56	94.44%
Frozen VIX	+8.41pt	-8.57%	15/15	56/56	96.24%
NEW47D	+11.85pt	-15.38%	15/15	56/56	96.28%
NEW Equal2P	+7.98pt	-6.31%	15/15	56/56	96.34%

LOYOでは全architectureで15/15、LOQOでは56/56で P1-Late>0、Late risk-on→defensive edge<0、Late risk-on→defensive whipsaw率>50%、P1 defensive→risk-on edge>0 が維持された。

COVID除外でP1-Late効果は縮小するが符号は維持: OLD14D +4.34pt、Frozen +3.39pt、NEW47D +4.73pt、Equal2P +2.35pt。したがってCOVIDが効果を増幅したが、COVID単独起源ではない。

Bootstrap 95% CIではP1-Late差は4本ともわずかに0を跨ぐ。一方Late risk-on→defensiveはOLD14D/Frozen/NEW47Dで95% CIが完全に負、Equal2Pのみ上限がわずかに正。したがって最も強いmechanistic statementは「P1が魔法のcalendar phase」ではなく「遅いphaseでrisk-on→defensiveへ反応することに負edgeがある」。

43. R130～R134 統合判定 / Governance Freeze

SUPPORTED: Quarterly P1の優位は複数architecture、LOYO、LOQOで再現し、state-persistence / anti-whipsaw mechanismと整合する。

PARTIALLY SUPPORTED: P1優位の全量を単一mechanismで説明できるわけではない。era別では2012-2015にP1-Lateが負のarchitectureもあり、普遍的calendar anomalyとは扱わない。

REJECT / DO NOT DO: R130～R134を見た後でrisk-on→defensive delayの閾値・日数・phaseをDEV内再探索してProductionへ昇格させない。これはpost-hoc optimizationになる。

Production: Frozen 4F EWを維持。VIX sleeveは現行Frozen VIX、Quarterly P1（Jan/Apr/Jul/Oct）のまま。R130～R134はProduction変更ではなく、現行timingを支持するmechanism evidenceとしてFreezeする。

注記: R134 ZIPのexactness表にはDEV cutoffが2026-07-01と表示されるが、親R133 event ledgerを449行完全保持し、研究ガバナンス上のDEV cutoff正本は2026-07-31。本Masterでは2026-07-31を正本とし、次回実装時に表示表記を統一する。

44. Bam先生 Source / Comment Registry（将来参照用）

目的: Bam先生の公開コメント・ユーザー共有コメントを、将来の研究で再利用できるauthoritative priorとして保持する。ただし公開されていないexact ruleを推測で事実化しない。Master v1.6のevidence hierarchyどおり、authoritative comment、published series、My reconstructionを分離する。

44.1 Four Gods lineage / architecture

記録済み authoritative lineage: 青龍←DM2、朱雀←DM3、白虎←DM6、玄武←DM7。特徴の異なる低相関DMを理論ベースで再構築してシン四神を作り、それらを素材にルールベースの動的portfolioへ組む、という思想。

関連思想: 青龍/朱雀/白虎をoffense、玄武をshieldとして分離し、offense (spear) 選択とspear-vs-shield allocationを別layerで扱う。

44.2 DM2 / multi-lookback comments

ユーザー共有のBam先生コメントとして保存:

DM2はmulti-lookbackを基本使用。12Mは強く、4~5Mに別のpeak、10Dにも情報価値がある。複数期間は単純な等分asset portfolioだけを意味せず、momentum signalをscalar化して合成し柔軟にon/offする考え方がある。absolute momentumとrebalance timingも結果を変える。

別のユーザー共有コメントとして保存: CPCV約1,000件で見た傾向を約5万件へ拡充しても同様に、「3期間 > 2期間 > 1期間」という傾向が観察された。異なるlookbackを組み合わせることで同一銘柄でも相関が下がり、時間軸の違いをportfolio化することが『真の時間的分散』になる、という説明。

44.3 このコメントとMy検証の関係

My研究では、10D / 4-5M / 10-12Mの複数peak地形とlookback間の不完全相関は確認され、temporal diversification mechanism自体は支持された。一方、candidate-count調整・OOS・falsification後にはgenericな『3P>2P>1P』の普遍則は確認されず、Long controlのreplacementには至らなかった。したがってBam先生コメントは重要なpriorだが、My strategyへの自動的なpromotion根拠にはしない。

44.4 Open Bam reverse-engineering gaps

OPEN-HIGH: exact DM2 scalar/operator、exact GS architecture。OPEN-MEDIUM: Tier A/B/C/D、exact DM2とabsolute momentum / rebalance timingのinteraction。新しい一次情報が得られた場合は、このRegistryへ原文・出典・日付・適用可能な研究仮説を追記する。

44.5 運用ルール

今後Bam先生の新コメントが共有されたら、(1)可能な限り原文、(2)共有日/出典、(3)事実・解釈・仮説の区分、(4)既存Runとの関係、(5)新規検証を許す一次情報か、をMasterへ追記する。既存Rejected branchをコメントだけで再開せず、exact ruleなど本当に新しいprimary informationがある場合のみpre-specified studyとして開く。

Master v1.7 end state: Production=Frozen 4F EW。Byakko/VIX Production=Frozen VIX Quarterly P1。R130~R134 timing branch=FROZEN MECHANISM FINDING。R129 Family Role Audit=RESERVED / UNEXECUTED。Bam Comment Registry=ACTIVE REFERENCE。

Master Research Bible v1.8 追補

Byakko / VIX Layer 0 Defensive-Destination Research - R135～R138

更新日: 2026-09-12 / DEV cutoff: 2026-07-31

本追補はMaster v1.7を完全継承し、ユーザーがStreamlit上で実行して返した公式ZIP

R135～R138を正本として追加する。Productionルールは変更しない。R135～R138は、白虎単体Fの内部構造、すなわちLayer 0におけるdefensive / risk-off時の退避先研究である。

重要なLayer定義: Layer 0 = 各単体F（青龍・朱雀・白虎・玄武）の内部戦略。Layer 1 = 4つのFを組み合わせるFoF / 4F。したがってR135～R138はLayer 1研究ではない。今後はこの定義を正本として統一する。

研究順序の更新: Layer 0研究を各Fについて可能な限り完了・Freezeした後に、Layer 1（4F / FoF）の統合・配分・rescue role研究へ進む。従って、R138直後に予定していた4F role auditは保留可能であり、Layer 0完了後に実施する。R129の予約状態も変更しない。

v1.8時点の白虎Layer 0結論: Production Frozen VIXのGLD固定は維持。R138の「Frozen VIX + Quarterly P1 + defensive時 GLD/XLU 126D relative-momentum Top1」をSUPPORTED / PROMISING CHALLENGERとして凍結する。TLT branchはREJECT。

45. R135 - Fixed Defensive Destination Audit

目的は、白虎本体のVIX判定、Quarterly P1、risk-on側、lookback、内部weightを変えず、defensive destinationだけをGLD（control） / TLT / XLUへ固定置換して比較すること。対象architectureはOLD14D、Frozen VIX、NEW47D、NEW Equal2P。Exactness 8/8 PASS。

Architecture	GLD CAGR	TLT CAGR	XLU CAGR	GLD Sortino	TLT	XLU
OLD14D	35.70%	26.16%	30.34%	1.400	1.135	1.220
Frozen VIX	44.55%	38.91%	43.55%	1.495	1.379	1.450
NEW47D	52.24%	43.98%	45.45%	1.462	1.330	1.342
NEW Equal2P	50.06%	42.40%	46.55%	1.521	1.386	1.442

4 architecture × CAGR/Sharpe/Sortinoの全比較でGLDが首位。固定退避先としてTLT・XLUへ単純置換する仮説はREJECT。GLD固定をcontrolとしてRETAINする。

ただし固定eraではregime dependenceが確認された。例えばFrozen VIXの2012-2015 CAGRはGLD 2.07%、TLT 13.86%、XLU 17.32%。一方で2020-2023はGLD 82.48%に対しTLT 77.38%、XLU 67.35%。この非一様性が、動的destination selectorを検証する根拠となった。

R135判定: GLD fixed = RETAIN / CONTROL。TLT fixed = REJECT。XLU fixed = REJECT。Dynamic defensive destination = TEST WARRANTED。

46. R136 - Relative-Momentum Defensive Selector

GLD/TLT/XLUの中からrelative momentum Top1を選択。Primaryは事前固定6M = 126 trading days。3M/12Mはsensitivityのみ。Absolute momentum gate、Top2、weight最適化、lookback grid searchは行わない。Exactness 10/10 PASS。

Architecture	GLD固定 CAGR	6M Top1 CAGR	Δ CAGR	Δ Sharpe	Δ Sortino
OLD14D	35.70%	31.90%	-3.80pt	-0.089	-0.107
Frozen VIX	44.55%	46.78%	+2.23pt	+0.045	+0.060
NEW47D	52.24%	50.22%	-2.02pt	-0.032	-0.026
NEW Equal2P	50.06%	50.58%	+0.52pt	+0.002	+0.016

Frozen VIXのみ明確に改善し、NEW Equal2Pはほぼ中立、OLD14D/NEW47Dは悪化。従って「白虎一般では3資産relative-momentum退避が優位」という普遍則は確認されない。一方、Frozen VIX専用Challengerとしてmechanism auditへ進む価値が生じた。

R136判定: GLD/TLT/XLU 126D Top1 = Frozen VIXではSUPPORTED CHALLENGER、白虎一般ではNOT CONFIRMED。Production変更なし。

47. R137 - Defensive Selector Mechanism / Asset Attribution Audit

R136 Primary 126Dを固定したまま、Frozenだけ改善した理由をasset attribution、quarter/year attribution、leave-one-outで分解。新しいparameter探索はゼロ。Exactness 10/10 PASS。

Architecture	Selector edge	TLT contribution	XLU contribution
OLD14D	-32.34%	-24.08%	-10.88%
Frozen VIX	+23.17%	-8.56%	+34.69%
NEW47D	-16.85%	-15.37%	-1.75%
NEW Equal2P	+4.68%	-7.79%	+13.52%

Frozen改善の主因はTLTではなくXLU。TLT contributionは4 architectureすべてnegative。FrozenではXLU +34.69%に対しTLT -8.56%。従って3資産selectorの改善を「TLTを含む分散効果」と解釈してはいけない。

ただしR136 Frozenのpositive edgeには2013Q2 +26.55%という大きな成功quarterへの依存があった。最大quarterを除くLOQではGLD比 -2.67%、2013年を除くLOYOでは -5.04%。従って、この時点ではProduction昇格に十分なrobustnessではない。

R137判定: Mechanism CONFIRMED。TLT branch = REJECT。新しいpost-R137仮説として「GLD/XLU 126D Top1」を独立falsificationする根拠あり。Production変更なし。

48. R138 - GLD / XLU Two-Asset Selector Falsification

R137で得た新しい一次的mechanism evidenceを受け、Frozen VIXのみ、Quarterly P1固定、126D固定、GLD vs XLU relative-momentum Top1を事前仕様として検証。新規optimizationなし。Exactness 11/11 PASS。

Series	CAGR	Sharpe	Sortino	MaxDD	Calmar
GLD control	44.55%	1.192	1.495	-49.66%	0.897
R136 3-asset 6M	46.78%	1.237	1.555	-49.66%	0.942
R138 GLD/XLU 6M	47.66%	1.251	1.572	-49.66%	0.960

GLD固定比でCAGR +3.11pt、Sharpe +0.059、Sortino +0.077。R136 3資産版に対してもCAGR +0.89pt、Sharpe +0.014、Sortino +0.017。TLTを削除して単純化したにもかかわらず、performanceとrisk-adjusted performanceがともに改善した。

Robustness	R136 3-asset	R138 GLD/XLU
LOQ positive rate	55/56 = 98.2%	56/56 = 100%
LOYO positive rate	14/15 = 93.3%	15/15 = 100%
LOQ min remaining edge	-2.67%	+5.87%
LOYO min remaining edge	-5.04%	+1.75%

R136最大の弱点だった2013依存が大幅に改善。2013Q2を除いても+5.87%、2013年を丸ごと除いても+1.75%のGLD比edgeを維持した。R136でTLTが選ばれた10 signalは、R138では5回GLD、5回XLUへ置換された。

49. R138 Robustness / Falsification Detail

R138は強いが、万能ではない。stress/eraごとの反証材料を保持する。

Stress episode	R138 active total return vs GLD
2015-16 stress	+2.26%
2018Q4	-6.05%
COVID shock/rebound	0.00%
2022 bear/rates	+2.40%

2018Q4ではGLD固定が明確に優位であり、TLT除外後も問題は残った。これはGLDに留まるべき局面でXLUを選ぶselector固有の失敗として保存する。

Era	GLD CAGR	R138 CAGR	Interpretation
2012-2015	2.07%	17.44%	大幅改善
2016-2019	48.41%	42.49%	GLD固定に劣後
2020-2023	82.48%	83.68%	小幅改善
2024-DEV	48.33%	47.44%	GLD固定に小幅劣後

従って「GLD/XLU

selectorは全eraでGLD固定より優れる」という主張はしない。支持されるのは、全DEVでのheadline改善と leave-one-quarter/year robustness、そしてTLT削除による単純化・頑健性改善である。

R138正式判定: GLD/XLU 126D Top1 = SUPPORTED / PROMISING CHALLENGER。Production Frozen
VIXのGLD固定はまだ変更しない。

50. R135～R138 統合結論 - 白虎 Layer 0 Defensive Destination

Branch	Status	Reason
GLD fixed	PRODUCTION / CONTROL	全固定destination比較で最良。現行Productionを維持。
TLT fixed	REJECT	固定置換で悪化。
XLU fixed	REJECT	固定置換で悪化。
GLD/TLT/XLU 126D Top1	NOT PROMOTED	Frozenでは改善するがTLT contributionは一貫してnegative。
GLD/XLU 126D Top1	SUPPORTED CHALLENGER	R138でheadline・LOQ・LOYO robustness改善。

Frozen Challenger仕様: 白虎本体はFrozen VIX、rebalance timingはQuarterly P1のまま。defensive状態でのみ、signal close時点の過去126 trading-day returnをGLDとXLUで比較し、Mom_XLU > Mom_GLDならXLU、それ以外はGLD。その他の白虎内部ロジックは変更しない。

Do not retune: R138を見た後に3M/9M/12M、threshold、GLD/XLU weight、Top2、absolute momentum gate、追加ETF等をDEV内探索してChallengerを改変しない。126D / GLD / XLU / Top1を現時点のFrozen Challenger仕様として保持する。

51. Layer Governance Update - Layer 0 → Layer 1

本研究系列の途中でLayer用語の曖昧さが判明したため、v1.8で明示的に正規化する。

Layer	正本定義	例
Layer 0	単体F内部のstrategy construction	青龍、朱雀、白虎、玄武それぞれのsignal / asset / timing / defensive logic
Layer 1	完成した複数Fを組み合わせたFoF / 4F	青龍+朱雀+白虎+玄武、EWまたは将来のrule-based allocation
Layer 2+	必要な場合のみ定義する上位portfolio layer	現時点では未確定。推測で仕様化しない。

従ってR135～R138はすべてLayer 0 / 白虎内部研究である。『defensive側をLayer 1と呼ぶ』表現は誤りであり、本Masterでは採用しない。

研究順序: まずLayer 0で各単体Fの研究を完了し、Production / Frozen Challenger / Rejectedを整理する。その後にLayer 1へ進み、完成したF同士の相関、rescue role、stress contribution、weighting、dynamic allocation等を研究する。

R128など過去に実施済みの4F replacement / rescue auditは既存Layer 1 evidenceとして保持するが、今後の新規研究順序をLayer 0優先へ変更する。R138直後に想定していた4F role auditは、Layer 0完了後まで延期してよい。

R129: Research Registry上の「4F Family Role Audit」 予約は維持し、R135～R138で上書きしない。実行時期はLayer 0 completion後に再評価する。

52. v1.8 End State / Frozen Registry

項目	v1.8 正式状態
Production 4F	Frozen 4F EW = MOM/MR/VIX/TEL-DM 各25%（変更なし）
Byakko Production	Frozen VIX + Quarterly P1 + defensive destination GLD fixed
Byakko timing mechanism	R130～R134 = FROZEN MECHANISM FINDING
Byakko defensive destination	R135～R138 = Layer 0 research
TLT branch	REJECT
R138 challenger	Frozen VIX / Quarterly P1 / GLD-vs-XLU / 126D / Top1
R138 status	SUPPORTED / PROMISING CHALLENGER; Production未昇格
DEV cutoff	2026-07-31
Forward OOS	2026-08 onward; DEV retuning禁止
Layer research order	Layer 0 completion → Layer 1 FoF/4F
R129	RESERVED / UNEXECUTED

次段階では、白虎Layer 0にまだ事前に定義された未完了研究がある場合はそれを優先する。他Fも含めLayer 0を完成・Freezeした後、Layer 1で4F/FoFとしての統合検証へ進む。R138の4F組込みは、それまで保留しても研究上問題ない。

Master v1.8 end state: Productionは不変。R138は白虎Layer 0のFrozen Challenger。Layer 1研究はLayer 0完了後に実施する。

Master Research Bible v1.9 追補

Byakko / VIX Layer 0 Completion - R139～R140

更新日: 2026-09-12 / DEV cutoff: 2026-07-31

本追補はMaster v1.8を完全継承し、修正版公式R139および公式R140を追加する。R140は白虎Layer 0の最終governance synthesisであり、新しいstrategy / lookback / threshold / weight / phase / asset探索を含まない。

最終結論: Byakko Layer 0 = FREEZE / COMPLETE FOR DEV。ProductionはFrozen VIXを維持。R138 GLD/XLU 126D Top1をPrimary new Layer-0 Challengerとして凍結。NEW Equal2PはSecondary Frozen Challenger、R118 AsymConfirmはmechanism-supported research challenger / forward OOS onlyとして保存する。

53. R139 - Frozen VIX Internal Sleeve Role Audit

目的はProduction Frozen VIX内部のOld15 / Old20 / New20の役割を、weight最適化を行わず診断すること。修正版R139はExactness 12/12 PASS。Production official seriesを正本としてidentityを固定し、sleeve再構成の微小なsource-precision driftは別監査とした。

Sleeve	CAGR	Sharpe	MaxDD	Formal role
Old15	38.07%	1.128	-44.58%	Diversifier / rescue
Old20	36.56%	1.088	-48.81%	Complementary stabilizer
New20	48.98%	1.125	-58.55%	Return / rebound engine
Production 25/25/50	44.55%	1.192	-49.66%	Integrated Frozen VIX

相関はOld15-Old20 daily 0.774、Old15-New20 0.694、Old20-New20 0.901。Old15が最も異質で、New20は最も高い単体CAGRと最深DDを持つ。

Target sleeve	Peer stress	Mean daily edge vs peer
Old15	bottom 1%	+3.42%
Old15	bottom 5%	+2.51%
Old20	bottom 1%	+1.68%
Old20	bottom 5%	+1.09%
New20	bottom 1%	-1.57%
New20	bottom 5%	-0.76%

Peer-stress auditはOld15のrescue/diversification roleを支持し、New20がstress rescueではなくgrowth/rebound側であることを支持する。

Removed sleeve	LOO CAGR	LOO Sharpe	LOO MaxDD	Interpretation
Old15	45.37%	1.138	-53.15%	CAGRは上がるがportfolio quality悪化
Old20	46.92%	1.202	-51.96%	incremental roleは最も弱く見えるが削除・再配分はしない
New20	38.21%	1.177	-44.58%	成長力を大きく失う

R139判定: Frozen VIX internal diversification mechanism = SUPPORTED。Old15 = diversifier/rescue、Old20 = complementary stabilizer、New20 = return/rebound engine。25/25/50には構造的意味がある。ただしR139はdiagnosticであり、結果を見てweightを再最適化しない。

54. R140 - Byakko Final Layer-0 Freeze Audit

R140はR118～R139の正式evidenceを統合する白虎Layer 0最終監査。公式R138と修正版R139を直接親とし、Master v1.8に保存されたR118～R134 evidence chainを継承。Exactness 12/12 PASS。DEV cutoff 2026-07-31、Optimization = NONE、ProductionChange = NONE automatically。

Candidate	Class	CAGR	Sharpe	MaxDD	Final status
Production Frozen VIX	PRODUCTION	44.55%	1.192	-49.66%	RETAIN PRODUCTION
R118 AsymConfirm OLD	RESEARCH CHALLENGER	43.78%	1.186	-33.15%	Mechanism-supported / OOS only
NEW Equal2P	FROZEN TEMPORAL CHALLENGER	-	-	-	Retain; not Production
R138 Frozen + GLD/XLU 126D	PRIMARY NEW CHALLENGER	47.66%	1.251	-49.66%	FREEZE / future OOS + Layer-1 test

R138はLayer 0ではProduction GLD controlを上回り、LOQ 56/56 = 100%、LOYO 15/15 = 100%、minimum remaining edgeはLOQ +5.87%、LOYO +1.75%。それでもProductionへ自動昇格しない。R138はまだ新しいLayer-1 / FoF role testを完了しておらず、Production Frozen VIXには既存のR128 rescue/complementarity evidenceがあるためである。

NEW Equal2Pはstandalone championとして保存するが、R127の4F replacementはREJECTEDであり、R128が失われたrescue complementarityを説明している。R118 AsymConfirmはmechanism evidenceを保持する一方、最終robustnessは10 gates中6 PASSであり、DEVでの追加tuningを行わない。

55. Formal Evidence Chain

Runs	Topic	Formal implication
R118-R122	OLD asymmetric architecture	Mechanism supported; final robustness insufficient. OOS only.
R125-R128	NEW Equal2P / portfolio role	Standalone optimization != portfolio optimization; Production rescue role retained.
R130-R134	Quarterly timing / whipsaw	Canonical Quarterly P1を維持。late P2/P3 R→D whipsaw mechanismをFreeze。
R135-R138	Defensive destination	GLD fixed control; TLT rejected; GLD/XLU 126D becomes frozen Layer-0 challenger.
R139	Internal sleeves	Old15 rescue, New20 growth, Old20 stabilizer; 25/25/50 remains frozen.
R140	Final governance	Byakko Layer 0 DEV research complete/frozen.

56. Branch Closure Registry

Branch	Decision	Reason
Production Frozen VIX	OPEN only for forward OOS / new primary info	Production retained; mechanism and internal role documented.
R118 AsymConfirm	NO MORE DEV TUNING	Mechanism supported; robustness insufficient.
NEW Equal2P	NO MORE DEV TUNING	Standalone champion; prior Layer-1 replacement rejected.
R138 GLD/XLU 126D	NO MORE DEV TUNING	Exact GLD/XLU / 126D / Top1 spec frozen.
TLT fixed / dynamic	CLOSE / REJECT	Weak fixed result; negative contribution; removal improved robustness.
XLU fixed	CLOSE / REJECT	Value exists conditionally, not as fixed replacement.
Rebalance cadence / phase	CLOSE	R130-R134 support canonical Quarterly P1.
Lookback / threshold / weight / ETF expansion	CLOSE FOR DEV	Would be post-hoc mining.
Layer 1 / 4F role test	DEFER	Begin after remaining family Layer 0 programs are complete.

57. Final Governance Decision / Byakko Layer 0 End State

Decision item	v1.9 formal decision
Byakko Layer 0	FREEZE / COMPLETE FOR DEV
Production	Frozen VIX retained
Primary new challenger	R138 GLD/XLU 126D Top1
Secondary challenger	NEW Equal2P
Mechanism research challenger	R118 AsymConfirm - forward OOS only
Production internal weights	Old15 25% / Old20 25% / New20 50% - frozen
Timing	Quarterly P1 Jan/Apr/Jul/Oct - frozen
DEV cutoff	2026-07-31
Forward OOS	2026-08 onward
Next research order	Finish remaining families at Layer 0 -> then Layer 1 FoF/4F
R129	Reserved / unexecuted; do not overwrite

Reopening rule: 白虎Layer 0をDEVで再開するのは、forward OOSの十分な蓄積、またはBam先生等からexact ruleのようなgenuinely new primary informationが入った場合に限定する。既存結果を見て新しいlookback・threshold・weight・phase・ETFを探索することは再開理由にしない。

Master v1.9 end state: 白虎/VIXはLayer 0研究を正式完了。Productionは不変。R138は凍結Challengerとして保存。次はLayer 1へ急がず、残る単体FのLayer 0研究を完了する。

Master Research Bible v1.10 追補

Seiryu / MOM Layer 0 Final Freeze - R141

更新日: 2026-09-12 / DEV cutoff: 2026-07-31

本追補はMaster v1.9を完全継承し、ユーザー実行の公式R141 ZIPによる青龍/MOM Layer 0最終監査を追加する。R141は既存の公式evidenceを統合するgovernance synthesisであり、新しいparameter、lookback、weight、threshold、asset、architecture探索を行わない。

R141 Exactness: 13/13 PASS。 最終判定は Seiryu Layer 0 = FREEZE / COMPLETE FOR DEV。 Production Frozen Operational 11Mを維持し、既存challenger群をforward OOSへ固定する。

58. R141 - Seiryu Final Layer-0 Freeze Audit

Build: R141-SEIRYU-FINAL-LAYER0-FREEZE-AUDIT-v1.0。ScopeはMaster Research Bibleの公式evidenceのみを使うLayer 0 final governance synthesis。DEV cutoffは2026-07-31、2026-08以降はforward OOS。

Exactness item	Observed	Result
DEV cutoff	2026-07-31	PASS
Production CAGR / Sharpe	28.03% / 1.266	PASS
R101 CAGR	30.45%	PASS
R108 CAGR / Sharpe	33.02% / 1.432	PASS
Balanced champion CAGR / Sharpe	34.54% / 1.484	PASS
Aggressive QLD CAGR	45.25%	PASS
TQQQ max-firepower CAGR	61.12%	PASS
3xMOM 12-1 CAGR	49.18%	PASS
No automatic promotion	Frozen Operational 11M retained	PASS
Layer 0 freeze	FREEZE / COMPLETE FOR DEV	PASS

公式R141では合計13 checksがすべてPASS。これにより、青龍Layer 0の終了判定とProduction不変更を同時に固定する。

58.1 Candidate Status Matrix

Strategy	Class	CAGR	Sharpe	MaxDD	Final status
Frozen Operational 11M	PRODUCTION	28.03%	1.266	-21.34%	RETAIN PRODUCTION
R101 3vs1 x Frozen	FROZEN CHALLENGER	30.45%	1.394	-21.34%	Forward OOS
R108 Full5P x Frozen ASH	PREDECLARED CHALLENGER	33.02%	1.432	-21.34%	Forward OOS
Full5P x QQQ/QLD 50:50 x GLD	POST-HOC CHAMPION	34.54%	1.484	-21.34%	FREEZE / Forward OOS
Full5P x QLD x GLD	AGGRESSIVE TIER	45.25%	1.431	-32.23%	Archive / Forward OOS
Full5P x TQQQ x GLD	MAX-FIREPOWER TIER	61.12%	1.297	-51.23%	Archive / Forward OOS
3xMOM 12-1	ARCHIVED HIGH-OCTANE	49.18%	1.077	-51.23%	SAVE; no standalone promotion

Balanced post-hoc championはFrozen Operational 11Mと同じMaxDD -21.34%の公式共通パネルでCAGR 34.54%、Sharpe 1.484、Sortino 3.014、Calmar 1.619。しかしpost-hoc discoveryであるためProductionへ自動昇格しない。R108は最強のpredeclared challengerだがArchitecture Gates 14/17であり、同様にProduction置換ではなくforward OOS対象とする。

59. Seiryu Formal Evidence Chain

Run range	Question	Formal conclusion	State
R41-R64	Ordinary MOM / lookback terrain	Long 10-12M structural; Frozen11M retained; Ultra/Mid not promoted	OFFICIAL / CLOSED
R65-R68	3xMOM branch	12-1 high-octane; extra CAGR largely leverage/beta; no standalone promotion	OFFICIAL / CLOSED
R75-R80	Peak-3P / temporal diversification	Temporal diversification real; generic 3P>2P>1P not confirmed	OFFICIAL / CLOSED
R89-R101	Pure MOM re-baselining / 3vs1	R101 3vs1 x Frozen = Frozen Challenger; Gate 8/9	OFFICIAL / FROZEN
R102-R106	Cross-leverage temporal mechanism	Long 11-13M robust; Full5P supports risk-on re-entry lag compensation	MECHANISM FROZEN
R107-R113	Full5P / implementation / defense / championship	R108 strongest predeclared; GLD fixed superior; balanced Full5P post-hoc champion	OFFICIAL / FROZEN
R141	Final Layer-0 governance	Production retained; challenger hierarchy frozen; DEV research complete	OFFICIAL / FINAL

時間分散そのものはmechanismとして支持される一方、generic 3P > 2P > 1Pは候補数補正/OOS後には普遍的に確認されていない。Full5Pの固定時間軸は10D / 54D / 6M_EX1 / 12M_EX1 / 24M_RAWであり、これをR141後に再最適化しない。

59.1 Branch Closure Registry

Branch	Decision	Governance reason
Frozen Operational 11M	KEEP	Production control
R101 3vs1 x Frozen	NO MORE DEV TUNING	Forward OOS only
R108 Full5P x Frozen ASH	NO MORE DEV TUNING	Forward OOS only
Full5P QQQ/QLD 50:50 GLD	NO MORE DEV TUNING	Post-hoc champion; exact spec frozen
3xMOM 12-1	CLOSE FOR DEV	Archived high-octane variant
Ultra/Mid satellite weighting	CLOSE FOR DEV	Incremental residual alpha not robust
1P/2P/3P generic search	CLOSE FOR DEV	Generic superiority not confirmed
Lookback fine sweep	CLOSE FOR DEV	No 54D->57D, 12M->13M, 24M->25M retuning
Vote weighting / threshold	CLOSE FOR DEV	Post-hoc mining risk
QQQ/QLD allocation ratio	CLOSE FOR DEV	50:50 challenger spec frozen
Additional defensive assets	CLOSE FOR DEV	BIL/IEF/XLU/GDX falsification complete

60. Open Primary-Information Gaps and Final Governance

以下は未解決だが、DEVで推測して埋めない。新しい一次情報が到来した場合のみ、事前固定された新研究として再開候補にできる。

Open question	Priority	Governance
Exact Bam DM2 scalar/operator	OPEN-HIGH	Do not infer; wait for exact primary rule
Exact GS architecture / DM2_GS construction	OPEN-HIGH	No further DEV reverse engineering without new primary info
Tier A/B/C/D exact definitions	OPEN-MEDIUM	Await exact primary definition
Exact DM2 x absolute momentum / rebalance timing	OPEN-MEDIUM	Await exact primary information

60.1 Final Decision

Decision item	Formal decision
Seiryu Layer 0	FREEZE / COMPLETE FOR DEV
Production	Frozen Operational 11M retained
Primary predeclared challenger	R108 Full5P x Frozen ASH - forward OOS
Post-hoc balanced champion	Full5P x QQQ/QLD 50:50 x GLD - forward OOS
Secondary challenger	R101 3vs1 x Frozen - forward OOS
Aggressive tiers	Full5P x QLD/TQQQ x GLD archived as separate risk budgets
Archived leverage variant	3xMOM 12-1 - do not promote standalone
Temporal mechanism	Full5P majority / risk-on re-entry lag compensation supported
DEV cutoff	2026-07-31
Forward OOS	2026-08 onward; no DEV retuning
Reopening rule	Only genuinely new primary information or sufficient forward-OOS evidence
Next research order	Finish remaining families at Layer 0, then Layer 1 FoF/4F

Integrated Layer governance after R141: Byakko/VIX Layer 0 and Seiryu/MOM Layer 0 are both formally FREEZE / COMPLETE FOR DEV. Existing Layer 1 evidence remains preserved, but new Layer 1 / FoF research remains deferred until the unfinished standalone families complete Layer 0.

Master v1.10 end state: Production strategy specifications remain unchanged. R141 is a governance close, not a strategy promotion. Seiryu's stronger backtest challengers are frozen exactly as documented and move only to forward OOS / future Layer-1 role testing.

Master Research Bible - My 4F / Four Gods / v3.45

Version 1.11 - Suzaku Layer 0 Completion Addendum (R142-R150)

基礎文書: v1.10 (2026-09-12)。本版はv1.10をそのまま保持し、公式user-run ZIPで確定した朱雀/TEL-DM R142-R150、179-series標準ガバナンス、最終FREEZE判断を追記する。

Source-of-truth rule: 正式な研究証拠はユーザーがStreamlitを実行して生成したofficial ZIPのみ。DEV cutoffは2026-07-31。2026-08以降はforward OOSであり、DEVパラメータ再調整には使用しない。Production 4Fは変更しない。

Layer taxonomy: TEL内部のTreasury Gate / Equity SelectorはそれぞれStage A / Stage Bと呼ぶ。研究taxonomy上は朱雀内部研究全体がLayer 0、4F/FoF結合がLayer 1である。

1. 新しい標準Lookback Coverage Governance

今後、広域lookback landscapeを行うLayer-0研究では、戦略固有の数学的理由がない限り、次の179-series coverageを標準benchmarkとする。

Family	Coverage	Series
Trading-day	1D-60D	60
Monthly RAW	1M-60M RAW	60
Monthly EX1	2M-60M EX1	59
Total		179

RAW: month-end close(t) / close(t-NM) - 1。EX1: 直近1Mを除外し、close(t-1M) / close(t-NM) - 1。R148ではこの定義をTELのStage AとStage Bの双方に同一適用した。

この標準は探索漏れを防ぐためのcoverage標準であり、fine tuningや任意のパラメータ探索を許可するものではない。すでにFREEZE済みで同等coverageの公式証拠がある研究を、形式統一だけを理由に再実行しない。

注: R145 READMEには当時のMOM履歴認識が記録されているが、その後の公式系譜回収でPure MOM R89/R90にも同型の179-series full temporal landscapeが存在することを確認した。本v1.11では後続の回収結果をガバナンス上の現行理解として採用する。

2. Suzaku / TEL-DM Canonical Control

Production control = Frozen TEL 3M_RAW. Stage A: TMV 3M > TMF 3MならTMV。そうでなければStage Bへ進み、TQQQ 3M > TECL 3MならTQQQ、else TECL。月末closeでsignalを決定し、翌月最初のtradable openから新holdingを適用する。

R142-R150を通じてProduction変更は一度も行っていない。研究目的はTEL内部の時間構造と時間分散を検証し、Layer -0 challengerを固定することだった。

3. R142-R146: Terrain Discovery -> Independent Modes

R142 - 1-24M Lookback Grid Atlas. Stage A/Bに同じcalendar-month lookbackを適用。初期terrainとしてFast 3-4M、Slow 17-20Mを確認。Frozen 3Mはcanonical identityを維持。

R143 - Fast/Slow Structure Audit.

3Mと17Mは非冗長だが、17M単体の普遍的優位は確立せず。時間分散を固定条件でfalsifyする根拠を得た。

R144 - 3M+17M provisional blend. 時間分散を予備的に支持したが、daily arithmetic blendに相当するexecution semantics懸念が後から判明。QUALIFIED archiveとし、正しいmonthly rebalanceを明示したR147がexecution evidenceをsupersedeする。

R145 - Extended Coverage Atlas. calendar 1-60Mと固定trading-day panelを調査。戦略的主要ピークはFast 3-4M、Slow-1 19-20M、Slow-2 30-31M。30Mが最も強いSlow候補。38M付近にはminor/local re-riseがあり、「局所極大が数学的に3つだけ」とは解釈しない。

R146 - Multi-Peak Structure Falsification. 3M / 19M / 30Mを固定。3M-30Mは月次return corr約0.625、holding disagreement約53.3%、Stage disagreement約40.1%で十分に非冗長。一方19M-30Mはcorr約0.841、same holding約72.5%、same Stage約78.4%。よって実質はFast $\approx$ 3M / Slow family $\approx$ 19-30Mの2 temporal modesで、Slow代表は30M。

4. R147: Fixed 1P vs 2P vs 3P Falsification

R145/R146でsleeveを固定し、portfolioを事前固定: 1P=3M 100%、2P=3M+30M 50:50、3P=3M+19M+30M equal thirds。毎月最初のtradable openでweightをresetし、月内は自然drift。daily arithmetic mixではない。

Portfolio	CAGR	Sharpe	Sortino	MaxDD	Calmar
1P 3M	35.815%	0.852	1.164	-70.160%	0.510
2P 3M+30M EW	41.477%	0.965	1.242	-72.177%	0.575
3P 3M+19M+30M EW	39.594%	0.942	1.203	-72.850%	0.544

Primary result: 2P vs 1PはLOYOでCAGR / Sharpe / Sortino / Calmarの全4主要指標が15/15勝利。一方3P vs 2PはCAGR 0/15、Sharpe 2/15、Sortino 0/15、Calmar 0/15。

したがってTELではgeneric仮説 $3P > 2P > 1P$ をREJECT。支持されたのは「lookback本数を増やすこと」ではなく、非冗長なFastとSlowの組合せによるtemporal diversification。2P 3M+30M monthly 50:50 EWをPrimary Layer-0 Challenger候補として固定した。

5. R148: Standard 179-Series + EX1 Audit

R148で朱雀も標準179-seriesを正式実行。Exactness 12/12 PASS。目的は広域terrain coverageとRAW-vs-EX1の直接比較であり、fine tuningではない。

59組のRAW-vs-EX1直接比較ではEX1がCAGR 37/59、Sharpe 36/59、Sortino 36/59、Calmar 37/59で勝利。平均的な改善傾向はあるが、効果は強くhorizon-dependent。

Horizon	RAW -> EX1 interpretation
3M	EX1で大幅悪化。Fast TELではrecent 1M情報が重要。
19M	EX1で改善。R148 landscapeでは強いEX1 candidate。
30M	EX1で悪化。Slow代表30Mでもrecent 1M除外は不利。

したがって「直近1Mを除外すればTEL全体が改善する」という一般則は支持されない。正しい結論はrecent 1M情報の価値はlookback horizonによって反転する、である。

6. R149: EX1 Temporal Structure Falsification

R148から3M_RAW / 19M_EX1 / 30M_RAWを固定し、第三の独立temporal modeが存在するかを検証。Exactness 12/12 PASS。portfolio blendやweight searchは行っていない。

3M_RAW vs 19M_EX1は月次corr約0.630、holding disagreement約50.6%、Stage disagreement約34.8%でFast 3Mとは非冗長。一方19M_EX1 vs 30M_RAWはcorr約0.832、holding agreement約71.9%、Stage agreement約79.0%で高冗長。

Standalone common-period	CAGR	Sharpe	Sortino	MaxDD	Calmar
3M_RAW	35.82%	0.852	1.164	-70.16%	0.510
19M_EX1	35.95%	0.852	1.101	-74.19%	0.484
30M_RAW	42.00%	0.920	1.170	-74.19%	0.566

19M_EX1 vs 30M_RAWのLOYO勝利はCAGR 1/15、Sharpe 1/15、Sortino 2/15、Calmar 1/15。19M_EX1はregime-dependent mechanismとして興味深いが、30M_RAWに対する第三の独立modeとは認定しない。

7. R150: Final Suzaku Layer-0 Freeze Audit

R150は新規探索を行わないgovernance synthesis。Official R142-R149を統合し、Exactness 13/13 PASS。最終判断はSUZAKU LAYER 0 FREEZE / COMPLETE FOR DEV。

Candidate	Final decision	Role
Frozen TEL 3M_RAW	RETAIN PRODUCTION	Canonical Production/control
3M_RAW + 30M_RAW Monthly 50:50 EW	FREEZE AS PRIMARY CHALLENGER	Primary Layer-0 Challenger
3M+19M+30M EW	REJECT / ARCHIVE	2Pに劣後
19M_RAW	ARCHIVE	Slow-family alternative / redundant
19M_EX1	ARCHIVE / MECHANISM ONLY	EX1 mechanism; 30Mと高冗長
R144 3M+17M blend	QUALIFIED ARCHIVE	semantic caveat; R147 supersedes

8. Closed Branches / Reopen Conditions

Branch	Status / Reopen only if
3M / 30M / 19M_EX1周辺fine lookback	CLOSED - new primary information or sufficient forward OOS
3M:30M weight optimization	CLOSED - new primary information or sufficient forward OOS
19M / 19M_EX1 third sleeve	CLOSED - independent information contentを示す新一次情報
Generic more-horizons-is-better	REJECTED FOR TEL - independent rationaleを持つpre-specified architectureのみ
Stage A x Stage B 2D lookback	CLOSED FOR DEV - asymmetric DM3 operatorの新一次情報
Threshold / selector / vote / adaptive weight	CLOSED FOR DEV - new primary source or forward-OOS justification

このFREEZEは「永久に研究禁止」を意味しない。DEVで同じデータを使った再最適化を止めるための統計的ガバナンスであり、十分なforward OOSまたは新しい一次情報が入った場合のみ、事前規定した研究として再開する。

9. Layer-0 Status After R150

Family	Status	Final Layer-0 control / challenger note
青龍 Seiryu / MOM	FREEZE / COMPLETE (R141)	Production Frozen11M; R108 primary predeclared challenger
白虎 Byakko / VIX	FREEZE / COMPLETE (R140)	Production Frozen VIX; R138 primary new Layer-0 challenger
朱雀 Suzaku / TEL-DM	FREEZE / COMPLETE (R150)	Production 3M_RAW; 3M+30M monthly 50:50 EW primary challenger
玄武 Genbu	OPEN / NEXT	Layer-0 completion remains before new Layer-1 work

Research order remains: 玄武Layer 0を完了・FREEZEした後に、新しいLayer 1 / FoF / 4F結合研究へ進む。既存Layer-1 evidenceは保存するが、Layer-0未完了のまま新しいLayer-1探索を増やさない。

10. R150 Provenance / Integrity

R150 official registry: build R150-SUZAKU-FINAL-LAYER0-FREEZE-AUDIT-v1.0; parents R142-R149; DEV cutoff 2026-07-31; ProductionChange NONE; Exactness 13/13 PASS.

Pinned official ZIP	SHA256
R147	af65fea21e10fc54e5114713fb7193938f8c36f1ea65f3678b944449588a15e7
R148	3651b88ec953ce0b0e64f57bdf180fe05052785a03ddc72e0b4747eae6f4448
R149	098055473c22090e506705b3614b557b315bbc252174a056c3cba85ac2e8b7e3

Reserved research ID governance: R129 remains reserved/unexecuted as “ 4F Family Role Audit ” unless explicitly changed. R142-R150 do not overwrite it.

Master Research Bible v1.12

Genbu / Pure MR Layer 0 Research Addendum

Official evidence incorporated: R151 PATCH1 through R157. DEV cutoff remains 2026-07-31. This addendum preserves all 73 pages of v1.11 unchanged and appends the Genbu research lineage.

Governance status. Production is unchanged. The current DEV Primary Layer-0 Challenger is **Pure MR 6D / 11M_RAW / 24M_RAW 2-of-3 Majority**. R155-R157 are mechanism/post-hoc evidence and do not authorize DEV retuning.

Research lineage

Run	Status	Purpose / result
Initial R151	INVALID	Implementation audit failed; terrain is not official evidence.
R151 PATCH1	14/14 PASS	Standard 179 atlas; Fast/Mid/Slow terrain.
R152	14/14 PASS	Three effective temporal modes.
R153	15/15 PASS	Fixed 1P/2P/3P; 11M adds robust value.
R154	14/14 PASS	2-of-3 Majority establishes DEV Primary Challenger.
R155	12/12 PASS	Split-vote audit; directional minority asymmetry.
R156	14/14 PASS	Mirror override falsification; information != allocation confidence.
R157	14/14 PASS	Stateful veto raises CAGR but damages risk quality.

Canonical Pure MR architecture

Signal asset SPXL. At weekly rebalance, SPXL trailing return $\geq$ matched historical RF $\rightarrow$ XLU; otherwise $\rightarrow$ TQQQ. Weekly cadence and next-close-to-close execution remain fixed. Production MR family remains unchanged.

R151 PATCH1 - Standard 179 Lookback Atlas

The initial R151 is explicitly INVALID. R151 PATCH1 passed all 14 exactness checks, including Frozen504D holding identity (mismatch 0, N=821) and matched-RF identity.

Family	Coverage	Official terrain
D	1D-60D	6D wins CAGR/Sharpe/Sortino/Calmar; secondary structure near 31-32D and 47-50D.
RAW	1M-60M	11M_RAW wins CAGR; 30M_RAW wins Sharpe/Sortino; 24M_RAW wins Calmar.
EX1	2M-60M	15M_EX1 wins CAGR/Sharpe/Sortino; 21M_EX1 wins Calmar.

Series	CAGR	Sharpe	Sortino	MaxDD	Calmar
6D	38.38%	0.911	1.052	-63.66%	0.603
11M_RAW	39.11%	0.999	1.261	-61.45%	0.636
24M_RAW	33.60%	1.120	1.465	-33.14%	1.014
30M_RAW	31.86%	1.171	1.592	-32.74%	0.973
15M_EX1	37.02%	1.127	1.424	-58.23%	0.636

Frozen504D lineage

Series/scope	CAGR	Sharpe	Sortino	MaxDD	Calmar	Agreement
Frozen504D/common	32.26%	1.084	1.415	-33.14%	0.973	99.8782%
24M_RAW/common	33.60%	1.120	1.465	-33.14%	1.014	99.8782%
Frozen504D/native	28.75%	1.052	1.376	-33.14%	0.867	99.8782%
24M_RAW/native	29.80%	1.083	1.421	-33.14%	0.899	99.8782%

Interpretation: Frozen504D is embedded in the broad Slow MR terrain. 24M_RAW is used as the clean Frozen-near Slow representative for research notation; this is not a Production retune.

R152 - Temporal Structure Audit

Fixed representatives: 6D (Fast), 11M_RAW (Mid), 24M_RAW (Slow/Frozen-near), 30M_RAW (Slow/high-Sharpe). No new search.

Pair	Monthly corr.	Holding agreement	Interpretation
6D vs 11M	0.427	59.6%	Distinct Fast vs Mid.
6D vs 24M	0.385	58.6%	Strong temporal non-redundancy.
6D vs 30M	0.359	58.1%	Strong temporal non-redundancy.
11M vs 24M	0.587	77.1%	Mid partly bridges Slow.
11M vs 30M	0.469	73.0%	Mid remains distinct.
24M vs 30M	0.726	91.8%	Same Slow family, not fourth mode.

Conclusion: three effective temporal modes: Fast=6D, Mid=11M_RAW, Slow=24-30M_RAW. Slow anchors risk quality, 6D carries distinct regime opportunity, and 11M bridges the two.

R153 - Fixed 1P / 2P / 3P Diversification

1P=24M_RAW; 2P=6D+24M_RAW 50:50; 3P=6D+11M_RAW+24M_RAW equal thirds. Weekly portfolio rebalance with natural drift, not daily arithmetic blending.

Portfolio	CAGR	Sharpe	Sortino	MaxDD	Calmar
1P 24M	29.98%	1.088	1.426	-33.14%	0.904
2P 6D+24M EW	35.14%	1.079	1.333	-49.22%	0.714
3P 6D+11M+24M EW	37.67%	1.135	1.462	-43.67%	0.863

Comparison	CAGR wins	Sharpe wins	Sortino wins	Calmar wins
2P vs 1P	17/17	7/17	2/17	1/17
3P vs 1P	17/17	15/17	15/17	4/17
3P vs 2P	17/17	17/17	17/17	16/17

Key finding: 11M is not redundant. Adding Mid to Fast+Slow robustly improves growth and risk-adjusted metrics. Generic 3P > 2P > 1P is not true across every metric.

R154 - 2-of-3 Majority vs Equal Weight

Each fixed lookback votes XLU or TQQQ. Majority holds the 2-of-3 winner 100%; underlying signals are unchanged.

Rule	CAGR	Sharpe	Sortino	MaxDD	Calmar
24M Control	29.80%	1.083	1.421	-33.14%	0.899
3P EW	37.67%	1.135	1.462	-43.67%	0.863
3P Majority	40.79%	1.121	1.461	-36.17%	1.128

Majority vs EW: LOYO CAGR 16/17 wins and Calmar 17/17 wins. Majority preserves much of the return gain while restoring drawdown quality.

DEV decision: 6D / 11M_RAW / 24M_RAW 2-of-3 Majority is the Genbu DEV Primary Layer-0 Challenger. 3P EW remains a secondary reference.

Vote structure: about 47.5% unanimous and 52.5% split 2-to-1. Majority state is XLU about 83.7% and TQQQ about 16.3%.

R155 - Split-Vote Minority Role Audit

Only exactly 2-to-1 weekly states are studied. Fixed forward outcomes: 1W=5TD, 2W=10TD, 4W=20TD, measured as realized TQQQ minus XLU.

Minority lookback	TQQQ win @4W	XLU win @4W	Functional interpretation
6D	56.3%	19.2%	Fast opportunity detector; XLU minority is whipsaw-like.
11M	61.5%	15.4%	Intermediate confirmation / bridge; strongest aggregate minority role.
24M	74.1%	35.7%	Persistent regime anchor; strong TQQQ result but latest-era concentrated.

Pattern	N	1W mean edge	2W mean edge	4W mean edge
6D minority TQQQ	216/215	+0.42%	+0.92%	+1.56%
11M minority TQQQ	78	+0.36%	+1.27%	+4.29%
24M minority TQQQ	27	+2.23%	+2.92%	+5.79%
6D minority XLU	26	-3.91%	-7.63%	-14.69%
11M minority XLU	13	-0.91%	-4.50%	-8.68%
24M minority XLU	70	-3.18%	-4.46%	-6.05%

Finding: minority TQQQ retains forward-return information across all three horizons, while minority XLU is generally weak. This is mechanism evidence, not a DEV rule change.

R156 - Directional Minority Asymmetry Falsification

Two fixed mirror rules test the R155 asymmetry with no search. This is post-hoc mechanism characterization only.

Rule	CAGR	Sharpe	Sortino	MaxDD	Calmar
24M Control	29.80%	1.083	1.421	-33.14%	0.899
Baseline Majority	40.79%	1.121	1.461	-36.17%	1.128
TQQQ Minority Override	48.05%	1.006	1.215	-72.60%	0.662
XLU Minority Override	17.20%	0.824	1.076	-31.87%	0.540

Metric	TQQQ Override vs Majority LOYO	Meaning
CAGR	17/17	Robust return-enhancing information.
Sharpe	1/17	Full override damages risk-adjusted quality.
Sortino	0/17	Downside quality deteriorates.
Calmar	0/17	Return gain is bought with excessive drawdown.

Conclusion: minority TQQQ is weak risk-on evidence, not sufficient confidence for 100% TQQQ allocation. The mirror XLU override broadly fails, supporting directional asymmetry.

R157 - Stateful Structural Majority Falsification

A weaker use of minority TQQQ was tested: under XLU majority, 11M- or 24M-minority TQQQ can hold the prior structural state; 6D-minority TQQQ has no veto.

Rule	CAGR	Sharpe	Sortino	MaxDD	Calmar
24M Control	29.80%	1.083	1.421	-33.14%	0.899
Baseline Majority	40.79%	1.121	1.461	-36.17%	1.128
Structural Majority	43.75%	1.086	1.387	-61.45%	0.712

Structural differs from baseline on only 11.8% of weekly votes, yet TQQQ exposure rises from 16.3% to 28.1%. The 97 veto episodes comprise 70 cases with 11M minority TQQQ and 27 with 24M minority TQQQ.

LOYO Structural vs Majority: CAGR 15/17 wins, but Sharpe 2/17, Sortino 2/17 and Calmar 1/17. The same core result as R156 reappears: minority TQQQ information raises return, but using it to delay risk-off weakens drawdown control.

Decision: Stateful Structural Majority is not adopted. Further DEV optimization of minority overrides, stateful vetoes, vote thresholds or conviction weights is closed.

Genbu Layer 0 Synthesis and Layer 1 Handoff

Component	Frozen research role	Evidence-based interpretation
6D	Opportunity detector / ultra-fast sensor	Distinct short-horizon information; TQQQ minority can be useful, XLU minority is weak.
11M_RAW	Intermediate confirmation / bridge	Adds robust value between Fast and Slow and helps form cross-horizon confirmation.
24M_RAW	Persistent regime anchor	Frozen-near Slow representative; 99.8782% agreement with 504D and strong risk-quality role.
2-of-3 Majority	Cross-horizon confirmation threshold	Requires two temporal modes before converting risk-on information into full TQQQ exposure.

Current freeze recommendation

Production: unchanged Frozen MR family.
DEV Primary Layer-0 Challenger: Pure MR 6D / 11M_RAW / 24M_RAW 2-of-3 Majority.
Secondary reference: 3P EW.
Mechanism evidence: R155, R156, R157.
Closed for DEV: new lookback search, weight optimization, vote-threshold optimization, minority override, stateful veto and related Majority tuning.

Layer 1 metadata to preserve

Preserve Genbu **final direction** (TQQQ/XLU), **vote strength** (0/3, 1/3, 2/3, 3/3 TQQQ) and **vote composition** (which of 6D, 11M and 24M voted each side). This keeps minority information without altering Layer 0.

Predeclared Layer 1 hypothesis

Cross-horizon confirmation x cross-mechanism confirmation: test whether Genbu 1/3 TQQQ becomes more informative when independently confirmed by Seiryu, Byakko or Suzaku. This belongs in Layer 1, not in renewed Genbu DEV optimization.

Governance boundary

DEV remains frozen at 2026-07-31; 2026-08 onward is Forward OOS. R155-R157 cannot retroactively promote a new Genbu rule. The next clean step is a final Genbu Layer-0 Freeze Audit, after which all four Layer-0 families can be treated as complete before new Layer-1 integration research.

Master Research Bible - v1.13 Addendum

Genbu completion (R158-R167) and first Layer-1 structural audits (R129, R168)

Governance. This addendum preserves the first 82 pages of v1.12 unchanged. All numerical conclusions below are grounded in user-run official ZIP evidence. DEV cutoff remains 2026-07-31. Production 4F remains Frozen EW: MOM/MR/VIX/TEL-DM = 25% each. No post-cutoff evidence is used to retune Layer-0 rules.

Executive update

Genbu Layer 0 is now formally complete. R167 passed 21/21 exactness checks and froze the DEV Primary Layer-0 Challenger as SPXL Pure MR using 6D / 11M_RAW / 24M_RAW, weekly 2-of-3 Majority, TQQQ risk-on and XLU risk-off. With Seiryu R141, Byakko R140, Suzaku R150 and Genbu R167 all frozen, the research program has crossed from Layer 0 construction into Layer 1 family-combination research.

R129 then mapped cross-mechanism family roles without optimizing weights. R168 tested marginal contribution by removing one family at a time from a 25% EW diagnostic portfolio. The evidence supports retaining all four families as distinct sources of portfolio function, while raising a new Layer-1 question: whether equal 25% weights are structurally ideal. That question is not answered by R129/R168 and remains open.

R158-R160 - cadence, immediacy and tail protection

R158 Rebalance Cadence Audit - 14/14 PASS. The fixed 6D/11M/24M Majority signal was tested at Weekly, Monthly, Bimonthly, Quarterly and Semiannual evaluation. Weekly delivered about CAGR 41.27%, Sharpe 1.127, MaxDD -36.17%, Calmar 1.141. Slower cadences lost CAGR and Calmar persistently. Same-date signal agreement was 100%; the loss came from delayed reaction rather than a different signal. Weekly monitoring generated only about 94 holding switches across 821 evaluations, so weekly cadence is best interpreted as state monitoring rather than weekly trading.

R159 Signal Decay / Delayed Execution - 14/14 PASS. Delaying the weekly Majority target by 1, 2 or 4 weekly observations damaged performance. 0W: CAGR ~41.22%, Sharpe 1.128, Calmar 1.140; +1W: 31.03%, 0.967, 0.813; +2W: 22.60%, 0.799, 0.384; +4W: 24.84%, 0.868, 0.537. Each delayed variant lost to 0W in all 17 LOYO samples for CAGR, Sharpe, Sortino and Calmar. Immediate weekly response is therefore a persistent property of the mechanism.

R160 Risk-Off Tail Protection - 14/14 PASS. TQQQ->XLU switches were not attractive as simple average-return timing. Instead, they acted as asymmetric insurance. In the predeclared crisis-like subset where old TQQQ fell $\leq -10\%$ over the next 4W (N=9), XLU-TQQQ mean edge was +17.74 percentage points at 4W and XLU won 100%. In non-tail events (N=38), the 4W edge was -6.47 points. The risk-off state therefore accepts false-alarm opportunity cost to avoid a minority of severe leveraged-equity losses.

R161-R164 - asset choice and failed extensions

R161 Risk-Off Asset Substitution - 15/15 PASS. Keeping the signal architecture fixed, XLU, GLD, SPY and QQQ were compared as the risk-off holding. XLU remained best overall risk-adjusted: CAGR 40.79%, Sharpe 1.121, MaxDD -36.17%, Calmar 1.128. GLD was the stronger true-crisis hedge (4W crisis-like GLD-TQQQ edge +21.55 points vs XLU +17.74), but paid more in non-tail false alarms. XLU is therefore the defensive-equity middle ground, not merely a lower-beta substitute.

R162 Confidence-Tiered Defense - 15/15 PASS, challenger rejected. A post-hoc tiered rule used XLU at exactly 2/3 risk-off and GLD at 3/3. Full DEV improved modestly, but the mechanism premise failed: crisis-like frequency was 18.94% at exactly 2/3 versus only 15.07% at 3/3. The apparent gain was heavily driven by 2023-DEV. Vote unanimity is not monotonic confidence; no GLD tier was adopted.

R163 Minority-Pattern Tail Mechanism - 14/14 PASS. Exactly-2/3 risk-off states are heterogeneous. The 6D-minority-TQQQ pattern was most tail-rich (4W TQQQ $\leq -10\%$ in 20.28% of cases), while the 24M-minority pattern had only 27 observations and all occurred in 2023-DEV. No pattern-specific trading rule was authorized. Minority horizon is retained as metadata only.

R164 TQQQ/TECL 3M Relative-Momentum Selector - fixed falsification. The Suzaku Stage-B 3M_RAW selector was inserted only on Genbu risk-on weeks. Challenger CAGR rose from 40.792% to 41.071%, MaxDD improved from -36.17% to -35.32%, and Calmar from 1.128 to 1.163, but Sharpe was essentially unchanged/slightly lower (1.1201 vs 1.1209). The improvement was too small/inconsistent to justify added architecture overlap with Suzaku. The selector was not adopted; Genbu risk-on remains fixed TQQQ.

R165-R166 - attribution and transition mechanism

R165 State Contribution / Counterfactual Attribution - 14/14 PASS. Genbu spends only about 16.36% of daily observations in TQQQ and 83.64% in XLU, yet the short risk-on state supplies the large majority of cumulative log-return contribution. Actual Genbu: CAGR 40.79%, Sharpe 1.121, MaxDD -36.17%, Calmar 1.128. Always-TQQQ had similar CAGR (~39.30%) but catastrophic MaxDD (~-81.66%) and Calmar ~0.481; always-XLU CAGR was only ~10.43%. Thus TQQQ is the return engine and XLU is primarily the tail-risk circuit breaker.

Switch attribution reinforced the asymmetry. XLU->TQQQ switches showed positive TQQQ-vs-XLU forward advantage that strengthened with horizon (about +2.70% at 1W and +6.89% at 4W, with 4W win rate ~70%). TQQQ->XLU switches did not show a broad average-return edge, consistent with insurance rather than alpha timing.

R166 Signal Stability / Vote-State Transition - 15/15 PASS. Across 821 weekly observations, TQQQ occupied 134 weeks (16.3%) and XLU 687 (83.7%). TQQQ runs were short (mean 2.85 weeks, median 2, 46.8% one-week runs); XLU runs were persistent (mean 14.31 weeks, maximum 167). There were 94 state switches and 33 one-week flips. Despite this apparent noisiness, R159 already falsified waiting one week, so no hysteresis was added.

At switch weeks the 6D horizon was the dominant mover: it changed on about 89.4% of TQQQ->XLU switches and 76.6% of XLU->TQQQ switches, versus 40.4%/48.9% for 11M and only 12.8%/10.6% for 24M. Strict 0->1->2->3 progression was nearly absent. Final mechanism: 24M = persistent regime anchor; 11M = intermediate bridge/confirmation; 6D = fast transition/opportunity sensor; 2-of-3 = cross-horizon confirmation threshold; weekly evaluation = rapid state-change monitor.

R167 - Genbu Final Layer-0 Freeze

Official status: GENBU LAYER 0 FREEZE / COMPLETE FOR DEV. Exactness 21/21 PASS. No new search was performed. R167 canonically reconstructed the selected architecture, compared it with controls, froze closed branches, and defined Layer-1 metadata.

Strategy	CAGR	Sharpe	Sortino	MaxDD	Calmar
Genbu Majority	40.79%	1.121	1.461	-36.17%	1.128
24M_RAW control	29.80%	1.083	1.421	-33.14%	0.899
Frozen504D	28.75%	1.052	1.376	-33.14%	0.867

LOYO versus 24M_RAW: CAGR 17/17, Sharpe 14/17, Sortino 14/17, Calmar 17/17 wins. Versus Frozen504D: CAGR 17/17, Sharpe 17/17, Sortino 16/17, Calmar 17/17. A recent-era exception remains: 2023-DEV favored 24M_RAW on headline CAGR/Calmar, and this was explicitly not used for retuning.

Frozen Genbu challenger specification. Signal asset SPXL. Horizons 6D / 11M_RAW / 24M_RAW. For each horizon, SPXL trailing return $\geq$ matched historical RF produces a risk-off vote. Two or more risk-off votes $\Rightarrow$ XLU; otherwise $\Rightarrow$ TQQQ. Evaluate weekly on the first trading day, using information through the prior trading day; new holding is active next close-to-close. DEV cutoff 2026-07-31. Production 4F change: NONE.

Layer-1 handoff metadata. Preserve FinalDirection, TQQQ vote strength 0-3, and 6D/11M/24M vote composition. These are descriptive inputs for future cross-mechanism studies, not authorization for post-hoc routing.

Four Gods - Layer 0 completion checkpoint

Family	Layer-0 completion	Primary research role
Seiryu / MOM	R141 - COMPLETE	Momentum / return-quality family
Byakko / VIX	R140 - COMPLETE	Volatility-state / growth-diversification family
Suzaku / TEL-DM	R150 - COMPLETE	Treasury-equity layered dual momentum; independent return engine
Genbu / MR	R167 - COMPLETE	Cross-horizon confirmation; defensive base + short attack

All four Layer-0 families are now frozen for DEV research. This closes the strategy-construction phase. Further work should treat the four completed families as fixed building blocks unless a new governance decision explicitly reopens a Layer-0 branch. The next research domain is Layer 1: how completed mechanisms interact at portfolio level.

Production remains the original Frozen 4F EW (25% each). The Layer-0 challengers and Layer-1 diagnostics are research comparators, not automatic production replacements.

R129 - 4F Family Role Audit

Purpose. R129 was the reserved Family Role Audit and was executed only after all Layer-0 families were frozen. It did not optimize weights, create a router, search thresholds, select Top-N families, reopen lookbacks, or change Production. Exactness passed 16/16 after a non-research implementation patch.

Frozen L0 representative	CAGR	Sharpe	MaxDD	Calmar
Seiryu R108	31.61%	1.358	-23.21%	1.362
Byakko R138	47.20%	1.165	-39.32%	1.200
Suzaku 3M+30M	41.29%	1.024	-54.20%	0.762
Genbu Majority	40.84%	1.299	-25.11%	1.626

A 25% EW diagnostic portfolio across the four frozen representatives produced approximately CAGR 44.27%, Sharpe 1.640, MaxDD -22.39%, Calmar 1.977 (formalized again in R168). Effective Bets increased from 2.646 over the full common sample to 3.135 in the 4F-EW worst-20% months. This is an important structural result: diversification did not collapse in the stress subset.

Monthly return correlations showed meaningful differentiation. Seiryu-Suzaku was relatively close (~ 0.557), while Suzaku-Genbu was nearly uncorrelated (~ 0.041). Tail-rescue analysis also showed complementary behavior, especially between Suzaku and Genbu. Genbu frequently remained defensive when other families were risk-on, consistent with its Layer-0 mechanism. R129 therefore supports the premise that the four families are cross-mechanism building blocks rather than four versions of the same trade.

R168 - Leave-One-Family-Out / Marginal Contribution Audit

Exactness 18/18 PASS. Control = four frozen Layer-0 representatives at 25% EW. Each challenger omitted exactly one family and equally weighted the remaining three. This was a marginal-contribution falsification, not a family-selection rule and not authorization to move Production to 3F.

Portfolio	CAGR	Sharpe	Sortino	MaxDD	Calmar
4F EW control	44.27%	1.640	2.283	-22.39%	1.977
Minus Seiryu	48.01%	1.592	2.124	-22.12%	2.171
Minus Byakko	41.54%	1.631	2.013	-29.06%	1.430
Minus Suzaku	42.07%	1.544	2.362	-22.00%	1.912
Minus Genbu	43.70%	1.469	1.894	-27.77%	1.574

Genbu. Removing Genbu materially damaged risk-adjusted quality. In 15 LOYO samples, the 4F control beat Minus-Genbu 15/15 for Sharpe, Sortino and Calmar (CAGR 12/15). This confirms Genbu as a persistent portfolio-quality / tail anchor.

Byakko. Removing Byakko reduced CAGR and worsened MaxDD/Calmar. LOYO: 4F won 15/15 for CAGR, Sortino and Calmar, and 11/15 for Sharpe. Byakko contributes both growth and portfolio-level drawdown quality.

Suzaku. Removing Suzaku reduced full-sample CAGR and Sharpe. LOYO: 4F won 15/15 CAGR and Sharpe, but only 2/15 Sortino and 13/15 Calmar. Suzaku is an independent, regime-dependent return engine rather than a uniformly defensive sleeve.

Seiryu. Removing Seiryu increased full-sample CAGR and Calmar, but the 4F control beat Minus-Seiryu 15/15 for Sharpe and 15/15 for Sortino while losing CAGR 0/15. Seiryu therefore behaves as a return-quality diversifier: it can dilute raw CAGR while persistently improving distributional quality. This does not justify deleting Seiryu; it raises a Layer-1 weight question.

Current synthesis and next research boundary

The evidence through R168 changes the Layer-1 question. The question is no longer whether four distinct families exist or whether combining them can improve risk-adjusted behavior. Both are supported. The open question is whether equal 25% weights are the most robust way to express those four functions.

Current family-role synthesis

Family	Portfolio role after R129/R168
Seiryu	Return-quality diversifier; persistent Sharpe/Sortino contribution, possible CAGR drag at 25%
Byakko	Growth + drawdown contributor; strong CAGR/Calmar marginal value
Suzaku	Independent, regime-dependent return engine; strong complement to Genbu
Genbu	Risk-adjusted quality / tail anchor; most persistent Sharpe/Sortino/Calmar contribution

Governance boundary for the next study. A future weight study should begin with a coarse, predeclared landscape around EW, not point optimization. It should ask whether a broad plateau exists and whether directional tendencies (for example, modest Seiryu underweight or Genbu/Byakko overweight) persist across LOYO, rolling windows and fixed eras. R168-derived hypotheses must be treated as post-hoc hypotheses and falsified prospectively within DEV governance; they are not conclusions.

Do not: reopen Layer-0 lookbacks; optimize weights to a single best point; create a router from R129/R168 descriptive states; promote a 3F subset because of one full-sample metric; or change Production without a separately governed Layer-1 freeze process.

End of v1.13 addendum. Source-of-truth hierarchy remains: Master Research Bible for governance/history; official user-run ZIPs for formal empirical evidence; latest app.py for implementation. Production 4F remains unchanged.

Master Research Bible - v1.14 Addendum

Layer-1 Spear/Shield research completion (R169-R181) and DEV freeze boundary

Governance. This addendum preserves the first 91 pages of v1.13 unchanged. All empirical conclusions below are grounded in user-run official Streamlit ZIP evidence. DEV cutoff remains **2026-07-31**. Production remains Frozen 4F EW: Seiryu / Byakko / Suzaku / Genbu = 25% each. R180 is a frozen Layer-1 primary challenger, not Production. R181 is a post-freeze experimental extreme challenger and cannot automatically supersede R180.

Executive update. Layer-1 DEV research is now complete. The mainline architecture is Spear = Seiryu + Byakko + Suzaku at 1/3 EW, Shield = Genbu. Current Production identity is exactly 75% Spear + 25% Shield. R180 freezes a conditional Shield reinforcement rule: normally 75/25; when prior Spear 1M return is negative AND prior 12M Shield return exceeds prior 12M Spear return, use 50/50. R181 separately records a one-shot 100% Shield extreme hypothesis under the same A-state. No further DEV weight search is authorized.

Current status

Domain	Status	Frozen / retained object
Layer 0	COMPLETE	Seiryu R141; Byakko R140; Suzaku R150; Genbu R167
Layer 1 - Spear choice	COMPLETE	Mainline remains 3-Spear EW
Layer 1 - Spear/Shield allocation	COMPLETE FOR DEV	R180 Conditional Shield Reinforcement Strong
Production	UNCHANGED	4F EW: 25% each
Post-freeze experiment	FROZEN EXPERIMENTAL	R181 A-state -> 100% Shield
Forward OOS	STARTS 2026-08	No DEV retuning from forward results

Source-of-truth hierarchy remains: Master Research Bible for governance/history; official user-run ZIPs for formal empirical evidence; latest app.py for implementation. Failed runs and assistant-side calculations are not formal evidence.

R169 - Coarse 4F weight landscape (exploratory derivative branch)

R169 PATCH1 passed 20/20 exactness checks. The invalid initial R169 grid is not evidence. PATCH1 used exactly EW plus 12 one-step 12.5 percentage-point transfers; it was a landscape/robustness audit, not point optimization.

Portfolio direction	CAGR	Sharpe	MaxDD	Calmar	Interpretation
4F EW	44.27%	1.640	-22.39%	1.977	Control
Seiryu 12.5 -> Genbu 37.5	45.67%	1.652	-20.22%	2.258	Strong balanced direction
Seiryu 12.5 -> Byakko 37.5	46.48%	1.585	-19.72%	2.357	Return/Calmar trade-off
Suzaku 12.5 -> Genbu 37.5	43.50%	1.649	-15.71%	2.769	Post-hoc risk-quality champion

The neighborhood was not flat. Genbu-overweight directions deserved follow-up, but R169 is explicitly retained as an **exploratory derivative static-weight study**. It is not the Bam mainline and does not authorize Production weight changes.

Bam 'Spear and Shield' source correction

The structural mapping recovered from the prior source material is: **Seiryu / Suzaku / Byakko = Spear (offense)**, and **Genbu = Shield**. Spear choice and Spear-vs-Shield allocation are separate research layers. 'Shield fixed' must not be interpreted as all four Gods fixed at 25%. No exact historical Bam numeric Shield allocation has been recovered; therefore no such number is asserted here.

This correction changes the mainline research question from arbitrary four-way weight optimization to: (1) how to express the Spear block, then (2) how to allocate between Spear and Shield.

R170 - Spear/Shield two-block identity and role

R170 passed 22/22 exactness checks. Spear was frozen as Seiryu + Byakko + Suzaku, 1/3 EW; Shield as Genbu. The current 4F EW control is algebraically identical to **75% Spear-EW + 25% Shield**. This 75/25 ratio is an identity of current Production, not a claim about Bam's historical allocation.

Block	CAGR	Sharpe	Sortino	MaxDD	Calmar
Spear EW	43.70%	1.469	1.894	-27.77%	1.574
Shield / Genbu	40.84%	1.299	1.739	-25.11%	1.626
75/25 = 4F EW	44.27%	1.640	2.283	-22.39%	1.977

Adding 25% Shield to Spear improved return quality and drawdown behavior. In Spear-negative months Shield averaged about +0.26%, and in Spear worst-10% months Shield averaged about -0.66% versus roughly -9.41% for Spear. Rolling 60M, the 75/25 control beat Spear on Sharpe and MaxDD in 100% of windows.

Interpretation: Genbu is not a simple inverse hedge. It is a defensive counter-reversal / loss-attenuation mechanism with different state transitions. The correct research order became Spear choice first, Shield allocation second.

R171-R175 - Spear choice branch

R171 (20/20 PASS): simple prior strongest Spear selection was not supported. Prior-leader edge versus 3-Spear EW was negative at 1M, 3M and 6M, and only +0.07%/month at 12M. The simple Top1 momentum Spear selector branch was closed.

R172 (23/23 PASS): Bam-inspired 'acceleration' and 'Kawarimi' structure was tested descriptively. 6M acceleration and 6M/12M strongest+weakest showed signal structure, but recent-era reversals prevented promotion.

R173 (22/22 PASS): the Kawarimi edge was decomposed into family-composition baseline and residual rank timing. Rank timing was genuine (about 58% of the 6M edge and 64% of the 12M edge) but era instability remained; high-dispersion gains were mostly composition, so no dispersion router was built.

R174-R175 - Fixed portfolio falsification and Spear mainline closure

R174 passed 23/23 exactness checks and compared four predeclared portfolios on the same 154-month sample.

Portfolio	CAGR	Sharpe	Sortino	MaxDD	Calmar	Decision
3-Spear EW	45.77%	1.495	1.915	-27.77%	1.648	Control
6M Kawarimi	47.78%	1.369	1.598	-36.80%	1.299	Reject quality
12M Kawarimi	48.58%	1.416	1.815	-27.82%	1.746	Only survivor
6M Accel Top	48.40%	1.200	1.642	-27.56%	1.756	Reject standalone

12M Kawarimi was the only survivor but was not dominant: it lost badly in 2021-2024 despite strong gains in other eras. This triggered R175 rather than adoption.

R175 passed 23/23. The failure mechanism was clear: when 12M Kawarimi failed, the excluded **middle-ranked family** frequently became the next-month winner. Middle won 57.1% of failure months and 90.3% of severe-failure months. The effect was not simply a weak-family problem; both selected extremes could lose to the middle.

Crucially, no stable t-1 predictor of that middle rebound was found. Family identity and dispersion patterns were regime-specific and unsuitable as rescue routers. Therefore no post-hoc rescue rule was created.

Formal Spear conclusion: simple strongest, 6M Kawarimi, 6M Acceleration Top, high-dispersion routing and middle-identity rescue are closed. 12M Kawarimi remains a research challenger only. Bam-mainline Spear returns to and freezes at 3-Spear EW.

R176 - Spear/Shield allocation descriptive structure

R176 PATCH1 passed 26/26 exactness checks; the initial qcut implementation failure is not evidence. No allocation rule was created. The study asked whether fixed t-1 state variables explain next-month Shield-minus-Spear relative value.

Pre-state	Next Shield-Spear edge	Shield win rate	Comment
Prior Spear 1M loss	+0.97%/mo	55.1%	Primary damage axis
Prior Spear 3M loss	+1.15%/mo	-	Secondary comparator
Prior 1M bottom20	+1.97%/mo	61.3%	Exploratory only
Deep Spear drawdown	+1.54%/mo	56.4%	Descriptive only
High 12M Spear dispersion	+1.73%/mo	59.0%	Exploratory only
12M Shield-leading	+0.31%/mo	53.6%	Continuation-like long horizon
12M Spear-leading	-0.36%/mo	41.2%	Opposite sign

The 12M relative-momentum axis was robust in LOYO but reversed in the latest fixed eras, so it was not promoted alone. R176 predeclared only two primary axes for interaction testing: **prior Spear 1M damage** and **12M Shield-vs-Spear relative momentum**.

R177 - Independent axes, strong interaction

R177 passed 26/26 exactness checks. Damage1M and ShieldLeading12M were almost independent (phi approximately 0.029), yet their interaction was large.

State	Definition	N	Next Shield-Spear	Shield win
A	Loss + Shield leading	23	+3.68%/mo	73.9%
B	Loss + Spear leading	26	-1.43%	38.5%
C	Non-loss + Shield leading	46	-1.38%	43.5%
D	Non-loss + Spear leading	59	+0.11%	42.4%

Difference-in-differences interaction was about +6.60%/month. LOYO: A positive 14/14 and interaction positive 14/14. Rolling 60M: A positive 100% and interaction positive 100%. This established the frozen mechanism state for the next test: A = prior Spear 1M < 0 AND prior 12M Shield > prior 12M Spear.

R178 - First dynamic allocation falsification

R178 passed 26/26 exactness checks. The A-state was inherited unchanged. Only two predeclared reinforcement doses were tested; there was no weight grid.

Portfolio	A-state allocation	CAGR	Sharpe	Sortino	MaxDD	Calmar
Control	75 Spear / 25 Shield	46.67%	1.678	2.309	-22.39%	2.085
Moderate	62.5 / 37.5	47.91%	1.723	2.379	-20.58%	2.329
Strong	50 / 50	49.11%	1.758	2.461	-18.76%	2.618

Strong versus Control improved CAGR by +2.44 percentage points, Sharpe +0.081, Sortino +0.151, MaxDD by +3.63 points, and Calmar +0.533. LOYO Strong wins: CAGR 14/14, Sharpe 14/14, Sortino 14/14, Calmar 14/14, MaxDD improvement 13/14.

The dose response across the two predeclared reinforcement levels was positive, but 50/50 is not an optimized weight. The correct conclusion was support for the conditional Shield-reinforcement mechanism, not discovery of an optimum.

R179 - Mechanism attribution / failure audit

R179 passed 24/24 exactness checks. R178 Strong was held fixed. The A-state occurred in 23 of 154 months (14.94%), with 16 entries and 7 continuation months.

A-state diagnostic	Result
A wins	17/23 = 73.9%
Mean incremental return	+0.921% per A month
Median incremental return	+0.830%
Entry mean increment	+1.090%
Continuation mean increment	+0.534%
LOYO positive mean increment	14/14
Top-1 contribution share	27.2%
Top-3 contribution share	77.8%

Loss magnitude, 12M relative-momentum gap, family identity, entry/continuation and run-position diagnostics did not justify new filters. 2025-DEV had a small negative mean A increment, driven importantly by a violent Spear rebound episode. This is a structural failure mode of increasing Shield exposure, not a basis for an ex-post rescue rule.

R179 decision: PASS with caution. Mechanism supported; small-N and event-contribution concentration retained as explicit forward-OOS cautions.

R180 - Final Layer-1 Freeze Audit

R180 passed **29/29 exactness checks**. No parameter was changed. The study consolidated identity, Full DEV, LOYO, fixed eras, rolling windows, stress, A-event dependence, transitions, recent behavior, research lineage and closed branches.

Frozen Primary Challenger specification

Field	Frozen value
Name	Conditional Shield Reinforcement Strong
Spear	Seiryu R108 + Byakko R138 + Suzaku 3M/30M, 1/3 EW
Shield	Genbu R167 Majority
Normal / Non-A	75% Spear / 25% Shield
A-state	Prior Spear 1M < 0 AND prior 12M Shield > prior 12M Spear
A-state allocation	50% Spear / 50% Shield
DEV cutoff	2026-07-31
Forward OOS	2026-08 onward
Production change	NONE

Final DEV metrics

Portfolio	CAGR	Vol	Sharpe	Sortino	MaxDD	Calmar
Control 75/25	46.67%	24.91%	1.678	2.309	-22.39%	2.085
R180 Frozen 50/50	49.11%	24.71%	1.758	2.461	-18.76%	2.618
Delta	+2.44pt	-0.20pt	+0.081	+0.151	+3.63pt	+0.533

LOYO: CAGR 14/14, Sharpe 14/14, Sortino 14/14, MaxDD 13/14, Calmar 14/14. Rolling 36M win rates: CAGR 94.1%, Sharpe 86.6%, Sortino 93.3%, MaxDD improvement 69.7%, Calmar 94.1%. Rolling 60M: CAGR 100%, Sharpe 96.8%, Sortino 100%, MaxDD improvement 80.0%, Calmar 100%.

Formal R180 decision: PASS - Layer-1 Primary Challenger FREEZE / COMPLETE FOR DEV. Production remains Frozen 4F EW. The challenger must be run prospectively from 2026-08 without DEV retuning.

R180 fixed-era and stress interpretation

Era	Control CAGR	R180 CAGR	Delta CAGR	Control Sharpe	R180 Sharpe	Control DD	R180 DD
2013-2016	32.24%	35.04%	+2.80pt	1.426	1.568	-11.99%	-11.15%
2017-2020	64.63%	68.91%	+4.28pt	1.706	1.793	-22.39%	-18.76%
2021-2024	42.90%	44.51%	+1.61pt	2.027	2.005	-8.87%	-8.87%
2025-DEV	44.74%	44.37%	-0.37pt	1.700	1.839	-7.00%	-5.93%

The latest era has slight CAGR drag but improved Sharpe, Sortino, MaxDD and Calmar. This is not classified as mechanism collapse. The A-state is sparse, and one strong Spear rebound can materially affect a short recent window.

Control-defined stress in R178/R180 also improved under reinforcement. The worst historical month itself was not necessarily rescued, confirming that A is not a universal crash detector. Its purpose is conditional relative allocation between two frozen blocks, not market-crash prediction.

Closed DEV branches after R180

Branch	Status
Simple prior-strongest Spear selector	CLOSED
6M Kawarimi / 6M Acceleration Top	CLOSED
12M high-dispersion router	DO NOT BUILD
Middle-identity rescue router	DO NOT BUILD
12M Kawarimi as mainline Spear	NOT ADOPTED
Shield fine-grid / optimum weight search	CLOSED / PROHIBITED
Loss-magnitude threshold	CLOSED / descriptive only
12M gap threshold	CLOSED / descriptive only
Family-identity A filter	CLOSED / descriptive only
Entry-vs-continuation A filter	CLOSED / descriptive only
New lookback / vote / router on DEV	PROHIBITED

R181 - Post-freeze Extreme Shield 100% one-shot falsification

After R180 was frozen, one additional hypothesis was predeclared and tested exactly once: keep Non-A at 75/25, but in the already-frozen A-state use **0% Spear / 100% Shield**. No intermediate weights, thresholds or new state definitions were searched. R181 passed **26/26 exactness checks**.

Portfolio	CAGR	Vol	Sharpe	Sortino	MaxDD	Calmar
Control 75/25	46.67%	24.91%	1.678	2.309	-22.39%	2.085
R180 A->50/50	49.11%	24.71%	1.758	2.461	-18.76%	2.618
R181 A->0/100	53.51%	26.15%	1.788	2.651	-15.83%	3.380

R181 versus R180: CAGR +4.40pt, Sharpe +0.029, Sortino +0.191, MaxDD +2.93pt improvement, Calmar +0.762, but Vol +1.44pt.
LOYO versus R180: CAGR 14/14, Sharpe 13/14, Sortino 14/14, MaxDD 14/14, Calmar 14/14.

Because the Control-to-50/50 change adds 25 points of Shield while Control-to-100% adds 75 points, the A-month incremental return is algebraically exactly 3x. Observed mean A increment therefore rose from +0.921% to **+2.762%**; the win rate stayed 17/23 (73.9%). Best A increment versus Control was +17.30pt, but worst A increment was **-19.07pt**. Extreme allocation magnifies both correct and incorrect A-state bets.

R181 fixed-era caution

Era	R180 CAGR	R181 CAGR	R181-R180	R180 Calmar	R181 Calmar
2013-2016	35.04%	40.17%	+5.14pt	3.142	4.241
2017-2020	68.91%	77.38%	+8.47pt	3.673	4.888
2021-2024	44.51%	47.50%	+2.99pt	5.017	5.353
2025-DEV	44.37%	42.02%	-2.35pt	7.483	5.827

The latest era is a genuine caution: unlike R180, R181 loses not only CAGR but also recent risk quality. The known failure mode is a violent Spear rebound after A-state activation; 100% Shield magnifies that miss. No DEV rescue filter is authorized.

Formal R181 status: PASS / FREEZE AS POST-FREEZE EXPERIMENTAL EXTREME CHALLENGER. It is NOT Production, does NOT replace R180, and does NOT reopen DEV weight search. No 75%, 87.5%, or other intermediate Shield weights may now be searched on DEV.

Final architecture after R181

The research program now has three explicitly separated portfolio objects. Keeping these labels separate is essential to avoid accidental promotion of a post-freeze experiment.

Object	Normal / Non-A	A-state	Status
Production - Frozen 4F EW	75 Spear / 25 Shield	75 / 25	Production unchanged
R180 Frozen Primary Challenger	75 / 25	50 / 50	Layer-1 frozen primary challenger
R181 Experimental Extreme	75 / 25	0 / 100	Post-freeze experimental challenger

A-state definition (frozen)

Prior Spear 1M return < 0 AND prior 12M Shield return > prior 12M Spear return. All state inputs are t-1 information. No magnitude threshold, gap threshold, family filter, run-position filter or new lookback is part of the rule.

Research interpretation

R169 suggested that more Genbu exposure could be structurally useful. R170 established the correct two-block architecture. R171-R175 showed that active Spear selection did not robustly beat 3-Spear EW and closed the Spear-choice branch. R176-R177 identified a sparse, nearly orthogonal two-axis A-state where Shield relative value was unusually strong. R178 converted that state into a predeclared allocation falsification; R179 diagnosed its mechanism and concentration; R180 froze the 50/50 primary challenger. R181 then tested one extreme dose after the freeze and found stronger long-run DEV metrics but a meaningful recent-era reversal and much larger event-level miss magnitude.

The correct conclusion is not that 100% Shield is 'optimal'. The correct conclusion is that the same frozen A-state exhibits a strong historical dose-response, while the extreme dose carries larger state-classification risk. The only scientifically clean way to distinguish R180 from R181 now is prospective data.

Forward OOS protocol - effective immediately

Boundary: DEV ends 2026-07-31. All data from 2026-08-01 onward are forward OOS. Neither good nor bad forward results may be used to alter the frozen DEV rules and then be reclassified as OOS.

Track prospectively	Rule
Production	4F EW, 25% each
R180	Normal 75/25; A 50/50
R181	Normal 75/25; A 0/100
A-state	Frozen R177 definition only
Retuning	None from 2026-08+ outcomes
Evaluation	Record monthly state, returns, relative deltas, drawdown, transitions and event outcomes
Promotion	Requires separate governance; DEV superiority alone is insufficient

Forward monitoring should emphasize A-events because the three objects are identical outside A. Important prospective questions are: A-state frequency; R180 and R181 incremental return at each A-event; rebound misses; run persistence; cumulative OOS delta; drawdown; and whether the historical 17/23 Shield-win tendency survives without rule changes.

Do not optimize the forward sample. A future promotion decision should be based on a predeclared governance standard (minimum elapsed time / event count, robustness criteria, and acceptable concentration), not on whichever rule has the best cumulative return at an arbitrary date.

Production replacement governance

R180 and R181 are research comparators. Neither is automatically entitled to replace Production. Any future Production change should be a separately governed decision that explicitly considers prospective OOS evidence, operational complexity, turnover/state transitions, concentration, tail behavior, and the cost of being wrong when Spear rebounds.

R169-R181 official evidence registry

Run	Exactness	Formal role / conclusion
R169 PATCH1	20/20	Exploratory coarse 4F weight landscape; static-weight derivative branch
R170	22/22	Spear/Shield identity and role; 4F EW = 75/25
R171	20/20	Simple prior-strongest Spear selector rejected
R172	23/23	Acceleration / Kawarimi structural audit
R173	22/22	Rank timing vs family composition decomposition
R174	23/23	Fixed Spear portfolio falsification; 12M Kawarimi only survivor
R175	23/23	Kawarimi failure mechanism; Spear mainline returns to 3-Spear EW
R176 PATCH1	26/26	Allocation descriptive structure; initial implementation failure invalid
R177	26/26	Damage x 12M relmom interaction; A-state frozen
R178	26/26	Conditional Shield reinforcement supported; Strong 50/50 best predeclared dose
R179	24/24	Mechanism attribution; PASS with concentration caution
R180	29/29	Final Layer-1 freeze; Primary Challenger frozen
R181	26/26	One-shot 100% Shield; post-freeze experimental challenger frozen

Invalid / non-evidence runs explicitly retained in history

R169 initial grid was invalid because the implementation admitted unintended portfolios; PATCH1 is the official R169 evidence. R176 initial run failed due to duplicate qcut label handling; PATCH1 is the official R176 evidence. These implementation failures did not alter the research hypotheses and are not empirical evidence.

Official ZIPs remain the performance source of truth. This addendum summarizes them; it does not replace the underlying CSV/manifest evidence.

Frozen research contract after v1.14

Item	Value
Master version	v1.14
DEV cutoff	2026-07-31
Forward OOS start	2026-08-01
Production	Frozen 4F EW, 25% each
Layer-1 Primary Challenger	R180 Conditional Shield Reinforcement Strong
Post-freeze Experimental	R181 Extreme Shield 100%
Latest implementation	r181/app.py
Latest app SHA-256	64ddd826920186bf0eaeaa85f9a121f373f8ef9a56901f615562b9fe68c2a877

Non-negotiable frozen rules

1. Do not reopen Layer-0 lookbacks or frozen family construction because of Layer-1 outcomes.
2. Do not search additional Shield weights on DEV after R181.
3. Do not add loss-magnitude, relmom-gap, family-identity, entry/continuation, dispersion or rebound filters to A-state on DEV.
4. Do not call 50/50 or 100% an optimized allocation. They are predeclared/frozen research doses.
5. Do not promote R181 over R180 from DEV evidence alone.
6. Do not change Production without a separate promotion-governance decision.
7. Preserve 2026-08+ as prospective OOS and record results without retuning.

Current synthesis

Four independent Layer-0 families remain intact. Spear selection did not justify abandoning equal weighting across Seiryu, Byakko and Suzaku. The strongest Layer-1 finding is instead conditional allocation between that diversified Spear block and Genbu Shield. A sparse two-axis A-state identifies historical months where additional Shield exposure had strong relative value. R180 converts that finding into the frozen primary challenger; R181 records the extreme-dose hypothesis without contaminating the R180 freeze.

Research phase boundary: Layer-0 COMPLETE. Layer-1 DEV COMPLETE. The next phase is prospective forward-OOS monitoring and separately governed Production-replacement evaluation, not further DEV optimization.

End of v1.14 addendum. The first 91 pages are the unchanged v1.13 Master. This addendum extends the official research history through R181.

Master Research Bible - v1.15 Addendum

My4F / Four Gods / v3.45 Research

Scope: R182 PATCH1 through R185 | 2026-09-14

Governance. Official evidence is user-run result ZIP only. The first 104 pages of v1.14 are preserved unchanged. Production remains Frozen 4F EW; R180 remains the Frozen Primary Challenger. DEV cutoff = 2026-07-31; 2026-08 onward is Forward OOS.

Critical correction: Initial R182 is INVALID / NOT EVIDENCE. It declared monthly family-sleeve rebalance but applied start-month family weights to daily family returns, implicitly creating daily family rebalancing. Only R182 PATCH1 is official.

Architecture after R185: evidence increasingly supports 3+1: Seiryu / Byakko / Suzaku form the Spear construction block; Genbu is a distinct Shield block. Spear construction and Spear-vs-Shield allocation are separate Layer-1 decisions.

R182 PATCH1 - Minimum Variance Structural Audit

24/24 PASS. 252 common trading days; sample covariance; t-1 only; monthly rebalance; long-only; sum=1; no cap; no expected-return input; no lookback/shrinkage search; no R180 A-state combination.

Strategy	CAGR	Sharpe	Sortino	MaxDD	Calmar
EW_4F	46.50%	1.668	2.304	-22.39%	2.077
SpearMinVar_Shield25	46.04%	1.671	3.117	-14.53%	3.168
Hierarchical_MinVar	42.41%	1.542	2.530	-18.34%	2.312
Direct4F_MinVar	42.19%	1.543	2.525	-19.33%	2.182

Result: SpearMinVar + Shield25 survives as the risk-quality challenger. Hierarchical MinVar and Direct 4F MinVar are weak/rejected. MinVar belongs inside Spear, not across Spear/Shield or all four families.

SpearMinVar + Shield25: CAGR 46.04%, Sharpe 1.671, Sortino 3.117, MaxDD -14.53%, Calmar 3.168. The result strengthens the interpretation that Genbu is not merely a fourth peer return sleeve.

R183 - Spear MinVar x Frozen A-State

25/25 PASS. R182 PATCH1 252D Spear MinVar is combined with the frozen R180 A-state. A-state remains calculated from Original 3-Spear EW only. Non-A = 75/25; A = 50/50.

Strategy	CAGR	Sharpe	Sortino	MaxDD	Calmar
Control_Original75_25	46.50%	1.668	2.304	-22.39%	2.077
R180_Frozen_Original_A50_50	48.95%	1.749	2.455	-18.76%	2.609
R182_SpearMinVar_Shield25	46.04%	1.671	3.117	-14.53%	3.168
R183_Combined_MinVarSpear_A50_50	47.54%	1.755	3.134	-14.53%	3.272

R183 vs Control: CAGR +1.04pt, Sharpe +0.0868, Sortino +0.8307, MaxDD +7.86pt improvement, Calmar +1.195.

R183 vs R180: CAGR -1.41pt but Sharpe +0.0061, Sortino +0.679, MaxDD +4.23pt improvement, Calmar +0.663. Formal role: Risk-Quality Challenger, not R180 replacement.

Mechanism: A-state R183-R180 mean increment about +0.781%/month, while Non-A is about -0.262%/month. The CAGR drag is concentrated in normal-state always-on MinVar.

R184 - A-State-Only Spear MinVar One-Shot

22/22 PASS. Explicit post-hoc one-shot inspired by R183 decomposition. Non-A = R180 exactly (Original Spear 75 / Shield25). A = MinVar Spear50 / Shield50. No additional state, covariance, cap, shrinkage, threshold, or weight search.

Strategy	CAGR	Sharpe	Sortino	MaxDD	Calmar
Control_Original75_25	46.50%	1.668	2.304	-22.39%	2.077
R180_Frozen	48.95%	1.749	2.455	-18.76%	2.609
R183_AlwaysMinVarSpear	47.54%	1.755	3.134	-14.53%	3.272
R184_AOnlyMinVar	50.84%	1.754	2.562	-15.83%	3.211

R184: CAGR 50.84%, Sharpe 1.754, Sortino 2.562, MaxDD -15.83%, Calmar 3.211. Versus R180: CAGR +1.89pt, Sharpe +0.0049, Sortino +0.107, MaxDD +2.93pt improvement, Calmar +0.602.

LOYO vs R180: CAGR 14/14, Sharpe 12/14, Sortino 14/14, MaxDD 14/14, Calmar 14/14. Formal role: Post-hoc Experimental Balanced Challenger. It does not automatically supersede R180.

R185 - Shield100 vs Defensive Spear50

24/24 PASS. Direct falsification only. No new weight. R181 and R184 have identical Non-A behavior. A-state compares R181 = Shield100 against R184 = MinVar Spear50 + Shield50.

Strategy	CAGR	Sharpe	Sortino	MaxDD	Calmar
Control_Original75_25	46.50%	1.668	2.304	-22.39%	2.077
R180_Frozen	48.95%	1.749	2.455	-18.76%	2.609
R181_ExtremeShield100	53.37%	1.779	2.647	-15.83%	3.371
R184_AOnlyMinVar	50.84%	1.754	2.562	-15.83%	3.211

Full DEV: R181 exceeds R184 by +2.53pt CAGR, +0.0253 Sharpe, +0.0848 Sortino, and +0.160 Calmar; MaxDD is identical at -15.83%.

LOYO R184 vs R181: CAGR 0/14, Sharpe 1/14, Sortino 0/14, MaxDD improvement 2/14, Calmar 0/14. Across 23 A-events, R181-R184 averages +1.060% per A-month, median +0.492%, win rate 13/23.

Long-run DEV therefore favors complete Shield retreat once A fires. This strengthens the 3+1 Spear/Shield interpretation, but does not prove that Shield100 is an optimum.

R185 Era Robustness and Recent Reversal

Era	R180	R181 Shield100	R184 A-only MinVar
2013-2016	34.06%	39.30%	34.22%
2017-2020	68.91%	77.38%	73.13%
2021-2024	44.51%	47.50%	44.92%
2025-DEV	44.46%	42.11%	48.84%

R181 has higher CAGR than R184 in 2013-2016, 2017-2020, and 2021-2024. 2025-DEV reverses the ranking: R181 CAGR 42.11%, Sharpe 1.612, MaxDD -7.21%, Calmar 5.840; R184 CAGR 48.84%, Sharpe 1.888, MaxDD -6.35%, Calmar 7.696.

This matches the known Extreme-Shield failure mode: an A-state followed by a sharp Spear V-rebound can create large opportunity loss. In R185 A-events, the best R181-R184 month is +10.90pt and the worst is -16.36pt.

Decision: the disagreement between long-run DEV and recent DEV should be resolved by untouched Forward OOS, not by testing Shield75/87.5 or another post-hoc state filter.

Layer-1 Structural Synthesis after R185

Spear block: Seiryu, Byakko, Suzaku are distinct attack/return engines. Covariance-aware construction can be useful within this block.

Shield block: Genbu behaves differently. Direct 4F MinVar and Spear-vs-Shield MinVar are weaker than preserving Genbu as a separate functional allocation block. R185 further shows that, in the frozen A-state, complete retreat to Genbu outperformed retaining a defensive MinVar Spear over long DEV.

Current architecture: Layer 1A = Spear construction; Layer 1B = Spear-vs-Shield allocation. This empirically aligns with the recorded Bam concept: Seiryu / Suzaku / Byakko = Spear; Genbu = Shield. It does not establish any unrecovered historical Bam numeric allocation.

Closed for DEV: Shield75/87.5 interpolation; alternative A filters; 126D/504D MinVar; covariance shrinkage; weight caps; EW/MinVar mixtures; additional state-specific MinVar branches. No further DEV weight grid after R181.

Frozen Status and Forward-OOS Contract

Production: Frozen 4F EW - unchanged.

R180 Frozen Primary Challenger: Normal Original Spear75 / Shield25; A -> Original Spear50 / Shield50.

R181 Frozen Experimental Extreme / Structural Shield Challenger: Normal75/25; A -> Shield100.

R183 Risk-Quality Challenger: Always 252D long-only MinVar Spear; Normal75/25; A50/50; A-state still from Original Spear.

R184 Experimental Balanced Challenger: Non-A = R180; A -> MinVar Spear50 / Shield50.

Forward OOS: 2026-08 onward. Compare Production / R180 / R181 / R183 / R184 unchanged. Forward results must not be used to retune DEV definitions.

Primary unresolved OOS question: when frozen A fires, does R181's long-run DEV advantage from complete Genbu retreat persist, or does R184's recent advantage from retaining 50% defensive MinVar Spear persist?

Evidence chain: R182 initial INVALID -> R182 PATCH1 24/24 PASS -> R183 25/25 PASS -> R184 22/22 PASS -> R185 24/24 PASS.

Master Research Bible - v1.16 Addendum

My4F / Four Gods / v3.45 Research
Research Engineering Consolidation + Bam Layer North Star
Updated 2026-09-14

本Addendumはv1.15の112ページを変更せず保持し、その後に確定した研究運用原則・長期アーキテクチャ目標を追記する。R185までの投資研究結果・Frozen status・Forward OOS contractは変更しない。

1. v1.16で変わるもの / 変わらないもの

変わらないもの：Production = Frozen 4F EW、R180 = Frozen Primary Challenger、R181/R183/R184の各status、DEV cutoff = 2026-07-31、2026-08 onward = Forward OOS、R182 initial = INVALID / NOT EVIDENCE、closed DEV branches、user-run result ZIP only = official evidence。

今回変わるもの：研究をL3まで安全に継続するためのResearch Engineeringを体系化し、Bam本来のLayer taxonomyを長期North Starとして明示する。これは研究工程の改善であり、投資strategyの再最適化ではない。

2. Current investment-research checkpoint

Object	Status	Frozen definition / role
Production	FROZEN	Seiryu / Byakko / Suzaku / Genbu = 25% each
R180	Frozen Primary Challenger	Non-A Original Spear75 / Shield25; A Original Spear50 / Shield50
R181	Experimental Extreme	Non-A 75/25; A Genbu Shield100
R183	Risk-Quality Challenger	252D long-only MinVar Spear always; frozen Original-Spear A-state
R184	Post-hoc Balanced Challenger	Non-A = R180; A MinVar Spear50 / Shield50
R185	COMPLETE / latest DEV study	Long-run DEV favors R181; 2025-DEV favors R184; resolve only in untouched Forward OOS

R185以降のPrimary unresolved question：frozen A-state発火時に、R181のGenbu100という長期DEV優位がForward OOSでも持続するか、それとも直近DEVで優位だったR184のdefensive MinVar Spear50保持が優位となるか。Forward OOSは判定材料であり、retuning材料ではない。

3. Bam Layer taxonomy and long-term North Star

今後はBam本来のLayer nomenclatureと、R185までMy4F内部で便宜的に使ってきたLayer表現を明確に分離する。過去文書の呼称を黙って書き換えない。

Bam-lineage target taxonomy

- L0：個別ETF/個別メカニズムのモメンタム戦略（四神など）。
- L1：複数L0を動的に統合・ローテーションするFoF。
- L2：複数L1をさらに統合するNested FoF / FoF of FoF（奥義）。
- L3：十分に独立した2つの完成L2奥義を50:50で保有するpair。

Long-term North Star：My4F研究は、十分なfalsificationとprospective evidenceを通過した2つの独立L2を最終的に50:50で保有するL3を長期目標とする。ただしL3を作ること自体を目的化しない。二つ目のL2が必要条件を満たさなければ作らない。

R185までの「Layer 1A = Spear construction / Layer 1B = Spear-vs-Shield allocation」はMy4F内部の便宜的なportfolio-research decompositionとして保持する。Bam L1/L2/L3とは別概念であり、今後のMasterでは必ず区別して記載する。

4. Frozen Strategy Alpha Falsification direction

今後の既存Frozen object評価では、CAGR / Sharpe / Sortino / MaxDD / Calmarだけでなく、strategyを変更しないdiagnostic/falsification axisとして、 β -adjusted alpha、OOS alpha、expanding/walk-forward alpha、regime別alpha、multi-metric alphaを検討対象にできる。

重要：これらはProduction / R180 / R181 / R183 / R184の性質を測るためのfalsification suiteであり、DEVでwinnerを選び直すoptimization toolではない。regime結果から新routerを作る、weight/A-state/lookbackを変える、closed branchを再開することは禁止。

5. Research Engineering operating model

ユーザー側の研究フローは従来通りとする。高度化するのはassistant側の研究管理・監査・再現性である。

User-facing canonical workflow

- ・ユーザーが検証を依頼、または「次は何を検証するか」を相談する。
- ・Research Designer + Skeptical Criticが、必要性・Governance・falsifiability・complexityを事前評価する。
- ・承認された場合のみapp.pyを実装する。basenameは常にapp.py。提出前に `python -m py_compile app.py` をPASSさせる。
- ・ユーザーがStreamlitで実行しresult ZIPを返す。user-run ZIPのみofficial evidence。
- ・Independent Audit / Evidence JudgeがZIPを監査し、PASS/REJECT/INVALIDを判定する。
- ・必要な次研究へ進む。節目でMasterを更新する。

Masterは毎study更新を必須としない。Freeze/Complete、研究フェーズ変更、複数studyで一まとまりの結論が形成された時点など、研究上の節目で更新する。

6. Role separation

Role	Primary responsibility
Research Designer	研究質問、事前仕様、pass/reject条件を定義
Skeptical Research Critic / 軍師	そもそも研究すべきかを実装前にfalsify
Implementer	承認済み仕様を実装。科学的変更をbug fixへ混入させない
Independent Auditor	Logic/Numerical, Interface/Schema, Evidence/Governanceなど直交したfailure modeを監査
Evidence Judge / Curator	official evidence昇格、Lesson/Gate/Graphのcanonical promotionを管理

7. Mandatory pre-implementation gates

- Research Necessity Gate：未解決の研究質問が本当に存在するか。
- Governance Gate：Frozen spec、closed DEV、Forward-OOS anti-retuningに違反しないか。
- Falsifiability Gate：実行前にPASS/REJECT/disconfirming evidenceを言語化できるか。
- Complexity Gate：追加機構が具体的failure modeを解決し、複雑性に見合うか。
- Mandatory Retrieval Gate：Current Frozen specs、closed branches、関連Failure Cases、ACTIVE lessons/gates、Evidence hierarchyを取得してから実装へ進む。
- Stub/Completeness Gate：compileやartifact存在だけでなく、TODO/pass/dummy validation/未接続audit等がないか。

IDEA/PROPOSAL段階では探索の自由度を保つ。一方、OFFICIAL EVIDENCE CANDIDATEやFREEZE/COMPLETEへの昇格は厳格なGateで強制する。「守ってください」という注意書きではなく、可能な限り仕組みでblockする。

8. Research memory architecture

研究記憶は、単一の長大な会話コンテキストに依存せず、次の三層で外部化する。

- Raw Evidence / Research Event Store：official ZIP、manifest、validation/audit、FAIL/INVALID、Freeze/Promotion decision、重要proposalを原形に近く保持。
- Causal Research Graph：study、hypothesis、failure、fix、lesson、gate、supersession、falsification、evidence promotionをtemporal provenance付きで接続。
- Semantic Retrieval Layer：異なる表現でも同一spec/lesson/failureへ到達できる検索層。

Hot/CoreにはCurrent/Frozen/governanceの最小地図を置き、Cold/DetailにMaster・official ZIP・Failure Casebook・詳細auditを保持する。古いという理由だけで重要度を下げず、Status / dependency / provenanceを優先する。

Canonical knowledgeは「everyone reads, one writer」の思想でtransactionalに昇格する。AI推論はAUTO-INFERRED/CANDIDATEのまま保存し、Curator/Evidence reviewなしにCANONICALへ昇格させない。

9. Temporal provenance, evidence classes, and decision lineage

重要研究イベントでは少なくとも、仮説/仕様が成立・Freezeされた時点、official evidenceが得られた時点、Master/Ledgerへrecord/promoteされた時点を分離する。目的は「結果を見る前にFrozenされていたか」を後から監査可能にすること。

Evidence/source reliability

Class	Meaning
E0 Official Evidence	user-run official result ZIP
E1 Canonical Record	最新Master等に正式収録されたcurrent record
E2 Frozen Spec / Decision	事前定義された仕様・Freeze/decision record
E3 Audit Finding	独立監査結果。official performance evidenceとは区別
E4 Hypothesis / Inference	研究仮説、assistant interpretation
E5 External Research-Engineering Reference	Bam先生コメント等。工程設計の参考でありMy4F投資成績の証拠ではない

Decision RecordsはADOPTED / REJECTED / DEFERREDを理由付きで保持する。closed DEV branchesは「忘れた」のではなく「意図的に閉じた」ことを追跡可能にする。

10. Failure Casebook and Root-Cause Cascade

抽象的Lessonだけでなく、重要FAIL/INVALIDの具体的経緯を保存し、関連タスクのfresh-context workerへ選択的にpreloadする。

Failure Case schema

- What happened / intended specification / actual behavior
- Root Cause
- Why existing defenses failed
- How detected
- Correction
- Evidence consequence/status
- Resulting Immunization/Gate
- Related lesson/principle
- Temporal provenance and later utility

Canonical case - R182 initial : monthly family-sleeve rebalanceを宣言したが、start-month family weightsをdaily family returnsへ直接適用し、実質daily family rebalancingとなった。Initial R182はINVALID / NOT EVIDENCE。PATCH1のみofficial evidence。この事故からdeclared-vs-implemented rebalance parityを具体的に監査する。

FAIL/INVALID時はImplementer errorだけで終わらず、Implementation → Specification → Documentation → Gate/Validation → Knowledge Retrieval/GovernanceまでRoot-Cause Cascadeを行う。一つの事故から複数の改善candidateが出てよいが、実装・昇格はatomicに行う。

11. Ralph / Dream / Immunize continuous-learning loop

- Orient : Current state / Frozen / Governanceを取得。
- Gather : 新しいofficial ZIP、audit、run eventを取得。
- Consolidate : 重複・矛盾・temporal provenanceを整理。
- Prune : Hot/Coreから不要な詳細を外しColdへ。
- Immunize : validated research-process failureを再発防止Gate候補へ変換。
- Curator/Evidence review後、approved improvementをatomicにcanonical knowledgeへ昇格。
- fresh-contextで再検証し、Lesson/Gateが実際に役立ったかutilityを再測定。

Lesson lifecycleはACTIVE / MERGED-SUPERSEDED / RETIRED等を持ち、履歴は削除しない。参照回数だけでなくUseful / Harmful-or-false-alarm等の実効性を測る。

Critical boundary : Research PROCESSは自己改善してよい。Frozen INVESTMENT STRATEGYは自己最適化してはならない。Forward OOSを使ってA-state、weight、lookback、covariance、router等を自動変更しない。

12. Research Ontology / Dependency & Impact Graph

Material changeが起きた場合、変更objectからdownstream dependencyを多段探索し、必要なparity/regression/audit/documentation checksを自動導出する。検査項目をagentの記憶に依存させない。

Example impact chain

Seiryu frozen series implementation correction → Original Spear → frozen A-state identity → R180 / R181 / R184 → Forward-OOS comparison continuity → Master records

各edgeはAUTO-INFERRED/CANDIDATE → validation/Curator → CANONICALのlifecycleを持つ。未知語・新概念はPROVISIONAL CONCEPTとして扱い、意味が確定してからcanonical化する。単純な出現回数だけで昇格させない。

13. Research Telemetry & Spiral Optimization

Research Engineeringが重くなりすぎないように、実際のuser-run path（app.py → official ZIP）に軽量telemetryを埋め込む。

- frozen-data / cached-series load time
- signal and portfolio construction time
- cache hit / miss
- parity / regression / validation duration and status
- warning / fallback usage
- artifact and ZIP generation status

Optimization loop：Measure end-to-end → dominant bottleneckを1つ特定 → atomic improvement → exact parity/regression → PASSならend-to-end再計測 → 必要箇所のみinstrumentを深める → repeat。

Frozen済みLayer-0系列、validated official artifacts、cache済み中間データを再利用し、完了済み研究を毎回再計算しない。失敗した高速化やtelemetry bugも Failure Casebookへ残す。

Priority order：Correctness / Evidence Integrity > Reproducibility / Auditability > Research Throughput。高速化が挙動を変える、validationを弱める、Gateを迂回する場合は不採用。

14. Fresh-context recovery target

研究品質を『現在のChatGPTがR1-R185を覚えているか』に依存させない。新しいチャット/agentでも、durable artifactsからCurrent Frozen specs、Evidence hierarchy、closed branches、重要Failure Cases、最新official evidence、次の研究質問を復元できることを完成条件とする。

Minimum recovery bundle

- Compact Core Research Contract / current-state map
- Latest Master Research Bible
- Latest relevant user-run official ZIP(s)
- Frozen Registry + Decision Records
- Research Lessons / Audit Ledger
- Task-relevant Failure Casebook
- Dependency/Provenance map and Forward-OOS status

15. Current forward plan after v1.16

- A. Production / R180 / R181 / R183 / R184を変更せずForward OOSで追跡する。
- B. R185のlong-run vs recent reversalは新DEV filterで解決せず、untouched Forward OOSで観察する。
- C. 必要に応じ、既存Frozen objectsを変更しないalpha falsification suiteをdiagnosticとして実施する。
- D. Bam本来のL0/L1/L2/L3 taxonomyとMy4F内部の便宜的taxonomyを明確に区別し、長期的には独立したL2奥義×2のL3 50:50をNorth Starとする。
- E. L3を急がない。次Layerへ進む前にResearch Necessity / Governance / Falsifiability / Complexity Gatesを通す。
- F. Master更新は研究上の節目で実施し、user-facing workflowは従来通り『依頼 → app.py → user-run ZIP → audit/analysis → 次研究 → 節目でMaster』を維持する。

16. Non-negotiable one-line rules

- User-run ZIP only = official evidence.
- Failed implementation run = NOT evidence.
- Bug fix ≠ research-spec change.
- Forward OOS ≠ retuning dataset.
- Closed DEV branch stays closed unless a new explicit governance decision reopens it.
- Production and Challenger are separate.
- R180 remains Frozen Primary Challenger until a separate promotion/governance decision.
- Self-growing Research Infrastructure ≠ Self-optimizing Investment Strategy.
- A fresh agent must retrieve before it reasons.
- Measure before optimizing; parity before accepting speed.

End of v1.16 Addendum

This addendum changes research infrastructure and long-term architectural direction only. It does not alter R185 official evidence, Frozen strategy definitions, Production status, or the Forward-OOS anti-retuning contract.

Master Research Bible - v1.17 Addendum

My4F / Four Gods / v3.45

North Star Correction

Updated 2026-09-14

本Addendumはv1.16の投資研究結果・Frozen status・Forward OOS contractを一切変更せず、v1.16で強く表現しすぎた長期North Starのみを訂正する。

1. Superseded statement

v1.16では、Bam-lineageのL0 → L1 → L2 → L3を長期North Starとして示し、独立したL2奥義を2つ構築して50:50で保有するL3を最終到達点のように記載した。この表現はユーザーの意図より狭く、L3構造そのものを目的化する危険があるため、本v1.17でsupersedeする。

2. Correct North Star

My4Fの長期目標は「L3を作ること」ではない。

目標は、Bam先生のL0/L1/L2/L3体系と同等レベルの、研究アーキテクチャの深さ、戦略間の独立性、頑健性、falsification品質、prospective validation、再現性、Evidence Governanceを備えたシステムへ到達することである。

BamのLayer体系は、研究成熟度・システム品質を測るbenchmark / reference architectureとして扱う。My4Fの最終構造は、Evidenceが支持するならBamのL3と同型でもよいが、異なる構造でもよい。

3. L3の位置づけ

L3 = 2つの独立L2を50:50で保有する構造は、今後も有力なreference architecture / possible destinationとして保持する。しかし、以下を明確に禁止する。

- ・L3を完成させるためだけに第二のL2を無理に作ること。
- ・50:50という形を先に固定し、それに合う候補をDEVから探すこと。
- ・Layer数を増やすこと自体を研究進歩とみなすこと。
- ・Forward OOSを利用してL3へ近づくようweight / router / state / lookbackをretuneすること。

4. What counts as progress toward the North Star

今後の進歩はLayer番号ではなく、次の性質で評価する。

- ・独立したreturn engine / functional roleが実証されているか。
- ・単一期間・単一metric・単一regimeに依存せずfalsificationを生き残るか。
- ・DEVでFreezeされた仮説がuntouched Forward OOSで評価されているか。
- ・βや単純なrisk exposureだけでは説明できない付加価値があるか。
- ・組み合わせによってrisk quality / tail behavior / robustnessが改善するか。
- ・複雑性の増加が具体的failure modeまたは独立性の追加によって正当化されるか。
- ・fresh-contextでも仕様・Evidence・失敗系譜を再現できるResearch Engineeringがあるか。

5. Relationship to current R185 state

Production = Frozen 4F EW、R180 = Frozen Primary Challenger、R181/R183/R184のstatus、R185のlong-run vs recent disagreement、DEV cutoff = 2026-07-31、2026-08 onward = Forward OOSはすべて変更しない。

現在の3+1 Spear/Shield構造は、それ自体がBam体系の特定Layerへ無理に押し込まれるものではない。まずMy4F固有のEvidenceに基づくarchitectureとして保持し、その後の研究でより高次の統合が自然に必要な場合のみ進める

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6. Canonical wording for future Masters / handoffs

North Star: My4F aims to reach a Bam-class level of research architecture and robustness - comparable depth of hierarchical construction, independence, falsification quality, prospective validation, and evidence governance. Bam's L0→L1→L2→L3 is a benchmark and possible reference architecture, not a mandatory endpoint. My4F may converge to L3-like structure only if its own evidence naturally supports it.

この記述はv1.16内の「独立したL2×2→L3 50:50を長期目標とする」という強い表現をsupersedeする。v1.16のその他のResearch Engineering追加事項はCURRENTのまま維持する。

End of v1.17 Addendum

Master Research Bible - v1.18 Addendum

My4F / Four Gods / v3.45

Frozen Strategy Robustness / Anti-Overfitting Falsification Stack

Updated 2026-09-14

Source inspiration: バム先生「DM-signalの検証について話そう。投資戦略をつくる10のステップ。オーバーフィッティングを信じるな。」(2026-09-09)。本AddendumはResearch Engineering guidanceとして採用する。外部記事はMy4Fの投資成績に関するofficial evidenceではない。

1. Purpose and governance boundary

本Addendumの目的は、将来の新規研究におけるselection bias / overfitting / researcher look-aheadをより系統的に壊しにいくためのfalsification stackを定義することにある。

現時点のProduction / R180 / R181 / R183 / R184はR185後すでにReal Forward OOS段階に入っている。本Stackを理由にDEVへ戻ってweight、lookback、A-state、covariance、router、regime filterを再探索しない。closed DEV branchesも再開しない。

2. Canonical 10-step falsification stack

Stage	My4F interpretation
1. Massive Grid Search	単一点のbest parameterを探すのではなく、事前に正当化されたparameter space全体の形状とplateauを見る。
2. Parameter-Space Robustness / Alpha Positive Rate	全valid candidateのうち、 β -adjusted α CAGR > 0である割合を確認する。Championだけでなく、edgeがparameter spaceのどの程度に広がっているかを見る。
3. IS / OOS	通常のIn-Sample / Out-of-Sampleは必要だが十分ではない。OOSまで含む多数候補選択によりselection biasは残り得る。
4. Multi-Window Walk-Forward	複数のhistorical learning-window長で選択→将来評価を繰り返し、特定training-window依存を疑う。
5. Beta-Adjusted Alpha	market betaで説明できる部分を除去してもstrategy固有の付加価値が残るかを評価する。Beta adjustmentでFAILする戦略が存在してよい。
6. Expanding Robustness	rolling window幅の恣意性に依存していないか、利用可能historyを拡張していく形でも確認する。
7. Regime Robustness	bull / bear / inflation / crash / rebound等のregime別挙動をdiagnosticに確認する。ただし結果からpost-hoc routerを作らない。
8. Vintage OOS	研究環境そのものを過去の時点へ戻し、その時点で利用可能だった情報だけで選択・研究し、その後をOOS評価する。Temporal Provenanceと連携する。
9. Reject / Adopt	一部結果が良くても追加検証で棄却してよい。Idea→Test→Good→More Tests→Rejectを正常な研究結果として残す。
10. Real Prospective Forward	strategy / code / parametersを固定後、研究者にも未知の未来データを順次受け取り、retuningなしで評価する。最終的に最も強い検証層。

Canonical sequence: Massive Grid Search → Parameter-Space Robustness → IS/OOS → Multi-Window Walk-Forward → Beta-Adjusted Alpha → Expanding Robustness → Regime Robustness → Vintage OOS → Reject/Adopt → Real Prospective Forward

3. Alpha Positive Rate - My4F adaptation

Alpha Positive Rateは、parameter landscape全体の β -adjusted α CAGRの符号を見るdiagnosticである。

Alpha Positive Rate = count(valid candidates with α CAGR > 0) / count(all valid candidates)

ただしMy4Fでは、これを理由に新しいgridを後付けで作ることを禁止する。適用対象は以下に限定する。

- 将来の新規研究で、実行前にparameter family / coverageがprospectively定義されている場合。
- 過去のofficial researchで既に全candidate landscapeが保存されており、Frozen strategyを変更せずdiagnostic re-analysisできる場合。
- Layer0等のFrozen branchを再探索せず、既存Evidenceからfalsificationだけを追加できる場合。

Alpha Positive Rateはcandidate選択器ではない。高いPositive Rateが出てもProduction promotionを自動化せず、低ければfalsification evidenceとして扱う。

4. Vintage OOS as researcher-level anti-look-ahead

Vintage OOSは単なるtrain/test splitではなく、「その時点の研究者が何を知り得たか」を復元する。My4FのTemporal Provenance / Decision Recordsと直接接続する。

Material vintage studyでは少なくとも次を固定する。

- Vintage date / information cutoff
- その時点で利用可能だったdata / strategy candidates / documentation
- その時点より後に得られた研究知識を注入しないこと
- candidate selection rule / rejection rule
- subsequent OOS evaluation period
- 後世のMaster知識をどこまでmaskしたかのaudit record

これにより「現在の研究者が過去全体を知った状態で良いcandidateを発見した」問題を弱める。ただしVintage OOS自体も過去データ上の再構築であり、Real Forwardの代替ではない。

5. Multi-Window Walk-Forward guardrail

複数のtraining-window長を使う場合、window setは実行前に定義する。各windowの結果を見て都合の良い期間だけ採用しない。

現在のR180-R184に対してwindowを変え、良いwindowを探してstrategyを再設計することは禁止。使用するならFrozen hypothesisのhistorical robustnessを壊しにくいdiagnosticに限定する。

6. Regime diagnostics without regime optimization

Regime別結果は『どの環境で何が壊れるか』を理解するために使う。結果を見て新しいregime filter / V-rebound predictor / Shield router / A-state modifierをDEVへ追加してはならない。

特にR185の2025-DEV reversalについて、Regime分析を用いてR181/R184を救済する新ルールを作ることは禁止。解決手段はuntouched Forward OOSである。

7. Where this applies now

Current R185 family: Production / R180 / R181 / R183 / R184はReal Prospective Forwardが主戦場。過去へ戻ってStackを再最適化用途に使わない。

Future new strategy research: 研究開始時点から本Stackを設計へ組み込み、各層でFAIL可能性を明示する。

Existing frozen candidate landscapes: official ZIP等に保存済みのcandidate universeがある場合のみ、strategyを変更せずAlpha Positive Rateやbeta-adjusted falsificationを追加診断できる。

8. Relationship to v1.17 North Star

v1.17で定義したNorth Star - Bam-class level of research architecture and robustness -

を、本Stackは直接補強する。重要なのはLayer数ではなく、どれだけ独立した逃げ道を潰してもEdgeが残るかである。

したがって、My4FがBam先生と同等レベルへ近づいたかは、L3に到達したかではなく、parameter robustness、OOS discipline、 β -adjusted alpha、walk-forward、vintage reconstruction、reject culture、Real Forward等を含む研究品質で評価する。

9. Integration with existing Research Engineering

- Skeptical Critic：Stackのどのfailure modeを今回の研究が検証するのかを実装前に確認。
- Mandatory Retrieval Gate：既存grid / closed DEV / prior OOS / vintage provenanceを取得。
- Temporal Provenance：hypothesis freeze dateとevidence availability dateを分離。
- Failure Casebook：OOS selection、window cherry-pick、regime rescue等の事故を具体例として保存。
- Dependency & Impact Graph：falsification追加がどのdownstream recordへ影響するかを追跡。
- Evidence Judge：新しいfalsification結果をstrategy changeではなくEvidence updateとして判定。
- Ralph / Immunize：発見したresearch-process failureのみをGate改善へ戻す。market outcomeをstrategy parameterへ自動反映しない。

10. Non-negotiable interpretation

本Stackの目的は『良いstrategyを見つけること』よりも、『一見良いstrategyを壊せるだけ壊して、それでも残るものだけを生き残らせること』である。

Real Forwardが悪化した場合に過去へ戻って救済することは、本Stackの思想と逆である。Forwardで悪ければ、その事実自体がEvidenceとなる。

Status: ACTIVE Research Engineering standard from v1.18 onward. External Bam source remains guidance, not official My4F performance evidence.

Master Research Bible - v1.19 Addendum

My4F / Four Gods / v3.45 Research

Permanent Streamlit Application Preservation Contract | 2026-09-14

1. Purpose of this addendum

本Addendumはv1.18の125ページを変更せず保持し、今後のStreamlit app.py更新で失ってははいけない恒久的なアプリ仕様を明文化する。これはResearch Engineering / application governanceの更新であり、Production、Layer-0、R180-R185その他の投資strategy statusやperformance evidenceを変更しない。

2. Permanent archive UI policy

「最新の検証のみでよい」とは、検証アーカイブ画面に表示・実行する過去研究セクションを最新研究1件だけにする、という意味である。アプリ全体を最新研究だけの単機能appへ縮小することを意味しない。

Canonical rule: 検証アーカイブUI = 最新検証のみ。

それ以外の研究基盤・仕様・監査・再現機能は従来どおり保持する。

3. Specifications that must remain in app.py

- Frozen Core / v3.45 / Four Godsおよび各Frozen family仕様。
- Production、Primary Challenger、その他current challengerのcanonical定義とstatus。
- closed DEV branches、DEV cutoff、Forward OOS anti-retuning contract。
- Evidence hierarchy、official user-run ZIP only rule、INVALID / NOT EVIDENCE履歴。
- official ZIP-derived frozen series、embedded evidence、必要なfrozen inputs。
- 再現ロジック、共通関数、データ取得、cache、validated intermediates、performance計算基盤。
- Forward OOS monitoring、live/current signal、必要なPerformance / Self Backtest / PV Performance等の非archive機能。
- registry / manifest / provenance / audit lineage / exactness checks / dependency information。
- 最新研究を正しく実行・監査・再現するために必要な過去研究情報。ただし過去検証UIそのものは再掲しない。

Master Research Bible - v1.19 Addendum

My4F / Four Gods / v3.45 Research

Permanent Streamlit Application Preservation Contract | 2026-09-14

4. Implementation rule for every future research app

新しいRnを追加するときは、直前のfull-spec app.pyを親として使い、検証アーカイブの旧Rn UI blockだけを最新Rnへ置換する。非archiveページ、shared infrastructure、Frozen仕様、embedded official evidence、monitoring/reproduction機能を削除してはならない。軽量化は重複計算・UI archive削減・cache再利用で行い、研究基盤を削ることで行わない。

5. Promotion and validation gate

- app.py basenameを維持し、配布前に `python -m py_compile app.py` をPASSさせる。
- 最新検証ZIPに、registry / exactness / manifest / research contract等のmachine-readable audit情報を残す。
- 実装bug修正とresearch-spec変更を分離し、failed runをofficial evidenceにしない。
- このapp preservation contractは今後のMaster / handoff / fresh-context recoveryでACTIVE ruleとして必ず取得する。

6. Non-negotiable interpretation

今後、Streamlit appの軽量化や検証UI整理を理由として、Frozen specification、official evidence lineage、Forward OOS monitoring、再現ロジック、共通計算基盤を削除してはならない。削除対象は原則として古い「検証アーカイブUI」のみである。

Future app rule:

Full-spec app.pyを保持したまま、検証アーカイブだけを最新Rnへ更新する。

研究基盤を削った単機能app.pyはcanonical successorとして扱わない。

Status: ACTIVE application-governance standard from v1.19 onward. No investment-strategy parameters are changed by this addendum.

Master Research Bible - v1.20 Addendum

My4F / Four Gods / v3.45 Research

Historical falsification program / transition to Real Forward | 2026-09-14

1. Purpose and status of the R186-R191 phase

本Addendumはv1.19の127ページを変更せず保持し、R186-R191で実施したHistorical Falsification / Anti-Ove rfitting programを公式Masterへ統合する。R186-R191はResearch Engineeringとretrospective falsificationの一連のphaseであり、Production 4F EW、各Layer-0 FREEZE / COMPLETE、R180 Primary Challenger、R181/R183/R184 challenger statusを変更しない。

Phase conclusion: retrospective falsification diagnosticsは主要項目まで実施済み。

次の優先順位はDEVでの追加救済・再探索ではなく、2026-08 onwardのReal Prospective Forward OOS。

R188-R191で弱点が見えた場合も、lookback / weight / threshold / routerを後付けしない。

2. Evidence status

- R186 Historical Falsification Coverage Audit: Exactness 14/14 PASS。
- R187 Four Layer-0 Historical Falsification Preparation: Exactness 14/14 PASS。
- R188 Existing Layer-0 Parameter-Space Alpha Audit: Exactness 16/16 PASS。
- R189 Four-Family Frozen Representative Beta-Adjusted Alpha Audit: Exactness 18/18 PASS。
- R190 Four-Family Vintage OOS Pilot: Exactness 16/16 PASS。
- R191 Four-Family Expanding & Regime Robustness Audit: Exactness 16/16 PASS。
- すべてユーザー実行Streamlit ZIPのみをofficial evidenceとして扱う。Assistant-side local calculationはofficial evidenceではない。

3. Unchanged canonical governance

- DEV cutoff = 2026-07-31。2026-08 onward = Forward OOS。
- Production = Frozen 4F EW 25% each。変更なし。
- Layer-0: Seiryu R141 / Byakko R140 / Suzaku R150 / Genbu R167 = FREEZE / COMPLETE。
- R180 = Frozen Primary Challenger。R181 = Frozen Experimental Extreme。R183 = Risk-Quality Challenger。R184 = Post-hoc Experimental Balanced Challenger。
- Closed DEV branchesを再開しない。Forward OOSをDEV retuningに使わない。

Master Research Bible - v1.20 Addendum

My4F / Four Gods / v3.45 Research

R186-R187 falsification coverage and fair-method assignment | 2026-09-14

4. R186 - Historical Falsification Coverage Audit

R186は戦略最適化ではなく、Master / app / official evidence lineageに対して、どのanti-overfitting testが既に存在し、何が不足しているかを棚卸ししたcoverage audit。ResearchObjects=22、SourceRunsReferenced=99。Exactness 14/14 PASS。Production、R180 status、R182 initial INVALID、closed DEV guardrailsをすべて維持した。

5. R187 - Four Layer-0 Historical Falsification Preparation

R187は4 Familyへ同じ「厳しさ」を課しつつ、存在しないcandidate universeを後付けで作らないfairness ruleを確立した。equal rigor does not require identical tests。

Family	Historical universe	R187 assignment
Seiryu / MOM	R89/R90 pre-existing 179	Parameter-space alpha eligible
Byakko / VIX	R118-R140 frozen lineage	Synthetic 179-grid prohibited
Suzaku / TEL-DM	R148 pre-existing 179	Parameter-space alpha eligible
Genbu / MR	R151 PATCH1 179 atlas	Parameter-space alpha eligible

6. Permanent fairness / evidence rule

- Parameter-Space Alpha Positive Rateは、R187以前に既に存在したcandidate universeだけに適用する。
- Byakkoのたぐいに対称性目的の179-gridを新造してはならない。ByakkoはFrozen representative / lineageでbeta-adjusted alpha、regime、vintage、Real Forwardを受ける。
- Genbu terrainはInitial R151を除外し、R151 PATCH1のみofficial evidenceとして使用する。
- R187は新しいstrategy parameterを1つも生成しない。

Master Research Bible - v1.20 Addendum

My4F / Four Gods / v3.45 Research

R188 parameter-space alpha breadth | 2026-09-14

7. R188 - Existing Layer-0 Parameter-Space Alpha Audit

R188はSeiryu / Suzaku / Genbuの既存179候補だけを対象に、SPYとQQQを事前固定benchmarkとしてbeta-adjusted residual alphaのbreadthを検証した。Alpha Positive = Residual Alpha CAGR > 0。winner selection / promotionなし。Byakko synthetic gridなし。

Family	Benchmark	Positive	Rate	Median residual alpha CAGR
Seiryu	SPY	179 / 179	100.0%	+6.14%
Seiryu	QQQ	175 / 179	97.8%	+4.38%
Suzaku	SPY	97 / 179	54.2%	+1.72%
Suzaku	QQQ	87 / 179	48.6%	-0.74%
Genbu	SPY	128 / 179	71.5%	+3.79%
Genbu	QQQ	118 / 179	65.9%	+4.25%

8. R188 interpretation

- Seiryu: parameter-space-wide plateauが非常に強い。SPY 100%、QQQ 97.8% positiveで、single best lookback依存の説明を強く反証。
- Genbu: 全時間軸一様ではないが、RAW / EX1の中長期terrainに広いpositive alpha。Fast D領域まで万能という意味ではない。
- Suzaku: breadthは明確に弱い。特にQQQ基準では179候補の過半がpositiveに届かず、system-wide parameter robustnessは支持されない。
- R188から新しいlookback、3M/30M近傍、weight、Stage A×B searchを始めない。弱いbreadth自体をEvidenceとして保存する。

R188 key distinction: parameter-space breadth と Frozen representativeの強さは別問題。R189でFrozen objectそのものを4神同条件で検証する。

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My4F / Four Gods / v3.45 Research

R189 frozen representative beta-adjusted alpha | 2026-09-14

9. R189 - Four-Family Frozen Representative Alpha Audit

R129で固定済みの4 representativeを166 common months (2012-10 - 2026-07) で比較。SPY / QQQ beta を除去しても4 FamilyすべてResidual Alpha CAGR > 0。

Frozen representative	SPY residual alpha CAGR	QQQ residual alpha CAGR	Beta SPY / QQQ
Seiryu R108	+15.70%	+13.36%	0.94 / 0.83
Byakko R138	+15.13%	+11.76%	2.10 / 1.74
Suzaku 3M+30M	+21.50%	+19.85%	1.15 / 0.94
Genbu R167	+22.92%	+21.87%	1.01 / 0.80

10. Residual independence

- SPY residual correlation: Seiryu-Genbu $\approx +0.004$ 、Suzaku-Genbu ≈ -0.168 、Byakko-Suzaku ≈ -0.056 。
- QQQ residual correlation: Seiryu-Genbu ≈ -0.029 、Suzaku-Genbu ≈ -0.166 、Byakko-Suzaku ≈ -0.081 。
- R129で見えた4-Family diversificationは市場beta差だけではなく、alpha residual levelでも相当部分が残る。
- R189は4神すべてがindependent return engineである仮説を強化するが、DEV上のautomatic promotionを許可しない。

11. R188 + R189 combined reading

特にSuzakuは重要。R188ではparameter-space breadthが弱い一方、R189のFrozen 3M+30M自体はSPY / QQ Q両方で強いpositive residual alpha。したがって「TEL-DMはどのlookbackでも強い」ではなく、「Frozen 3M+30M objectには強いalphaがあるが、parameter-space全体へ一般化できない」がcanonical interpretation。

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My4F / Four Gods / v3.45 Research

R190 frozen-object Vintage OOS | 2026-09-14

12. R190 - Four-Family Vintage OOS Pilot

R190はfull historical decision-process replayではない。現在のFrozen representativeを過去のfixed cutoffsへ持ち込み、各cutoffまでの情報だけでbetaを推定し、その後24M OOSへ固定するFrozen-object historical-cutoff falsification。

Predeclared vintages: 2016-12 -> 2017-18、2019-12 -> 2020-21、2022-12 -> 2023-24。SPY + QQQ。
betaはIS onlyで推定しOOS中はrefitしない。

Family	SPY positive vintages	QQQ positive vintages	Worst OOS residual alpha CAGR SPY / QQQ
Seiryu R108	3/3	3/3	+10.77% / +2.58%
Byakko R138	3/3	3/3	+11.92% / +3.38%
Suzaku 3M+30M	1/3	1/3	-31.63% / -39.52%
Genbu R167	3/3	3/3	+4.54% / +4.11%

13. Suzaku temporal failure is official evidence

- 2017-18: Suzaku OOS residual alpha positive (SPY +23.29%、QQQ +5.93%)。
- 2020-21: negative (SPY -4.36%、QQQ -21.10%)。
- 2023-24: sharply negative (SPY -31.63%、QQQ -39.52%)。
- この結果を見て3M/30M、weight、filter、regime routerを変更することは禁止。failure自体を保存する。

14. R190 interpretation

Seiryu / Byakko / Genbuは3/3 × 両benchmarkでpositive。Suzakuだけtemporal persistenceに明確な疑義。R 189 full-sample alphaの強さだけでこのfailureを打ち消してはならない。Real Prospective ForwardはR190より強いEvidence layerとして未使用のまま保持する。

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My4F / Four Gods / v3.45 Research

R191 expanding / regime robustness | 2026-09-14

15. R191 - Expanding Robustness

common startを固定し、endpointだけを2016-12 / 2018-12 / 2020-12 / 2022-12 / 2024-12 / 2026-07
へ拡張。endpoint選択・最良期間探索なし。

Family	SPY positive endpoints	QQQ positive endpoints	Worst expanding alpha CAGR SPY / QQQ
Seiryu	6/6	6/6	+1.82% / +1.38%
Byakko	5/6	5/6	-7.01% / -4.24%
Suzaku	5/6	4/6	-2.39% / -1.97%
Genbu	6/6	6/6	+15.71% / +17.11%

16. R191 - Regime Robustness

Regimeはprior SPY information onlyで固定。Stress = prior drawdown $\leq$ -10%。NegativeTrend = not Stress and prior 12M SPY return $<$ 0%。Normal = その他。Unclassifiedは除外。Full-DEV betaを1回だけ固定し、そのresidualをregime別に記述した。

Regime	Months	Interpretation
Normal	146	Baseline / dominant sample
Stress	16	Prior drawdown $\leq$ -10%
NegativeTrend	4	Very small N; interpretation limited

- Seiryu: 3/3 regimes positive mean residual for both SPY/QQQ。
- Genbu: 3/3 regimes positive; Stress mean residual annualized $\approx$ +87.3% SPY / +84.1% QQQ。Shield role とindependent alphaを同時に補強。
- Suzaku: 3/3 regimes positive despite R190 Vintage failures。単純なNormal / Stress / NegativeTrend分類ではtemporal instabilityを説明できない。
- Byakko: NegativeTrendだけmean residual negative (SPY -13.34%、QQQ -0.73%) だがN=4。新routerの根拠にはならない。

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My4F / Four Gods / v3.45 Research

Canonical synthesis after R191 / phase freeze | 2026-09-14

17. Family-level evidence synthesis

Family	R188 breadth	R189 frozen	R190 vintage	R191 expanding/regime	Canonical reading
Seiryu	Strong breadth	Strong frozen alpha	3/3	6/6; all regimes +	Most consistently supported
Byakko	N/A by design	Strong frozen alpha	3/3	5/6; NegTrend weak N=4	Good, lineage-based
Suzaku	Weak / mixed	Strong frozen alpha	1/3	5/6 SPY, 4/6 QQQ; all regimes +	Temporal instability
Genbu	Moderate-strong, long-horizon	Strongest frozen alpha	3/3	6/6; all regimes +	Very robust / independent

18. Canonical conclusion after R186-R191

- Seiryu: parameter breadth + Frozen alpha + Vintage OOS + Expanding + Regimeの全層で最も一貫して支持。
- Genbu: 中長期parameter terrain、Frozen alpha、Vintage、Expanding、Stress robustness、residual independenceが整合。Shieldであると同時に独立return engine。
- Byakko: synthetic gridを作らずにFrozen-object falsificationを通過。NegativeTrend N=4の弱点候補は記録のみ。
- Suzaku: Frozen 3M+30Mのfull-sample alphaは強いが、parameter breadthとVintage OOSで弱点。R191 regimeでは単純regime explanationが支持されず、temporal instabilityを持つ有効return engine候補としてFrozen観察を継続。
- 4 Familyすべてを「同じ形で強い」とは扱わない。異なるfailure profileを保持すること自体がanti-overfitting evidence governanceの成果。

19. Phase transition: Real Prospective Forward first

R186-R191 retrospective falsification phase = COMPLETE / FREEZE FOR NOW.

Next priority = untouched 2026-08 onward Real Forward OOS for Production / R180 / R181 / R183 / R184 and frozen Layer-0 representatives.

No DEV rescue of Suzaku, Byakko NegativeTrend, or any other observed weakness. New strategy research must begin as a new predeclared hypothesis, not as a patch to these outcomes.

20. Master / app carry-forward requirements

- v1.19 Permanent Streamlit Application Preservation Contract remains ACTIVE.
- Future app.pyは直前full-spec appを親にし、検証アーカイブUIだけ最新Rnへ更新する。
- R186-R191 official ZIP findings、failure profile、temporal provenance、closed DEV guardrailsを手off / fresh-context retrieval対象にする。
- Bam-class North Starは研究architecture / falsification quality / prospective validation / evidence governanceの水準であり、mandatory L3ではない。

Master Research Bible - v1.21 Addendum

My4F / Four Gods / v3.45 Research

R192-R201 Layer1 representative selection, falsification, and frozen legacy architecture transplant | 2026-09-15

Purpose. v1.20までのR186-R191 retrospective falsification phaseを変更せず、その後に完了したR192-R201の正式Evidenceを追記する。既存134ページは不変。Production / 既存Forward OOS objectsも自動変更しない。

Evidence rule. 本追補で正式Evidenceとして扱うのは、ユーザー実行ZIPのうちexactness gateを全PASSしたrunのみ。R193初回/PATCH1、R195初回/PATCH1、R201初回/PATCH1はNOT EVIDENCEとして履歴のみ保持する。

このフェーズは二段階に分かれる。R192-R200は主として「Layer1に使う4 family representativeをどう選ぶか」を検証し、R201はそこで選ばれた新representativeへ、旧R180/R181/R183/R184 architectureを一切retuneせず固定移植した。

1. R192 - ERC risk architecture falsification

25% x 4F EWを、Direct ERC-4FおよびHierarchical ERCへ置換するone-shot falsification。252D covariance、monthly rebalance、cap/shrinkage/state/routerなし。Exactness 19/19 PASS。

Variant	CAGR	Vol	Daily MaxDD	Monthly MaxDD	Sharpe
Frozen 4F EW	39.02%	22.92%	-36.77%	-19.49%	1.556
Direct ERC-4F	34.45%	20.77%	-30.00%	-13.40%	1.533
Hierarchical ERC	33.23%	20.95%	-31.14%	-12.72%	1.477

Decision: ERC branch REJECT / COMPLETE for adoption. DD低下はあるがCAGR/Sharpeを犠牲にし、EWを上回らない。window/cap/shrinkage/EW blendで救済しない。

この結果により、以後のrepresentative比較は基本的に25% x 4F EWという単純構造を保ったまま行う。

2. R193-R194 - Equal2P transplant and candidate robustness

R193 PATCH2 (17/17 PASS) はByakko Equal2P-47D/20MへGLD/XLU 126D defensive destinationをone-shot移植。Equal2P GLD ControlのCAGR 50.07% / Sharpe 1.198に対し、GLD/XLU126 challengerは51.32% / 1.209。LOQ 55/56、LOYO 14/15。ただしpost-hoc prospective candidateでありDEV promotionなし。

R194 (10/10 PASS) はSeiryu Full5P-Balanced / QLD / TQQQとR193をtemporal robustnessへ通した。Seiryu 3候補はいずれもExpanding 6/6、Vintage 3/3、LOYO 22/22。R193はExpanding active delta 6/6だがVintage active delta 1/3で、時間方向の不均一性が明確になった。

3. R195 - Active Layer1 Registry Final Robustness Gate

R195 PATCH2（19/19 PASS）でActive Layer1 Candidateを14本に固定。全14候補がfull-sampleでSPY/QQQ双方にpositive beta-adjusted residual alphaを持った一方、Vintage OOSでfamilyごとの差が顕在化した。

Family / Candidate	Canonical status	Vintage SPY	Vintage QQQ
Seiryu Full5P-ASH	Robust Core	3/3	3/3
Seiryu Full5P-Balanced	Robust Qualified	3/3	3/3
Seiryu Full5P-QLD	Robust Qualified	3/3	3/3
Seiryu Full5P-TQQQ	Temporal-Qualified	3/3	1/3
Byakko FrozenVIX	Robust Qualified	3/3	3/3
Byakko Equal2P-47D/20M	Temporal-Qualified	2/3	1/3
Byakko FrozenVIX-GLD/XLU126	Robust Core	3/3	3/3
Byakko Equal2P-GLD/XLU126	Temporal-Qualified	2/3	1/3
Suzaku TEL-3M	Temporal-Qualified	1/3	1/3
Suzaku TEL-30M	Temporal-Qualified	2/3	0/3
Suzaku TEL-3M/30M	Temporal-Qualified	1/3	1/3
Genbu MR-24M	Robust Qualified	3/3	3/3
Genbu MR-TriEW-6D/11M/24M	Robust Qualified	3/3	3/3
Genbu MR-Majority-6D/11M/24M	Robust Core	3/3	3/3

Status semantics freeze: Robust Core / Robust Qualified / Temporal-Qualified。 Temporal-QualifiedはREJECTではない。full-sample alphaを持つがfixed-Vintageで発現が不均一なactive candidateであり、portfolio complementarityで価値を持ち得る。R195 artifactの“ TEMPORAL WARNING ”はhuman-facing taxonomyではTemporal-Qualifiedへ読み替えるが、研究結果自体は変更しない。

4. R195B - Byakko Standard-179 landscape gap closure

ByakkoだけR188相当の179-lookback landscapeが欠けていたため、post-freeze retrospective auditとして6 landscapes x 179 = 1,074系列を検証。Exactness 21/21 PASS。

Benchmark	Positive	Total	Positive rate	Median residual alpha CAGR
SPY	1074	1074	100.00%	11.62%
QQQ	1034	1074	96.28%	8.29%

Canonical anchorsのlocal +/-5 grid-cell neighborhoodもSPY/QQQとも11/11 positive。よってEqual2Pの弱さを「47D/20Mの一点当てparameter overfit」と説明する根拠は弱くなった。より自然な解釈はparameter robustだがtemporal instabilityがある、である。

5. R196 - Frozen-object Multi-Window Walk-Forward

R195の固定14候補を一切refitせず、2014-02..2026-07 common eraで12M x 9、24M x 8、36M x 7 = 24 windows。betaのみ各originのexpanding ISでfreeze。Exactness 21/21 PASS。

Candidate	SPY residual +	QQQ residual +	Layer1 Δ Sharpe +	Median Δ4F Sharpe
Seiryu Balanced	95.8%	95.8%	70.8%	+0.0119
Seiryu QLD	95.8%	91.7%	41.7%	-0.0093
Seiryu TQQQ	79.2%	66.7%	33.3%	-0.0514
Byakko FrozenVIX	79.2%	75.0%	75.0%	+0.0097
Byakko Equal2P-47D/20M	58.3%	58.3%	41.7%	-0.0526
Suzaku TEL-3M	33.3%	25.0%	29.2%	-0.0409
Suzaku TEL-3M/30M	41.7%	25.0%	Primary	-
Genbu TriEW	95.8%	95.8%	33.3%	-0.0643
Genbu Majority	87.5%	87.5%	Primary	-

R196でBalancedのLayer1 contribution、FrozenVIXのtemporal advantage、Suzakuのtemporal weakness、Genbu TriEWのstandalone alphaとMajorityのportfolio roleの違いが明確になった。

6. R197 - Combination Complementarity Audit

問いを「単体で最強か」から「他3神へ価値を追加するか」へ変更。raw/residual correlation、Other-3 tail rescue、fixed replacement、worst-window rescue、era、effective rank、LOFOを固定条件で評価。Exactness 20/20 PASS。

Seiryu Balanced: full-era Δ Sharpe +0.0064、 Δ Calmar +0.0523。R196 Layer1 Δ Sharpe positive 70.8%。新Seiryu代表として一貫したportfolio contribution。

Byakko FrozenVIX: current R138 Primaryに対しfull-era Δ Sharpe +0.0108、effective-rank +0.0251。worst6 window Sharpe positive 83.3%。単体Robust CoreのR138より、Layer1 combination representativeとして元のFrozenVIXが有力になった。

Suzaku TEL-3M: Other-3 Primaryとの相関0.258と低く、full-era replacementではSharpe -0.0185だがCalmar +0.1279、worst6 Calmar positive 83.3%。temporal alphaは弱いだがdiversifier/tail-rescue specialistとして意味を持つ。

Genbu: LOFOでGenbu除去時のSharpe悪化が最大（ Δ Sharpe -0.1835）。TriEWはstandalone alphaが強いがreplacementでは Δ Sharpe -0.0333、 Δ Calmar -0.2896。MajorityのShield/portfolio roleを維持。

7. R198 - Byakko & Suzaku Representative Head-to-Head

R196の同一24 WF windows上でfamily内を直接対決。Exactness 21/21 PASS。

Family	Candidate	CAGR beaten	Sharpe beaten	Calmar beaten	4F Sharpe beaten
Byakko	FrozenVIX	3/3	3/3	3/3	3/3
Byakko	Equal2P-47D/20M	0/3	1/3	1/3	1/3
Byakko	FrozenVIX-GLD/XLU126	2/3	2/3	2/3	2/3
Byakko	Equal2P-GLD/XLU126	0/3	0/3	0/3	0/3
Suzaku	TEL-3M	1/2	1/2	1/2	0/2
Suzaku	TEL-30M	0/2	0/2	0/2	1/2
Suzaku	TEL-3M/30M	2/2	2/2	2/2	2/2

Interpretation: ByakkoはFrozenVIXがLayer1 frontrunner。SuzakuはR197で3Mのtail特性が目立ったが、直接対決ではTEL-3M/30Mが総合代表を維持。

8. R199 - Layer1 Representative Championship

Current Primary 4F = ASH / FrozenVIX-GLD/XLU126 / TEL-3M/30M / Majority と、R194-R198から事前固定したChallenger = Balanced / FrozenVIX / TEL-3M/30M / Majority を25% EWで直接比較。Exactness 25/25 PASS。

Portfolio	CAGR	Sharpe	MaxDD	Calmar
Current Primary	47.24%	1.679	-22.39%	2.110
New Challenger	47.23%	1.697	-21.85%	2.162

24 WFではChallengerのCAGR win 50.0%、Sharpe win 75.0%、MaxDD improvement 62.5%、Calmar win 62.5%。SPY residual win 79.2%、QQQ residual win 83.3%。Challenger自身のresidual positiveはSPY 91.7%、QQQ 95.8%。

Sharpe win rateは12M 66.7% → 24M 75.0% → 36M 85.7%と長いwindowほど強まった。full-sample CAGRをほぼ維持しつつrisk efficiencyが改善したため、新4代表EWはDEV-selected Layer1 challengerとして支持された。

9. R200 - Byakko Final Portfolio-Level Falsification

Freeze前の最後の穴として、他3神をBalanced / TEL-3M30M / Majorityに固定し、ByakkoだけA FrozenVIX、B Equal2P-47D/20M、C Equal2P-GLD/XLU126へ変更。Exactness 27/27 PASS。

Portfolio	CAGR	Sharpe	MaxDD	Calmar
A FrozenVIX	43.42%	1.561	-21.85%	1.987
B Equal2P-47D/20M	44.45%	1.548	-21.85%	2.035
C Equal2P-GLD/XLU126	44.85%	1.553	-21.85%	2.053

Full sampleのCAGR/CalmarではEqual2P系が高い。しかし24 WFでBがAに勝つ率はSharpe 37.5%、SPY residual 33.3%、QQQ residual 25.0%。CはSharpe 29.2%、SPY residual 29.2%、QQQ residual 20.8%。

Final Byakko reading: Equal2Pは高performanceかつparameter robustだが時間的発現が不均一。Layer1 representativeはFrozenVIX。Equal2P-47D/20MはTemporal-Qualified high-return alternativeとして保存し、削除・Rejectはしない。

10. Layer1 representative selection freeze after R200

Family	Layer1 representative	Why	Other preserved roles
Seiryu	Full5P-Balanced	WF / portfolio contribution	ASH Robust Core; QLD/TQQQ aggressive alternatives
Byakko	FrozenVIX	temporal reproducibility / 4F role	R138 Robust Core; Equal2P Temporal-Qualified
Suzaku	TEL-3M/30M	family head-to-head overall winner	TEL-3M tail-rescue specialist
Genbu	MR-Majority-6D/11M/24M	Shield / portfolio role	TriEW strong standalone alpha

この時点のunderlying New Layer1 Controlは Balanced / FrozenVIX / TEL-3M/30M / Majority = 25% EW。ただしProduction promotionではなく、DEV-selected and subsequently frozen challengerという位置付け。

11. R201 - Frozen Legacy Architecture Transplant

R129-R185で発見・凍結済みの3 Spear + 1 Shield architectureが旧representative固有の偶然かを検証するため、新representativeへ固定移植。A-state、75/25、50/50、Shield100%、252D MinVar、t-1を一切retuneしない。

Object	Normal	A-state
New_EW_Control	25% x 4	same
New_R180	Spear EW 75 / Shield 25	Spear EW 50 / Shield 50
New_R181	Spear EW 75 / Shield 25	Shield 100%
New_R183	252D MinVar Spear 75 / Shield 25	MinVar Spear 50 / Shield 50
New_R184	Spear EW 75 / Shield 25	MinVar Spear 50 / Shield 50

Implementation history: initial R201はnew representative monthly parity 3/4 FAILでNOT EVIDENCE。PATCH1はR196正式月次系列4/4 parityを達成したがMinVar daily bridgeの約1e-15 floating-point差でhard gate FAILしNOT EVIDENCE。PATCH2は戦略計算を変えずnumerical tolerance 1e-12で監査し、27/27 ALL PASS。正式EvidenceはPATCH2のみ。

12. R201 PATCH2 official results

Object	CAGR	Sharpe	Sortino	MaxDD	Calmar	Δ Sharpe vs EW
New EW	47.23%	1.697	2.348	-21.85%	2.162	-
New-R180	49.97%	1.787	2.599	-17.74%	2.817	+0.090
New-R181	54.98%	1.831	2.756	-13.94%	3.944	+0.134
New-R183	48.08%	1.767	3.051	-14.15%	3.399	+0.070
New-R184	51.97%	1.794	2.730	-14.36%	3.620	+0.098

Full DEVでは旧architecture 4本すべてがNew EWのSharpeを上回った。New-R181はCAGR +7.75pp、Sharpe +0.134、MaxDD約7.90pp改善、Calmar +1.782。

Object	LOYO CAGR win	LOYO Sharpe win	LOYO Calmar win	60M Sharpe win	60M Calmar win
R180	13/13	13/13	13/13	96.7%	100%
R181	13/13	13/13	13/13	89.0%	100%
R183	10/13	13/13	12/13	75.8%	79.1%
R184	13/13	13/13	13/13	98.9%	100%

この再現は「新代表に合わせてarchitectureを再最適化した」結果ではない。代表はR194-R200という別系列で先に選択され、旧architectureはFrozen仕様のまま移植された。よって旧Layer1効果が旧代表固有の偶然だけでは説明しにくい、という独立性の高い補強Evidenceになる。

13. Regime dependence also reproduced

ただしR181をDEV winnerとして単独昇格させない。旧R185と同様、full DEVではR181が強い一方、recentでは性格が変わる。

Era	EW Sharpe	R180	R181	R183	R184
2013-2016	1.353	1.498	1.612	1.392	1.606
2017-2020	1.760	1.867	2.029	1.889	1.848
2021-2024	2.062	2.039	1.896	1.813	2.057
2025-DEV	1.710	1.851	1.620	2.318	1.883

2025-DEVではR181がEWにも劣り、R183/R184が強い。この結果を見てR181/R184 router、Shield比率、A-state threshold、MinVar window等をDEVで再探索することは禁止。これはReal Forward OOSが答えるべき未解決問題として保存する。

14. Canonical architecture after R201

My4Fの現在の研究構造は次のように整理する。

Level	Canonical object / role
Layer 0	各Seiryu / Byakko / Suzaku / Genbu return engine
Layer 1a - Representative selection	Balanced / FrozenVIX / TEL-3M30M / Majority
Layer 1a Control	New 4F EW = 25% x 4
Layer 1b - Frozen architecture challengers	New-R180 / New-R181 Extreme Shield / New-R183 Risk-Quality / New-R184 Balanced
Structural hypothesis	Seiryu + Byakko + Suzaku = Spear; Genbu = Shield

Canonical interpretation: R201は3 Spear + 1 Shield architecture自体にrepresentative交換後も再現する構造的効果がある可能性を支持した。ただし「R181が真の最適解」とは判定しない。R180/R181/R183/R184はそれぞれFrozen challengerとして共存させる。

15. Freeze / no-retuning rules after R201

- 1) Production Frozen4F EWは変更しない。
- 2) 既存Forward cohort: Production / R180 / R181 / R183 / R184は2026-08以降のReal Forward OOSを継続し、retuneしない。
- 3) 新representative cohort: New EW / New-R180 / New-R181 / New-R183 / New-R184も仕様をfreezeし、以後DEVでwinner selectionやrouter rescueをしない。
- 4) Equal2P、TEL-3M、TriEW等の非代表候補もEvidence lineageから削除しない。
- 5) Temporal-QualifiedはREJECTではない。
- 6) R192で閉じたERC、旧closed DEV branches、A-state threshold tuning、Shield weight tuning、R181/R184 router、MinVar window/cap/shrinkage探索を再開しない。
- 7) 2026-08以降のForward OOSは、R194-R201の選択・調整に使わない。

16. Phase conclusion

R192-R201で、単なるheadline performance比較から、representative selection -> temporal robustness -> parameter breadth -> multi-window WF -> complementarity -> family head-to-head -> portfolio championship -> legacy architecture transplantまで進んだ。最も重要な発見は、単体最強とportfolio representativeは異なること、および新representativeへ交換しても3 Spear + 1 Shield architectureの改善が広く再現したことである。

この時点でLayer1のDEV探索フェーズはFREEZE FOR NOW。次のEvidence priorityはReal Prospective Forward OOSであり、DEV上の追加rescue optimizationではない。

Master Research Bible v1.22 - R202 PATCH1 TQQQ Benchmark & Application Integration

Addendum p.1

17. R202 PATCH1 - Frozen Layer1 Architecture vs TQQQ Benchmark Audit

Purpose: R201 PATCH2で凍結したNew Layer1 architecture 5系列を変更せず、実TQQQ Buy & Holdと同一common monthly sampleで比較するbenchmark-only audit。TQQQに合わせたstate、weight、MinVar、representative、router等のretuningは禁止。Forward OOSは使用しない。

Canonical evidence: r202_patch1_frozen_layer1_architecture_vs_tqqq_benchmark_audit.zip / Exactness 17/17 ALL PASS. Common sample = 2014-02..2026-07, N=150 months. Benchmark = actual TQQQ adjusted Close; synthetic 3x seriesは使用しない。DEV cutoff = 2026-07-31。

Object	CAGR	Vol	Sharpe	Sortino	MaxDD	Calmar
TQQQ	38.02%	57.03%	0.854	0.937	-79.09%	0.481
New-EW	47.23%	24.85%	1.697	2.348	-21.85%	2.162
New-R180	49.97%	24.63%	1.787	2.599	-17.74%	2.817
New-R181	54.98%	26.05%	1.831	2.756	-13.94%	3.944
New-R183	48.08%	24.04%	1.767	3.051	-14.15%	3.399
New-R184	51.97%	25.35%	1.794	2.730	-14.36%	3.620

Headline: DEV sampleでは全New Layer1 objectがTQQQより高いCAGR、低いVol、低いMaxDD、高いSharpe/Sortino/Calmarを示した。特にNew-R181はCAGR 54.98%、Sharpe 1.831、MaxDD -13.94%、Calmar 3.944。これは将来優位の証明ではなく、凍結仮説のDEV benchmark evidenceとして扱う。

Master Research Bible v1.22 - R202 PATCH1

Addendum p.2

18. Rolling / LOYO robustness vs TQQQ

Object	36M CAGR win	36M Sharpe/Calmar/DD	60M CAGR win	60M Sharpe/Calmar/DD
New-EW	55.65%	100% / 100% / 100%	80.22%	100% / 100% / 100%
New-R180	59.13%	100% / 100% / 100%	87.91%	100% / 100% / 100%
New-R181	71.30%	100% / 100% / 100%	94.51%	100% / 100% / 100%
New-R183	56.52%	100% / 100% / 100%	79.12%	100% / 100% / 100%
New-R184	60.87%	100% / 100% / 100%	90.11%	100% / 100% / 100%

36M = 115 windows、60M = 91 windows。全5 objectがSharpe、Calmar、DD改善で全window TQQQを上回った。New-R181のCAGR win rateは36M 71.30%、60M 94.51%。

LOYO = 13。全5 object共通で、CAGR win 12/13 (92.31%)、Sharpe 13/13、Calmar 13/13、DD improvement 13/13。したがってheadline resultが単一の特定年だけで成立しているという説明は弱い。

Important interpretation: Rolling/LOYOはDEV内robustnessでありReal Prospective Forwardではない。R202結果を理由にLayer1 architectureを再調整してはならない。

Master Research Bible v1.22 - R202 PATCH1

Addendum p.3

19. Corrected beta-adjusted alpha reporting

R202初版ではOLS residualを複利した値をResidualAlphaCAGRと表示しており、interceptを含むOLS residualの平均が原理上ほぼ0になることと整合しないreportingだった。初版の当該alpha表示はcanonical evidenceにしない。PATCH1ではstrategy return/state/weights/windowsを一切変更せず、alpha reportingのみ修正した。

Object	Beta to TQQQ	Monthly OLS alpha	Annualized alpha	Beta-adjusted CAGR
New-EW	0.3360	+2.1496%	+29.07%	+27.60%
New-R180	0.3195	+2.3706%	+32.47%	+30.82%
New-R181	0.2864	+2.8126%	+39.49%	+36.95%
New-R183	0.3007	+2.3194%	+31.67%	+30.03%
New-R184	0.3197	+2.4929%	+34.38%	+32.54%

OLSResidualMeanは全objectで約 10^{-17} で、切片を含むOLSの性質と整合。TQQQ betaは約0.29-0.34に留まり、DEV sample上ではNew Layer1群を単なるTQQQ/Nasdaq leveraged betaだけで説明しにくい。ただしbeta-adjusted alphaも同じDEV期間上の推定値であり、将来保証ではない。

Master Research Bible v1.22 - Application State & Handoff

Addendum p.4

20. Application integration after R202 PATCH1

Current full-spec app.pyはPermanent Streamlit App Preservation Contractを維持し、validation/archive UIはlatest R202 PATCH1のみを表示する。Production、Frozen specs、evidence lineage、Forward OOS、signals、Performance等の非archive infrastructureは保持する。

Performance pageに以下5系列を追加済み: Layer1 New-EW / Layer1 New-R180 / Layer1 New-R181 / Layer1 New-R183 / Layer1 New-R184。R201 PATCH2で凍結したofficial monthly seriesを利用し、個別/同時表示、資産曲線、Annual Returns、Rolling、Risk、Metrics等の既存Performance visualizationへ統合。DEV historyは2014-02..2026-07までで、2026-08以降を補完しない。Production/live signalは変更しない。

Current app.py SHA256: 8804d68aedccd35758cc3f326a913c784f98563731a950c16954d00418adbf12

Canonical R202 PATCH1 ZIP SHA256: 411b8134ed57010d605661768a955f72d492420316dd0418a693d58fcff92e84

21. Freeze state after R202

R202 PATCH1 = COMPLETE / FREEZE. Layer1 DEV explorationはFREEZE FOR NOWを維持。Production Frozen4F EWは不変。既存Forward cohortとNew representative cohortはretuneしない。TQQQ benchmark resultを使ったrescue optimization、state/weight tuning、representative再選択、MinVar window/cap/shrinkage探索を禁止する。次の最優先EvidenceはReal Prospective Forward OOS。

Canonical Layer1a: Seiryu Full5P-Balanced / Byakko FrozenVIX / Suzaku TEL-3M/30M / Genbu MR-Majority-6D/11M/24M. Control = 25% x 4 EW.

Canonical Layer1b: New-R180 / New-R181 Extreme Shield / New-R183 Risk-Quality / New-R184 Balanced. Structural hypothesis = Seiryu + Byakko + Suzaku = Spear; Genbu = Shield. R181をoptimalとは呼ばない。

Master Research Bible v1.23 Addendum

My4F / Four Gods / v3.45 Research | R203-R215 PATCH1 | 2026-09-15

22. Scope and authority of this update

This addendum updates Master v1.22 through the official user-run R215 PATCH1 evidence. Master v1.22 remains intact as the parent Source of Truth; this addendum appends the post-R202 research lineage and current canonical state.

- DEV cutoff remains 2026-07-31. 2026-08 onward remains Real Forward OOS and must never be used to retune frozen DEV rules.
- Production remains Frozen4F EW. No R203-R215 result automatically promotes a challenger to Production.
- Only user-run Streamlit ZIPs with exactness ALL PASS are official evidence. Failed R209, R210, R212, R213 and initial R215 runs are NOT EVIDENCE.
- Permanent Streamlit App Preservation Contract remains ACTIVE: full-spec app.py is the parent; only validation/archive UI is latest-only; basename app.py is fixed; UI/navigation must not be redesigned as part of research updates.

23. R203-R206 - Legacy R181 historical falsification program

Study	Official status	Canonical result
R203	18/18 ALL PASS	Exact Legacy R181 formal start = 2013-10; common inputs start 2012-10 because A-state requires prior 12M. Clean N=154. CAGR 53.508%, Vol 26.152%, Sharpe 1.7878, MDD -15.830%, Calmar 3.3801. Exact 2008 replay impossible under actual-ETF-only.
R204	26/26 ALL PASS	23 A-state months: Entry 16 + Continuation 7. R181 vs Control wins 17/23. A-state mean increment +2.762%/mo. 2015-16 and COVID decomposition supports sparse conditional overlay interpretation.
R205	26/26 ALL PASS	All 6 R181 failures were positive-Spear rebound/opportunity-cost misses; no both-negative failure. Entry wins 12/16; Continuation 5/7; 11/16 A-state sequences beat Control.
R206	25/25 ALL PASS	16 sequences: 11 payout / 5 premium. R181 vs R180 linked relative edge +45.2023%; benefit/cost ratio 2.7367; removing best sequence still +29.58%; 16/16 LOSO remain positive.

Interpretation: R181 DEV advantage is not explained by one isolated crisis, but it is sparse and conditional. This does not establish future superiority or Production eligibility.

Master Research Bible v1.23 Addendum

Gekiko target benchmark and official-source reconciliation

24. Target strategy: “シン四つ目-激攻”

The user-designated target strategy is “シン四つ目-激攻” (Gekiko), originally Bam-sensei Layer 2. The primary benchmark source is the user-provided Monthly Trade History Return column. R207 ETF reconstruction remains an independent reconstruction audit, not the preferred source after R208.

Formal Four Gods common DEV comparison remains 2014-02..2026-07 (N=150). Gekiko observations from 2026-08 onward are Forward OOS and are excluded from DEV comparison.

25. R207-R209 PATCH1 - growth engine attribution

Study	Status	Key evidence
R207	20/20 ALL PASS	ETF reconstruction: Gekiko CAGR 69.85%, Vol 41.26%, Sharpe 1.498, MDD -31.13%, Calmar 2.244. New-R181 CAGR 54.98%, Sharpe 1.831, MDD -13.94%, Calmar 3.944. Gekiko wins LOYO CAGR 13/13; R181 wins Sharpe/DD/Calmar 13/13.
R208	21/21 ALL PASS	Official Monthly Trade vs reconstruction: corr 0.9643, MAE 2.3441%/mo. Official common-sample Gekiko CAGR 71.20%, Vol 43.57%, Sharpe 1.458, MDD -33.68%, Calmar 2.114. Pure+Mixed Risk-On = 68/150 months but about 87.1% of total log wealth.
R209 PATCH1	25/25 ALL PASS	Full Risk-On = 68 months. Official arithmetic +10.0796%/mo; RiskOnEW +9.7548%. Not-Full-Risk-On RiskOnEW -0.1527%/mo. Entry and exit attribution shows strong timing value. 2014-02 boundary is continuation; true historical Entry count in common window is 13.

Canonical lesson: Gekiko growth is concentrated in selected risk-on months. Defense contributes positively overall. This motivates studying sparse attack overlays without copying or fitting Gekiko final positions.

26. Hierarchical interpretation correction

Because Gekiko is Bam-sensei Layer 2, final output positions should not be assumed to be directly recoverable from a one-layer price rule. The correct conceptual model is market information -> latent/upper-layer state -> Layer-2 branch -> final position. R210-R213 therefore falsify simple direct-rule explanations; they do not prove that no hidden signal exists.

Master Research Bible v1.23 Addendum

R210-R214 - Entry falsification and Layer-2 topology

27. R210-R213 - simple pre-state explanations fail

Study	Official status	Canonical conclusion
R210 PATCH1	17/17 ALL PASS	True Entry N=13; 2014-02 artifact excluded. Prior 1/3/6/12M price features are heterogeneous; simple absolute/dual momentum threshold is not supported. QQQ history incomplete and excluded from conclusions.
R211	17/17 ALL PASS	Entry pathways: A=3M DefensiveOnly 6/13; B=M-1 DefensiveOnly but not streak 4/13; C=M-1 Hybrid 3/13. Immediate prior is never RiskOnly. Multiple natural pathways exist.
R212 PATCH2	14/14 ALL PASS	Entry13 vs eligible non-entry controls65 from same DefensiveOnly/Hybrid starting state. Simple pre-state levels do not naturally separate entries from controls.
R213 PATCH1	16/16 ALL PASS	30 base + 30 Delta1M features. Immediate acceleration also fails to cleanly separate Entry from controls. Direct one-layer momentum-level/acceleration explanation is not supported.

28. R214 - Layer-2 hierarchical state-machine / transition topology

Official R214 ZIP: 15/15 ALL PASS. Common sample 2014-02..2026-07, N=150; 149 transitions; six fixed output states; no unknown states; true Full Risk-On Entries=13.

State	Months	Runs	Mean run	Max run
Pure Risk-On	48	10	4.80m	12
Mixed Risk-On	20	7	2.86m	6
XLU Defensive	40	10	4.00m	11
GLD Defensive	21	13	1.62m	4
Hybrid Risk+Defensive	15	11	1.36m	2
Mixed Defensive	6	4	1.50m	2

95/149 transitions (63.8%) are self-transitions. Superstate counts: Risk-On->Risk-On 54; Defensive->Defensive 47; Defensive->Risk-On 10; Hybrid->Risk-On 3. The 13 true Entries are therefore Defensive/Hybrid -> Risk-On transitions. 28/36 directed six-state edges were observed; unobserved edges are not declared impossible.

Interpretation: final Layer-2 output has strong persistence and asymmetric topology with multiple entry paths. This supports hierarchical/state-machine study, but does not identify or prove hidden branch logic.

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R215 PATCH1 - Four Gods Power-Up / Sparse Risk-On Booster

29. R215 initial run - NOT EVIDENCE

Initial R215 failed because required frozen objects were unavailable: the implementation referenced non-existent or under-specified Layer1 risk-on globals. Fail-closed worked correctly. The run is NOT EVIDENCE and is superseded for research execution by R215 PATCH1.

30. R215 PATCH1 research-spec correction

R215 PATCH1 explicitly removes the invented dependency and uses only frozen executable objects already present in the full-spec app. Exactness = 16/16 ALL PASS. Common DEV sample = 2014-02..2026-07, N=150; Forward months in analysis = 0; frozen A-state count = 23.

- H0 Control: official R201 PATCH2 New-R181.
- A-state identity: $\text{abs}(\text{New-R181} - \text{New-R180}) > 1\text{e-}12$; expected 23/150. A-state has supremacy and suppresses the booster.
- H1_TURBO: frozen Turbo ON -> fixed 10% TQQQ sleeve; otherwise R181.
- H2_PROMOTION100: frozen Promotion100 ON -> fixed 10% TQQQ sleeve; otherwise R181.
- H3_TURBO_PERSIST2: frozen Turbo ON in current and prior month -> fixed 10% TQQQ sleeve; otherwise R181.
- Booster return construction in active month: $0.90 * \text{R181} + 0.10 * \text{TQQQ}$. No threshold/lookback/weight grid, classifier, clustering, router fitting, optimization, or 2026-08+ tuning.

Strategy	CAGR	Vol	Sharpe	MaxDD	Calmar	Active
R181	54.98%	26.05%	1.831	-13.94%	3.944	-
H1_TURBO	56.23%	26.41%	1.841	-13.94%	4.033	32/150 (21.3%)
H2_PROMOTION100	55.44%	26.16%	1.836	-13.94%	3.976	15/150 (10.0%)
H3_TURBO_PERSIST2	55.48%	26.17%	1.836	-13.94%	3.980	14/150 (9.3%)

Master Research Bible v1.23 Addendum

R215 PATCH1 robustness and canonical interpretation

31. H1_TURBO robustness vs New-R181

Audit	Result
Active-month outcome	32 active months; 23 wins / 9 losses vs R181; mean active-month increment about +0.3605%.
36M rolling	115 windows: CAGR improvement 108/115 (93.9%); Sharpe 92/115 (80.0%); Calmar 94/115 (81.7%).
60M rolling	91 windows: CAGR improvement 91/91 (100%); Sharpe 81/91 (89.0%); Calmar 82/91 (90.1%).
LOYO caution	2014, 2017, 2019 have worse Sharpe/Calmar; 2024 has CAGR -0.47pt vs R181. 2022 and 2026 have no effect.
Full-period MDD	Unchanged at -13.94% in this DEV sample; this is not a guarantee that future drawdown cannot worsen.

32. Canonical interpretation after R215 PATCH1

R215 PATCH1 is a POSITIVE DEV FALSIFICATION SIGNAL for the specific structural hypothesis that a sparse, already-frozen Turbo attack overlay can raise growth without discarding R181 A-state defense. H1_TURBO is the strongest of the three predeclared challengers on headline CAGR/Sharpe/Calmar and broad rolling robustness.

This does NOT promote H1 to Production and does NOT establish H1 as optimal. The result remains DEV evidence. The 10% sleeve must not be tuned to 5/15/20% on the same DEV sample, and Turbo thresholds/lookbacks must not be retuned from this result.

H2 shows high selectivity but smaller aggregate growth improvement. H3 does not support requiring two-month persistence as a necessary confirmation layer. These are descriptive conclusions within the predeclared R215 PATCH1 comparison, not invitations to open a new parameter search.

33. Next research question

Preferred next study: R216 - H1 Turbo Booster Historical Falsification / Failure-Mechanism Audit. Decompose all 32 H1 activation months and test concentration, era dependence, event dependence, 23-win/9-loss failure structure, and leave-one-event/sequence robustness. The purpose is falsification of the fixed H1 structure, not weight or signal optimization.

- Keep 10% TQQQ sleeve fixed.
- Keep frozen Turbo definition fixed.
- Keep frozen R177 A-state supremacy fixed.
- Do not use 2026-08+ Forward OOS for tuning or challenger redesign.
- If H1 survives R216, it may be frozen as a Four Gods Power-Up challenger; promotion requires separate governance.

Master Research Bible v1.23 Addendum

Current canonical state / application governance / provenance

34. Current canonical state after R215 PATCH1

Domain	Canonical state
Production	Frozen 4F EW unchanged.
Layer1a	Seiryu Full5P-Balanced / Byakko FrozenVIX / Suzaku TEL-3M/30M / Genbu MR-Majority-6D/11M/24M.
Layer1b frozen challengers	New-R180 / New-R181 Extreme Shield / New-R183 Risk-Quality / New-R184 Balanced. Do not call R181 optimal.
Power-Up candidate	H1_TURBO = R181 + fixed 10% TQQQ sleeve only when frozen Turbo ON and A-state OFF. Positive DEV signal; not yet frozen Production challenger pending falsification/governance.
Gekiko role	Target/teacher benchmark; primary source = official Monthly Trade History Return. Do not fit Four Gods to Gekiko final output.
DEV / Forward	DEV ends 2026-07-31. 2026-08+ is Real Forward OOS and cannot alter frozen DEV rules.

35. Application governance - non-negotiable

- Use the immediately prior FULL-SPEC app.py as parent.
- Keep existing UI design/navigation. Do not redesign UI as part of validation updates.
- Validation/archive page shows latest study only; replace old validation UI block rather than accumulating old study buttons.
- Preserve Frozen specs, official evidence lineage, Production, Forward OOS, current signals, Performance, Self Backtest/PV, caches, manifests, provenance, audit and reproducibility infrastructure.
- Basename is always app.py. Before distribution, python -m py_compile app.py must PASS.
- Latest validation ZIP must contain machine-readable registry/exactness/research contract/manifest where applicable.
- Distinguish implementation bugfix from research-spec change. Failed run is NOT EVIDENCE.

36. Provenance snapshot

Parent Master v1.22 SHA256: 2d8012ec71ffb8f796ca5915976544313343b79e63ca28e1fa24e6e2d3fb1c28

Official R214 ZIP SHA256: 38f2119166deceb666183b6fa6df944ece168b9e295e804f0ba1fea9e11d8199

Official R215 PATCH1 ZIP SHA256: 57c64fb687648916663b92caeb693f5e94f2fb72e5105733aa57bf5da2d539f6

Current R215 PATCH1 full-spec app.py SHA256: b848c7672f44414c1d0593d3ee8696bb61fe6f1e2031b31c033f74eb492ee718

Source hierarchy remains: Text authority = current Master Bible; implementation authority = current full-spec app.py; performance authority = official user-run Streamlit ZIP. If discrepancies appear, audit them; do not change strategy rules to fit results.

Master Research Bible v1.24 Addendum

My4F / Four Gods / v3.45 Research | R216-R231 Layer1 Structural Program | 2026-09-15

37. Scope and authority of this update

- This addendum updates Master v1.23 through the official user-run R231 evidence. All 153 pages of v1.23 remain unchanged and authoritative as the parent text; this addendum appends the Layer1 structural research lineage.
- DEV cutoff remains 2026-07-31. 2026-08-01 onward remains Real Forward OOS and was not used to tune R216-R231.
- Production remains Frozen4F EW. No R216-R231 result automatically promotes a Layer1 challenger, ensemble, state switch, or pair interaction into Production.
- Only user-run Streamlit ZIPs with exactness ALL PASS are official performance evidence. Failed R216 initial, R216 PATCH1, and R220 initial runs are explicitly NOT EVIDENCE.
- The permanent full-spec app.py preservation contract remains active: immediately prior full-spec app.py is the parent; application UI/navigation and operational infrastructure are preserved; only validation/archive UI is latest-study only.

R216 numbering correction. Master v1.23 section 33 contained a planned label "R216 - H1 Turbo Booster Historical Falsification." The user explicitly pivoted back to Layer1 before that planned study was executed and reused R216 for the Layer1 Candidate Combination Atlas. This addendum formally records the user-directed reassignment: official R216 is the Layer1 study lineage below. The unexecuted planned H1 label is historical planning text only and is not evidence.

38. Official evidence ledger - R216 through R231

Study	Official status	Canonical evidence / interpretation
R216 PATCH2	18/18 ALL PASS	144 fixed 25% EW Layer1 combinations from the R195 14-candidate registry. Initial R216 and PATCH1 failed and are NOT EVIDENCE.
R217	14/14 ALL PASS	Matched-context candidate identity/complementarity attribution. Majority strongest Genbu material; Balanced quality, TQQQ growth; FrozenVIX stability; Suzaku role split.
R218	16/16 ALL PASS	Five predeclared role-aware 25% EW constructions. Quality baseline = Balanced / FrozenVIX / TEL-3M/30M / Majority.
R219	17/17 ALL PASS	Quality vs Hybrid-SuzakuGrowth falsification. TEL-3M is regime-dependent tail-rescue/growth material, not a stable constant replacement.
R220 PATCH1	17/17 ALL PASS	TEL-3M tail-rescue mechanism audit. Frozen A Gate / allow_L2 show strongest positive conditional association; A-state shows inverse association. Initial R220 14/15 failed = NOT EVIDENCE.
R221	18/18 ALL PASS	Predeclared A Gate ON -> TEL-3M, OFF -> TEL-3M/30M challenger improves CAGR/Calmar/DD but slightly reduces Sharpe; remains DEV challenger, not promoted.
R222	17/17 ALL PASS	All 18 within-family fixed 50:50 candidate pairs. Strong standalone temporal diversification appears in Suzaku; Genbu pair smooths risk quality.
R223	18/18 ALL PASS	Suzaku 3M+30M 50:50 benefit does not carry through to the full Quality portfolio.
R224	18/18 ALL PASS	Genbu TriEW+Majority 50:50 does not beat Majority at portfolio level.
R225	18/18 ALL PASS	Byakko FrozenVIX+Equal2P 50:50 adds modest CAGR/Calmar but lowers Sharpe; insufficient replacement case.
R226	18/18 ALL PASS	Seiryu Balanced+TQQQ 50:50 gives stable CAGR premium but consistently weaker risk-adjusted quality.
R227	18/18 ALL PASS	Family marginal contribution / leave-one-family-out role map. Quantifies growth, convexity, tail-response and stability roles.
R228	19/19 ALL PASS	Six-pair nonlinear synergy map. Synergy is mainly risk-adjusted, not CAGR. Byakko+Genbu and Seiryu+Suzaku are primary positive structures.
R229	23/23 ALL PASS	Ex-ante SPY t-1 regime topology. Primary pair Sharpe synergy persists across Bull/Pullback; negative controls remain weak.
R230	23/23 ALL PASS	Temporal persistence audit. Byakko+Genbu Sharpe synergy is exceptionally persistent; Seiryu+Suzaku Sharpe synergy also persists.
R231	20/20 ALL PASS	Structural Interaction Map / evidence synthesis. Recomputes and integrates family roles, pair synergy, ensemble carry-through and candidate anchors.

39. R216-R219 - Layer1 candidate and role program

R216 PATCH2 - Active Layer1 Candidate Combination Atlas

The study used only the frozen R195 active registry: Seiryu 4 candidates, Byakko 4, Suzaku 3, Genbu 3. All 4 x 4 x 3 x 3 = 144 portfolios were fixed 25% EW. No new materials, weights, state logic, Forward OOS or Production changes were introduced.

Objective	Canonical result
Maximum CAGR	TQQQ x Equal2P-47D/20M x TEL-3M x Majority = 55.90% CAGR.
Maximum Sharpe	Balanced x FrozenVIX x TEL-3M/30M x Majority = 1.697 Sharpe.
Maximum Calmar	Balanced x Equal2P-47D/20M x TEL-3M x Majority = 2.347 Calmar.
Family structure	Majority broadly strongest Genbu material; TQQQ maximizes growth; Balanced maximizes quality; Byakko and Suzaku show explicit growth-vs-quality trade-offs.

R217 - Candidate Identity / Complementarity Attribution

- **Genbu Majority:** Full DEV 100% matched-context wins versus family alternatives on CAGR, Sharpe and Calmar; LOYO all 13 years remained all-comparator superior. Mean marginal effects: CAGR +2.02pt, Sharpe +0.0375, Calmar +0.189.
- **Seiryu TQQQ:** growth/firepower role. Full DEV CAGR 100% matched-context wins; LOYO 13/13 all-comparator CAGR, but Sharpe/Calmar/DD broadly worse.
- **Seiryu Balanced:** quality role. Full DEV Sharpe/Calmar/MaxDD 100% wins; LOYO Sharpe 13/13 all-comparator.
- **Byakko FrozenVIX:** stability/Sharpe role. Equal2P-47D/20M is the stronger CAGR/Calmar material, but FrozenVIX is the more stable quality anchor.
- **Suzaku:** TEL-3M carries CAGR/Calmar/DD rescue characteristics; TEL-3M/30M carries Sharpe quality. No single unconditional Suzaku winner.

R218 - Role-Aware Portfolio Construction

Construction	CAGR	Sharpe	MaxDD	Calmar	Interpretation
Quality	47.23%	1.697	-21.85%	2.162	Balanced / FrozenVIX / TEL-3M/30M / Majority. Canonical Layer1 quality baseline.
Growth	54.93%	1.621	-28.87%	1.903	TQQQ / FrozenVIX / TEL-3M / Majority. High growth, lower quality.
Hybrid-SuzakuGrowth	47.36%	1.675	-20.93%	2.263	Looked attractive in DEV; required direct falsification.
Hybrid-SeiryuGrowth	54.76%	-	-30.02%	1.824	Growth premium with clear drawdown cost.

R219 - Quality vs Hybrid-SuzakuGrowth Falsification

The only difference was the Suzaku sleeve: TEL-3M versus TEL-3M/30M. HSG improved Full DEV CAGR by only +0.13pt, Calmar by +0.101 and MaxDD by +0.92pt, while Sharpe fell -0.022. LOYO showed HSG CAGR positive in 8/13 leave-out cases, while Sharpe was worse in 12/13. Fixed 2017-18, 2020-21 and 2023-24 vintages all favored Quality on CAGR, Sharpe and Calmar. The canonical interpretation is that TEL-3M is a regime-dependent tail-rescue/growth material, not a stable unconditional replacement for TEL-3M/30M.

Canonical baseline after R219: Layer1 Quality remains Balanced / FrozenVIX / TEL-3M/30M / Majority, 25% EW.

40. R220-R221 - TEL-3M tail-rescue mechanism and fixed challenger

R220 PATCH1 - Frozen-state mechanism audit

Initial R220 failed exactness 14/15 because operational state construction was unresolved; it is NOT EVIDENCE. PATCH1 reused the exact proven R215 state construction pathway and resolved the existing frozen signals. No new state, threshold, router or Forward data was introduced.

Frozen condition	ON mean portfolio delta vs Quality	OFF mean	ON-OFF	Interpretation
A Gate	+0.216%/mo	-0.221%/mo	+0.438%/mo	Strongest clean mechanism signal; 60M ON-OFF positive in all 91 windows.
allow_L2	+0.183%	-0.162%	+0.345%	Similar positive structure; 60M ON-OFF positive in all 91 windows.
Turbo	+0.289%	-0.075%	+0.364%	Positive mean but more sparse / less clean win-rate structure.
Promotion100	+0.375%	-0.031%	+0.406%	Positive but only 17 active months; sparse.
A-state	-0.294%	+0.071%	-0.365%	Inverse result; does not support switching TEL-3M during A-state.
VIX30 Brake	-0.250%	+0.030%	-0.280%	No support for TEL-3M switch under this brake condition.

R221 - Predeclared A Gate switch falsification

Metric	Quality Control	A-Gate Challenger	Delta
CAGR	47.23%	48.92%	+1.69pt
Sharpe	1.697	1.687	-0.009
MaxDD	-21.85%	-20.62%	+1.22pt improvement
Calmar	2.162	2.372	+0.210
Worst Month	-13.94%	-13.53%	+0.41pt improvement

- Rule was fixed before R221 execution: A Gate ON -> TEL-3M; otherwise TEL-3M/30M. Other three families remained unchanged at 25% each.
- A Gate was ON in 81/150 months. Active months: 43 positive delta, 33 negative delta, mean +0.216%/mo.
- Rolling 60M: CAGR improvement in 90/91 windows, Calmar improvement in 85/91, MaxDD improvement in 65/91.
- LOYO: CAGR and relative terminal wealth positive in 13/13. Calmar and MaxDD improved in 12/13. Sharpe improved only 2/13.
- Leave-one-A-Gate-sequence-out: dCAGR remained positive in 20/20. Tail/DD improvement has material COVID-sequence dependence.

Governance: R221 is a strong DEV challenger, not a Production promotion. It is post-R220 and not independent OOS. Further DEV tuning of its condition is prohibited.

41. R222-R226 - Intra-family temporal diversification and portfolio carry-through

R222 scanned all 18 fixed within-family 50:50 pairs. The main standalone result was genuine temporal diversification in Suzaku: TEL-3M + TEL-30M 50:50 beat both constituents on Full DEV CAGR, Sharpe, Calmar and MaxDD. Genbu TriEW + Majority also improved several standalone risk-quality metrics. R223-R226 then tested whether those family-level effects survived inside the full Layer1 Quality portfolio.

Family	Fixed ensemble	Portfolio-level result	Canonical decision
Seiryu	Balanced + TQQQ 50:50	CAGR +3.78pt vs Balanced, but Sharpe -0.031, Calmar -0.199, MaxDD worsened 4.14pt. CAGR premium was 13/13 LOYO but risk-quality loss also persistent.	Do not replace Balanced; this is a growth/firepower blend, not diversification alpha.
Byakko	FrozenVIX + Equal2P-47D/20M 50:50	CAGR +0.58pt, Calmar +0.026, Sharpe -0.0069, MaxDD unchanged. LOYO CAGR 13/13 positive but Sharpe improved only 2/13.	Keep FrozenVIX as Quality anchor; ensemble is only a light growth tilt.
Suzaku	TEL-3M + TEL-30M 50:50	Standalone temporal diversification was strong, but in full Quality portfolio: CAGR -0.19pt, Sharpe -0.0068, Calmar -0.0095 vs TEL-3M/30M. LOYO 13/13 worse across major metrics.	Standalone ensemble benefit does not carry through. Keep TEL-3M/30M Quality anchor.
Genbu	TriEW + Majority 50:50	Full portfolio CAGR -0.54pt, Sharpe -0.0092, Calmar -0.162, MaxDD worsened 1.50pt vs Majority. LOYO CAGR/terminal wealth 13/13 worse.	Keep Majority alone.

R222-R226 synthesis: same-family 50:50 averaging can improve a family in isolation, but the marginal benefit often disappears once the other three families are present. Layer1 decisions must therefore be made on portfolio-level contribution, not standalone candidate metrics. No intra-family 50:50 ensemble earned a Quality-baseline replacement.

42. R227 - Family marginal contribution / role map

R227 removed one 25% family sleeve at a time by replacing it with 0% monthly return while leaving the other three sleeves at their original 25% weights. The LOFO portfolios are diagnostics, not tradable alternatives.

Family	CAGR contribution	Sharpe contribution	Calmar contribution	MaxDD contribution	Structural role
Seiryu	+10.00pt	+0.060	-0.052	-5.03pt	Growth / Firepower
Byakko	+14.38pt	-0.015	+0.645	-0.19pt	Long-horizon Calmar / Crisis Convexity
Suzaku	+14.38pt	+0.118	+0.203	-5.08pt	Growth + regime-dependent tail rescue
Genbu	+13.24pt	+0.181	+0.524	-1.10pt	Stability / Sharpe / Portfolio Quality

- **Genbu:** strongest risk-adjusted stabilizer. 60M rolling Sharpe improvement in 88/91 windows and Calmar improvement in 91/91; LOYO Sharpe and Calmar positive 13/13.
- **Byakko:** largest Calmar contribution and major long-horizon growth contribution, but not a Sharpe-improvement sleeve in isolation.
- **Suzaku:** growth/Sharpe engine with the largest drawdown cost in aggregate; nevertheless can become a rescue sleeve in specific stress regimes.
- **Seiryu:** clean growth accelerator; contributes CAGR and some Sharpe, but is not a risk-control sleeve.

43. R228-R230 - Pairwise nonlinear synergy, regime topology and persistence

R228 defined pair synergy only on nonlinear portfolio metrics. Raw monthly return interaction is mechanically zero under fixed 25% EW. For each pair: Synergy = joint metric contribution from the two sleeves minus the sum of their two single-sleeve metric contributions.

Pair	CAGR synergy	Sharpe synergy	Calmar synergy	MaxDD synergy	Structural interpretation
Seiryu + Byakko	-0.58pt	+0.051	+0.159	~0	Mild positive risk-adjusted synergy.
Seiryu + Suzaku	-0.57pt	+0.104	+0.494	+3.70pt	Persistent Sharpe synergy; Calmar/DD synergy regime-dependent.
Seiryu + Genbu	-0.70pt	+0.050	+0.114	~0	Mild positive synergy.
Byakko + Suzaku	-1.08pt	-0.109	-1.107	-8.51pt	Persistent anti-synergy.
Byakko + Genbu	-0.78pt	+0.185	+0.001	~0	Strongest persistent Sharpe synergy.
Suzaku + Genbu	-1.20pt	-0.018	-0.298	-3.66pt	Low correlation but weak/non-persistent nonlinear synergy.

R229 - Regime topology

Pair	Bull Sharpe synergy	Stress/Pullback	Bear	Recovery	Interpretation
Seiryu+Suzaku	+0.090	+0.080	+0.170	-0.006	Risk-adjusted synergy survives Bull and Pullback; Pullback Calmar/DD synergy is especially strong.
Byakko+Genbu	+0.086	+0.165	+0.948	-0.137	Sharpe synergy in Bull/Pullback and very large in small-sample Bear regime.
Byakko+Suzaku	-0.024	-0.299	-0.159	+0.533	Anti-synergy across the major regimes; Recovery sample is only 7 months.
Suzaku+Genbu	+0.040	+0.004	negative	-	Near-zero/non-synergistic rather than a strong persistent anti-pair.

R229 regime counts: Bull 96 months; Stress/Pullback 37; Bear 10; Recovery 7. Bear and Recovery are therefore small-sample diagnostics and must not be over-interpreted. The regime definition used only SPY information available at t-1 and zero boundaries; no threshold search or router creation was performed.

R230 - Temporal persistence

- **Byakko + Genbu:** Full DEV Sharpe synergy +0.185; rolling 36M positive 114/115 and rolling 60M positive 91/91; fixed temporal slices 12/12 positive. Pre-2020 +0.053, Post-2020 +0.275. This is the most persistent synergy structure.
- **Seiryu + Suzaku:** rolling 36M Sharpe synergy positive 85/115 and 60M positive 91/91; fixed slices 12/12 positive. Pre-2020 +0.123, Post-2020 +0.069. Calmar/DD synergy is materially more regime/time dependent.
- **Byakko + Suzaku:** negative control remains strongly negative: rolling 60M Sharpe synergy positive 0/91 and MaxDD positive 0/91.
- **Suzaku + Genbu:** not reliably positive despite very low return correlation; low correlation alone does not imply nonlinear portfolio synergy.
- **CAGR synergy:** remains negative almost everywhere. Primary-pair value is improved risk-adjusted efficiency, not extra growth beyond additive sleeve contributions.

Structural conclusion: pair interaction is not a basis for dynamic routing. It is an explanatory map of how fixed 25% sleeves combine nonlinearly through compounding, volatility and drawdown geometry.

44. R231 - Layer1 Structural Interaction Map / evidence synthesis

R231 recomputed the frozen Layer1 structures in one study without creating a new strategy, optimizer, router or promotion rule. Exactness 20/20 ALL PASS.

Portfolio	CAGR	Sharpe	MaxDD	Calmar
Layer1 Quality	47.23%	1.697	-21.85%	2.162

Candidate role anchors

Family	Quality anchor	Alternative role material	Frozen interpretation
Seiryu	Full5P-Balanced	Full5P-TQQQ	Balanced = Sharpe/Calmar/DD quality; TQQQ = CAGR/firepower.
Byakko	FrozenVIX	Equal2P-47D/20M	FrozenVIX = Sharpe/stability; Equal2P = CAGR/Calmar challenger.
Suzaku	TEL-3M/30M	TEL-3M	TEL-3M/30M = Sharpe quality; TEL-3M = growth / regime-dependent tail rescue.
Genbu	MR-Majority	MR-TriEW	Majority = broad robustness; TriEW = smoothing material but no portfolio-level replacement.

Structural interaction matrix

	Seiryu	Byakko	Suzaku	Genbu
Seiryu	Growth / Firepower	Mild + synergy	Persistent Sharpe synergy; Calmar/DD regime-dependent	Mild + synergy
Byakko	Mild + synergy	Long-horizon Calmar / Crisis Convexity	Persistent anti-synergy	Strong persistent Sharpe synergy
Suzaku	Persistent Sharpe synergy; Calmar/DD regime-dependent	Persistent anti-synergy	Growth + regime-dependent tail rescue	Low corr; weak/non-persistent synergy
Genbu	Mild + synergy	Strong persistent Sharpe synergy	Low corr; weak/non-persistent synergy	Stability / Sharpe / Portfolio Quality

Integrated interpretation

- Family standalone metrics are insufficient. Layer1 candidate decisions must be based on marginal contribution inside the full Four Gods portfolio.
- Intra-family 50:50 averaging did not produce a superior Quality baseline in any family once full-portfolio carry-through was tested.
- The strongest persistent positive nonlinear interaction is Byakko + Genbu on Sharpe; Seiryu + Suzaku also has durable Sharpe synergy.
- Byakko + Suzaku is a persistent anti-synergy pair, but this does not imply either sleeve should be removed: each has distinct marginal value in the four-sleeve system.
- The Layer1 architecture is therefore best described as four distinct roles plus two important interaction structures, not as a single winner-take-all candidate ranking.

45. Current canonical state after R231

Domain	Canonical state
Production	Frozen 4F EW unchanged. R216-R231 are research evidence only.
Layer1 Quality baseline	Seiryu Full5P-Balanced / Byakko FrozenVIX / Suzaku TEL-3M/30M / Genbu MR-Majority-6D/11M/24M; 25% each.
Layer1 growth alternatives	TQQQ is the Seiryu growth/firepower material; TEL-3M is the Suzaku growth/tail-rescue material. Neither is an unconditional Quality replacement.
R221 state challenger	A Gate ON -> TEL-3M else TEL-3M/30M remains a strong DEV challenger. Not promoted; do not retune on DEV.
Intra-family ensembles	No 50:50 family ensemble is adopted into Quality. Family-only diversification gains were not sufficient at portfolio level.
Pair interaction map	Diagnostic only. Byakko+Genbu = strongest persistent Sharpe synergy; Seiryu+Suzaku = persistent Sharpe synergy; Byakko+Suzaku = persistent anti-synergy.
DEV / Forward	DEV ends 2026-07-31. 2026-08 onward is Real Forward OOS and remains untouched by R216-R231 tuning.
Next research direction	Layer1 is structurally mature enough to pause. Natural next branch: Layer1 role / interaction connection to existing frozen Layer2 states, with no retroactive DEV optimization.

46. Active Layer1 registry - frozen R195 candidate universe

Family	Candidate	Frozen status
Seiryu	Full5P-ASH	Robust Core
Seiryu	Full5P-Balanced	Robust Qualified
Seiryu	Full5P-QLD	Robust Qualified
Seiryu	Full5P-TQQQ	Temporal-Qualified
Byakko	FrozenVIX	Robust Qualified
Byakko	Equal2P-47D/20M	Temporal-Qualified
Byakko	FrozenVIX-GLD/XLU126	Robust Core
Byakko	Equal2P-GLD/XLU126	Temporal-Qualified
Suzaku	TEL-3M	Temporal-Qualified
Suzaku	TEL-30M	Temporal-Qualified
Suzaku	TEL-3M/30M	Temporal-Qualified
Genbu	MR-24M	Robust Qualified
Genbu	MR-TriEW-6D/11M/24M	Robust Qualified
Genbu	MR-Majority-6D/11M/24M	Robust Core

Temporal-Qualified does not mean REJECT. It denotes a candidate whose historical robustness is more time-dependent and whose role must be assessed in portfolio context.

47. Official artifact provenance - R216-R231

Study	Official user-run ZIP SHA256
R216 PATCH2	a601c901d7912df677f890c757255596339c7f643f2ea1fa6ecdd757296b628e
R217	c71dc96256e764d24c3b64a83e76300bdf6c8374cdcdf99d5416104022e49d3a
R218	29de04c7fbbfa5d39d8d9b4fefe2122a7fa2df1aa8082da7c7ea97413dee715f
R219	5c82d6ffb7795073c3ebcccbfa20a36938c64d1fd9143605921e00270e541677
R220 PATCH1	3765d88e61ebbc15dfb3fd0e088f99b3925b797664f4cfab377450b661cf793
R221	2a8793c2c88240b5313aa54b81595df1bad95399611a1ea9d150b50da62c30af
R222	7a696b68d9e48c784219850de672a963d42ddbc6f3ad9b6d5add2bb123b1751c
R223	0ffc9fb189488f6ba0ab0dd8c344e4ecacd4a7cc5ef7476a25006f92849a71f2
R224	7456534c971c1153cd138e1d03a3d5cf040f37f8c64e84e5de51aca47ce48e72
R225	ea4b9375a72d662e4011cace4f352d265e65668d840844c83546768e23416d03
R226	874892fd40ad3240fc1e788b688166090c28476587faa4ea9fa3cf7e611fa1b1
R227	ea7302b851eae313416ae9ec6e5bfd95d3c240d15fe79a330398b8f2a82a0c17
R228	42f06e5335099d3fd7d8d850dbb5ed81bb4384c868a3c35cbc9466f4b395fbc
R229	76c920d34b35c8c896c5e5f4931d84e7d7d486fdee6640c2e7719d81404963cd
R230	015ef24b252fe6b64ea623b95cc67691886a3fcd90ef349e809ccf0550b550f
R231	34d380cb3bd0dab31a5bee84a05e4a4fc109964b107bb1d0cfb31d300d86414f

Current R231 full-spec app.py SHA256: **1dbb7ea80f984a3bdd088a61fda9f37c9b5ca500b0bb2dcefb501fb683db909f**.

Source hierarchy remains: textual authority = current Master Research Bible; implementation authority = current full-spec app.py; performance authority = official user-run Streamlit ZIP. If discrepancies appear, audit them; do not silently change strategy rules to fit results.

48. Application governance - unchanged and non-negotiable

- Use the immediately prior FULL-SPEC app.py as parent.
- Basename is always app.py. Do not rename it.
- Do not redesign UI layout, navigation, Production pages, operational controls, or provenance infrastructure during research updates.
- Validation/archive page shows only the latest study UI, replacing old study buttons while preserving underlying infrastructure.
- Preserve Frozen specs, Production, Forward OOS, current signal, Performance, Self Backtest/PV, caches, manifests, provenance, audit and reproducibility infrastructure.

- Before distribution, `python -m py_compile app.py` must PASS.
- Distinguish implementation bugfix from research-spec change. Failed run is NOT EVIDENCE.
- Official evidence requires user-run ZIP and exactness ALL PASS. Assistant-side calculations are debugging only.
- No new Layer1 weight grids are authorized by this addendum. Pair interactions and family roles are explanatory evidence, not an implicit optimization license.

End of v1.24 addendum. Parent v1.23 pages 1-153 are preserved verbatim before this addendum.

Master Research Bible v1.25 Addendum

My4F / Four Gods / v3.45 Research | R232-R236 + Teacher Frontier Reference | 2026-09-15

49. Scope and authority of this update

- This addendum updates Master v1.24 through official user-run R236 evidence. All 161 pages of v1.24 remain preserved verbatim as the parent Source of Truth; this addendum appends R232-R236 and the teacher-strategy reference frontier observed during the same research session.
- DEV cutoff remains 2026-07-31. 2026-08-01 onward remains Real Forward OOS and must never be used to retune DEV rules.
- Production remains Frozen4F EW. No R232-R236 result automatically promotes a challenger or router to Production.
- Only user-run Streamlit ZIPs with exactness ALL PASS are official performance evidence. Teacher-strategy metrics and assistant-side common-period calculations are REFERENCE ONLY and are not Four Gods Evidence.
- No new Layer1 weight grid is authorized. R233 and R234 use predeclared fixed constructions; R236 uses one predeclared one-month state-transition pulse with no threshold, lookback, weight or duration search.

50. Official evidence ledger - R232 through R236

Study	Exactness	Canonical interpretation
R232	26/26 ALL PASS	Four predeclared Layer1 composites. Attack-Pair-Diversified adds robust CAGR but worsens risk quality; no Quality replacement.
R233	20/20 ALL PASS	Fixed 75% Quality + 25% Attack satellite. Mild, robust CAGR tilt; risk-quality does not improve.
R234	21/21 ALL PASS	Fixed pair-barbell 75% QualityPair + 25% AttackPair. Stronger CAGR and Full-DEV Calmar/MDD, but Sharpe lower.
R235	20/20 ALL PASS	COVID/crisis-removal falsification. R234 CAGR edge persists, but Full-DEV Calmar/MDD superiority is COVID-sensitive and reverses when COVID support is removed.
R236	32/32 ALL PASS	Post-A re-entry pulse is valid Evidence but primary hypothesis FAILS: only +0.14pt CAGR vs R181, with near-flat risk metrics.

Canonical status after R236: Layer1 Quality remains the cleanest quality baseline; R181 remains the strongest established high-efficiency Four Gods control; R233 is a mild growth tilt; R234 is a stronger CAGR challenger with episodic crisis convexity; the simple Post-A re-entry pulse is rejected as a meaningful growth upgrade.

51. R232 - Layer1 predeclared composite construction

R232 used only the frozen R195 14-candidate universe and preserved 25% top-level family weights. It introduced no new candidate, no weight grid, no router, no Forward OOS use and no Production change.

Portfolio	CAGR	Vol	Sharpe	MaxDD	Calmar	Worst Month
Quality	47.23%	24.85%	1.697	-21.85%	2.162	-13.94%
Defense-Pair-Diversified	47.25%	25.19%	1.678	-23.35%	2.024	-13.82%
Attack-Pair-Diversified	51.14%	27.28%	1.665	-25.40%	2.014	-16.46%
Multi-Representative	51.12%	27.72%	1.643	-26.85%	1.904	-16.34%
Democratic Four Gods	48.98%	26.72%	1.640	-25.75%	1.902	-16.54%

- Attack-Pair-Diversified vs Quality: CAGR +3.91pt, but Sharpe -0.031, Calmar -0.148 and MaxDD worse by 3.55pt.
- Attack CAGR advantage is exceptionally persistent: rolling 36M 115/115, rolling 60M 91/91, LOYO 13/13 positive. Risk-quality improvements do not persist: LOYO Sharpe/Calmar/MaxDD 0/13.
- Multi-Representative and Democratic constructions do not solve the same trade-off; they add complexity while further weakening risk-adjusted quality.

R232 conclusion: the Layer1 candidate pool contains a robust CAGR booster structure, but simple fixed averaging does not create a superior all-objective portfolio. Attack-Pair-Diversified is a growth material, not a replacement for Quality.

52. R233 - Quality core + fixed Attack satellite

R233 predeclared one blend only: 75% Quality + 25% R232 Attack-Pair-Diversified. No weight grid was permitted. The resulting effective candidate weights are Balanced 21.875%, TQQQ 3.125%, FrozenVIX 25%, TEL-3M/30M 21.875%, TEL-3M 3.125%, Majority 25%.

Metric	Quality	R233	Delta
CAGR	47.23%	48.22%	+0.99pt
Vol	24.85%	25.42%	+0.57pt
Sharpe	1.697	1.691	-0.006
MaxDD	-21.85%	-22.74%	-0.89pt
Calmar	2.162	2.120	-0.042
Worst Month	-13.94%	-14.57%	-0.63pt

- CAGR improvement is robust: rolling 36M 115/115, rolling 60M 91/91, LOYO 13/13.
- Risk-quality superiority is absent: LOYO Sharpe, Calmar and MaxDD improvement are all 0/13.

R233 conclusion: valid mild growth tilt. Do not extend into a new DEV weight grid. It is useful as a fixed reference point between Quality and stronger growth constructions.

53. R234 - Pair-structured Layer1 construction

R234 reorganized existing representatives into two fixed sleeves: AttackPair = 25% each of Balanced, TQQQ, TEL-3M/30M and TEL-3M; QualityPair = 50% FrozenVIX + 50% Majority. Two predeclared top-level constructions were evaluated. Pair-EW is exactly identical to the R232 Attack portfolio. The focal Pair-Barbell is 25% AttackPair + 75% QualityPair.

R234 Pair-Barbell effective weights: FrozenVIX 37.5%, Majority 37.5%, Balanced 6.25%, TQQQ 6.25%, TEL-3M/30M 6.25%, TEL-3M 6.25%. No pair-weight search or alternative split grid was run.

Portfolio	CAGR	Vol	Sharpe	MaxDD	Calmar	Worst Month
Quality	47.23%	24.85%	1.697	-21.85%	2.162	-13.94%
Pair-EW / R232 Attack	51.14%	27.28%	1.665	-25.40%	2.014	-16.46%
Pair-Barbell	50.49%	27.96%	1.613	-19.63%	2.572	-15.00%

- Pair-Barbell vs Quality: CAGR +3.26pt, MaxDD +2.22pt improvement, Calmar +0.411, but Sharpe -0.083 and Worst Month 1.05pt worse.
- LOYO: CAGR positive 13/13; Sharpe 0/13; Calmar and MaxDD 12/13. Rolling 60M: CAGR 86/91 positive; Calmar and MaxDD 72/91; Sharpe 42/91.
- AttackPair vs QualityPair return correlation = approximately 0.344, supporting genuine sleeve diversification, but diversification does not imply Sharpe superiority.

Initial R234 interpretation was a promising CAGR + Calmar challenger. R235 materially narrows this interpretation: its risk-side advantage is not structurally stable outside the COVID-supported sample geometry.

54. R235 - R234 COVID dependence / crisis-removal falsification

R235 did not redesign R234. It froze the R234 Pair-Barbell and Quality control, then tested whether Full-DEV risk advantages survived removal of COVID-related periods. Excision samples are conditional diagnostics, not contiguous tradable backtests.

Sample	dCAGR	dSharpe	dMaxDD	dCalmar
Full DEV	+3.26pt	-0.083	+2.22pt	+0.411
Pre-COVID 2014-02..2019-12	-0.55pt	+0.027	+0.97pt	+0.214
Post-2020 2021-01..2026-07	+1.95pt	-0.382	-10.77pt	-2.558
Leave COVID Core 2020-02..06	+1.20pt	-0.156	-6.95pt	-1.197
Leave full 2020	+0.65pt	-0.192	-6.95pt	-1.125
Leave 2020-02..2021-12	+0.98pt	-0.189	-6.95pt	-1.090

55. R235 synthesis - persistent growth, episodic crisis convexity

- The CAGR premium survives every primary COVID-removal diagnostic, so the R234 growth edge is not solely a COVID artifact.
- The Full-DEV Calmar and MaxDD advantage does not survive COVID removal. It reverses materially: leave-COVID-Core dCalmar -1.197 and dMaxDD -6.95pt; leave-2020 dCalmar -1.125 and dMaxDD -6.95pt.
- Rolling 60M windows overlapping COVID Core: CAGR positive 64/64; Calmar and MaxDD positive 59/64. Non-COVID 60M windows: CAGR 22/27 positive, but Calmar and MaxDD only 13/27; mean dCalmar -1.133 and mean dMaxDD -4.95pt.
- COVID Core contains only 5 months. It generated about +19.1% relative wealth, while the remaining 145 months still generated about +10.4% relative wealth. The growth premium is broad enough to survive outside COVID, but risk superiority is crisis-episode dependent.

Canonical post-R235 interpretation: R234 Pair-Barbell = persistent CAGR challenger with episodic crisis convexity, NOT a structurally superior Calmar/MDD portfolio. Quality remains the cleanest Layer1 risk-adjusted baseline. R233 remains the milder growth tilt; R234 is the stronger growth tilt.

56. Teacher strategy frontier - reference only, not Four Gods Evidence

During this session the user supplied Bam-sensei teacher-strategy metrics and Monthly Trade histories. These sources are retained as structural reference material. They do not enter official Four Gods performance Evidence, and hidden asset identities in the 奥義-分身 family remain undisclosed.

Teacher strategy	Published CAGR	Sharpe	MaxDD	Calmar	Role
奥義-分身-激攻	75.32%	1.63	-40.63%	1.75	Maximum firepower
奥義-分身-常勝	71.03%	1.86	-22.83%	3.09	Growth-quality frontier
奥義-分身-鉄壁	52.87%	1.44	-12.94%	3.67	Drawdown-control frontier
シン四つ目-激攻	70.43%	1.48	-33.19%	2.05	Visible regime-switch attack
シン四つ目-常勝	45.35%	1.24	-59.90%	0.71	Historically unstable in supplied table
シン四つ目-鉄壁	44.89%	1.34	-19.75%	2.16	Defensive visible regime-switch

Important: “奥義-分身” and “シン四つ目” versions with the same label must never be conflated. Their observed return geometry is materially different.

57. Teacher common-period benchmark - 2014-02..2026-07

For comparison with Four Gods DEV, displayed Monthly Trade returns were clipped to the exact 150-month DEV window and recomputed with one common monthly formula. Because site returns are displayed rounded to 0.01%, these figures are approximate reference calculations, not official Evidence. For シン四つ目-激攻, official R208 remains the performance authority.

Strategy	CAGR	Vol	Sharpe	MaxDD	Calmar	Worst Month
奥義-分身-激攻	80.70%	41.09%	1.666	-40.63%	1.986	-31.21%
奥義-分身-常勝	73.46%	31.59%	1.925	-22.83%	3.218	-16.63%
奥義-分身-鉄壁	57.03%	32.54%	1.554	-12.24%	4.658	-11.98%
シン四つ目-激攻 (rounded display)	71.05%	41.58%	1.507	-33.18%	2.141	-26.52%
R181 official DEV	54.98%	26.05%	1.831	-13.94%	3.944	-13.94%
Layer1 Quality official DEV	47.23%	24.85%	1.697	-21.85%	2.162	-13.94%
R234 Pair-Barbell official DEV	50.49%	27.96%	1.613	-19.63%	2.572	-15.00%

- 奥義-分身-常勝 is the most informative growth-quality teacher frontier: approximately 73.46% CAGR and 1.925 Sharpe at -22.83% MDD on the common displayed-return window.
- 奥義-分身-鉄壁 occupies a different frontier: approximately 57.03% CAGR with -12.24% MDD and 4.66 Calmar, but lower Sharpe and materially higher Vol than R181.
- R181 is already highly competitive on low-volatility efficiency: 54.98% CAGR, 1.831 Sharpe, -13.94% MDD. The largest apparent gap versus the teacher frontier is not basic defense but growth capture / re-risking efficiency.

Research implication: do not respond by merely adding constant leverage or repeatedly tuning static Layer1 weights. The teacher frontier suggests that timing of attack intensity and recovery capture may matter more than another unconditional blend.

58. Teacher-derived hypothesis boundary

The teacher data can motivate falsifiable structural hypotheses but may not be copied into the Four Gods Evidence chain. Hidden teacher assets are not inferred, and teacher returns are not used as labels or optimization targets. R236 therefore tests only an internally available Four Gods state-transition hypothesis.

59. R236 - R181 Post-A Re-entry Pulse Falsification

R236 tested one predeclared temporal hypothesis motivated by the teacher frontier: perhaps R181 is already sufficiently defensive, and the missing value is faster growth re-entry after an A-state episode. The study intentionally used no teacher returns and no teacher asset identities.

Frozen rule: Control = Layer1 New-R181. Trigger only when $A[t-1] = 1$ and $A[t] = 0$. For that one month only, replace R181 return with the already-frozen R234 Pair-Barbell return. All other months remain exactly R181. Pulse duration = exactly one month. Breadth confirmer = NONE. No threshold, weight, lookback or duration search.

Metric	R181 Control	R236 Pulse	Delta
CAGR	54.98%	55.13%	+0.14pt
Vol	26.05%	26.20%	+0.15pt
Sharpe	1.831	1.825	-0.006
MaxDD	-13.94%	-13.94%	0.00pt
Calmar	3.944	3.954	+0.010
Worst Month	-13.94%	-13.94%	0.00pt
Relative terminal wealth	-	-	+1.17%

Pulse event statistics: 16 pulse months; mean pulse delta +0.103%/month; median +0.166%; hit rate 50%; best +3.13pt; worst -2.32pt; cumulative pulse relative wealth +1.17%. Mean prior A-run length = 1.44 months.

60. R236 predeclared verdict

Criterion	Observed	Threshold	PASS
dCAGR	+0.14pt	$\geq +3.00\text{pt}$	NO
dSharpe	-0.006	≥ -0.05	YES
dMaxDD	0.00pt	$\geq -3.00\text{pt}$	YES
Primary hypothesis	-	All primary criteria	FAIL

R236 is official Evidence because exactness is 32/32 ALL PASS. The primary hypothesis nevertheless FAILS because the growth uplift is far below the predeclared +3.00pt CAGR requirement. A-state exit alone is not a sufficiently informative re-risking trigger.

61. R236 robustness and failure mechanism

- Rolling 36M: dCAGR positive 82/115, dSharpe nonnegative 69/115, dMaxDD nonnegative 77/115. Mean dCAGR only +0.31pt/year.
- Rolling 60M: dCAGR positive 76/91, but dSharpe nonnegative only 40/91; dMaxDD nonnegative 72/91. Mean dCAGR only +0.36pt/year.
- The weak CAGR uplift is not one isolated failure. The pulse fires infrequently and has only 50% event hit rate, so A-state exit does not cleanly separate genuine rebound months from false restarts.
- Because risk metrics remain almost unchanged, the failure is not excessive aggression. The failure is insufficient selectivity / information content in the trigger.

Canonical lesson: “re-entry matters” is not falsified in general. What is falsified is the simple rule “A-state just ended, therefore increase growth exposure for one month.” Any next re-entry study must add an independently frozen confirmation signal rather than tuning pulse size or duration.

62. Next permitted research direction after R236

Preferred next hypothesis: fixed offensive-breadth confirmation layered on the same Post-A transition. Example concept: A[t-1]=ON and A[t]=OFF, plus pre-existing Seiryu/Suzaku offensive agreement, then use the already-frozen growth challenger; otherwise remain R181. The confirmation definition must be fixed from existing frozen signals before execution.

- Do not grid-search confirmation thresholds.
- Do not tune pulse duration beyond the one-month R236 result on the same DEV sample.
- Do not tune R233/R234 blend weights.
- Do not use teacher returns as labels, classifiers, optimization targets or hidden-state supervision.
- Do not use 2026-08 onward Forward OOS for DEV redesign.

The research question is now narrower: can a pre-existing cross-family confirmation distinguish true recovery re-entry from false A-state exits while preserving R181 risk quality?

63. Current strategy-role map after R236

Object	Current canonical role
Frozen4F EW	Production. Unchanged.
R181	High-efficiency Four Gods control; strong Sharpe/Calmar and shallow MDD.
Layer1 Quality	Canonical Layer1 risk-adjusted baseline.
R233	Mild fixed growth tilt; robust CAGR increment, no risk-quality superiority.
R234 Pair-Barbell	Stronger persistent CAGR challenger with episodic crisis convexity; not structurally superior on Calmar/MDD.
R236 Pulse	Rejected as meaningful growth upgrade; retain as negative evidence against unconfirmed Post-A re-entry.

64. Official artifact provenance - R232 through R236

Artifact	SHA256
Parent Master v1.24	d3f954c341078ab857e2e6e5e0c4b1b1ea7e8e6974f2cc7ac25de5baaded6869
R232 official ZIP	b38559b6573277cbf991e41406b0f4e81b76bf434812cab1cd35bbf7c96e99b8
R233 official ZIP	4b3e8f680e16368cf1908ab49de926a851087dde8d72aa2a3eb6393efd7b96e4
R234 official ZIP	3c0380b203261cb79cd87272d487b39a71d8b7d3320efe2e88a60761cac78665
R235 official ZIP	c3b2cb9320b48465127b3665b82413819577f289530309b816cfe7d5137eed0b
R236 official ZIP	996bd1ed562a1a5acf91a9996c620652bccc8a38ef67639b7df154f6269822e7
Current R236 full-spec app.py	5c9821f666db793893d7d335ddd3c36526ec74289e7d24cc9987f02d7e4781fe

Source hierarchy remains: textual authority = current Master Research Bible; implementation authority = current full-spec app.py; performance authority = official user-run Streamlit ZIP with exactness ALL PASS. Teacher benchmarks are reference-only and never override official Four Gods Evidence.

65. Application and research governance - unchanged and non-negotiable

- Use the immediately prior FULL-SPEC app.py as parent. Basename is always app.py.
- Do not redesign UI layout/navigation. Validation/archive page shows only the latest study UI while preserving all operational infrastructure.
- Preserve Production, Frozen specs, official evidence lineage, Forward OOS, current signal, Performance, Self Backtest/PV, caches, manifests, provenance, audit and reproducibility infrastructure.
- Before distribution, python -m py_compile app.py must PASS.
- Distinguish implementation bugfix from research-spec change. Failed run = NOT EVIDENCE.
- No new Layer1 weight grid after R215. Fixed challenger weights may be falsified but not optimized on the same DEV sample.
- 2026-08-01 onward is Real Forward OOS and cannot be used for DEV retuning.

66. v1.25 canonical Source-of-Truth snapshot

Master v1.25 supersedes v1.24 as the textual Source of Truth through R236. Pages 1-161 are the exact v1.24 parent; the appended pages record R232-R236, teacher-frontier reference observations, the R236 negative result, and the narrowed next-study boundary. No Production promotion is made by this update.

End of v1.25 addendum. Parent v1.24 pages 1-161 are preserved verbatim before this addendum.

Master Research Bible v1.26 Addendum

My4F / Four Gods / v3.45 Research | R237-R259 | 2026-09-16

67. Scope and authority of this update

- This addendum extends Master v1.25 through official user-run Evidence up to R259. Parent pages 1-169 are preserved verbatim and remain authoritative for all earlier lineage.
- DEV cutoff remains 2026-07-31. 2026-08-01 onward remains Real Forward OOS and must not be used to retune DEV rules.
- Production remains Frozen4F EW. No result in R237-R259 constitutes Production promotion.
- Only user-run Streamlit ZIPs with exactness ALL PASS are official performance Evidence. Failed runs remain NOT EVIDENCE.
- Permanent Layer1 convention remains monthly rebalancing; target month t allocation may use only information known through t-1. Rebalance frequency is fixed and must not be optimized unless explicitly changed.
- From R257 onward, standard performance comparison is CAGR / Vol / Sharpe / Sortino / MDD / Calmar. Sortino convention is MAR=0 at monthly frequency; downside deviation = $\sqrt{\text{mean}(\min(r,0)^2)}$; annualized Sortino = $\text{mean}(r)/\text{downside_deviation}*\sqrt{12}$.

68. R237-R243 - Layer1 ERC branch

Study	Exactness	Canonical result / role
R237 PATCH1	28/28 PASS	Direct 4F ERC12 and SpearERC75+Shield25 tested. Spear-ERC + separate Genbu Shield is the cleaner architecture.
R238	37/37 PASS	R181 architecture + Spear ERC12: 55.96% CAGR / 25.98% Vol / 1.860 Sharpe / -12.03% MDD / 4.653 Calmar. Risk-Quality variant, not replacement.
R239	51/51 PASS	ERC6/12/24 all pass triple risk-quality. ERC6: 61.09% CAGR / 25.63% Vol / 2.011 Sharpe / -11.71% MDD / 5.219 Calmar on common N=126.
R240	64/64 PASS	6/12/24 ERC ensemble: 60.28% / 25.88% / 1.974 / -11.92% / 5.056. Robust multihorizon candidate.
R241 PATCH2	91/91 PASS	Branch synthesis confirms ERC dominates MinVar as this branch; ERC6 full-sample risk-quality leader, Ensemble robust multihorizon representative.
R242	39/39 PASS	ERC6 vs Ensemble persistence: ERC6 wins full-sample all four; LOYO 11/11 four-way; rolling60 64/67 four-way; Sharpe/MDD/Calmar 67/67.
R243	37/37 PASS	Frontier synthesis: ERC6 61.09/25.63/2.011/-11.71/5.219; teacher Iron Wall 64.26/32.64/1.693/-12.24/5.250. Larger research gap is growth-quality direction.

R243 interpretation: ERC6 is the Layer1 ERC branch Primary Performance Representative, but not Production. It sits near the low-MDD / high-Calmar frontier while preserving strong growth.

69. R244-R250 - firepower, Trend Momentum, and Three-Spear Ninpo

Study	Status	Key result
R244	21/21 PASS	TECL secret holding unproven. Conditional TECL firepower supported when prior Seiryu and Suzaku are both positive.
R245	26/26 PASS	Conditional TECL overlay on ERC6: 62.47% CAGR / 25.78% Vol / 2.035 Sharpe / -11.89% MDD / 5.255 Calmar. Growth tilt; strict Pareto FAIL.
R246	41/41 PASS	Conditional Seiryu Balanced->Full5P-TQQQ: 64.32% CAGR, but Sharpe 1.939, MDD -15.83%, Calmar 4.064. Firepower supported; risk-quality denied.
R247	46/46 PASS	Dynamic stance prototype badly failed: 54.03% CAGR / 35.64% Vol / 1.405 Sharpe / -32.98% MDD / 1.638 Calmar. Close this specific attack construction.
R248	47/49; NOT EVIDENCE	Two count-expectation failures. Indicative only; never use as official Evidence.
R249 PATCH1	24/24 PASS	God-level signed-R2 Trend Momentum 12M/skip1M failed to rank next-month god returns: mean cross-sectional Spearman +0.011; Rank1 avg 3.54% vs Rank4 4.05%.
R250	33/33 PASS	Three-Spear Ninpo. Strongest-only momentum variants fail; Kowarimi (strongest+weakest) is the only clear CAGR improver, but Primary Growth-Quality FAIL.

R250 fixed architecture and canonical metrics

Seiryu / Byakko / Suzaku are the three Spears; Genbu is the separate 25% Shield. Quality representatives are fixed: Seiryu Full5P-Balanced, Byakko FrozenVIX, Suzaku TEL-3M/30M, Genbu MR-Majority-6D/11M/24M.

Rule	CAGR	Vol	Sharpe	MDD	Calmar
Bunshin	53.21%	25.13%	1.843	-21.85%	2.436
Gatekeeper	46.12%	25.18%	1.645	-21.85%	2.111
Oikaze	47.79%	35.91%	1.268	-45.82%	1.043
Nukimi	35.74%	39.00%	0.976	-43.60%	0.820
Kasoku	52.50%	35.28%	1.378	-29.76%	1.764
Kowarimi	56.69%	27.65%	1.779	-23.54%	2.408

R250 Primary FAIL. The key surviving clue is structural: strongest-only momentum did not help, whereas strongest+weakest materially increased CAGR. This directly motivates R251-R259.

70. R251-R252 - Kawarimi mechanism and Weakest-Spear Reversal

R251 - Kawarimi Structural Audit

R251 is official Evidence (29/29 ALL PASS). Exact identity: Kawarimi - Bunshin = 12.5%*(realized Strongest - realized Middle) + 12.5%*(realized Weakest - realized Middle).

Component	Mean contribution / observation
Kawarimi - Bunshin	+0.240% per month
Momentum side	+0.069% per month
Contrarian side	+0.171% per month

Mechanism classification: CONTRARIAN-DOMINANT. Roughly seven tenths of the historical active return came from the weakest-vs-middle leg, not from the strongest-vs-middle leg.

R252 - Weakest-Spear Reversal Audit

R252 is official Evidence (19/19 ALL PASS) and all four predeclared Primary conditions PASS.

Diagnostic	Official result
Weakest - Middle mean	+1.365%/month
Weakest - Strongest mean	+0.813%/month
LOYO positive	11/11
Rolling60 positive	65/67 = 97.0%
Severity Spearman	+0.157
Next-month mean: Strongest / Middle / Weakest	3.81% / 3.26% / 4.62%

Family dependence is decisive: Suzaku weakest -> Weakest-Middle +3.27%/month; Byakko weakest -> +1.86%; Seiryu weakest -> -0.97%. Therefore the reversal is supported, but not as a universal rank-3 factor. It is strongest in Suzaku and Byakko and absent in Seiryu.

When the weakest spear prior-12M return is actually negative, the reversal is much stronger: Weakest-Middle +5.52%/month and Weakest-Strongest +5.99%/month across 20 months. Severity is only a secondary descriptor; rank/family state is the main signal.

71. R253-R255 - From family-aware contrarian to Shield redistribution

R253 - Family-Aware Contrarian

Trigger only when Byakko or Suzaku is prior-12M weakest. Weight then = Weakest 37.5%, remaining two Spears 18.75% each, Genbu 25%. Seiryu-weakest months remain Bunshin. Official Evidence: 28/28 ALL PASS.

Strategy	CAGR	Vol	Sharpe	MDD	Calmar
Bunshin	53.21%	25.13%	1.843	-21.85%	2.436
R253	55.38%	25.31%	1.890	-24.11%	2.297

Primary FAIL: CAGR and Sharpe improve, but MDD worsens by 2.26 points and Calmar falls. Trigger-level reversal survives, but direct overweighting is not enough.

R254 - Family-Aware Middle Suppression

On the same Byakko/Suzaku-weakest trigger, fixed weights = Strongest 25 / Middle 0 / Weakest 50 / Genbu 25. Official Evidence: 28/28 ALL PASS.

Strategy	CAGR	Vol	Sharpe	MDD	Calmar
R254	60.02%	27.48%	1.870	-24.93%	2.408

Primary FAIL, but Middle suppression is strongly supported as a growth mechanism: CAGR rises +6.81 points versus Bunshin. Sending all removed Middle capital to the Weakest is too aggressive.

R255 - Contrarian / Shield Redistribution

Two predeclared structures only; no grid. Split = Strongest 25 / Middle 0 / Weakest 37.5 / Genbu 37.5. Shield = Strongest 25 / Middle 0 / Weakest 25 / Genbu 50. Official Evidence: 30/30 ALL PASS.

Strategy	CAGR	Vol	Sharpe	MDD	Calmar	Primary
Bunshin	53.21%	25.13%	1.843	-21.85%	2.436	Control
R255 Split	58.42%	26.64%	1.881	-21.36%	2.735	PASS
R255 All->Shield	56.55%	26.51%	1.842	-17.80%	3.177	FAIL

R255 Split is the first new family-aware contrarian construction to improve growth, Sharpe, MDD, and Calmar simultaneously versus Bunshin. Formal role: Layer1 broad challenger. The all-to-Shield variant is strongly defensive but slightly misses Sharpe.

72. R256-R257 - Robustness and Sortino adoption

R256 PATCH1 - R255 Split Robustness / Failure-Mode Audit

Initial R256 was NOT EVIDENCE because parity tolerance was implemented too tightly; PATCH1 is official Evidence with 23/23 ALL PASS. Research specification did not change.

- Both trigger families have positive mean active contribution.
- 3 of 4 fixed eras are positive.
- 9 of 11 calendar years are positive.
- Largest single-year contribution to cumulative log advantage = 41.7%, below the 50% concentration ceiling.
- Full-sample R255 Primary PASS is preserved.

Structural Robustness therefore PASS. Weakness remains concentrated around 2023 and a long recent rolling window: 2023-09 through 2026-07 produced a 35-window run of Growth-Quality FAIL in the 60M diagnostic. This is not enough to invalidate R255, but it defines the main regime-risk question.

R257 - Sortino / downside-risk audit

R257 is official Evidence (19/19 ALL PASS). Standard reporting expands permanently to CAGR / Vol / Sharpe / Sortino / MDD / Calmar. Sortino uses monthly MAR=0 convention specified in Section 67.

Metric	Bunshin	R255 Split	Difference
Sortino	4.956	5.225	+0.269
LOYO Sortino difference positive	-	11/11	Persistent
Rolling60 Sortino difference positive	-	43/67	64.2%

Interpretation: R255 strength is more visible on downside-risk-adjusted efficiency than on total-volatility efficiency. However, the recent rolling weakness does not disappear: in the 2023-09 to 2026-07 window run, Sortino is positive more often than Sharpe, but not consistently. Loss-month diagnostic: Bunshin is negative in 32 months and R255 active averages +0.782% in those months; when R255 itself loses (29 months), average loss is -4.63% versus Bunshin -3.99%. R255 tends to lose less often, but its losing months can be somewhat deeper.

73. R258 - Regime mechanism and ex-ante diagnostic

R258 PATCH1 is official Evidence (28/28 ALL PASS). The initial run failed from an implementation-only boolean-index alignment bug and is NOT EVIDENCE. PATCH1 changed no research rule.

On R255 trigger months, active return is decomposed exactly as: $12.5\% \cdot (\text{Weakest-Middle}) + 12.5\% \cdot (\text{Genbu-Middle})$. These are named Contrarian leg and Shield leg.

Sample	Active / trigger month	Contrarian leg	Shield leg
Full DEV	+0.474%	+0.317%	+0.157%
2022-2023	-0.180%	-0.181%	+0.001%

Formal mechanism conclusion: the 2022-2023 weakness is primarily Contrarian-leg failure, not Shield-leg failure. Within that era the damage is concentrated in 2023; 2022 itself remains strong. Byakko-weakest triggers stay positive on average, whereas Suzaku-weakest triggers are the main failure concentration.

Ex-ante diagnostic results

State / feature	Result
Weakest prior12 < 0: mean active	+1.362%/month
Weakest prior12 >= 0: mean active	+0.214%/month
Weakness gap vs next-month active Spearman	+0.131

Both predeclared diagnostics PASS. The key insight is counterintuitive: the problem is not that the weakest spear is too weak. Deeply negative prior12 states have stronger reversal. False contrarian risk rises when the spear is only relatively weakest while still absolutely positive, especially in Suzaku-like 2023 states. This evidence justifies one separate predeclared gate test, not threshold mining.

74. R259 - Weakest-negative gate

R259 is official Evidence (28/28 ALL PASS). One fixed gate only: trigger R255 Split when Byakko or Suzaku is prior-12M weakest AND that weakest spear has prior-12M total return < 0. Otherwise hold Bunshin. No threshold grid was run.

Strategy	CAGR	Vol	Sharpe	Sortino	MDD	Calmar
Bunshin	53.21%	25.13%	1.843	4.956	-21.85%	2.436
R255 Split	58.42%	26.64%	1.881	5.225	-21.36%	2.735
R259 Gate	56.57%	26.26%	1.858	5.229	-21.85%	2.589

R259 Primary PASS versus Bunshin. Versus R255 it trades away CAGR (-1.85 points), Sharpe (-0.023), and Calmar (-0.145) while improving Sortino only slightly (+0.004) and lowering Vol modestly. Therefore it is not a total replacement for R255.

Gate statistic	Official result
Original R255 trigger months	84
R259 gate-ON months	19
Gate-ON mean active	+1.362%/month
Discarded prior12>=0 trigger months	65
Discarded group mean active	+0.214%/month

Role separation: R255 Split = broad challenger. R259 = high-conviction / conservative variant. The gate successfully concentrates higher-conviction reversal months and materially fixes the 2023 false-contrarian episode, but it also discards many still-positive trigger months and therefore lowers long-run growth.

75. Canonical Layer1 implications after R259

- R255 Split is the principal new finding of the R251-R259 line: family-aware Weakest reversal + Middle suppression + equal redistribution to Weakest and Genbu.
- R259 is a valid high-conviction variant, not a replacement for R255. Its prior12<0 gate is supported as a concentration mechanism, not as a globally superior architecture.
- ERC6 remains the primary risk-quality representative of the ERC branch. R255 occupies a different structural branch: contrarian + Shield redistribution.
- Production remains Frozen4F EW. Neither R255 nor R259 is promoted by this addendum.
- Do not reopen DEV threshold mining around prior12<0, weakness-gap cutoffs, family-specific thresholds, or era-specific gates. Doing so after observing R259 would materially increase post-hoc optimization risk.
- The scientifically cleaner next phase is frozen Forward OOS tracking of R255 and R259 from 2026-08-01 onward, without retuning DEV rules.

76. Updated research governance - anti-overfit boundary

- R255 and R259 are frozen as named challengers for Forward observation unless a separately predeclared, non-DEV-derived research question is approved.
- No new weight grid after R215 remains in force. R255/R259 weights are fixed research constructions, not optimized weight grids.
- No post-hoc search over negative-return thresholds (for example -5%, -10%, +5%), weakness-gap thresholds, family-specific thresholds, or era-dependent conditional logic should be treated as confirmatory Evidence on the same DEV sample.
- R258 diagnostic findings are explanatory. R259 is the one confirmatory gate derived from that diagnostic. Further DEV subdivision of the 65 discarded months is exploratory at best and should not be used to create a new canonical rule.
- Forward OOS is the preferred discriminator between R255 broad participation and R259 high-conviction concentration.

77. Current canonical performance snapshot

Strategy / role	CAGR	Vol	Sharpe	Sortino	MDD	Calmar	Status
Production Frozen4F EW	historical canonical	-	-	-	-	-	Production unchanged
Quality Bunshin Control	53.21%	25.13%	1.843	4.956	-21.85%	2.436	Control
R181 legacy challenger*	61.46%	26.83%	1.942	not retro-added	-13.94%	4.408	Common N=126 reference
ERC6	61.09%	25.63%	2.011	not retro-added	-11.71%	5.219	ERC Primary Performance Representative
R255 Split	58.42%	26.64%	1.881	5.225	-21.36%	2.735	Layer1 broad challenger
R259 Gate	56.57%	26.26%	1.858	5.229	-21.85%	2.589	High-conviction conservative

* R181 figures shown here are the common 2016-02..2026-07 N=126 R241/R243 reference, not a new rerun. Sortino was not retroactively inserted into old official Evidence.

78. Updated Source-of-Truth snapshot

Master v1.26 supersedes v1.25 as the textual Source of Truth through R259. Parent v1.25 pages 1-169 are preserved verbatim before this addendum. The appended pages record the complete official lineage from R237 through R259, including failed/non-evidence branches where relevant, the permanent Sortino reporting convention, and the anti-overfit boundary after R259.

Canonical current research state: Production = Frozen4F EW; ERC branch representative = ERC6; family-aware contrarian broad challenger = R255 Split; high-conviction conservative variant = R259 Gate. Forward OOS begins 2026-08-01 and must remain untouched by DEV retuning.

79. Provenance and implementation authority

Performance authority remains the official user-run Streamlit ZIP with exactness ALL PASS. Implementation authority remains the current full-spec app.py. The Master is the textual research authority. Failed runs, assistant-side calculations, and indicative values from non-PASS artifacts are not official Evidence.

End of v1.26 addendum. Parent v1.25 pages 1-169 are preserved verbatim before this addendum.

Master Research Bible v1.27 Addendum

My4F / Four Gods / v3.45 Research | Layer1 Challenger Sortino Completion | 2026-09-16

80. Scope and authority of this update

- This addendum extends Master v1.26 without changing any strategy rule, weight, trigger, lookback, rebalance frequency, DEV cutoff, or Production status.
- The purpose is to complete the current Layer1 challenger comparison with Sortino under the permanent R257 convention: monthly $MAR=0$; downside deviation = $\sqrt{\text{mean}(\min(r,0)^2)}$; annualized Sortino = $\text{mean}(r)/\text{downside_deviation}*\sqrt{12}$.
- R255 Split and R259 Gate Sortino values were already official Evidence. R181 and ERC6 Sortino values below are calculated from the official R243 common-period monthly return series (2016-02 through 2026-07, N=126). This is a derived comparison metric from official return evidence, not a new strategy test and not a retroactive change to prior study conclusions.

81. Current Layer1 challenger comparison with Sortino

Strategy / role	CAGR	Vol	Sharpe	Sortino	MDD	Calmar
Bunshin Control	53.21%	25.13%	1.843	4.956	-21.85%	2.436
R181 legacy challenger	61.46%	26.83%	1.942	5.884	-13.94%	4.408
ERC6 ERC branch representative	61.09%	25.63%	2.011	7.017	-11.71%	5.219
R255 Split broad challenger	58.42%	26.64%	1.881	5.225	-21.36%	2.735
R259 Gate high-conviction variant	56.57%	26.26%	1.858	5.229	-21.85%	2.589

82. Interpretation of the completed downside-risk comparison

- ERC6 Sortino = 7.017. It has the strongest downside-risk-adjusted value in the current common-period challenger set, reinforcing its role as the ERC branch risk-quality representative.
- R181 Sortino = 5.884. R181 remains strong on growth and regime-defense, while ERC6 has higher Sharpe, higher Sortino, shallower MDD, and higher Calmar on the same N=126 period.
- R255 Split and R259 Gate retain a distinct contrarian / Shield-redistribution role. Their Sortino values (5.225 and 5.229) are both above Bunshin 4.956, but below R181 and ERC6 on the common comparison.
- R259 only marginally exceeds R255 in Sortino while giving up CAGR, Sharpe, and Calmar. Its role therefore remains high-conviction / conservative concentration rather than broad replacement.
- No Production promotion is implied. Production remains Frozen4F EW; R255 and R259 remain frozen for Real Forward OOS observation from 2026-08-01 onward.

83. Canonical role map after Sortino completion

Object	Canonical role
Production Frozen4F EW	Production unchanged
R181	Legacy regime-defense challenger
ERC6	ERC branch primary risk-quality representative
R255 Split	Family-aware contrarian + Shield redistribution broad challenger
R259 Gate	High-conviction / conservative R255 variant

84. Updated Source-of-Truth snapshot

Master v1.27 supersedes v1.26 as the textual Source of Truth. Parent v1.26 pages 1-178 are preserved verbatim. This addendum adds only the completed Sortino comparison for the active Layer1 challenger set and does not reopen DEV optimization.

Master Research Bible v1.28 - R260 Five-Factor Residual Return Audit Addendum

Authority: appended to v1.27. Parent pages 1-179 remain unchanged. R260 is diagnostic factor attribution only; it does not create, tune, or promote a strategy.

80. R260 official status and frozen design

R260 PATCH4 completed with Exactness 29/29 ALL PASS and is official Evidence. Fixed factors: MKT-RF / SMB / HML / MOM / AQR TSMOM. Beta is estimated from the immediately preceding 36 months only and then fixed for the next 12 OOS months. OOS blocks are non-overlapping and advance by 12 months. Missing-value zero filling, incomplete-window shortening, factor search, and window search are prohibited. DEV remains capped at 2026-07; Real Forward OOS from 2026-08 onward is not used.

The realized common history supports seven complete OOS blocks, 2019-02 through 2026-01, N=84 months. Remaining DEV months are not used to shorten an eighth block.

81. Strategy-level five-factor-adjusted performance

Object	Raw CAGR	Adj CAGR	Adj Sharpe	Adj Sortino	Adj MDD	Adj Calmar
Bunshin	56.23%	38.06%	1.623	4.607	-13.28%	2.865
R181	69.77%	53.10%	1.815	5.665	-12.95%	4.101
ERC6	69.48%	55.29%	1.954	6.276	-10.99%	5.033
R255 Split	61.99%	43.82%	1.576	3.975	-23.42%	1.871
R259 Gate	61.07%	43.75%	1.676	4.812	-12.45%	3.514

All five adjusted CAGRs remain positive. Bunshin retains 38.06% adjusted CAGR. ERC6 retains 55.29% adjusted CAGR with adjusted Sharpe 1.954, Sortino 6.276, MDD -10.99%, and Calmar 5.033. R181 retains 53.10% adjusted CAGR. This means only that the fixed five-factor linear exposures do not explain away the observed returns; it does not prove true alpha.

82. Active return versus Bunshin

Active object	Raw ann.	Adjusted ann.	Adj positive windows
R181 minus Bunshin	8.93%	11.38%	7/7
ERC6 minus Bunshin	8.17%	12.66%	6/7
R255 minus Bunshin	4.20%	4.80%	6/7
R259 minus Bunshin	3.54%	4.56%	7/7

Architecture-specific active returns remain positive after factor removal: R181 - Bunshin +11.38% annualized (7/7 positive OOS windows), ERC6 - Bunshin +12.66% (6/7), R255 Split - Bunshin +4.80% (6/7), and R259 Gate - Bunshin +4.56% (7/7). For R181 and ERC6, adjusted active annualized return exceeds raw active annualized return in this sample.

83. Research interpretation

R260 independently strengthens the current Layer1 interpretation without changing any allocation rule. ERC6 remains the ERC branch Primary Performance / Risk-Quality Representative. R181 remains a distinct legacy regime-defense challenger. R255 Split remains the broad family-aware contrarian challenger; its Weakest-reversal / Middle-suppression active return is not eliminated by the included MOM and TSMOM factors. R259 remains the high-conviction conservative variant.

The correct claim is narrow: fixed linear exposures to MKT-RF, SMB, HML, MOM, and AQR TSMOM do not explain away these returns. R260 does not identify causal alpha and does not control for all possible risks. Gold, rates/bonds, volatility/convexity, leveraged-ETF path dependence, nonlinear/state-dependent beta, and cross-asset timing remain outside this model.

84. Governance after R260

R260 is diagnostic, not a license to mine additional factor sets, beta windows, thresholds, or strategy rules on DEV. R255 and R259 remain frozen named challengers. Production remains Frozen4F EW. The preferred discriminator remains Real Forward OOS from 2026-08-01 onward. Any later broader factor model must be separately predeclared and must not retroactively redefine R260.

85. Evidence provenance

Official evidence bundle: r260_five_factor_walkforward_residual_return_audit.zip. Exactness 29/29 ALL PASS. Factor-source audit records Kenneth French for MKT-RF, SMB, HML, RF and MOM, and the AQR Time-Series Momentum Factors Monthly workbook, sheet TSMOM Factors, for aggregate TSMOM. PATCH1-PATCH4 were implementation-only fixes; the research specification remained unchanged.

86. Updated Source-of-Truth statement

Master Research Bible v1.28 supersedes v1.27 as the latest textual Source of Truth. Pages 1-179 are inherited unchanged; this addendum appends R260 official Evidence and governance only.

Master Research Bible v1.29 - Layer1 Mechanism, Meta-Signal, and Strategy-Health Addendum

Authority: appended to v1.28. Parent pages 1-181 remain unchanged. This addendum records the completed R261-R274 research line, preserves the frozen Production/Challenger separation, and incorporates the attached official R273HB and R274 user-run ZIP evidence.

87. Scope and authority of this update

- DEV cutoff remains 2026-07-31. 2026-08-01 onward is Real Forward OOS and must not be used to retune DEV rules.
- Production remains Frozen4F EW. No R261-R274 result constitutes Production promotion.
- Current Layer1 challenger set remains R181, ERC6, R255 Split, and R259 Gate. Bunshin is the Layer1 control/reference.
- No new Layer1 weight grid is authorized. Diagnostic studies may explain or falsify a mechanism, but they do not silently create allocation rules.
- Official performance Evidence still requires a user-run Streamlit ZIP with Exactness ALL PASS. Failed runs remain NOT EVIDENCE.
- Strategy Health is a descriptive monitoring layer unless a separately predeclared predictive study validates a decision rule. R274 does not provide that validation.

88. R261 - Layer1 Active-Edge Orthogonality Audit

R261 asked whether the active returns of the current Layer1 challengers versus Bunshin were manifestations of one common edge or structurally distinct sources of return. Exactness 29/29 ALL PASS.

Active-edge pair	Pearson correlation	Interpretation
R181 vs ERC6	0.923	Very similar regime-defense / risk-quality edge
ERC6 vs R255 Split	0.316	Substantially distinct
R181 vs R255 Split	0.344	Substantially distinct
R255 Split vs R259 Gate	0.554	Related, but not identical

The four-edge PCA effective dimension was approximately 2.30; PC1 explained about 58.1% and PC1+PC2 about 87.2%. The canonical implication is that R181/ERC6 largely occupy one defense-quality direction, while the R255 branch contributes a distinct contrarian / Shield-redistribution direction.

89. R262 - Layer2 Composite Falsification

R261 motivated one predeclared primary composite only: 50% ERC6 + 50% R255 Split. No weight sweep was permitted. Two fixed context controls were 50% R181 + 50% R255 and 50% ERC6 + 50% R259. Exactness 27/27 ALL PASS.

Primary composite	CAGR	Sharpe	Sortino	MDD	Calmar
ERC6 / R255 50:50	60.04%	2.006	6.333	-13.00%	4.618

Despite strong absolute performance, the primary composite failed the parent-frontier noninferiority requirement because it did not dominate ERC6

itself across the fixed quality criteria. Therefore, low active-edge correlation is not sufficient evidence that blending improves the frontier. No 60/40, 70/30, or other follow-up weight search is authorized on DEV.

90. R263 - ERC6 Mechanism Decomposition

The first R263 implementation failed Exactness (19/23) and is NOT EVIDENCE. PATCH1 corrected implementation only, without changing the research specification, and completed 27/27 ALL PASS.

ERC6 minus Bunshin attribution	Mean active return / month
Seiryu	-0.056%
Byakko	-0.300%
Suzaku	-0.369%
Genbu	+1.162%
Total	+0.436%

The frozen A-state occurred in 19 of the 126 common months. ERC6 minus Bunshin averaged +3.198%/month in A-state and -0.054%/month outside A-state. In Bunshin down months, ERC6 active return averaged +1.685%/month with an 81.25% win rate. The dominant historical source of ERC6 superiority is therefore the A-state shift toward Genbu Shield, not ordinary-state Spear optimization.

Canonical mechanism statement: ERC6 is best interpreted as a defense-quality architecture whose active edge is concentrated in the frozen defense state. This is explanatory evidence only; it does not authorize new A-state thresholds, state subdivisions, or additional filters.

91. R264-R265 - Defense-edge falsification and time stability

R264 froze the R263 mechanism and tested whether the defense edge was explainable by one or a few exceptional events, drawdown-only geometry, or obvious era concentration. R264 completed 26/26 ALL PASS. The defense mechanism survived the planned falsification stack, but the event count remained small and therefore retained explicit uncertainty.

R265 then tested time stability / decay with the same frozen 19 defense events: early versus late event blocks, fixed eras, rolling 5-event and 10-event windows, event-order trend, jackknife diagnostics, and a 2024-2026DEV versus 2016-2023 comparison. No clear evidence of complete edge disappearance was established. The correct governance response is Forward OOS monitoring, not a new DEV rescue filter.

92. R266-R270 - Four-Gods signal and meta-signal program

R266 moved from portfolio-return comparison to common-scale signal geometry. PATCH1 completed 23/23 ALL PASS. Using the fixed 12M t-1 family signal scale, Byakko-Genbu signal correlation was about 0.790 and the signal-space PCA effective dimension was about 2.62. Simple directional majority was frequently unanimous and therefore carried limited discriminatory information.

R267 (22/22 ALL PASS) found residual / nonlinear structure that was not reducible to the dominant shared direction. The most important fixed observations were Seiryu residual

predictive association around -0.276 and the predeclared Growth/Reversal axis around -0.302.

R268 fixed one meta-signal only: $\text{PrimaryMeta} = -(z(\text{Seiryu}) + z(\text{Suzaku}))/2$. Exactness 20/20 ALL PASS. Against next-month Bunshin return, Spearman was about +0.302; the High-minus-Low tercile future-return spread was about +4.77%/month, with positive direction in 4/4 fixed eras, 10/10 LOYO blocks, and 54/54 rolling-60M windows. This is strong evidence that the frozen Four-Gods signal structure contains broad market-environment information.

R269 then made one predeclared routing test: use ERC6 when PrimaryMeta is in the fixed High tercile, otherwise Bunshin. Exactness 29/29 ALL PASS, but the Primary Decision Gate failed.

R270 therefore audited whether the same frozen PrimaryMeta predicted next-month ERC6 minus Bunshin active return directly. The resulting interpretation is narrow: the signal is informative about the broad return environment, but it is not validated as a reliable selector between ERC6 and Bunshin. Threshold rescue, quartile mining, and blend searches are prohibited.

93. R271-R273 - Teacher-inspired discrete signal architecture diagnostics

These studies use teacher-strategy ideas only as structural reference. They do not claim reconstruction of any unpublished teacher rule and do not use hidden asset identities as Four Gods Evidence.

Study	Frozen design	Formal role
R271	Four equal 25% God sleeves; each selects the strongest R195 candidate	24/24 ALL PASS, 12M signal only; no portfolio-return
R272 PATCH2	Four equal votes over shared TQQQ / TECL / XLU / GLD universe using 20/20/20/20	24/24 ALL PASS, 12M signal only; no portfolio-return
R273	Four leave-one-out synthesized sleeves; each combines 3 of the 4 from R272	24/24 ALL PASS, 12M signal only; no portfolio-return

The teacher reference pattern distribution used in R272 was 4-0 = 120/161, 2-2 = 34/161, 3-1 = 5/161, and 2-1-1 = 2/161. This remained reference-only and was not used as a fitting target. The main scientific use of R271-R273 is to study how temporal signal diversification and signal synthesis affect vote concentration, not to reverse engineer an unpublished strategy.

94. R273H-R273HA - Strategy Health monitoring line

R273H transplanted the already-existing Strategy Health methodology to frozen ERC6 without changing the methodology: 6/12/24M deterioration, erosion, Newey-West HAC treatment, REVIEW/WATCH logic, and p-bar consistency. R273HA extended the same monitor to Bunshin, R181, ERC6, and R255 Split. R259 was deliberately fail-closed because its official monthly lineage was not safely reconstructible in the monitor path; it therefore remained NOT AUDITED rather than being approximated.

The Health line is explicitly non-ranking and non-selection. Its purpose is to describe current strategy condition, not to choose a challenger or change Production.

95. R273HB - G1 / G2 / P / pbar Health-Dot Audit

R273HB is official user-run Evidence with Exactness 15/15 ALL PASS. It reuses the R273HA lineage and freezes four visual axes with no strategy/rule change and no optimization/search.

Portfolio	G1	G2	P	pbar
Bunshin	YELLOW	GREEN	MIXED / YELLOW	Strong / GREEN
R181	GREEN	GREEN	DETERIORATING / ORANGE	Strong / GREEN
ERC6	GREEN	GREEN	DETERIORATING / ORANGE	Strong / GREEN
R255 Split	GREEN	GREEN	GOOD / GREEN	Strong / GREEN

Frozen axis definitions: G1 = 12-point slope of erosion long-mean; G2 = 12M p_erosion; P = 6/12/24M deterioration classification; pbar = Primary 5-split absolute consistency. The current snapshot therefore shows P-only deterioration warnings for R181 and ERC6 while their G1, G2, and pbar remain green. That pattern alone is not a strategy-failure diagnosis.

96. R274 - Historical Health Predictive Falsification

R274 is official user-run Evidence with Exactness 15/15 ALL PASS. It reconstructs G1/G2/P/pbar historically using information available through each endpoint month only, then asks whether a first green -> yellow/orange transition predicts weaker returns over the predeclared next 12 months. Main inference uses all endpoints with HAC(11); robustness uses 12 fixed calendar offsets. No threshold search, parameter search, or allocation/rule change is permitted.

Portfolio	Endpoints	Events	Effect: future minus baseline	HAC p	Offset median	Offset negative share
Bunshin	19	2	-0.124%	0.508	-0.615%	50%
R181	19	2	0.605%	0.061	1.307%	0%
ERC6	19	2	0.703%	0.033	1.398%	0%
R255 Split	19	0				

G2 and pbar produced zero transitions for every audited strategy. R255 Split produced zero transitions on G1, P, and Composite as well. Composite warning events were therefore only N=2 for Bunshin, R181, and ERC6. Bunshin showed a small negative effect (-0.124%/month) with HAC p=0.508; R181 and ERC6 showed positive, not negative, effects (+0.605%/month, p≈0.061; +0.703%/month, p≈0.033). With only two events each, neither the intended warning rule nor an opposite contrarian rule is scientifically validated.

Formal R274 interpretation: PREDICTIVE HYPOTHESIS NOT SUPPORTED / SAMPLE-LIMITED. Health dots remain descriptive monitors. They must not be used to reduce allocation, stop a strategy, rotate challengers, or create an inverse signal on the same DEV evidence.

97. Updated canonical Layer1 state after R274

Object	Canonical role after R274
Production Frozen4F EW	Production unchanged
R181	Legacy regime-defense challenger
ERC6	ERC branch primary risk-quality representative; strongest current defense-quality mechanism evidence
R255 Split	Family-aware contrarian + Shield-redistribution broad challenger
R259 Gate	High-conviction / conservative R255 variant; Forward discriminator remains preferred
R268 PrimaryMeta	Structural market-environment signal; not an allocation router
R273HB Health dots	Descriptive monitoring only

The central post-v1.28 synthesis is therefore: Layer1 contains at least two materially different active-edge directions; ERC6 risk quality is historically dominated by the frozen defense state; Four-Gods signals contain real environment information; but neither edge blending, meta-signal routing, nor Health-state switching has earned a Production rule on DEV.

98. Governance after R274

1. Freeze Production = Frozen4F EW.
2. Freeze active Layer1 challengers R181 / ERC6 / R255 Split / R259 Gate as named objects; do not conduct a new DEV weight grid.
3. Do not rescue R262 with alternative blend weights.

4. Do not tune the ERC6 A-state or add post-hoc state filters from R263-R265 diagnostics.
5. Preserve R268 as structural signal Evidence, but do not mine new routing thresholds after R269 failure.
6. Preserve R271-R273 as architecture diagnostics only; do not fit to teacher frequency distributions or claim hidden-rule identity.
7. Preserve R273HB Health colors as monitoring only. R274 explicitly blocks conversion of those colors into an allocation/stop/rotation rule on the same DEV data.
8. Real Forward OOS from 2026-08-01 remains the preferred discriminator for challenger durability and any future health-warning usefulness.

99. Implementation and evidence provenance

Field	v1.29 snapshot
Textual parent	Master Research Bible v1.28, pages 1-181 unchanged
Implementation authority	app(20260916-235015).py
app.py bytes	3,607,989
app.py lines	59,821
app.py SHA-256	bea5b1964674be405555dc04cd771cd98bf8aa0eb5970db33312d9f2bd5141ec
Compile gate	python -m py_compile PASS
Attached official Evidence	r273hb_layer1_g1_g2_p_pbar_health(2).zip; r274_historical_health_predictive_falsification.zip
R273HB Exactness	15/15 ALL PASS
R274 Exactness	15/15 ALL PASS

The source hierarchy is unchanged: current Master Bible = textual authority; current full-spec app.py = implementation authority; successful user-run ZIP with Exactness ALL PASS = performance/evidence authority. If these disagree, audit the difference; do not alter strategy rules merely to force numerical agreement.

100. Updated Source-of-Truth statement

Master Research Bible v1.29 supersedes v1.28 as the latest textual Source of Truth. Pages 1-181 are inherited unchanged; this addendum records the R261-R274 line and the governance boundary reached after the historical Health predictive falsification. Production remains Frozen4F EW. No research in this addendum authorizes retroactive DEV retuning.

Master Research Bible v1.30 Addendum

Scope. This addendum preserves Master Research Bible v1.29 pages 1-187 unchanged and appends the R275-R278 research line: Attack75 Layer1 admission, the ERC6 + Attack75 defense-first hybrid, historical robustness, same-period Bam teacher benchmarking, and the resulting overfitting/governance assessment.

Evidence authority. User-run ZIPs with Exactness ALL PASS remain the performance/evidence authority. The textual Master is the research-rule authority. Production remains Frozen4F EW; no result below authorizes retrospective DEV retuning.

DEV / Forward boundary. DEV ends 2026-07-31. Data from 2026-08-01 onward are Real Forward OOS and must not be used to re-fit the Hybrid, Attack75, ERC6, or teacher-comparison rules.

101. R275 - 4F Attack75 admitted to the current Layer1 challenger registry

R275 was a comparison / active-edge / Health audit only. It introduced no strategy-rule, weight, lookback, threshold, or Production change. Exactness was 14/14 ALL PASS. The common comparison period was 2016-02..2026-07 (N=126).

Strategy	CAGR	Sharpe	Sortino	MaxDD	Calmar
R181	61.46%	1.942	5.884	-13.94%	4.408
ERC6	61.09%	2.011	7.017	-11.71%	5.219
R255 Split	58.42%	1.881	5.225	-21.36%	2.735
R259 Gate	56.57%	1.858	5.229	-21.85%	2.589
4F Attack75	64.01%	1.674	4.801	-21.83%	2.931

Attack75 active-return correlation versus the existing challengers was materially low/negative (approximately -0.38 vs R181, -0.42 vs ERC6, -0.27 vs R255 Split, and -0.02 vs R259 Gate). This supported an orthogonality hypothesis, not automatic promotion.

Canonical role after R275: 4F Attack75 = state-dependent Booster / firepower challenger. Production remained unchanged.

102. R276 PATCH1 - ERC6 + Attack75 Defense-first Growth Booster Hybrid

R276 tested a one-shot architecture with no search. The first build exposed a state-timing exactness defect; PATCH1 corrected implementation timing without changing the research hypothesis. PATCH1 Exactness was 17/17 ALL PASS.

Frozen hierarchy:

Priority	Rule
1. ERC6 defense state	If ERC6 frozen defense A-state is ON: Genbu 100%; Attack75 is blocked.
2. Normal ERC6 state	Use the frozen ERC6 base allocation.
3. Frozen Attack75 Turbo	If Turbo is ON outside ERC6 defense state: 75% to the existing TECL/TQQQ selector; residual 25% is distributed proportionally
4. No Turbo	ERC6 remains unchanged.

R276 common performance - 2016-02..2026-07 (N=126)

Strategy	CAGR	Vol	Sharpe	Sortino	MaxDD	Calmar	Worst M
Bunshin	53.21%	25.13%	1.843	4.956	-21.85%	2.436	-13.94%
ERC6	61.09%	25.63%	2.011	7.017	-11.71%	5.219	-11.71%
4F Attack75	64.01%	33.09%	1.674	4.801	-21.83%	2.931	-21.07%
ERC6 + Attack75	75.71%	30.29%	2.040	6.757	-20.59%	3.678	-20.59%

Primary falsification gates all passed: Hybrid CAGR minus ERC6 = +14.62 percentage points; Hybrid Sharpe minus Attack75 = +0.367; Hybrid MaxDD was 1.25 points shallower than Attack75.

State counts: ERC6 defense A-state 19 months; raw Turbo 27; A-state intersect Turbo 4 months (Attack blocked); effective Attack 23 months.

Health snapshot: Bunshin G1 yellow / G2 green / P MIXED / p-bar green; ERC6 green / green / DETERIORATING / green; Attack75 yellow / green / GOOD / green; Hybrid green / green / WATCH / green. Health remains monitoring-only and is not an allocation trigger.

103. R277 - Historical robustness and concentration falsification

R277 held the R276 PATCH1 strategy completely fixed. Exactness was 14/14 ALL PASS. It tested whether the Hybrid edge depended on a narrow era, one exceptional Attack month, or fragile resampling.

Diagnostic	Official result
Fixed eras with Hybrid CAGR > ERC6	4/4
Fixed eras satisfying original R276 primary trio	2/4
Effective Attack months with positive active return	18/23 = 78.3%
Leave-one-Attack neutralization preserving R276 trio	23/23
Best-10 Attack months removed: Hybrid CAGR - ERC6	+0.67 pt
Best-10 removal preserving original trio	PASS
12M circular block bootstrap original-trio pass rate	97.9% = 1958/2000

Fixed-era results

Era	Hybrid CAGR	ERC6 CAGR	dCAGR	Hybrid Sharpe	Attack75 Sharpe	Trio
2016-02..2018-12	53.38%	44.74%	+8.65pt	2.032	1.980	FAIL
2019-01..2021-12	132.79%	105.21%	+27.58pt	2.356	1.834	PASS
2022-01..2024-12	48.64%	47.16%	+1.48pt	1.927	2.154	FAIL
2025-01..2026-07	81.87%	47.22%	+34.66pt	2.031	1.352	PASS

Concentration audit: top-1 / top-3 / top-5 Attack-month contribution shares were 16.6% / 40.3% / 59.6% of net active return. Removing the 10 best Attack months reduced Hybrid CAGR from 75.71% to 61.76%, but the original R276 trio still passed.

Interpretation: the historical Hybrid edge is broad enough to survive severe concentration tests, but risk-adjusted superiority is not uniform across every era. The 2022-24 block is an explicit counterexample and must remain in the record.

104. Performance-page max-history implementation note

The Performance-page Hybrid series is now displayed from the earliest currently reconstructible month, 2014-08, through 2026-07. This is an implementation/display extension, not a retroactive expansion of R276/R277 official Evidence.

Interval	Status
2014-08..2016-01	Retrospective max-history reconstruction for Performance display.
2016-02..2026-07	Official R276/R277 comparison/evidence window.
2026-08 onward	Real Forward OOS; excluded from DEV retuning.

105. R278 - Same-period Hybrid vs Bam teacher strategies

R278 aligned all direct comparisons to 2016-02..2026-07 (N=126) and added a Bam-class robustness stack. Exactness was 19/19 ALL PASS. Ougi-Bunshin-Joushou and Ougi-Bunshin-Teppeki use user-supplied displayed Monthly Trade returns rounded to 0.01%; Shin-Yotsume-Gekiko is an allocation x ETF monthly-return reconstruction and is descriptive only.

Strategy	CAGR	Vol	Sharpe	Sortino	MaxDD	Calmar
ERC6 + Attack75	75.71%	30.29%	2.040	6.757	-20.59%	3.678
Ougi-Bunshin-Joushou	79.66%	31.31%	2.057	6.891	-22.83%	3.489
Ougi-Bunshin-Teppeki	64.26%	32.64%	1.693	6.851	-12.24%	5.250
Shin-Yotsume-Gekiko (recon)	76.62%	42.09%	1.573	4.136	-31.13%	2.461

Joushou is the strongest direct growth-quality benchmark in the supplied teacher series. On the same 126-month window it exceeded the Hybrid by 3.94 CAGR points, 0.017 Sharpe, and 0.134 Sortino; the Hybrid retained a 2.25-point shallower MaxDD and a 0.189 higher Calmar.

106. R278 Bam-style robustness results

R278 used fixed WF13, CPCV 8-choose-2 = 28 folds, LOYO, Rolling36/60, best-month removal, fixed stress, SPY beta-adjusted alpha, expanding robustness, and a 12M circular block bootstrap with 2,000 replications. No post-R276 parameter grid was allowed.

Hybrid wins versus Ougi-Bunshin-Joushou

Diagnostic	CAGR	Sharpe	Calmar
WF13	4/13	6/13	5/13
CPCV28	12/28	10/28	15/28
LOYO	0/11	5/11	10/11
Rolling60	1/67	27/67	41/67
12M block bootstrap	33.8%	44.1%	61.8%

Interpretation: Joushou CAGR superiority is broad, not a one-year artifact. The Hybrid should not be described as having surpassed Joushou. Its comparative advantage is drawdown/risk efficiency, not teacher-beating growth.

Hybrid wins versus Shin-Yotsume-Gekiko reconstruction

Diagnostic	CAGR	Sharpe	Calmar
WF13	7/13	9/13	9/13
CPCV28	12/28	22/28	22/28
LOYO	5/11	11/11	11/11
Rolling60	48/67	67/67	67/67
12M block bootstrap	46.95%	97.75%	91.75%

The reconstructed Gekiko series had nearly identical CAGR to the Hybrid, but materially higher volatility and drawdown. Because the Gekiko series is reconstructed from public allocation x ETF returns, this is descriptive evidence and must not be overstated as exact hidden-rule replication.

107. Stress and beta-adjusted diagnostics

Selected fixed stress episodes - cumulative return

Episode	Hybrid	Joushou	Teppeki	Shin-Yotsume-Gekiko recon
2018 Q4	+5.51%	+5.31%	-0.14%	+7.38%
COVID	+40.16%	+14.33%	+32.56%	-0.52%
2022 selloff	+4.93%	+29.56%	+11.44%	-6.32%
2025 selloff	+17.87%	+0.42%	+3.67%	+5.82%

The stress geometry differs materially. The Hybrid was strong in COVID and the 2025 selloff, while Joushou was exceptionally strong in the fixed 2022 selloff window. This is a mechanism clue, not a license to construct a new post-hoc router.

SPY beta-adjusted monthly OLS diagnostic

Strategy	SPY beta	Alpha / month	Arithmetic alpha / y	Alpha t	R2
ERC6 + Attack75	1.133	3.669%	44.02%	5.44	0.318
Ougi-Bunshin-Joushou	1.506	3.398%	40.77%	5.84	0.526
Ougi-Bunshin-Teppeki	1.370	2.814%	33.76%	4.13	0.400
Shin-Yotsume-Gekiko recon	1.794	3.172%	38.06%	3.65	0.413

The Hybrid combines lower SPY beta than Joushou/Gekiko with a large positive intercept in this simple regression. This is descriptive beta-adjusted evidence only; it is not a causal factor model and does not by itself prove transportable alpha.

108. Current overfitting / selection-bias assessment

The current evidence does not look like a simple parameter-fit artifact, but it also does not eliminate architecture/model-selection overfitting.

Risk source	Current assessment	Reason
Parameter overfitting	Low	ERC6 and Attack75 were already frozen; no new Hybrid weight/lookback/threshold grid; R276 used one fixed
Single-event concentration	Low to moderate	R277: 18/23 positive Attack months; leave-one-Attack 23/23; best-10 removal still passed; bootstrap trio 97
Time/regime fragility	Moderate	Original R276 trio passed only 2/4 fixed eras; 2022-24 is a documented weak regime for the Hybrid relative
Architecture selection bias	Moderate	The decision to combine ERC6 and Attack75 was made after extensive DEV research and after observing m
Researcher degrees of freedom	Moderate	Many historical candidate architectures existed before the Hybrid was proposed; family-wise selection bias
Real prospective evidence	Insufficient so far	The decisive test is the frozen 2026-08 onward Real Forward OOS period.

Canonical judgment: parameter overfitting is currently low; architecture/model-selection overfitting remains moderate. The correct response is not more DEV tuning. Freeze the Hybrid and accumulate Real Forward OOS.

Do not tune the 75% attack intensity, ERC6 defense A-state, VIX30 brake, inflection definition, selector, residual-weight rule, or new regime router from R277/R278 diagnostics. Any future change requires a separately versioned prospective hypothesis.

109. Canonical Layer1 registry after R278

Object	Canonical role	Status
Frozen4F EW	Production control	UNCHANGED
Bunshin	Quality/control baseline	Control
R181	Legacy regime-defense challenger	Frozen challenger
ERC6	Primary risk-quality representative	Frozen challenger
R255 Split	Broad family-aware contrarian + Shield redistribution	Frozen challenger
R259 Gate	High-conviction conservative R255 variant	Frozen challenger
4F Attack75	State-dependent Booster / firepower	Frozen challenger
ERC6 + Attack75	Defense-first Growth Booster Hybrid	Frozen robust challenger; forward OOS only

110. Evidence provenance and Source-of-Truth update

Field	v1.30 snapshot
Textual parent	Master Research Bible v1.29, pages 1-187 preserved unchanged
Official R275 evidence	r275_attack75_layer1_challenger_admission.zip - 14/14 ALL PASS
Official R276 evidence	r276_erc6_attack75_defense_first 2.zip - PATCH1, 17/17 ALL PASS
Official R277 evidence	r277_erc6_attack75_robustness.zip - 14/14 ALL PASS
Official R278 evidence	r278_hybrid_vs_bam_robustness.zip - 19/19 ALL PASS
Current DEV cutoff	2026-07-31
Real Forward OOS	2026-08-01 onward; no DEV retuning
Production	Frozen4F EW unchanged

Master Research Bible v1.30 supersedes v1.29 as the latest textual Source of Truth. This addendum records the R275-R278 line and the governance boundary reached after the Hybrid-vs-teacher robustness study. The current research direction is mechanism understanding and prospective observation, not retrospective Hybrid optimization.

The evidence hierarchy remains unchanged: current Master Bible = textual authority; current full-spec app.py = implementation authority; successful user-run ZIP with Exactness ALL PASS = performance/evidence authority. If these disagree, audit the discrepancy; do not alter strategy rules merely to force numerical agreement.

Master Research Bible v1.31 Addendum

Scope: R279-R282. This addendum preserves v1.30 pages 1-195 unchanged and adds the newest anti-overfitting, factor-alpha, nonlinear/timing, and crash-convexity evidence for the frozen ERC6 + Attack75 Hybrid.

109. Canonical state entering v1.31

Production remains Frozen4F EW. ERC6 + Attack75 remains a Frozen Layer-1 challenger with formal role Defense-first Growth Booster Hybrid. No R279-R282 result changes Production, Attack intensity, ERC6 defense A-state, VIX30 brake, selector, lookback, or residual-weight rules.

Run	Exactness	Purpose / official conclusion
R279	17/17 ALL PASS	Frozen Hybrid anti-overfitting stack completion: multi-window WF + pseudo-Vintage OOS
R280 v1.1	19/19 ALL PASS	Bam-style MKT/SMB/HML/MOM/TSMOM factor-alpha audit
R281	19/19 ALL PASS	Nonlinear market / downside / VIX timing exposure audit
R282	21/21 ALL PASS	Crash convexity / option-like payoff / event-concentration audit

Evidence rule: user-run ZIP with ALL PASS is the performance source of truth. Earlier R280 fail-closed run (factor panel available only through 2026-05 while exactness expected 126 months) is implementation evidence only, not empirical evidence. R280 v1.1 fixes the availability contract without changing strategy or factor specification.

110. R279 - Frozen Hybrid Anti-Overfitting Stack Completion

R279 extends the frozen Hybrid validation toward the Bam-style anti-overfitting stack without reopening parameter search. IS windows are fixed at 6/12/18/24/30/36/42/48/54 months; each OOS window is fixed at 12 months. Candidate comparison is ERC6 / Attack75 / Hybrid using IS Sharpe as the diagnostic selector.

IS	Hybrid selected	Hybrid CAGR > ERC6	Hybrid Sharpe > Attack75	Hybrid MDD shallower	R276 trio PASS
6M	10.0%	80.0%	70.0%	70.0%	60.0%
12M	33.3%	66.7%	66.7%	77.8%	55.6%
18M	22.2%	77.8%	77.8%	77.8%	66.7%
24M	37.5%	75.0%	75.0%	87.5%	62.5%
30M	37.5%	87.5%	87.5%	87.5%	75.0%
36M	42.9%	71.4%	71.4%	85.7%	57.1%
42M	71.4%	85.7%	85.7%	85.7%	71.4%
48M	66.7%	66.7%	66.7%	83.3%	50.0%
54M	100%	83.3%	83.3%	83.3%	66.7%

Interpretation: Hybrid is not consistently selected by very short IS windows, while selection frequency rises at longer IS windows; at 54M it is selected in 6/6 origins. This pattern is consistent with a long-horizon risk-quality advantage rather than a single short-lived peak, but it does not itself remove architecture-selection bias.

Pseudo-Vintage OOS

Cutoff	IS selection	Next 12M CAGR>ERC6	Sharpe>Attack75	MDD shallower	Trio PASS
2020-12	Hybrid	PASS	PASS	PASS	PASS
2022-12	Hybrid	PASS	FAIL	PASS	FAIL
2024-12	Hybrid	PASS	PASS	PASS	PASS

Aggregate: Hybrid selected 3/3; OOS CAGR > ERC6 3/3; OOS Sharpe > Attack75 2/3; shallower MDD 3/3; full trio 2/3. This is intentionally named pseudo-Vintage, not true Vintage OOS: the Hybrid architecture was discovered in 2026, and historical candidate registries/code snapshots/researcher-information states are insufficient to prove that an earlier researcher could have selected it.

111. R280 v1.1 - Bam-style Five-Factor Alpha Audit

Common factor window: 2016-02 through 2026-05, N=124, determined mechanically by the intersection of complete strategy returns, Kenneth French MKT-RF/SMB/HML/RF/MOM, and AQR aggregate TSMOM availability. This does not move the DEV cutoff; factor regressions simply stop at the last complete factor month.

Full model: excess strategy return = alpha + beta(MKT-RF) + beta(SMB) + beta(HML) + beta(MOM) + beta(TSMOM) + error. HAC/Newey-West lag 3 is used for inference.

Strategy	Annual alpha	HAC t	R ²	Status
ERC6 + Attack75	+45.94%	6.88	0.421	Strong residual alpha
奥義-分身-常勝	+43.39%	5.87	0.561	Teacher exact displayed monthly series
ERC6	+38.15%	6.71	0.418	Frozen risk-quality base
シン四つ目-激攻*	+37.75%	4.15	0.474	Reconstructed descriptive
奥義-分身-鉄壁	+35.58%	4.15	0.428	Teacher exact displayed monthly series
Attack75	+32.42%	5.24	0.488	Frozen Booster

* Reconstructed from public allocation x ETF returns; lower evidence strength than exact teacher monthly histories.

Hybrid nested factor result

Model	Hybrid annual alpha
MKT only	+45.29%
MKT + SMB + HML	+44.07%
+ MOM	+45.90%
+ TSMOM (full 5-factor)	+45.94%

Adding MOM and TSMOM does not materially erode Hybrid alpha. In this fixed linear factor model, the Hybrid is not well explained as simple leveraged exposure to common equity or momentum/trend premia.

Hybrid factor loadings

MKT beta 1.05; SMB -0.45; HML -0.69; MOM -0.42; TSMOM approximately -0.02. TSMOM loading is near zero and not a material explanatory channel in this specification.

Walk-forward / expanding

36M IS -> 12M OOS factor-adjusted Hybrid: adjusted CAGR approximately 64.1%, adjusted Sharpe approximately 1.96, MaxDD approximately -14.2%, Calmar approximately 4.53, with 7/7 positive OOS windows. Expanding full-factor alpha is positive in 8/8 endpoints.

112. R281 - Nonlinear Market / Timing Exposure Audit

R281 asks whether the R280 residual alpha is merely nonlinear beta or volatility timing. Fixed controls add MKT^2 , downside-market interaction, VIX monthly change, and fixed market/VIX regime summaries. VIX 30 is inherited from the frozen strategy; no threshold search is performed.

Hybrid result	Value
Full nonlinear/timing annual arithmetic alpha	+38.77%
Full nonlinear/timing annual compound alpha	+46.46%
HAC t	2.97
R^2	0.486
Residual vol	21.56%

Nested alpha path

Specification	Annual alpha
Full5	+43.77%
Full5 + MKT^2 only	+29.06%
Full5 + downside interaction only	+21.11%
Full5 + VIX change only	+40.75%
Full nonlinear/timing	+38.77%

The one-at-a-time nonlinear controls can reduce alpha substantially, but in the joint model the alpha remains large. Individual nonlinear/timing coefficients are not strongly significant: MKT^2 HAC t 1.56, downside interaction 0.93, VIX change 0.97. Multicollinearity and interaction among controls means the one-at-a-time alpha reductions should not be interpreted as unique causal attribution.

Regime behavior

Hybrid mean monthly return: Market Up approximately +7.52%; Market Down approximately -0.35%; $VIX > 30$ approximately +6.97%; $VIX \leq 30$ approximately +4.94%; Market Down & $VIX > 30$ (7 months) approximately +0.58%. Attack75, 奥義-分身-常勝, and reconstructed シン四つ目-激攻 are negative on average in that final joint stress bucket.

OOS residual audit

36M IS -> 12M OOS extended-factor residual: Hybrid adjusted CAGR approximately +36.59%, adjusted Sharpe approximately 1.25, positive windows 6/7. Expanding alpha is positive 8/8, with minimum annual alpha approximately +13.25% and median approximately +28.72%.

113. R282 - Crash Convexity / Option-like Payoff Audit

R282 asks whether the remaining alpha is a crash-convexity artifact or depends on extreme-event months. Fixed diagnostics include market tails at +/-5% and +/-10%, VIX monthly spike +50%, up/down conditional beta, downside semibeta, capture ratios, market-tail hinges, VIX-spike hinge, leave-one-crash-month-out, and event-growth concentration. No quantile or threshold search is allowed.

Hybrid full tail/convexity result	Value
Annual arithmetic alpha	+35.54%
Annual compound alpha	+41.94%
HAC t	2.53
R ²	0.494
Residual vol	~21.4%

Tail coefficients

Term	Beta	HAC t
MKT < -5% tail hinge	-0.51	-0.18
MKT > +5% tail hinge	+0.45	0.20
VIX spike > +50% hinge	-0.16	-1.78

No tail coefficient provides strong evidence that Hybrid returns are generated by a simple option-like extreme-event payoff.

Asymmetric participation

Metric	Hybrid
Up-market conditional beta	1.61
Down-market conditional beta	0.42
Downside semibeta	0.46
Up capture	2.06x
Down capture	0.13x
Mean monthly return - Up	+7.41%
Mean monthly return - Down	-0.51%

The observed shape is more consistent with state-dependent asymmetric participation - strong upside participation with reduced downside exposure - than with a crash-only payoff engine.

114. R282 event-concentration and leave-one-crash-month-out evidence

R282 classifies 14 fixed tail/VIX event months. Hybrid compounded contribution from those event months is approximately -32.6%, while the 109 non-event months produce the overwhelming cumulative growth. Event contribution to total log growth is approximately -6.9%. Therefore the high CAGR is not created by crash/event months; those months are, in aggregate, a drag rather than the source of the return engine.

Leave-One-Crash-Month-Out

Diagnostic	Result
Crash months tested	13
Alpha positive after each single-month removal	13/13
Minimum annual alpha	+30.89%
Median annual alpha	+35.58%
Minimum HAC t	2.24

This materially weakens the hypothesis that one special crash month is responsible for the residual alpha.

115. Hypotheses weakened by R277-R282

Candidate explanation	Current evidence
One/few extreme Attack months	Weakened by R277 best-month removal, leave-one-Attack-out, bootstrap
Short-window coincidence	Weakened by R279 multi-window WF and pseudo-Vintage OOS
Simple market beta	Weakened by R278/R280 beta-adjusted models
SMB/HML/MOM/TSMOM premia	Weakened by R280 five-factor audit
Nonlinear beta / VIX timing	Weakened by R281 joint nonlinear/timing model and OOS residuals
Crash convexity / option-like tail payoff	Weakened by R282 tail model and event concentration
Single crash month	Weakened by R282 LOCO 13/13 positive alpha

These are falsification results, not proof of “true skill.” Unobserved factors, nonlinear interactions, dynamic leverage path-dependence, implementation frictions, and research-selection effects can still matter.

116. Updated overfitting / selection-bias assessment

Risk source	v1.31 assessment	Reason
Parameter overfitting	Low	ERC6 and Attack75 frozen; no R279-R282 strategy parameter search or rescue tuning
Single-event concentration	Low	R277 and R282 removal/concentration tests do not show dependence on a few winning events
Known-factor masquerading	Low	R280 residual alpha remains large after MKT/SMB/HML/MOM/TSMOM
Nonlinear/timing masquerading	Low	R281 joint controls reduce but do not erase alpha; OOS residual remains positive in 6/7 windows
Crash/option-like masquerading	Low	R282 tail coefficients weak; event months are aggregate drag; LOCO robust
Time/regime fragility	Low-Moderate	Multi-window/pseudo-Vintage/expanding evidence is supportive, but fixed-era heterogeneity remains
Architecture-selection bias	Moderate	Hybrid architecture was proposed after extensive DEV research and many prior branches
Researcher degrees of freedom	Moderate	Family-wise selection bias is not removed by robustness testing of the selected object
Real prospective evidence	Insufficient	2026-08 onward remains the decisive frozen Real Forward OOS

Canonical judgment: the Hybrid no longer looks like a simple parameter-fit, factor-premium disguise, nonlinear-beta artifact, crash-convexity strategy, or single-event winner. The dominant remaining scientific concern is architecture/model-selection bias, which DEV diagnostics cannot fully remove.

Governance response: do not tune the Hybrid further on DEV. Freeze it and accumulate Real Forward OOS. Any future rule change must be a separately versioned prospective hypothesis and must not overwrite the frozen R276 PATCH1 object.

117. Current Hybrid interpretation and teacher-strategy context

The combined evidence supports the following descriptive interpretation: the Hybrid compounds primarily in ordinary/non-event periods, participates strongly in positive market states, and suppresses exposure in negative market states. Its return engine is not concentrated in crash months and is not well captured by the fixed linear common factors tested so far.

Teacher benchmark context

奥義-分身-常勝 remains the most important teacher growth-quality benchmark. On the R278 common period it retains more robust CAGR dominance over Hybrid in LOYO/rolling tests, while Hybrid has somewhat shallower MaxDD and higher Calmar. No v1.31 result changes the rule that Hybrid must not be described as having “beaten” 常勝 overall.

シン四つ目-激攻 remains a descriptive reconstructed comparison. Its common-period CAGR is similar to Hybrid, while Hybrid has materially stronger Sharpe/Sortino/MaxDD/Calmar and rolling risk efficiency. Because the monthly series is reconstructed rather than an exact teacher series, evidence strength remains lower.

Canonical Hybrid status

Field	Current state
Name	ERC6 + Attack75
Formal role	Defense-first Growth Booster Hybrid
Layer	Frozen Layer-1 challenger
Production status	NOT Production; Frozen4F EW remains Production Control
Official R276 common-period CAGR	75.7121%
Official R276 Sharpe	2.040
Official R276 Sortino	6.757
Official R276 MaxDD	-20.5866%
Official R276 Calmar	3.6777
Latest falsification sequence	R277 -> R278 -> R279 -> R280 -> R281 -> R282
Real Forward boundary	2026-08-01 onward; no DEV retuning

118. Latest research workflow / UI governance

User preference: the Streamlit verification UI should show only the latest verification module. Historical runs remain in code for lineage, reproducibility, and audit, but are hidden from the active validation UI unless explicitly reopened for forensic work.

Item	v1.31 rule
Active validation UI	Latest validation only
Historical research code	Preserve; do not delete for “lightweighting”
Official evidence	User-run ZIP + Exactness ALL PASS
Failed implementation run	Retain in lineage; never treat as empirical evidence
DEV cutoff	2026-07-31
Real Forward OOS	2026-08-01 onward
Production	Frozen4F EW
Hybrid	Frozen challenger; no post-R282 DEV tuning
Source-of-truth hierarchy	Master text / current full-spec app.py / ALL-PASS user ZIP performance

119. v1.31 scientific bottom line

R279-R282 materially strengthen falsification evidence for the frozen Hybrid without changing its rules. Time-direction tests, common-factor regressions, nonlinear/timing controls, and crash-convexity audits all leave substantial positive residual performance. This reduces several plausible overfitting explanations, but does not solve the historical selection problem created by choosing the architecture after extensive DEV research.

Therefore the next decisive evidence source is not another DEV optimization layer. It is frozen Real Forward OOS. Forward observations should be recorded without changing R276 PATCH1 rules. DEV analysis may continue only for mechanism understanding or predeclared independent falsification, not for rescuing or improving the frozen Hybrid.

This addendum is a textual research registry. Official ZIPs remain the performance source of truth; app.py remains the implementation source of truth.

Master Research Bible v1.32 Addendum

Mv4F / Four Gods / Frozen v3.45 - 2026-09-18

Scope: R283 Trend Clarity Independent Audit + operational application completion milestone. This addendum preserves v1.31 pages 1-204 unchanged. Production remains Frozen4F EW. ERC6 + Attack75 remains a Frozen Layer-1 challenger / Defense-first Growth Booster Hybrid.

113. Canonical state entering v1.32

No research result in this addendum promotes a new Production strategy or reopens DEV optimization. The Hybrid remains frozen. 2026-08 onward is Real Forward OOS and must not be fed back into DEV tuning.

Item	Canonical state
Production	Frozen4F EW - unchanged
Frozen Layer-1 challenger	ERC6 + Attack75
Hybrid role	Defense-first Growth Booster Hybrid
DEV cutoff	2026-07-31
Real Forward OOS	2026-08 onward
Trend Clarity	Independent diagnostic / path-quality candidate; not a trading rule
Application milestone	ERC6 + Attack75 operational page + Forward OOS monitor + current-date Performance views

114. R283 - Trend Clarity Independent Audit

R283 formalizes Trend Clarity (TC) as an independent audit rather than a production signal. The audit includes TC 2M/6M/12M, Signed TC, momentum, next-month return and MaxDD, multi-window consensus, momentum-crash comparison, a 2024-06 snapshot, Four Gods base diagnostics, and a 12-1 momentum top-subset TC-tercile to next-6-month test.

Diagnostic	Official interpretation
Mean IC, TC 12M	+0.034
Next-month MaxDD, TC12 Low	-4.46%
Next-month MaxDD, TC12 High	-3.85%
Momentum-crash resistance	Not established
Research conclusion	TC may contain path-quality / trend-clarity information, but evidence is insufficient for adoption as a trading rule.

Frozen decision: Trend Clarity research is paused after R283. No TC gate, weight, overlay, or production rule is introduced.

115. ERC6 + Attack75 - operational Forward OOS implementation

The dedicated top-level ERC6 + Attack75 page is now treated as the operational view of the frozen Hybrid. The page applies frozen rules to data available after the DEV cutoff without changing parameters.

Operational contract

Component	Frozen operational behavior
Rebalance cadence	Monthly
Action banner	Compares prior-month allocation with current-month allocation; shows HOLD or REBALANCE
ERC6	Uses frozen four-family construction and 6M covariance weighting
Defense A-state	Frozen defense-first rule; when active, Genbu 100%
Attack75	Frozen month-entry overlay; no DEV retuning
Forward data	2026-08 onward classified as Real Forward OOS
Failure handling	Do not fabricate missing family returns or allocations; audit status remains visible

The previous display-only 2026-07 endpoint has been replaced by a current operational month view. The displayed month advances with the live calendar/data pipeline, while 2026-07 remains the frozen evidence cutoff.

Forward OOS monitor

The monitor records the month, Real Forward OOS classification, DEV-return flag, family-engine state, and current Hybrid operational state. Forward observations are explicitly prohibited from returning to DEV optimization.

Family status cards now communicate the operational risk state of Seiryu, Byakko, Suzaku, and Genbu rather than merely saying LIVE.

116. Operational portfolio presentation

The Hybrid page separates what the investor actually holds from the internal strategy-family allocation.

Panel	Purpose
Portfolio Mix	Final ticker-level allocation used for implementation
Strategy Mix	Seiryu / Byakko / Suzaku / Genbu / Booster allocation
Health	G1 / G2 / P shown as color state only; raw values are intentionally hidden in the compact operational view
Monthly return	Latest MTD, prior month, 2026 YTD, then recent monthly cards vs SPY
State badges	Recent return cards identify NORMAL, BOOSTER (TQQQ/TECL), and A-STATE months

Donut charts use independent rendering scopes so the Portfolio Mix and Strategy Mix cannot overwrite one another. Slice labels/legends are designed for mobile readability.

2026-09 operational snapshot (captured during v1.32 completion)

Final asset	Weight
GLD	29.9%
SPXL	2.6%
TQQQ	5.2%
TMV	37.3%
XLU	25.0%

Strategy family	Weight
Seiryu	27.3%
Byakko	10.4%
Suzaku	37.3%
Genbu	25.0%

Snapshot state: ERC6_NORMAL; Defense A-state OFF; ERC6 Spear ACTIVE; Attack75 Turbo OFF; Genbu Defense NORMAL. This is an operational snapshot, not a new frozen research parameter set.

117. Performance page - longest-history and current-date contract

Performance views were corrected so strategy history is not artificially truncated to a fixed 2015 start or a 2026-07 endpoint.

View	v1.32 behavior
Annual Returns	Selected strategy uses its own longest available history
Single-strategy performance	Uses that strategy's longest available history
Multi-strategy comparison	Uses the longest common overlapping history required for a fair comparison
Start date	Not hard-coded to 2015-02
Latest date	Uses the latest available current data; current year is shown as provisional YTD/MTD
DEV cutoff	Still 2026-07 for research governance; not used as a UI truncation date

This distinction is deliberate: **research governance cutoff** and **operational display endpoint** are different concepts. The former prevents retrospective tuning; the latter should show the latest available data.

118. Application architecture / performance milestone

The previously monolithic application was split without deleting research lineage. The active UI/operational code remains in **app.py**, while historical research/archive code is stored in **research_archive.py** and loaded lazily when needed.

Item	State
Research lineage	Preserved; historical audit code not deleted
Archive loading	Lazy-loaded when the archive is needed
Forward engine	Cached to reduce repeated reconstruction
Frozen official data	Cached separately from live Forward work
Mobile target	iPhone-first layout; avoid horizontal overflow and oversized headings
Current app validation	py_compile PASS after latest Performance-page update
Latest app.py SHA-256	2ac7d2eba2bac507a8aa45628617f8cd9eb8909e75dc22f64f4f74c17ead8ae8

119. v1.32 freeze statement

This version is the interim completed application/research baseline as of 2026-09-18.

Frozen item	v1.32 decision
Production	Frozen4F EW remains Production
ERC6 + Attack75	Remain Frozen Layer-1 challenger / Defense-first Growth Booster Hybrid
Hybrid DEV tuning	Closed
Real Forward OOS	Continue from 2026-08 onward
Trend Clarity	R283 retained as independent audit; no trading-rule adoption
Operational app	Current ERC6 + Attack75 page and Performance-page behavior accepted as interim completion baseline
Future research	Must branch from this frozen state; do not silently rewrite historical evidence or operational definitions

Overfitting assessment carried forward from v1.31

Risk dimension	Assessment
Parameter overfitting	Low
Single-event concentration	Low
Known-factor masquerading	Low
Nonlinear/timing masquerading	Low
Crash/option-like masquerading	Low
Time/regime fragility	Low-Moderate
Architecture-selection bias	Moderate
Researcher degrees of freedom	Moderate
Real Forward evidence	Insufficient - accumulation continues

The central unresolved evidence gap is time: Real Forward OOS has only recently begun. No amount of additional DEV analysis substitutes for elapsed forward evidence.

End of v1.32 addendum.

Master Research Bible v1.33 Addendum

My4F / Four Gods / KIRIN | 2026-09-18

Scope: KIRIN naming freeze, R284-R286 execution robustness program, and operational 5% Execution Rule. v1.32 pages are preserved unchanged.

120. Naming freeze: KIRIN

KIRIN (■■■) is the official operational name for the strategy previously identified by the research code **ERC6 + Attack75**. Historical code names remain in source, ZIPs, and audit lineage for reproducibility. Formal role remains Four Gods Defense-first Growth Hybrid / Frozen Layer-1 challenger. Production remains Frozen4F EW.

121. R284-R286 execution robustness program

Run	Purpose	Official status
R284	5% grid at KIRIN family / Attack75-residual decision blocks	Exactness ALL PASS
R285	5% grid at aggregated final executable ticker targets	Exactness ALL PASS
R286	Pre-specified synthesis / governance freeze audit of R284 + R285	11/11 ALL PASS

Metric	Exact KIRIN	R285 final ticker 5%
CAGR	68.8768%	69.2895%
Sortino	3.0858	3.1932
MaxDD	-20.5866%	-20.8452%
Tracking Error	—	0.8342%
Mean abs monthly diff	—	0.1371%

Interpretation: the small apparent performance improvement is not the reason for adoption and must not be treated as an optimized edge. The evidence supports execution robustness: quantization introduces only small tracking error while preserving the frozen strategy's performance profile.

122. Frozen KIRIN 5% Execution Rule

At each monthly KIRIN rebalance: (1) calculate the Frozen KIRIN family allocation exactly; (2) expand frozen family holdings and Attack75 exposure into aggregated executable final ticker targets; (3) normalize the exact ticker targets; (4) quantize those final ticker targets to **5 percentage-point increments** using a largest-remainder allocation; (5) preserve total target weight at exactly 100%; (6) use the 5%-grid ticker weights as the official execution / Portfolio Mix targets.

The quantization is an **Execution Layer only**. It does not change Four Gods signals, family definitions, ERC6 covariance lookback, A-state, Attack75 Turbo, TECL/TQQQ selector, or family-native intramonth switching. Exact pre-quantization ticker targets are retained for audit.

Frozen order of operations

Four Gods returns -> 6M ERC -> A-state -> Turbo / TECL selector -> family allocation -> exact ticker holdings -> 5% execution grid

123. Governance and Forward OOS

Item	v1.33 state
DEV cutoff	2026-07-31 - unchanged
Real Forward OOS	2026-08 onward - excluded from R284-R286 selection/freeze evidence
Parameter optimization	NONE
Production	Frozen4F EW - unchanged
KIRIN research core	Frozen
KIRIN execution	5% final-ticker grid - Frozen by R286
Forward feedback to DEV	PROHIBITED

R286 governance tolerances were used as falsification limits, not searched to maximize performance. Future Forward OOS observations may evaluate implementation quality but must not retrospectively alter the DEV evidence.

124. Application operational contract

The KIRIN page now presents the 5%-grid final ticker targets as the primary **Portfolio Mix**. Exact ticker targets remain available in audit information. The Strategy Specification records the construction chain and frozen 5% Execution Rule. The validation archive UI shows only the newest validation while historical validation implementations remain in source for lineage.

The archive loader explicitly reloads `research_archive.py` on app reruns so a newly deployed validation module is not intentionally served from Python's module cache. A platform-level Streamlit worker or deployment cache may still require a platform restart if the hosting environment does not rerun the application code.

v1.33 is the canonical baseline after the KIRIN execution freeze.

Master Research Bible v1.34 Addendum

Scope: R287 Practical 5% Execution freeze and operational UI correction. v1.33 remains unchanged; this addendum supersedes only the KIRIN execution-layer rule frozen at R286.

120. R287 - Practical 5% Execution / Small-Position Pruning

R287 tested a predeclared operational rule on DEV only through 2026-07. Exact final-ticker sleeves below 5% are pruned to 0%; surviving sleeves remain on a 5% grid; residual 5% units are assigned only among survivors by largest fractional remainder; total allocation remains exactly 100%. Forward OOS was excluded and no parameter optimization was performed.

Metric	Exact KIRIN	R286 LR5	R287 Practical 5%
CAGR	68.88%	69.29%	69.12%
Sharpe	1.927	1.937	1.946
Sortino	3.086	3.193	3.270
MaxDD	-20.59%	-20.85%	-19.87%
Calmar	3.346	3.324	3.479
Tracking Error vs Exact	0	0.83%	1.13%
Mean abs monthly difference	0	0.137%	0.188%
Mean ticker count	4.08	—	3.61

Exactness: R287 12/12 ALL PASS. The empirical interpretation is robustness, not performance superiority. The rule is adopted for execution simplicity and elimination of small sleeves, not because DEV performance happened to improve.

121. KIRIN Practical 5% Execution - new frozen rule

Step	Frozen operational rule
1	Compute Frozen KIRIN exactly: Four Gods -> 6M ERC -> A-state -> Attack75 / selector -> family allocation.
2	Expand frozen family holdings to the aggregated exact executable ticker target.
3	Any exact ticker weight <5% is set to 0% and is not traded.
4	Only surviving tickers participate in the 5% grid.
5	Residual 5% units are assigned among survivors by largest fractional remainder until total = 100%.
6	Family-native intramonth switches remain frozen; no cross-family re-rounding intramonth.

R286 remains in the research lineage as the first frozen LR5 control. R287 supersedes R286 for live KIRIN execution. No 3%, 4%, 7.5%, or other pruning threshold search is opened.

122. Operational UI contract after R287

The KIRIN page now uses the same Practical 5% convention for the action display and the executable Portfolio Mix. Strategy Mix remains an internal/exact strategy-allocation view.

UI element	v1.34 behavior
This month's action	Change diagnostic displayed on a 5% scale; no raw 37.3% / 27.3% style action text.
Portfolio Mix	Authoritative executable final-ticker Practical 5% target.
Strategy Mix	Internal KIRIN family allocation; may retain exact percentages for diagnostic transparency.
Audit expander	Retains the pre-quantization exact ticker target.
Specification	States prune exact <5% -> survivor 5% grid -> 100%.

Illustrative current operational mapping from the previously displayed exact target:

Ticker	Exact target	R287 execution target
GLD	29.9%	30%
SPXL	2.6%	0%
TQQQ	5.2%	5%
TMV	37.3%	40%
XLU	25.0%	25%

This example explains the key R287 change: SPXL 2.6% is no longer promoted to a 5% position; it is pruned to zero and the freed capital is allocated only among surviving sleeves.

123. Governance state

Production: Frozen4F EW unchanged. **KIRIN:** Frozen Layer-1 challenger / Defense-first Growth Booster Hybrid.
Execution: R287 Practical 5% frozen. **DEV:** closed through 2026-07. **Real Forward OOS:** 2026-08 onward; no feedback into DEV.

Master Research Bible - v1.35 Addendum

My4F / Four Gods / v3.45 / KIRIN | 2026-09-18

Scope: v1.34 (213 pages) is preserved unchanged. This addendum records R288-R290 and the resulting live-execution/UI governance decisions. No prior historical page is rewritten.

Authoritative change: KIRIN canonical strategy remains unchanged. R289 Practical 10% becomes the live execution layer; R287 remains a high-fidelity shadow control. R288 Top-3 is retained as a non-adopted challenger. Threshold exploration stops at 10%.

R288 - Practical Top-3 Execution Audit

Predeclared rule: exact final ticker allocation -> retain only the three largest ticker sleeves -> zero rank 4+ -> quantize survivors to 5% grid -> preserve 100%. DEV cutoff 2026-07; Forward OOS excluded; no Top-2/Top-4 search.

Variant	CAGR	Sharpe	Sortino	MaxDD	Calmar	TE	Mean monthly diff	Mean holdings
Exact	68.88%	1.927	3.086	-20.59%	3.346	0	0	4.08
R287 <5%	69.12%	1.946	3.270	-19.87%	3.479	1.13%	0.188%	3.61
R288 Top-3	69.13%	1.944	3.266	-20.85%	3.316	4.40%	0.674%	2.65

Audit: 13/13 PASS. Performance did not collapse, but TE rose materially to 4.40%. Top-3 therefore behaves as a structural execution overlay rather than a simple rounding rule. It is not adopted for live execution.

R289 - Practical 10% Pruning Audit

Frozen candidate rule: exact final ticker allocation -> every exact ticker weight <10% becomes 0% -> renormalize surviving tickers -> 5% grid -> total 100%. No 7.5%, 12.5%, 15%, or alternative threshold search.

Variant	CAGR	Sharpe	Sortino	MaxDD	Calmar	TE	Mean monthly diff	Mean holdings	Turnover
Exact	68.88%	1.927	3.086	-20.59%	3.346	0	0	4.08	43.42%
R287	69.12%	1.946	3.270	-19.87%	3.479	1.13%	0.188%	3.61	44.48%
R288	69.13%	1.944	3.266	-20.85%	3.316	4.40%	0.674%	2.65	43.46%
R289	71.04%	1.951	3.163	-20.99%	3.384	3.72%	0.603%	2.77	45.87%

Audit: 13/13 PASS. R289 achieves almost Top-3-level simplification without a hard asset-count cap and with lower TE than R288. The higher DEV CAGR is explicitly not treated as evidence of execution alpha. Turnover is not reduced; the rationale is operational simplicity and removal of small sleeves, not fee reduction.

R290 - R289 Execution Robustness / Falsification Audit

R289 is held fixed. No threshold, grid, or Top-N search. DEV <= 2026-07 only; 2026-08 onward Real Forward OOS excluded. The audit attacks the frozen execution rule through chronological thirds, fixed stress windows, cost stress, divergence-tail concentration, and attack-sleeve pruning behavior.

Chronological thirds

Period	Months	Exact CAGR	R289 CAGR	Exact Sharpe	R289 Sharpe	Exact MaxDD	R289 MaxDD	TE
2014-08 - 2018-07	48	42.88%	42.62%	1.633	1.630	-9.47%	-9.47%	2.07%
2018-08 - 2022-07	48	103.76%	110.11%	2.144	2.207	-20.59%	-20.99%	2.96%
2022-08 - 2026-07	48	65.43%	66.98%	2.050	2.060	-6.15%	-5.09%	5.34%

The rule does not depend on a single favorable chronological block: P1 is slightly weaker than Exact while P2/P3 are stronger. The principal caution is that tracking error rises in the most recent third (5.34%), reinforcing that R289 is an execution overlay, not an identity-preserving rounding.

Transaction-cost stress

Cost / 100% turnover	Exact CAGR	R289 CAGR	Exact Sharpe	R289 Sharpe	R289 - Exact CAGR
0 bps	68.88%	71.04%	1.927	1.951	+2.16 pp
5 bps	68.46%	70.59%	1.919	1.943	+2.13 pp
10 bps	68.05%	70.15%	1.911	1.934	+2.10 pp
20 bps	67.22%	69.26%	1.895	1.917	+2.04 pp

Cost stress does not reveal a fragile edge, but this does not establish that R289 saves transaction costs: observed monthly turnover is slightly higher than R287. Operational simplicity remains the adoption rationale.

Divergence concentration and attack pruning

Metric	Observed
Maximum absolute monthly divergence	5.84 percentage points
95th percentile absolute divergence	2.22 percentage points
Top-5 months share of total absolute divergence	22.32%
Months R289 > Exact / R289 < Exact	60 / 61
Attack sleeve pruned months	81 of 144
Mean absolute diff when attack sleeve pruned	0.867% / month

R290 result: 13/13 PASS. No single-event or small-set concentration failure was detected. Attack sleeves are pruned frequently, so forward monitoring must explicitly track whether the execution layer changes the growth engine in live data.

KIRIN Execution Governance Freeze

Effective after R290

Layer / name	Definition	Status
麒麟「天界」	Canonical Exact KIRIN. Research/theoretical reference; no execution pruning.	Canonical / frozen
麒麟「現世」	R289 Practical Live: <10% exact ticker sleeves -> 0; renormalize survivors; 5% grid; 100%.	Live execution / frozen
R287	<5% prune + 5% grid. High-fidelity execution approximation.	Shadow control
R288	Top-3 + 5% grid.	Non-adopted challenger

Naming rule: ordinary UI uses only 麒麟「天界」 and 麒麟「現世」. Explanatory subtitles such as Canonical Exact or Practical Live are used only where definition/audit context is necessary.

Research stop rule: the 10% threshold is frozen. Do not inspect 7.5%, 12.5%, 15%, alternative grids, or Top-N variants to choose a better DEV result. Any future modification requires a new predeclared research question and must not use Real Forward OOS for tuning.

Forward governance: 2026-08 onward remains Real Forward OOS. Track 天界, 現世, and R287 shadow control in parallel. Forward observations may evaluate the frozen rule but must not be fed back into DEV optimization.

Operational app contract

The KIRIN page is updated to make the execution/research distinction operationally explicit while preserving exact internal data for audit.

Area	Contract
Portfolio Mix	麒麟「現世」 executable ticker allocation (R289).
Strategy Mix	麒麟「天界」 internal Four-Gods / canonical mix.
今月のアクション	Compare previous vs current 現世 asset sets; state whether holdings changed; show added/removed tickers and allocation rebalance.
Monthly Returns	天界 and 現世 are both independently calculated. 2026-08 onward is Real Forward OOS, not imputed from 天界.
Performance	天界 and 現世 selectable independently and comparable together.
Specification	Keep reconstruction lineage: Four-Gods reps, t-1 constraints, 6M covariance/ERC, A-state, Attack75, selector, ticker expansion, execution layer.

Real Forward calculation rule

For each month, 天界 uses the canonical Exact KIRIN ticker allocation. 現世 applies the already-frozen R289 transformation to that month's Exact allocation. Both are then evaluated independently using the app's Open->Open convention and current/latest price for MTD. This produces actual forward execution-series values rather than substituting or estimating 現世 from 天界.

Monitoring UI and Benchmark Contract

The operational display now emphasizes quick iPhone review without changing research definitions.

Monthly return summary

The summary displays latest MTD, previous month, and YTD for both 天界 and 現世. Benchmark YTD is also shown for SPY and TQQQ. SPY/TQQQ are benchmarks only and do not alter KIRIN construction or execution.

Recent 12 months

Each monthly card is optimized for mobile scanning. The month and NORMAL/A-STATE badge share the header. The primary row shows 麒麟: 天界 and 現世. A visually secondary BM row shows SPY and TQQQ. All four series use the same Open->Open comparison convention; the current month is MTD to the latest retrieved price.

Primary	麒麟「天界」	麒麟「現世」
Benchmark	SPY	TQQQ

Interpretation guardrails

- 1) 天界 vs 現世 differences are expected and can be material month to month. R290 shows a DEV tracking error of 3.72% overall and 5.34% in the latest chronological third. Therefore 現世 must not be described as a mere rounding of 天界.
- 2) The observed R289 DEV CAGR improvement is not a basis for expecting future outperformance. The formal conclusion is robustness of the practical execution layer under the tested DEV falsification program.
- 3) SPY and TQQQ are display benchmarks. Relative monthly or YTD performance against them is descriptive only; it is not a strategy-selection rule and does not feed back into KIRIN signals.
- 4) Exact internal strategy information remains preserved for audit even when the operational UI shows simplified live allocations.

v1.35 canonical status

Production: Frozen4F EW remains unchanged. KIRIN canonical: 麒麟「天界」 = Exact KIRIN. KIRIN live execution: 麒麟「現世」 = frozen R289 Practical 10%. Execution control: R287. Execution research: R290 closes the DEV execution-simplification program unless a genuinely new, predeclared hypothesis is introduced.

This addendum supersedes v1.34 only where the live KIRIN execution rule/status is concerned. All unrelated v1.34 research history remains authoritative and unchanged.

Master Research Bible v1.36 - Addendum

R291 Teacher Strategy Comparative Anatomy / My4F Fast Phase 1

Date: 2026-09-18 | Parent: v1.35 (217 pages)

v1.36の位置づけ

v1.35の既存217ページは研究正本として保持し、本Addendumを末尾に追加する。ProductionはFrozen4F EWのまま。麒麟「天界」はCanonical Exact、麒麟「現世」はFrozen R289 Practical 10% Live Execution。DEV cutoffは2026-07、2026-08以降はReal Forward OOSであり、Forward結果をDEVへ戻さない。

今回の追加事項

1) R291: 麒麟「現世」とバム先生Teacher Strategiesの共通期間比較を正式記録。2) Teacher seriesの証拠レベルを分離し、秘匿Position/ルールを推測しない。3) My4F Fast Phase 1: Streamlit研究正本を保持したまま、日常運用表示を高速Web frontendへ分離する設計を記録。4) 将来候補としてEdge Persistence / Jevプロトコル案をResearch Queueへ記録する。

Freeze statement

R291は比較解剖であり、Production promotion、KIRIN rule変更、threshold再探索を行わない。My4F Fastも表示層の変更であり、戦略ロジック変更ではない。

R291 exactness

項目	結果
Build	R291-KIRIN-GENZE-VS-BAM-TEACHER-COMPARATIVE-ANATOMY-v1.0
Exactness	13 / 13 ALL PASS
Common sample	2016-02..2026-07 / N=126
Teacher exact displayed series	2
Reconstructed teacher series	1
Published registry	6
Forward OOS	2026-08 onward excluded
Optimization / promotion	NONE

R291 - Common-period performance anatomy

全系列を同一126か月へ揃えた比較。数値の優劣をもって戦略のwinnerを決めない。目的はrisk-return geometryと時間方向の差を記述すること。

Strategy	Evidence	CAGR	Sharpe	Sortino	MaxDD	Calmar
麒麟「現世」	Frozen R289 exact	77.99%	2.058	3.280	-20.99%	3.715
奥義-分身-常勝	displayed monthly	79.66%	2.057	2.944	-22.83%	3.489
奥義-分身-鉄壁	displayed monthly	64.26%	1.693	3.662	-12.24%	5.250
シン四つ目-激攻	reconstructed	76.62%	1.573	2.388	-31.13%	2.461

常勝との比較

麒麟「現世」と奥義-分身-常勝は、共通期間CAGRが77.99%対79.66%、Sharpeが2.058対2.057と近い。一方、月次Pearson相関は0.684。現世のMaxDDは-20.99%、常勝は-22.83%、Sortinoは3.280対2.944。これは近い長期risk-adjusted performanceでもreturn pathが同一ではないことを示す。

鉄壁との比較

奥義-分身-鉄壁はCAGR 64.26%に対しMaxDD -12.24%、Calmar 5.250。現世との月次相関は0.690。Defense frontierのTeacher referenceとして有用だが、秘匿されたholdingsやルールの逆推定は行わない。

Evidence caveat

奥義-分身-常勝/鉄壁はユーザー表示の月次Return（0.01%丸め）を使用。シン四つ目-激攻はAllocation x ETFのdescriptive reconstructionであり証拠レベルが異なる。Published summaryのみのTeacher seriesを月次系列として扱わない。

R291 - Rolling / stress anatomy

Teacher	12M 現世win	36M 現世win	Pearson
奥義-分身-常勝	39.1%	27.5%	0.684
奥義-分身-鉄壁	63.5%	62.6%	0.690
シン四つ目-激攻*	53.0%	52.7%	0.522

* reconstructed series

Fixed stress windows

Episode	現世	常勝	鉄壁	シン四つ目-激攻*
COVID 2020 (3M)	+40.78%	+14.33%	+32.56%	-0.52%
2022 bear (12M)	+28.33%	+42.18%	+20.30%	-8.69%
2025-2026 recent (19M)	+166.28%	+130.45%	+81.97%	+268.02%

Interpretation freeze

COVIDでは現世、2022では常勝など、局面ごとの優位は一貫しない。2つ程度のcrisis windowから危機耐性の一般則を導かない。R291はTeacher benchmarkの位置関係を把握するためのcomparative anatomyであり、Teacherの秘密ロジックを推測する研究ではない。

Teacher benchmark roles

今後のreference用途として、奥義-分身-常勝はGrowth / Quality benchmark、奥義-分身-鉄壁はDefense frontier benchmarkとして扱う価値がある。ただしこれは役割上のreference designationであり、Production rankingやpromotionではない。

R291 governance conclusion

R291 = 13/13 PASS。KIRIN/Four GodsのProduction構造は変更なし。Forward OOS 2026-08+は比較DEVから除外。

My4F Fast - Phase 1 architecture freeze

目的

Streamlit研究アプリの機能・研究系譜を削らず、日常確認UIだけを高速化する。iPhoneでGitHub Pages版を試験し、ユーザーがStreamlit版より明確に高速と確認した。

二層構造

Layer	役割	正本
Research / Calculation	R1-R291, CPCV, Self Backtest, Frozen/Forward計算	Python / Streamlit
Operation UI	KIRIN, Allocation, Action, Health, Return, OOS表示	My4F Fast frontend
Bridge	確定結果を小さなsnapshotとして供給	JSON schema / future API

Non-negotiable principle

戦略計算ロジックをJavaScriptへ複製しない。Python側を唯一の計算正本とし、Fast frontendは確定したsnapshot/resultを表示する。これによりStreamlit版とFast版の計算乖離を防ぐ。

Current Phase 1 state

GitHub repository: my4f-fast。GitHub Pagesで公開し、Fast prototype -> Fast v1 -> Fast v2へ進行。Fast v1ではStreamlit正本とのUI差が大きかったため、現行KIRINページのiPhoneスクリーンショットをデザイン正本としてFast v2を作成。v2は今月のアクション、G1/G2/P、現世Execution Holdings、天界Four Gods、Forward OOS、4神risk state、KIRIN signal、月次/YTD、SPY/TQQQ、直近12か月、Strategy Specificationを再現対象とする。

Fast v2 artifact

Artifact	Value
File	my4f_fast_v2.zip
SHA-256	1e3cb1e0d07e46f01230cafae3019d9eabc156e84519efb8ab8cc954491fbd98
Status	UI / speed validation snapshot; not yet automatic live data

Next phase - Live Data Bridge / Edge Persistence queue

My4F Fast v3 - Live Data Bridge

Fast v2のUI確認後、次の本命はPython正本からFast版へ実データを自動供給するBridge。Fast側で計算せず、Pythonからkiri n_snapshot.json等を生成する。

Proposed frozen snapshot schema

Field group	内容
identity	as_of / month / mode / schema_version
allocation	genze_allocation / tenkai_four_gods / exact_ticker_target
action	prior/current / added / removed / action_text
diagnostics	health / risk_states / signals
governance	forward_oos / dev_cutoff / rule_version
performance	monthly_returns / ytd / SPY / TQQQ / recent_12m

Required audit

同一as-of時点でPython/Streamlit正本とFast snapshotの主要数値を照合する一致監査を追加する。JSON schemaをFreezeし、変更時はversionを上げる。Frozen historyは再計算せず再利用し、最新月だけ更新する設計を優先する。

Edge Persistence / Jev - Research Queue

バム先生の「PF edge persistenceを分類し、t+1/3/6/12でOOS評価する」案は、現行Strategy Health / Trend Clarityを将来実績でfalsifyする枠組みとして参考になる。Jev利用可否とは分離して、まずPhase 0 protocolを事前Freezeする候補とする。

Phase 0 freeze candidates

input features / classification categories / question or prompt / model-version / leakage prevention / horizons (1,3,6,12M) / outcome definitionsを結果閲覧前に固定。JevやLLM分類をProduction売買ルールへ直接接続せず、diagnostic / Edge Persistence Monitorとして検証する。

v1.36 final governance

Production = Frozen4F EW。麒麟「天界」 = Exact canonical。麒麟「現世」 = Frozen R289 live。R287 shadow、R288 non-adopted、10% threshold再探索なし。2026-08+ Real Forward OOSはDEVへ戻さない。Fast化は表示アーキテクチャ変更であり戦略変更ではない。

Master Research Bible v1.37 - Addendum

My4F Fast Live Data Bridge - End-to-End Operational Freeze

Date: 2026-09-19 | Parent: v1.36 (222 pages)

v1.36の既存222ページは研究正本として保持し、本Addendumを末尾に追加する。ProductionはFrozen4F EWのまま。麒麟「天界」= Canonical Exact、麒麟「現世」= Frozen R289 Practical 10% Live Execution。DEV cutoffは2026-07、2026-08以降はReal Forward OOSであり、Forward結果をDEVへ戻さない。

今回の更新は表示・配信インフラの完成であり、戦略ロジック、Frozen rule、研究結果、Production statusを変更しない。

132. My4F Fast Live Data Bridge - completed architecture

日常運用表示を高速化するため、研究・計算と表示を分離した。Python / Streamlitを唯一の計算正本とし、Fast frontendはPythonが確定したsnapshotのみを表示する。JavaScript側にKIRIN研究ロジックを複製しない。

Layer	Frozen operational contract
Research / Calculation	Python / Streamlit。Frozen4F、天界、現世、Health、risk、signals、returns、benchmarksを確定。
Bridge	kirin_snapshot.json。schema_version = kirin-fast-1.0。Python確定値のserializationのみ。
Publish	StreamlitからGitHubへsnapshotを自動Publish。通常運用ではsnapshotのみ更新。
Fast UI	GitHub PagesのMy4F Fast。snapshotを取得・監査し、表示のみ担当。
Governance	2026-08+ Real Forward OOS。Fast表示やForward結果をDEV最適化へ戻さない。

133. Snapshot audit / visible synchronization state

Fastページ上部にLive Data Bridge statusを常設した。ユーザーは画面だけで連携状態とデータ鮮度を判断できる。正常時は「Live Data Bridge - PASS」、最終更新JST、対象月、snapshot取得・schema・配分100%監査PASSを表示する。取得失敗またはschema / allocation audit不整合時はERRORとして可視化する。

Snapshotには生成時刻(JST)と監査情報を持たせる。現世execution allocationと天界Four Gods allocationは各100%を監査対象とする。Fast UIのPASS表示は単なる装飾ではなく、snapshot fetch、schema、allocation totalsの検証結果をユーザー向けに可視化する。

134. Incident record and recovery audit

Phase 1導入中、GitHub上のFast app.jsが「Unsupported Media Type」の1行だけに破損し、固定HTMLは表示される一方、ドーナツ、Health、risk、signals、returns等のJavaScript生成部分が空になる事象を確認した。snapshot自体は正常であったため、原因をdisplay renderer破損へ切り分けた。

復旧ではStreamlit側のGitHub write権限を用いたone-shot repairを実施し、画面上で「app.js repair: PASS」を確認。Fast実画面でも全動的セクションの復旧を確認した。その後one-shot repairは通常運用から撤去し、恒常的なGitHub書込み対象をkirin_snapshot.jsonのみに戻した。

Audit item	Result
Python -> snapshot generation	PASS
Snapshot schema kirin-fast-1.0	PASS
現世 execution total = 100%	PASS
天界 Four Gods total = 100%	PASS
GitHub snapshot auto publish	PASS
Fast snapshot fetch / render	PASS
Visible Live Data Bridge status	PASS
Visible latest-update timestamp (JST)	PASS
One-shot app.js repair	COMPLETED / REMOVED FROM NORMAL OPERATION
Strategy logic duplicated in JS	NO

135. End-to-End live refresh proof

実運用テストでFastの最終更新表示が 2026-09-19 08:42:56 JST から 08:47:07 JST へ更新されたことを確認。これにより、Streamlit/Python計算 -> snapshot生成 -> audit -> GitHub自動Publish -> Fast取得 -> 最新更新時刻表示、というEnd-to-End経路が実際に継続更新することを確認した。

判定: My4F Fast Live Data Bridge = END-TO-END PASS / operationally frozen.

136. Normal operating procedure

通常運用では、StreamlitのKIRINページを開いてPython側の最新計算を実行し、「My4F Fast Sync - PASS / GitHubへ更新済み」を確認する。その後My4F Fastを開き、「Live Data Bridge - PASS」と最終更新JSTを確認する。JSONの手動ダウンロード、手動GitHub upload、app.js修復は不要。

将来、Streamlitを開かず定刻自動更新する仕組みを追加する場合も、Pythonを唯一の計算正本とし、GitHub PAT等の秘密情報をpublic frontendへ置かず、Real Forward OOS governanceを維持する。

137. v1.37 freeze statement

v1.37時点でFast表示系のPhase 1およびLive Data Bridgeは完成扱いとする。Fast UIの見た目は現行iPhone-first仕様を基準として凍結し、不要な横スクロールを導入しない。本更新によるKIRINのルール変更、R289 threshold再探索、Production promotion、DEV再最適化はNONE。

Research Queueはv1.36から継承する。Jev / Edge Persistence等の新研究は、既存Frozen rulesおよび2026-08+ Real Forward OOSを汚染しない独立プロトコルとして事前定義してから実施する。

Master Research Bible v1.38 - Addendum

My4F Fast Auto Refresh — Direct-Python Operational Freeze

Date: 2026-09-19 | Parent: v1.37 (224 pages)

v1.37までの研究正本を保持し、本Addendumを末尾に追加する。Production = Frozen4F EW、麒麟「天界」 = Canonical Exact、麒麟「現世」 = Frozen R289 Practical 10% Live Execution、DEV cutoff = 2026-07、2026-08+ = Real Forward OOSというガバナンスは不変。本更新は配信運用の自動化であり、戦略ロジック、Frozen rule、研究結果、Production statusを変更しない。

138. Browser automation廃止とDirect-Python化

定刻自動更新の初期試行ではGitHub

ActionsからPlaywrightでStreamlit公開画面を開き、KIRINページを操作してsnapshot publishを誘発する方式を検証した。しかし、Streamlitの認証・sleep/readiness・DOM可視性に依存し、locator timeoutや「app does not exist / sign in」等で安定しなかった。これを恒常運用には採用しない。

最終方式ではブラウザ操作を完全に廃止し、GitHub Actions上で既存app.pyをStreamlit AppTestによりin-process実行する。main navigationの既存KIRINページを選択し、Python正本が確定するKIRIN計算・Bridge監査・publish処理をそのまま走らせる。Fast frontendへ戦略ロジックを移植・複製しない。

139. Frozen Auto Refresh architecture

Layer	Frozen operational contract
Scheduler	GitHub Actions / my4f-fast-auto-refresh.yml
Execution	kirin_fast_refresh.py → app.pyをin-process実行
Calculation SoT	既存Python / Streamlitのみ
Publish	既存Bridge / publisherからkirin_snapshot.jsonを更新
Frontend	My4F Fastはsnapshotを取得・監査・表示のみ
Secrets	MY4F_FAST_GITHUB_TOKENはGitHub Actions Secret。public frontendへ置かない
Artifacts	不要。Actions artifact uploadを通常経路に使用しない

140. End-to-End automatic refresh proof

2026-09-19、Direct-Python版GitHub Actionsをworkflow_dispatchで実運用テスト。run全体 Status = Success、refresh job = green PASS、総所要約1分28秒（refresh約1分23秒）を確認した。従来のPlaywright / Streamlit公開URL依存エラーは解消された。

同run後、My4F Fast上部の可視監査表示が「Live Data Bridge — PASS」、「最新更新: 2026-09-19 11:11:20 JST」、「Snapshot取得・schema・配分100%監査 PASS | 2026-09」と更新されたことを実画面で確認。したがって GitHub Actions → Python正本 → KIRIN計算 → R289現世 → snapshot監査 → GitHub publish → My4F Fast fetch/render → 最新更新時刻表示 の自動経路をEnd-to-End PASSとする。

141. Frozen schedule / normal operation

自動更新scheduleは`cron: 30 22 \* \* 1-5`（GitHub ActionsはUTC）を採用。日本時間では平日07:30 JSTに相当する。米国通常取引終了後に余裕を持って最新データを取り込む運用とする。GitHub Actions scheduled workflowは厳密な秒単位実行を保証しないため、実際の完了時刻はFast上部の「最新更新」JSTで監査する。workf low_dispatchによる手動実行も維持する。

通常運用ではユーザーがStreamlitを開く必要はない。平日朝のscheduled runが成功すればFastへ自動反映される。異常時はGitHub Actions run statusとFast上部のLive Data Bridge status / 最新更新時刻を一次監視点とする。

142. Auto Refresh v1.0 freeze statement

My4F Fast Live Data Bridge / Auto Refresh v1.0 = FROZEN。ブラウザ自動操作方式はdeprecated / non-adopted。Direct-Python方式をcanonical operational pathとする。Fast化・自動化によるKIRINルール変更、R289 threshold再探索、Production promotion、DEV再最適化はNONE。2026-08+ Real Forward OOSをDEVへ戻さない。

Master Research Bible v1.39 - Addendum

My4F Fast UI Consolidation - Annual Returns / Version Audit / Mobile Layout

Date: 2026-09-19 | Parent: v1.38 (226 pages)

v1.38までの226ページを研究正本としてそのまま保持し、本Addendumを末尾に追加する。Production = Frozen4F EW、麒麟「天界」= Canonical Exact、麒麟「現世」= Frozen R289 Practical 10% Live Execution、DEV cutoff = 2026-07、2026-08+ = Real Forward OOSという研究ガバナンスは不変。本更新はMy4F Fastの表示・運用UIの改良記録であり、戦略ロジック、Frozen rule、研究結果、Production statusを変更しない。

143. Fast Performance / Analytics UI - current display contract

Fastは引き続きPythonを唯一の計算正本とし、JavaScriptは表示のみを担当する。performance.history等、Pythonからsnapshotとして公開済みのcanonical return seriesを表示・集計することは許容するが、Fast側へ戦略選択・売買判定・Frozen ruleを複製しない。固定repo filenameは index.html / styles.css / app.js / kirin_snapshot.json / README.md を維持し、同名上書きで更新する。cache bustはindex.html内のquery stringで行う。

Research PerformanceおよびAnalytics系ページはiPhone-firstを継続する。ページ全体のhorizontal overflowは禁止し、必要な場合のみ表など局所領域に限定して横スクロールを許容する。ポートフォリオ選択chip / benchmark switchはコンパクト表示を標準とし、Research PerformanceのSelect Portfoliosは折りたたみ内に配置する。

144. Annual Returns - v41 target state

Annual Returns表はBalance列を廃止し、Year / 選択Portfolio Return / 選択Benchmark Return の3列構成とする。BenchmarkはUI選択に従いSPYまたはTQQQを表示する。BalanceはAnnual Returnsの比較目的には不要であり、return比較へ情報密度を集中する。

項目	Frozen display contract
列	Year / Selected Portfolio Return / Selected Benchmark Return
Balance	表示しない
Benchmark	SPY または TQQQ - 現在のBM選択に追随
Mobile width	Year 18% / Portfolio 41% / Benchmark 41% を基準
Page overflow	Annual表ではglobal horizontal scrollを発生させない

145. Visible update-number governance

Fast UI変更時は、ユーザーが実際に新buildを読み込んでいるか画面だけで判定できるよう、可視の更新番号を必ず1個だけ表示する。変更ごとに v42, v43 ... のように単調増加させ、同時に index.html の app.js?v=... / styles.css?v=... も更新する。旧version badgeをJavaScriptから別途生成してはならない。

v34-v40の修正過程では、index.html側の新badgeとapp.js側の旧v34 badgeが二重表示される事象、旧badgeの文字列のみ消した結果として空の緑枠が残る事象、旧badge処理を広く削除した際にmenu initializationへ影響した事象を確認した。これを受け、version proofはindex.html側の単一canonical markerへ集約する。

146. Incident chronology - v34 to v41

Build	Observed / change	Disposition
v34	Annual Returns 3列化。旧UI更新badgeをapp.js側でも生成。	Superseded
v35-v36	index側の新version markerと旧v34 markerが併存。Annual列幅をmobile向けに調整。	Superseded
v37	旧badge削除処理がmenu initializationへ影響し、menu open regression。	Rejected / superseded
v38-v40	menuを復旧しつつ旧badgeを非表示化・空要素化したが、空枠が残存。	Superseded
v41	旧badge生成関連statementを限定的に除去し、menu click handlerを保持。Annual 3列・幅調整を維持。	Current implementation baseline

v41 artifact-level verificationでは、旧 uiBadge / oldBadges / my4fUiVersion 生成参照がapp.jsから除去され、fastMenuButtonおよび btn.addEventListener('click', openMenu) が保持され、node --check がPASSした。これはコードレベルの検証であり、最終的なiPhone実画面確認はユーザー側表示を一次証拠とする。

147. Delivery / iPhone transport rule

GitHub上のcanonical filenameは絶対にversion付きへ変更しない。一方、配布ZIP filenameは識別しやすいversion付き名称としてよい。iPhoneでZIP展開・保存を安定させるため、ZIPは必ず2ファイル以上を含める。変更対象が1ファイルだけでも、DUMMY.txtまたはREADME等の無害な補助ファイルを同梱してよい。GitHubへは変更対象のcanonical fileだけを同名上書きする。

Fast更新時は、変更内容だけでなく「どのファイルをGitHubで上書きするか」を明示する。更新番号・cache query・実ファイルの整合を検証してから配布する。

148. v1.39 freeze statement

v1.39ではFast UI operational baselineをv41相当として記録する。Annual Returnsは3列化、mobile幅調整、single visible update markerを採用し、旧v34 badge生成系は廃止する。menu navigationは保持する。これらは表示層の変更のみであり、KIRIN / Frozen4F / R289 / Strategy Health / Trend Clarity / Real Forward OOSの研究定義を変更しない。

次回Fast変更はv42から開始する。次回研究番号は既存queueを継承し、R292 Phase 0 - Jev Edge Persistence Protocol Freezeが候補である。過去研究を再実行せず、Master・official result ZIP・frozen/cached seriesを再利用する。

Master Research Bible v1.40 — Addendum

My4F / Four Gods / KIRIN — Forward OOS • SOXL falsification • Fast Monthly Trade operational consolidation

Date: 2026-09-20 | Parent: v1.39

v1.39までの研究正本を変更せず保持し、本Addendumを末尾に追加する。Production = Frozen4F EW、麒麟「天界」 = Canonical Exact KIRIN、麒麟「現世」 = Frozen R289 Practical 10% Live Execution、DEV cutoff = 2026-07-31、2026-08+ = Real Forward OOSというガバナンスは不変。

149. Post-v1.39 governance / latest-validation policy

- 研究StreamlitのValidation

UIは「最新のみ」を表示・実行する。旧validationを画面上へ累積しない。ただし過去結果・系譜・EvidenceはMaster、official ZIP、コード内に保持する。

- Fastは引き続きPythonを唯一のstrategy calculation source of truthとし、JavaScriptは表示のみ。Python → JSON snapshot → GitHub → FastのBridgeを維持する。

- Fastの更新は直前の確定版をFrozen

Baselineとして最小差分patchで行い、ユーザーが指定していないUI・挙動・サイズ・表示内容を変更しない。固定repo filenameを維持し、cache bustはquery stringで行う。

150. R292 — Layer1 Double-OOS selector audit

Item	Frozen specification / result
Protocol	36m estimation → 12m OOS、12m step、non-overlap。IS Sortino0最大、tie = Calmar → CAGR → name。OOS中は選択固定。
Candidates	R181 / ERC6 / R255 Split / R259 Gate / 麒麟「現世」
Coverage	8 OOS windows = 96 months。W1 現世、W2-W5 ERC6、W6-W8 現世。選択頻度 ERC6 4/8、現世 4/8。
Selector	CAGR 74.2751%、Sharpe 2.0103、Sortino 7.4427、MaxDD -11.7054%、Calmar 6.3454。
Guardrail	Exactness / leakage 5/5 PASS。ただしretrospective candidate setであり、selection bias排除・true alphaの証明とはしない。

151. R293 / R294 — historical availability boundary and Real Forward monitor

- R293 Historical-Availability Fail-Closed Audit: 現在のLayer1候補は実際の研究・freeze availabilityが2026-09でDEV cutoff後のため、過去時点で利用可能だったと仮定するretrospective testは実施しない。eligible pre-DEV candidate/window = 0。結論は NOT TESTABLE RETROSPECTIVELY / FAIL-CLOSED。これはstrategy failureではなくgovernance boundary。

- R294 Real Forward OOS Monitor: 2026-08+のみを蓄積する設計。初回実行ではgovernance 6/6 PASSだがN=0。原因はresearch_archiveのFrozen seriesが2026-07で終了しており、Forward data pipelineが未接続だったため。戦略FAILとは解釈しない。

152. R295 / R296 — SOXL Turbo falsification branch

Variant	CAGR	Sortino	MaxDD	Calmar	SOXL months
Frozen KIRIN control	68.88%	6.09	-20.59%	3.35	0
R295 unrestricted SOXL	70.18%	4.66	-34.13%	2.06	15
R296 H1 6M+12M	67.19%	4.49	-34.13%	1.97	12
R296 H2 abs positive	69.94%	4.64	-34.13%	2.05	14
R296 H3 relative12	67.87%	4.52	-34.13%	1.99	13

- R295は既存6M XLK vs QQQ selectorへSOXXを同格追加するminimal-change test。Exactness 7/7 PASS。SOXLは15/29 Turbo monthsで選択され、選択月のmean incremental returnは+1.49pt、positive rate 66.7%だった一方、MaxDDが-34.13%へ悪化。
- R296は事前固定3 gate（H1 multi-horizon、H2 absolute、H3 relative12）を検証。Exactness 8/8 PASS。2019-05のSOXL固有downsideを全gateが通過し、downside geometryを回復できなかった。
- Decision: SOXL branchはnon-adopted challengerとしてarchive。Frozen 麒麟「現世」は変更しない。追加DEV optimizationは停止し、今後の最終的な頑健性証拠はReal Forward OOS accumulationとする。

153. Fast Monthly Trade — canonical history bridge

- Monthly Tradeページを追加。strategy selector、Month / Position Start / Position / Return、MTD表示を持ち、Python正本の月次returnおよびposition historyを表示する。Fast側でhistorical allocationを推測・再計算しない。
- v48: Python canonical Monthly Position History bridgeを追加。v49: position decimal fraction → percentage-point境界を修正。v50: completeness auditを追加し、欠損時は「未収録 / 正本データなし」を表示。
- v51–v52: Forward seamのposition bridgeを試行。実snapshot inspectionにより、KIRIN monthly trade historyが2026-07で停止し、snapshot自体にv50 audit fieldsがないことを確認。Fast frontendではなくoperational snapshot pathの問題と切り分けた。
- v53: 既存Python operational engineが計算済みの previous-month authoritative allocation `prev\_live` をMonthly Tradeへ直接接続。2026-08 麒麟「現世」Positionを正本データとして復元し、ユーザー実画面で成功確認。2026-09 current executionはsnapshot executionを使用。

154. Fast operational publish path — clarified

- Canonical auto-refresh path = GitHub Actions scheduler / workflow_dispatch → `kirin\_fast\_refresh.py` → `app.py` in-process (Streamlit AppTest) → finalized `kirin\_snapshot.json` → secure publisher → Fast repo → Fast display。
- `kirin\_fast\_publish.py` はfinalized snapshot bytesをGitHubへpublishするだけであり、strategy calculationは行わない。snapshot bytesが既存GitHub版と同一ならstatus=unchanged。
- Streamlitを手動で開くこととGitHub Fast snapshot更新は同義ではない。operational Actionsが使用するapp.py / refresh pathが正本と一致していることを監査対象とする。

155. Monthly Trade overlay visibility — v54 to v56

Fast build	Change	Governance
v54	Monthly Tradeの月ラベル下にA-STATE / BOOSTER等のevent badgeを追加。	Python stateをsnapshot経由で表示。Fast再判定なし。
v55	全portfolio対応。Frozen v3.45 / Attack75 / Promotion100 / 麒麟天界 / 麒麟現世で各strategy固有overlayを表示。	Frozen Core / Frozen 4F / SPY / TQQQはoverlayなし。
v56	overlay typeごとに色分け。A-STATE=赤、BOOSTER=紫、PROMOTION100=ゴールド系、その他=グレー。	表示層のみ。position / return / strategy logic変更なし。

Overlay labels currently standardized for display: A-STATE / BOOSTER TQQQ / BOOSTER TECL / PROMOTION100 TECL。種別color codingはMonthly Trade内にscopeし、既存ページのlayoutへ波及させない。iPhoneでglobal horizontal scrollingを発生させない。

156. Current Fast baseline / delivery state

- Latest Fast artifact at this Master update: v56 Overlay Color Coding。v55の全portfolio overlay表示を継承し、v56は色分けのみの最小差分。
- 固定repo filename = index.html / styles.css / app.js / kirin_snapshot.json / README.md。version付きfilenameをrepoへ増殖させない。配布ZIPはversion名可。
- visible update markerとindex.html cache queryをversionと一致させる。Fast headerの旧 `Fast v4.3` badgeは再導入しない。
- 直前の確定版をFrozen Baselineとし、次回変更はv56を起点に明示された箇所だけ変更する。

157. v1.40 freeze statement

v1.40は、v1.39以降のR292–R296およびFast Monthly Trade v42–v56系の運用変更を追記する。Production / Frozen strategy rules / R289 threshold / DEV cutoffは変更しない。R293のfail-closed boundary、R294のForward pipeline requirement、SOXL non-adoption、v53 Forward Position Bridge成功、v55全portfolio overlay、v56 overlay color codingを現時点のcanonical operational stateとして記録する。次回研究は過去研究を再実行せず、Master・official ZIP・frozen/cached series・Real Forward OOSを再利用する。

Master Research Bible v1.41 — Fast UI Addendum

My4F / Four Gods / KIRIN — Fast v57 – v70 operational UI consolidation

Date: 2026-09-20 | Parent: v1.40

v1.40までの研究正本・Production・Frozen strategy rules・DEV cutoff・Real Forward OOS governanceを変更せず保持する。本AddendumはFast表示層の更新履歴とFrozen Baseline運用を追記する。

158. Fast update governance — reaffirmed

- Fast更新では更新番号を必ず明示し、visible update markerとindex.html内のcache-busting queryを同じversionへ更新する。
- 直前の確定版をFrozen Baselineとする。ユーザーが明示した変更箇所以外のUI・仕様・挙動・サイズ・表示内容・計算ロジックを変更しない。
- 固定repo filenameを維持する。原則 index.html / styles.css / app.js / kirin_snapshot.json / README.md。version付きファイル名をrepo内に増やさない。
- Fast JavaScriptは表示層。strategy/backtest calculationのSource of TruthはPython。Fast UI変更でFrozen strategy ruleを変更しない。

159. Research Performance typography / layout — v57 – v59

Version	Frozen delta
v57	Research PerformanceのLOG / LINEAR / PERIOD MODE / Common / Fullのみを一段小さく調整。
v58	Comparison・1Y/2Y/3Y/5Y/10Y/ALLを少し拡大。secondary controlsとChart period説明を小さくし、グラフ軸・凡例を拡大。
v59	From selectorのみをSelect Portfolios枠外へ移動し、Comparison直前へ配置。ID/JS behaviorは維持。

160. Summary portfolio selector — v60 – v61

- v60: Summaryでperformance.historyに存在する全portfolioを選択可能にした。選択portfolio vs SPY/TQQQ Benchmarkのcommon-period summaryを表示。表header (Metric / portfolio / BM) は各cell中央。
- v61: Summaryの複数button gridをcompactなPortfolio dropdownへ変更。BM SPY/TQQQ、比較表、計算は変更なし。

161. Annual Returns / Drawdowns / global centering — v62 – v66

Version	Frozen delta
v62	Annual chartを拡大、DrawdownsのWorst cardsをcompact化。analytics tabs/button labelsとtable header/titleを中央揃え。body数値cellは従来配置を維持。
v63	Annual chartだけを一段拡大。plot area・axis/year text・legendを拡大。
v64	Annual chartの上下余白を縮小し、adaptive percentage y-axis (% tick + horizontal guide) を追加。
v65	v64の% axisを維持したままAnnual SVGの上下余白をさらに削減。tight chart balance。
v66	v65のtight proportionsを維持しつつ、Annual chart全体・axis・year labels・legendを一段拡大。

162. Metrics display corrections — v67 – v69

- v67: Metrics最下部 Market Capture chartのlegendを左右equal columns・同一vertical alignment・center配置へ統一。
- v68: 同legendでBenchmark側 (SPY等) だけに見えていた枠を除去し、main/BMを同じ表示形式へ統一。
- v69: MetricsのPortfolio dropdown selected textおよびoption textを中央揃え。BM tabs等に変更なし。

163. Monthly Returns all-portfolio dropdown — v70

- Monthly Returnsのmain strategyを固定 麒麟「現世」から、performance.historyに存在する全portfolio選択へ拡張。compact Portfolio dropdownを追加し、selected text / optionsを中央揃え。
- 選択portfolioのmonthly return、Benchmark return、差を表示し、table headerにも選択portfolio名を表示する。

・既存BM SPY/TQQQ switchを維持。portfolio選択はdisplay selectionであり、underlying monthly seriesやstrategy calculationを変更しない。

164. Current Fast Frozen Baseline

本Master更新時点の最新Fast artifactは v70 Monthly Returns Portfolio Dropdown。次回Fast変更はv70を直前Frozen Baselineとして開始し、更新番号はv71以降へ進める。明示された変更箇所以外はv70から変更禁止。FastのUI lineageはv56以降も連続的な最小差分patchとして扱い、過去版へ巻き戻して再構築しない。

165. Research state unchanged

Fast v57-v70は表示・navigation・typography・selectorの改善であり、Production = Frozen4F EW、麒麟「天界」= Canonical Exact KIRIN、麒麟「現世」= Frozen R289 Practical 10% Live Execution、R289 threshold 10%、DEV cutoff 2026-07-31、2026-08+ Real Forward OOS、SOXL non-adoption、latest-validation-only policyを変更しない。

Master Research Bible v1.42 - Fast Health Bridge / UI Governance Addendum

Date: 2026-09-21 | Parent: v1.41 (233 pages) preserved verbatim

Authority and scope

This addendum records implementation and governance work completed after v1.41 through My4F Fast v124. It does not change Production, strategy definitions, DEV results, or the DEV/Forward boundary. Production remains Frozen4F EW. KIRIN remains a Layer-1 challenger. DEV remains frozen through 2026-07-31; 2026-08-01 onward remains Real Forward OOS and must not be fed back into DEV tuning.

The principal architectural change is the Strategy Health bridge: Research/Python is the sole calculation source of truth, while Fast is display-only. Browser-side Health recomputation is prohibited.

v119 - Research -> Fast Strategy Health bridge

Item	Frozen implementation
Calculation authority	Research / Python only
Transport	kirin_snapshot.json -> strategy_health
Fast responsibility	Read and render only; no Health statistics are recomputed in JavaScript
Portfolio coverage	All portfolios present in the Research Performance snapshot are eligible automatically
Minimum history	<96 monthly observations -> INSUFFICIENT_DATA / fail-closed
Axes	G1 / G2 / P / p-bar using the existing frozen Strategy Health methodology
Canonical protection	Existing official R273HB snapshots remain canonical; expansion does not rewrite prior official evidence

Observed bridge behavior

When Fast v120 was deployed before the new Research snapshot had propagated, the Deterioration Monitor correctly displayed FAIL-CLOSED with zero portfolios. After the Research-generated snapshot arrived, the full portfolio Health table populated automatically. This is accepted behavior and demonstrates that Fast does not invent or reconstruct missing canonical Health data.

Packaging note

v120 added DUMMY.txt to both Research and Fast distribution ZIPs per deployment workflow. This is packaging-only and has no research meaning.

v1.42 Addendum - Strategy Health interpretation and UI contract

Frozen Health axes

Axis	Frozen meaning	Operational reading
G1	12-point slope of erosion long-mean	Direction / early structural warning: is the long-run edge drifting downward?
G2	12M p_erosion	Chronic erosion state: has the long-run edge already moved into a materially weaker level?
P	6 / 12 / 24M deterioration classification	Recent weakening across multiple horizons; the most responsive current-condition warning axis.
p-bar	Primary 5-split absolute consistency	Historical consistency across time partitions; context for whether the edge persists across eras.

Combined operational interpretation

Health is descriptive, non-ranking, and non-selection. The four axes are interpreted jointly. A P-only warning with G1/G2 and p-bar still green is read as recent weakening within an otherwise still-supported long-run structure. Concurrent G1/G2/P deterioration is more consistent with a structural-deterioration hypothesis and should trigger deeper audit, not an automatic Production change. Weak p-bar together with deterioration increases concern because historical time-consistency is also weaker.

Fast v123 therefore adds a portfolio-specific integrated interpretation in Health Detail. The interpretation is generated from the already-calculated Research snapshot states; it is explanatory UI logic, not a new performance calculation, optimizer, ranking, or allocation rule.

Governance wording

Health is non-ranking / non-selection. It describes current strategy condition and is not itself a Production-change signal.

Health Axis Guide interaction

From v122, each G1 / G2 / P / p-bar guide card can be tapped to reveal its practical operational interpretation. The guide remains compact when closed. This improves interpretability without changing any statistical definition.

v1.42 Addendum - Fast UI lineage v121-v124

v121 - Deterioration table mobile layout repair

Long portfolio names were overlapping the G1/G2/P/p-bar dot columns on iPhone. The portfolio-name column was widened and long names were constrained/wrapped so text cannot invade the Health-axis columns. No data or Health logic changed.

v122 - Guide, Governance, and annotation hierarchy

Health Axis Guide cards became tap-to-expand practical explanations. Governance text layout was repaired to prevent label/body overlap on mobile. The Deterioration Monitor subtitle was demoted visually to a small annotation: "Strategy Health - current deterioration signs monitored on G1/G2/P/p-bar; automatically linked from Research/Python canonical source; Fast does not recalculate Health."

v123 - Integrated portfolio interpretation

Health Detail now summarizes G1, G2, P, and p-bar jointly for the selected portfolio and displays an operational interpretation. The provenance sentence (Python canonical calculation, history length, no Fast recomputation) was reduced to annotation size. This is an interpretation layer only; underlying Health states remain snapshot-driven.

v124 - Dashboard title and data-provenance annotation

The Dashboard header was normalized to the same blue-bar page-title language used elsewhere. The title is now English: "Dashboard". Directly below it, a small annotation displays the data retrieval timestamps for BM and My Portfolio. Timestamps must come from snapshot metadata; Fast must not fabricate missing retrieval times.

Dashboard-only status contract

The top version badge and Live Data Bridge status remain Dashboard-only. They must not be reintroduced globally on other pages.

v1.42 Addendum - Frozen implementation rules carried forward

Source-of-truth hierarchy

1. Research/Python owns calculations and canonical strategy outputs. 2. Snapshot JSON transports frozen/calculated outputs to Fast. 3. Fast renders the snapshot and may provide explanatory UI, but does not recalculate strategy or Health statistics. 4. Missing canonical data remains fail-closed; no browser inference or approximation is allowed.

Version / deployment governance

Every modification increments the Fast version. Fixed filenames remain the deployment standard (index.html, styles.css, app.js, kirin_snapshot.json, README.md where applicable); cache busting is handled by query strings rather than proliferating versioned filenames. Changes should remain minimal patches against the immediately preceding Frozen Baseline. Unrequested UI, behavior, layout, data, or completed research must not be altered.

Research evidence boundary

The v119-v124 work is implementation/UI lineage, not new official performance evidence. No new R-number is assigned here. Any future claim that changes strategy conclusions, promotion status, or performance evidence still requires the established research gate: predeclared scope, user-run Streamlit evidence ZIP, exactness ALL PASS, successful Python compile, and explicit promotion into the Master.

Current state at v1.42

Domain	Current frozen state
Production	Frozen4F EW
KIRIN	Layer-1 challenger; 天界 = Canonical Exact KIRIN; 現世 = Frozen R289 Practical 10% Execution
DEV / Forward	DEV through 2026-07-31; Real Forward OOS from 2026-08-01
Strategy Health	Research/Python canonical calculation -> snapshot -> Fast display-only
Deterioration Monitor	All eligible Performance portfolios; <96M fail-closed; tap guide + integrated interpretation
Fast latest recorded	v124
Dashboard	Blue-bar English title + BM / My Portfolio retrieval-time annotation

End of v1.42 addendum. Parent v1.41 pages remain unchanged.